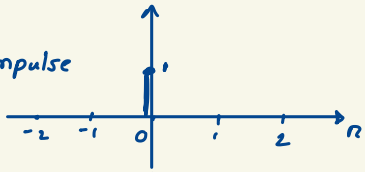
# PPV - 2/3

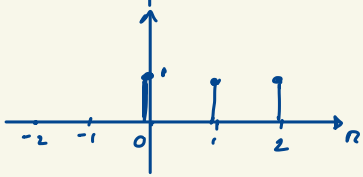
# Review of DSP

## Signals — Discrete-time / Digital Signals

Ex:- $\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$ - unit delta / Impulse

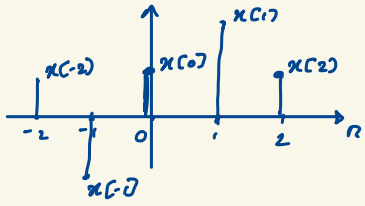



Ex:- $s[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$ - Unit step




- Any signal can be formerly (Since summation is $\infty$) expressed as a combination of $\delta[n]$ and its integer shifts.

$$x[n] = \sum_{m=-\infty}^{\infty} x[m] \cdot \delta[n-m]$$




- $u[n] = \sum_{m=-\infty}^{\infty} u[m] \cdot \delta[n-m] = \sum_{m=0}^{\infty} \delta[n-m]$

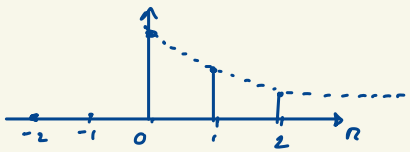
Ex:- Sinusoidal and Complex exponentials

$x_1[n] = A \cdot \cos(\omega_0 n + \theta)$ - real valued

$x_2[n] = \alpha \cdot e^{j\omega_0 n}$ - complex valued

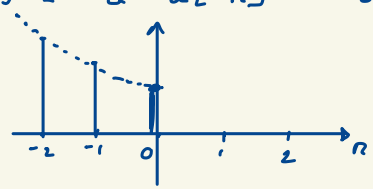
Ex: Exponentials

$x_1[n] = a^n \cdot u[n] \qquad 0 < |a| < 1$



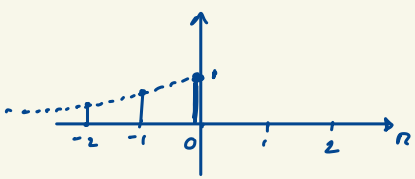

right sided signal.

$x_2[n] = a^n \cdot u[-n] \qquad 0 < |a| < 1$




left sided signal

$x_3[n] = a^{-n} \cdot u[-n] \qquad 0 < |a| < 1$




left sided signal

## The z Tranform

- The z transform of $x[n]$ is formerly (summation is infinite) defined as,

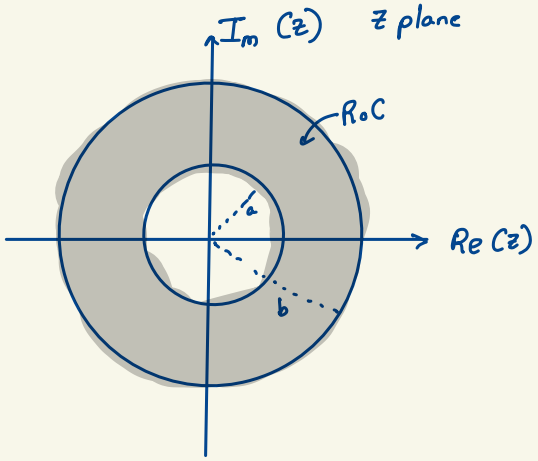
$$x[z] = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n} \qquad z \in \mathbb{C}$$

associated with a region of convergence (RoC) of form $a < |z| < b$ for some $a, b \in \mathbb{R}$

- RoC is defined as the set of all z for which

$$\sum_{n=-\infty}^{\infty} |x[n] \cdot z^{-n}| < \infty$$

absolute summability

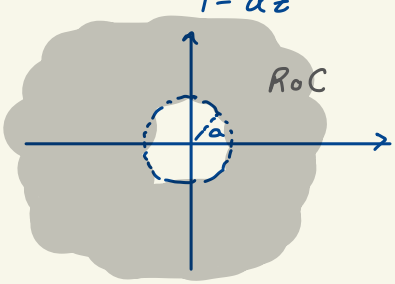



- Some signals are not "absolute summable" but infinite summation "formally" exists.

Ex: $e^{j\omega_0 n} \xleftrightarrow{DTFT} 2\pi \sin(\omega - \omega_0)$

Ex: $x[n] = a^n \cdot u[n] \quad a > 0$

$$X(z) = \frac{1}{1 - az^{-1}}, \quad RoC \quad |z| > a$$




Ex: Find the signal $x[n]$ where $X(z) = \frac{1}{1 - az^{-1}}$ but RoC is $|z| < a$

$$x[n] = a^n \cdot u[-n-1]$$

## Birthday Problem

Claim: with probability $> 80\%$ at least 2 of us have the same birthday

$A^c$ = all have different birthdays

$$P(A^c) = \frac{365 \times 364 \times \ldots \times 332}{365 \times 365 \times \ldots \times 365}$$

$$= \left(1 - \frac{1}{365}\right)\left(1 - \frac{2}{365}\right) \ldots \left(1 - \frac{33}{365}\right)$$

$$\leq e^{-0/365 - 1/362 - 2/365 \ldots - 33/365} \quad \ldots \; (1 - x \leq e^{-x})$$

$$= e^{\frac{-33 \times 34}{2 \times 365}}$$

$$=$$

## The inverse of Z transform

$$x[n] = \frac{1}{2\pi j} \oint_C X(z) \cdot z^{n-1} dz$$

- C: a counter clockwise closed path that includes the origin and is strictly in the RoC.

## The Discrete Time Fourier Transform (DTFT)

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\omega n} \qquad \forall \omega \in \mathbb{R}$$

$$= X(z)\Big|_{z=e^{j\omega}}$$

- DTFT Exists if the RoC of $X(z)$ includes the unit circle ($|z| = 1$)
- Same as $X(e^{j\omega})$
- DTFT is the Z transform evaluated at unit circle. But inverse of DTFT is much easier.

## Inverse DTFT

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(\omega) \cdot e^{j\omega n} d\omega$$

Much easier to compute than inverse Z transform

Scalar integral.

Ex: What is the DTFT of $x[n] = e^{j\omega_0 n}$ $\qquad X(\omega) = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\omega n}$

$$X(\omega) = 2\pi \, \delta(\omega - \omega_0)$$

Proof: Assume $X(\omega) = \delta(\omega - \omega_0)$ we are trying to find $x[n]$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(\omega) \cdot e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{2\pi} \delta(\omega - \omega_0) \, e^{j\omega n} d\omega \quad \cdots$$

$$= \frac{1}{2\pi} e^{j\omega_0 n}$$

Dirac Delta Rule

$$\int_{-\infty}^{\infty} \delta(\omega - a) \cdot f(\omega) \, d\omega = f(a)$$

This can be seen by noting that the delta function is zero everywhere except at $x = a$. Therefore, the integrand is zero everywhere and the only contribution from $f(x)$ will be from $x = a$. So, we can replace $f(x)$ with $f(a)$ under the integral. Since $f(a)$ is a constant, we have that

$$\int_{-\infty}^{\infty} \delta(x-a)f(x)dx = \int_{-\infty}^{\infty} \delta(x-a)f(a)dx$$
$$= f(a)\int_{-\infty}^{\infty} \delta(x-a)dx = f(a). \quad (9.4.2)$$

## Preservation of Inner Product

$$\langle x[n], y[n] \rangle := \sum_{n=-\infty}^{\infty} x[n] \cdot y^*[n]$$

DTFT ↕ DTFT

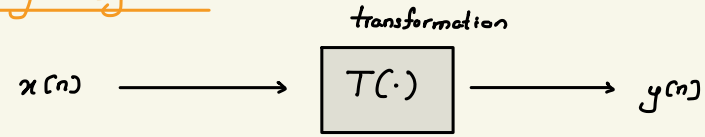
$$\langle X(\omega), Y(\omega) \rangle := \frac{1}{2\pi} \int_0^{2\pi} X(\omega) \cdot Y^*(\omega)\, d\omega$$

- Special case when $x[n] = y[n]$

$$\Rightarrow \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_0^{2\pi} |X(\omega)|^2 d\omega$$ Parseval's Identity

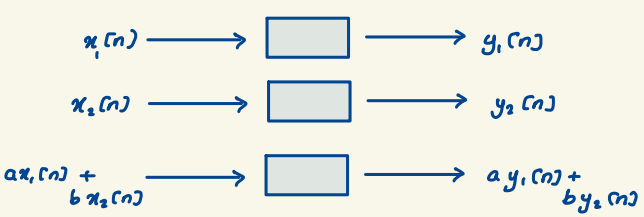
- Hence energy of a signal can be computed in both time and fourier domain.

## Characterising Systems



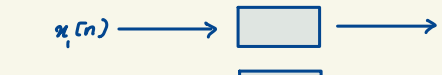

In general a system is characterised by specifying $y[n]$ for any $x[n]$.

### Assumption 1 - Linearity



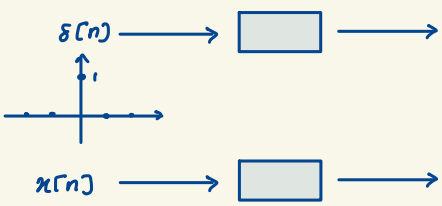

$\forall\ a_1, a_2 \in \mathbb{C}$
$\forall\ x_1[n], x_2[n]$

## Assumption 2 : Time - Invariant




$\forall N \in \mathbb{N}$
$\forall\ x_1[n]$.

## Linear Time Invariant Systems (LTI)




$$= \sum_{m=-\infty}^{\infty} x[n] \cdot h[n-m]$$

$$= x[n] * h[n]$$

- A LTI system can be fully characterised by its impulse response. $h[n]$

Proof : $x[n] = \sum_{m=-\infty}^{\infty} x[m] \cdot \delta[n-m]$

$$\Rightarrow y[n] = \sum_{m=-\infty}^{\infty} x[m] \cdot T(\delta[n-m]) \quad \text{- from linearity}$$

$$\Rightarrow y[n] = \sum_{m=-\infty}^{\infty} x[m] \cdot h[n-m] \quad \text{- from time invariance}$$

$\because T(\delta[n]) = h[n]$

## Transfer Function

In z transform domain

$$Y(z) = X(z) \cdot H(z) \qquad \sum_{n=-\infty}^{\infty} h[n] \cdot z^{-n}$$

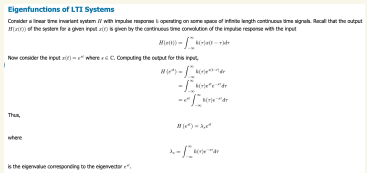
$$H(z) = \frac{Y(z)}{X(z)} \quad . \quad \text{Transfer function of the system.}$$

# Rational Transfer Functions

$$H(z) = \frac{\sum_{k=0}^{N} b_k \cdot z^{-k}}{\sum_{m=0}^{N} a_m \cdot z^{-m}}$$

a rational TF of order N if the function is irreducible.

Some properties hold for DTFT.



**Eigenfunctions of LTI Systems**

Consider a linear time invariant system $H$ with impulse response $h$ operating on some space of infinite length continuous time signals. Recall that the output $H(x(t))$ of the system for a given input $x(t)$ is given by the continuous time convolution of the impulse response with the input

$$H(x(t)) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

Now consider the input $x(t) = e^{st}$ where $s \in \mathbb{C}$. Computing the output for this input,

$$\begin{aligned} H(e^{st}) &= \int_{-\infty}^{\infty} h(\tau)e^{s(t-\tau)}d\tau \\ &= \int_{-\infty}^{\infty} h(\tau)e^{st}e^{-s\tau}d\tau \\ &= e^{st}\int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau \end{aligned}$$

Thus,

$$H(e^{st}) = \lambda_s e^{st}$$

where

$$\lambda_s = \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau$$

is the eigenvalue corresponding to the eigenvector $e^{st}$.

- Why are complex exponentials $e^{j\omega_0 n}$ important?

They are eigen functions of LTI systems.

$x[n] = e^{j\omega_0 n}$ → [ ] → $y[n] = \underbrace{H(\omega_0)}_{\in \mathbb{C}} \cdot e^{j\omega_0 n}$

When $e^{j\omega n}$ is given to a LTI system, the output is just scaled by $H(\omega_0)$.

- An LTI system is Causal if $h[n] = 0 \quad \forall n < 0$

$y[n] = \sum_{m=-\infty}^{\infty} x[m] \cdot h[n-m] \quad \forall n.$

For causality $y[n]$ should not depend on any input $x$ after $n$. $\therefore \forall n > m \quad h[n-m] = 0$

$\Leftrightarrow h[n-m] = 0 \quad \forall n > m \quad \Leftrightarrow h[n] = 0 \quad \forall n < 0$

- If system is LTI, what properly should $H(z)$ for causality?

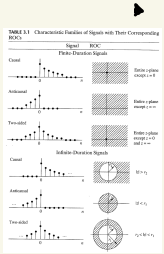


TABLE 3.1 Characteristic Families of Signals with Their Corresponding ROCs

| Signal | ROC |
|---|---|
| Finite-Duration Signals | |
| Causal | Entire z-plane except z = 0 |
| Anticausal | Entire z-plane except z = ∞ |
| Two-sided | Entire z-plane except z = 0 and z = ∞ |
| Infinite-Duration Signals | |
| Causal | $\lvert z \rvert > r_1$ |
| Anticausal | $\lvert z \rvert < r_2$ |
| Two-sided | $r_1 < \lvert z \rvert < r_2$ |

The ROC: $a < |z| < b$ if $b = \infty$ then the system is Causal.

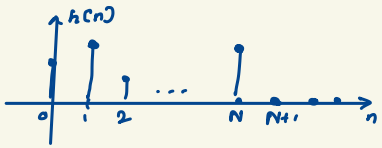
That means ROC should start from some value an go to $\infty$.

- An LTI system is called Finite Impulse Response (FIR) of order N, if

$$H(z) = \sum_{n=n_0}^{n_0+N} b_n \cdot z^{-n}$$

- A causal FIR system of order $N$, $(n_0 = 0)$

$$H(z) = \sum_{n=0}^{N} b_n \cdot z^{-n}$$




- An LTI system is called Infinite Impulse Response (IIR) if it is not FIR.

## Bounded Input Bounded Output (BIBO) Stability

- An LTI system is BIBO if for any $x[n]$ such that $x[n] < \infty \ \forall n$ we also have $y[n] < \infty \ \forall n$

- An LTI system is BIBO stable if

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

**Proof:** If $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$

Consider a bounded input $x[n] < B$.

$$\Rightarrow |y[n]| = \left| \sum_{m=-\infty}^{\infty} h[m]\, x[n-m] \right|$$

$$\Rightarrow < \sum_{m=-\infty}^{\infty} |h[m]| \cdot |x[n-m]| \quad \text{.... } x[n-m] \text{ is bounded } \because < B < \infty$$

$$\Rightarrow < B \cdot \sum_{m=-\infty}^{\infty} |h[m]| \quad \text{..... } \sum_{m=-\infty}^{\infty} |h[m]| < \infty$$

$$\Rightarrow < B \cdot \infty = \infty$$

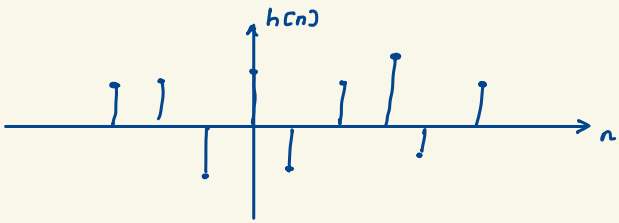
$\therefore$ for a bounded input, output is also bounded $\Rightarrow$ BIBO

## If the System is BIBO

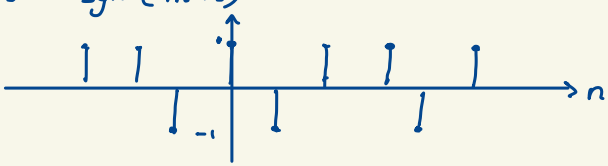
- To prove : If BIBO $\Longrightarrow \sum_{n=-\infty}^{\infty} |h[n]| < \infty$
- The counter positive : If $\sum_{n=-\infty}^{\infty} |h[n]| = \infty \Longrightarrow$ System is not BIBO

We are going to give a counter example where $h[n]$ summation is infinite but is BIBO unstable

Consider $h[n]$




Define input $x[n] := \text{Sgn}(h[-n])$




$|x[n]| = 1$

$$y[n=0] = \sum_{m=-\infty}^{\infty} x[m] \cdot h[0-m] = \sum_{m=-\infty}^{\infty} \text{Sgn}(h[-m]) \cdot h[-m]$$

$$= \sum_{m=-\infty}^{\infty} |h[-m]|$$

$$= \sum_{m=-\infty}^{\infty} |h[m]|$$

$$= \infty$$

■

- BIBO stability in $z$ Domain.

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

$$= \sum_{n=-\infty}^{\infty} |h[n] \cdot e^{-j\omega n}| < \infty \Longrightarrow e^{j\omega} \ \forall \ \omega \text{ is in ROC}$$

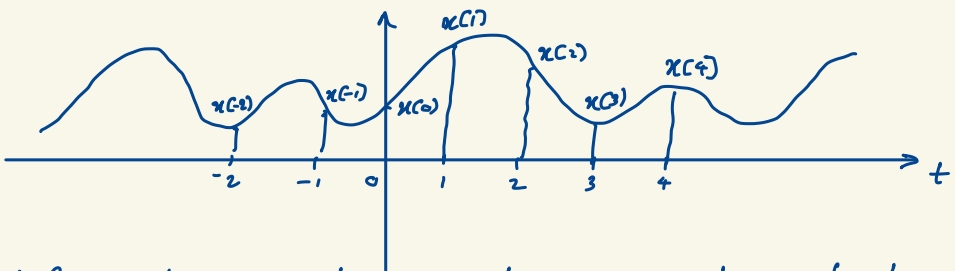
$\Longrightarrow$ ROC must contain unit circle.

## Continuos Time FT

$$X(\Omega) := \int_{-\infty}^{\infty} x(t) \cdot e^{-j\Omega t}\, dt \qquad \forall \Omega \in \mathbb{R}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) \cdot e^{j\Omega t}\, d\Omega$$

$x(t)$ is a continuos time. $\Omega = 2\pi f$ = angular frequency.




∴ By sampling a continuos signal we can get a discrete signal.

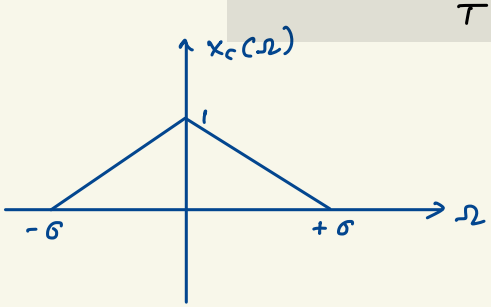
$$x[n] = x(nT) \qquad n \in \mathbb{Z}$$

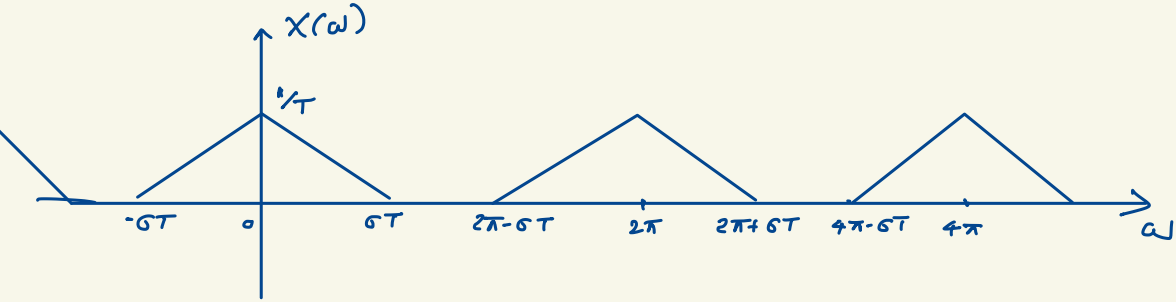
$$x(t) \xleftrightarrow{FT} X_c(\Omega)$$

$$x[n] \xleftrightarrow{DTFT} X(\omega)$$

- How is $X_c(\Omega)$ and $x(\omega)$ are related.

$$X(\omega) = \frac{1}{T} \sum_{K=-\infty}^{\infty} X_c\left(\frac{\omega - 2\pi k}{T}\right)$$






- $\Omega_s = \frac{2\pi}{T}$ sampling frequency

- If $\Omega_s \geq 2\sigma$, then there is no overlap between the copies (aliases) of the FT.

## Shanon - Nyquist Theorem

$$\text{If no aliasing} \Rightarrow \Omega_s \geq 2\sigma$$

- The time domain signal $x(t)$ can be fully recovered from its discrete samples $x[n] = x(nT)$ $n \in \mathbb{Z}$.

# Building Blocks of Multirate SP

## Shanon - Nyquist Theorem

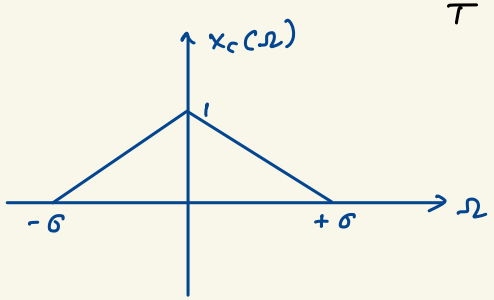
- If no aliasing $\Rightarrow \Omega_s \geq 2\sigma$
  The time domain signal $x(t)$ can be fully recovered from its discrete samples $x[n] = x(nT)$ $n \in \mathbb{Z}$.

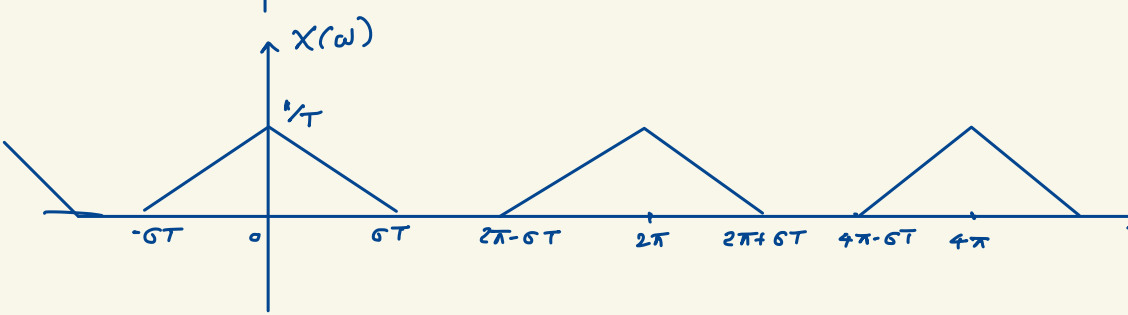
- If $x(t)$ has a FT $X_c(\Omega)$ supported in $[-\sigma, \sigma]$ for $\sigma > 0$, then if $\Omega_s = 2\pi/T \geq 2\sigma$, then $x(t)$ can be reconstructed from its discrete samples $x[n] = x(nT)$ $n \in \mathbb{Z}$

$$x(t) \xleftrightarrow{FT} X_c(\Omega)$$

$$x[n] \xleftrightarrow{DTFT} X(\omega)$$

$$X(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(\frac{\omega - 2\pi k}{T}\right)$$

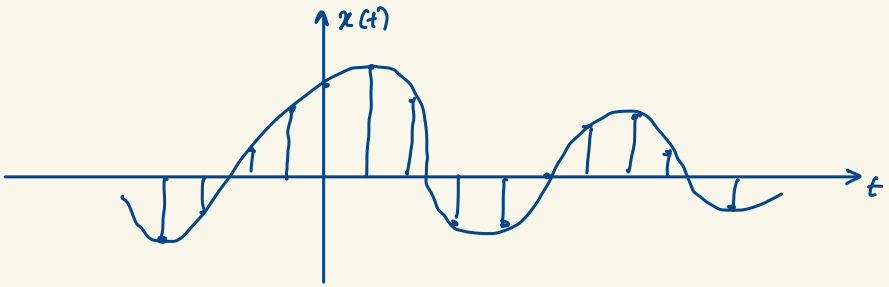


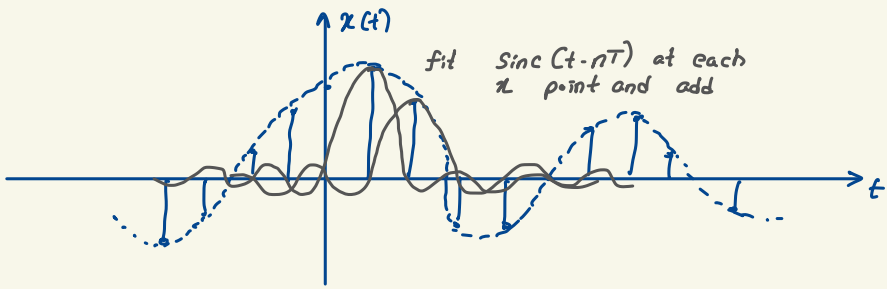



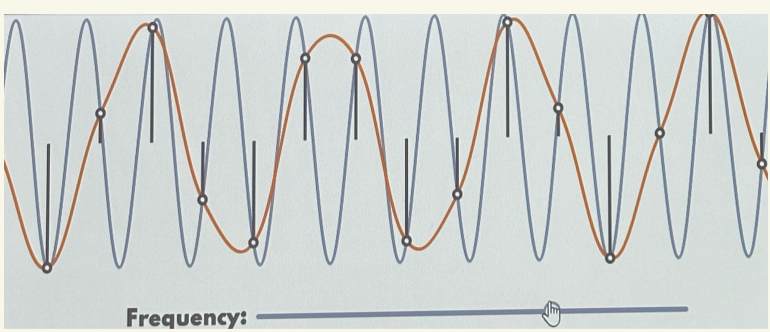
Reconstruct $x(t)$

$\text{Sinc}(t-nT)$

$$x(t) = \sum_{n=-\infty}^{\infty} x[n] \cdot \frac{\sin(\pi(t-nT))}{\pi(t-nT)}$$



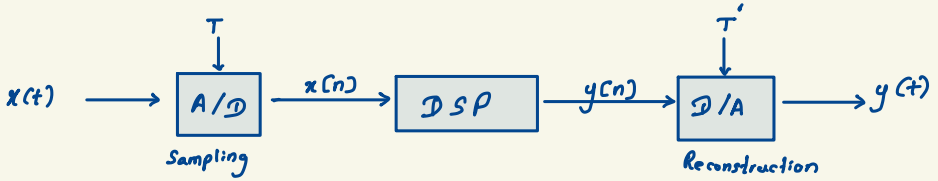




- When sooming out of a detailed image, due to compression the aliasing happen. (Copies of the same base region in high frequency areas.)

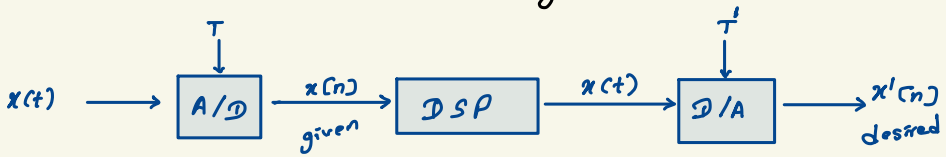
# PPV-4

# Single - Rate vs Multi - Rate DSP




- The sampling rate $2\pi/T$ is the same along the entire pipeline.
- Why we need multirate DSP
  - resizing images.
  - rate conversion from $T$ to $T'$
- The use of samples $x[n]$ of $x(t)$ acquired at $T$ intervals to samples $x'[n]$ acquired at $T'$ intervals.

Ex: Assume there is a CD at rate 44.1 kHz and Studio tapes at 48 kHz. How can we merge them.




going back to analog domain to achieve this and comming back to digital is very lossy. Hence use multirate DSP.

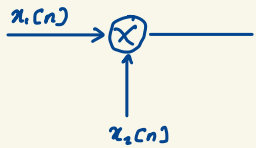
By adding DSP pipelines with the different rate of inter connect, we can,

1) Substatially reduce computational complexity.
2) Perform all digital SP without the need of D/A
3) Sample the signals belove Niquist rate and still retain all information.

# Building Blocks of Classic SP (Single-rate)

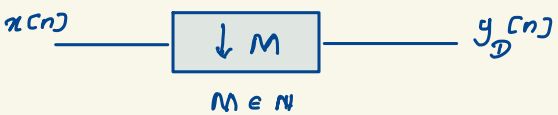
$x[n] \xrightarrow{a} y[n] = a\,x[n]$ scaling by $a$

$x[n] \xrightarrow{z^{-1}} y[n] = x[n-1]$ unit delay




New Building Blocks

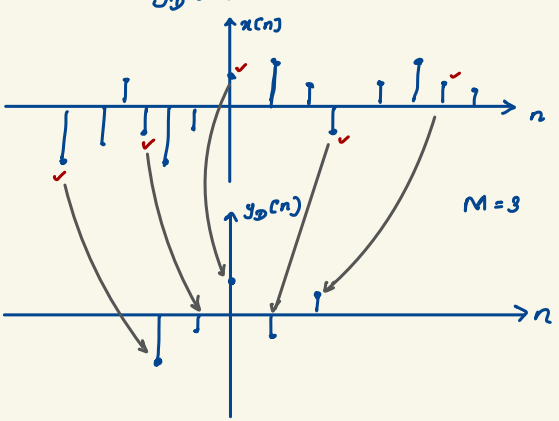
## Decimation



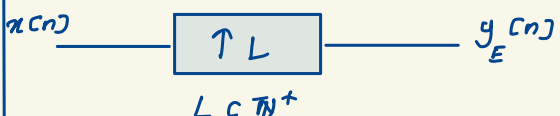

$M \in \mathbb{N}$

Decimation by a factor of $M$

$y_D[n] = x[Mn]$



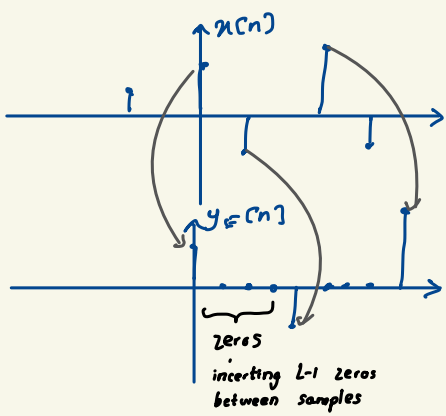

## Expantion




$L \subset \mathbb{N}^+$

Expansion by a factor of $L$

$$y_E[n] = \begin{cases} x\left[\frac{n}{L}\right] & \text{if } L \mid n \cdots (*) \\ 0 & \text{otherwise.} \end{cases}$$



# Expansion in the Z Domain

$$x[n] \xleftrightarrow{Z} X(z)$$

$$\text{expansion} \quad \Big\updownarrow \quad ?$$

$$y_E[n] \longleftrightarrow Y_E(z)$$

$$Y_E(z) = \sum_{n=-\infty}^{\infty} y_E[n] \cdot z^{-n}$$

$y_E[n]$ is non zero only when $L/n \Rightarrow n = kL \quad k \in N$

$$= \sum_{k=-\infty}^{\infty} y_E[kL] \cdot z^{-kL}$$

$$= \sum_{k=-\infty}^{\infty} x(k) \cdot (z^L)^{-k} \quad \cdots \because (*)$$

$$= X(z^L)$$

$$Y_E(z) = X(z^L) = X(L\omega)$$

- No scaling. Magnitude is same
- Frequency domain shrinks

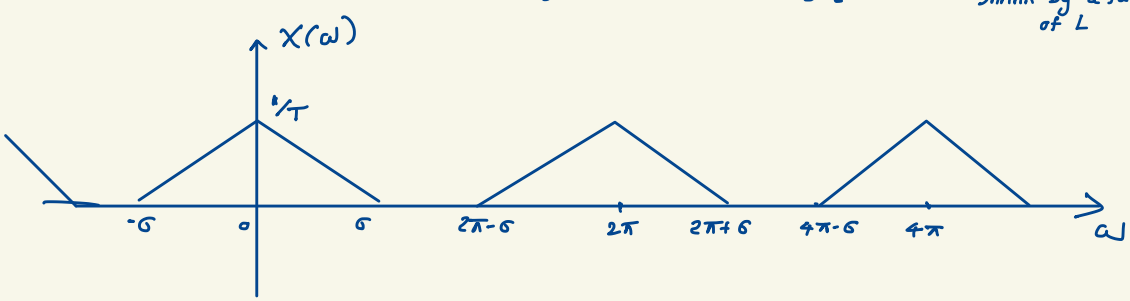
Ex: $X(z) = \dfrac{1 - 2z^{-1}}{1 + 2 \cdot z^{-1} - 3z^{-2}}$ and $L = 4$ then

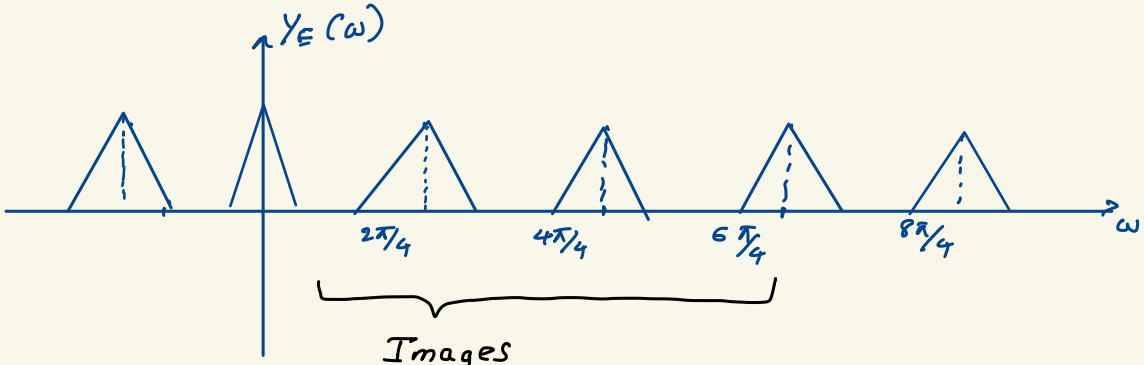
$$Y_E(z) = \frac{1 - 2z^{-4}}{1 + 2 \cdot z^{-4} - 3 \cdot z^{-8}} \; //$$

- Expansion in DTFT domain.

$$Y_E(\omega) = Y_E(z)\Big|_{z=e^{j\omega}} = X(z^L)\Big|_{z=e^{j\omega}} = X(\omega L)$$

Shrink by a factor of L






Expansion creates the imaging effect.

- All the original info is preserved. But images happen in between the previous shapes. These 'images' are the results of 'zeros' we added.

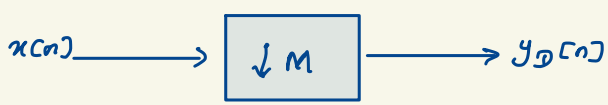
## Decimation in Z-Domain

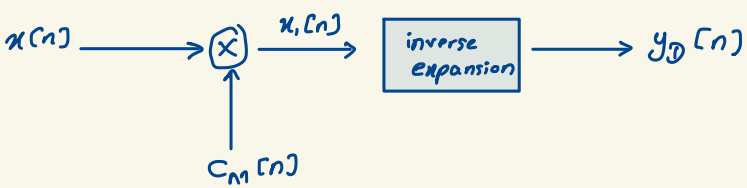
$$x[n] \xleftrightarrow{z} X(z)$$

Decimation by M: $x[n] \rightarrow y_D[n]$, $X(z) \rightarrow ?$

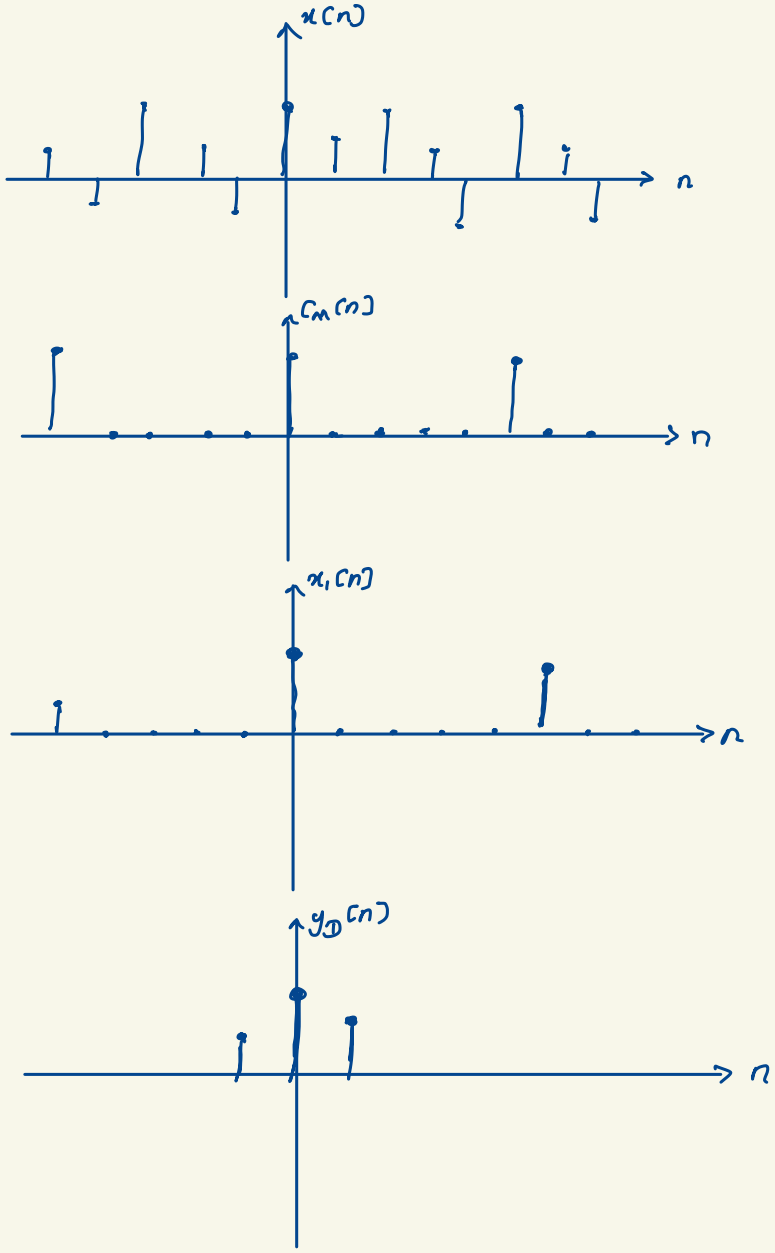
$$y_D[n] \xleftrightarrow{z} Y_D(z)$$

$$y_D[n] = x[Mn]$$




$$C_M[n] = \text{comb function} = \begin{cases} 1 & n \text{ is an integer multiple of } M \\ 0 & \text{otherwise.} \end{cases}$$




$Y_D(z) = X_1(z^{1/M})$ from expansion logic.

$$Y_D(z) = \sum_{n=-\infty}^{\infty} y_D(n) \cdot z^{-n} = \sum_{k=-\infty}^{\infty} x_1[kM] \cdot z^{-k}$$

($n$)

$$= \sum_{k=-\infty}^{\infty} x_1(n) \cdot z^{-k/M} = X_1(z^{1/M})$$

①

**Claim:** $C_m[n] = \frac{1}{M} \sum_{k=0}^{M-1} e^{i 2\pi k n / M} \quad \forall\, n \in \mathbb{Z}$

$$\begin{cases} n = lM \implies C_m[lM] = \frac{1}{M} \sum_{k=0}^{M-1} \underbrace{e^{\frac{i 2\pi k\, lM}{M}}}_{1} = \frac{1}{M} \cdot M = 1 \\ n \neq lM \implies \frac{1}{M}\left(a^0 + a^1 + a^2 + \dots + a^{M-1}\right) = \frac{1}{M} \frac{1-a^M}{1-a} = \frac{0}{\neq 0} = 0 \end{cases}$$

$$X_1(z) = \sum_{n=-\infty}^{\infty} x_1[n] \cdot z^{-n} = \sum_{n=-\infty}^{\infty} x[n] \cdot C_m[n] \cdot z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x[n] \cdot \left( \frac{1}{M} \sum_{k=0}^{M-1} e^{-i \frac{2\pi k n}{M}} \right) \cdot z^{-n}$$

$$= \frac{1}{M} \cdot \sum_{k=0}^{M-1} \sum_{n=-\infty}^{\infty} x[n] \left( e^{-i \frac{2\pi k}{M}} \right)^n \cdot z^{-n}$$

$$= \frac{1}{M} \sum_{k=0}^{M-1} \Bigg( \underbrace{\sum_{n=-\infty}^{\infty} x[n] \cdot \left( e^{-i \frac{2\pi k}{M}} \cdot z \right)^{-n}}_{X\left(e^{-i \frac{2\pi k}{M}} \cdot z\right)} \Bigg)$$

$w_M = e^{-i 2\pi / M}$ ..... $= \frac{1}{M} \sum_{k=0}^{M-1} X\left(w_m^k \cdot z\right)$ ...②

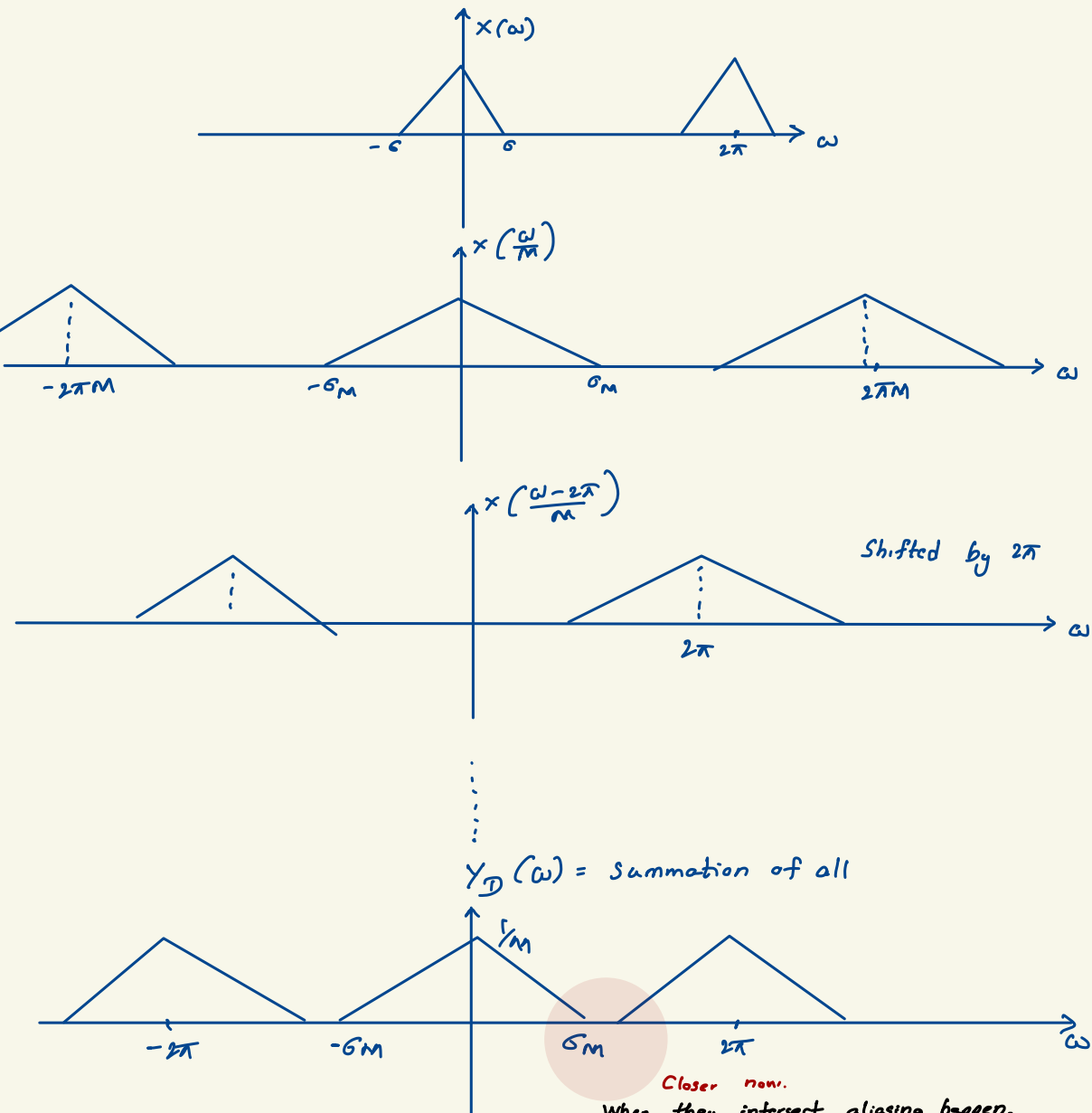
①, ②

$$Y_D(z) = X_1\left(z^{1/m}\right) = \frac{1}{M} \sum_{k=0}^{M-1} X\left(w_m^k \cdot z^{1/m}\right)$$

- Decimation in DTFT Domain

$$Y_D(\omega) = \frac{1}{M} \sum_{K=0}^{M-1} X\left(\frac{\omega - 2\pi k}{M}\right) = \frac{1}{M} \sum_{k=0}^{M-1} X\left(\frac{\omega - 2\pi k}{M}\right)$$

all copies are aliases

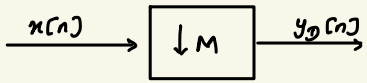


To avoid aliasing due to decimation $\Rightarrow$ $\sigma M \leq 2\pi - \sigma M$

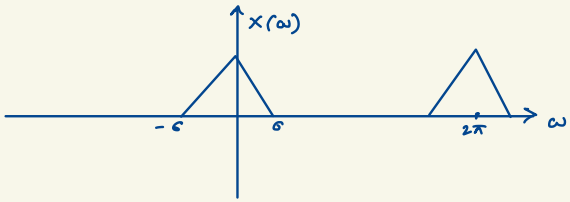
$\Rightarrow$ $\sigma \leq \frac{\pi}{M}$

# Multirate Building Blocks

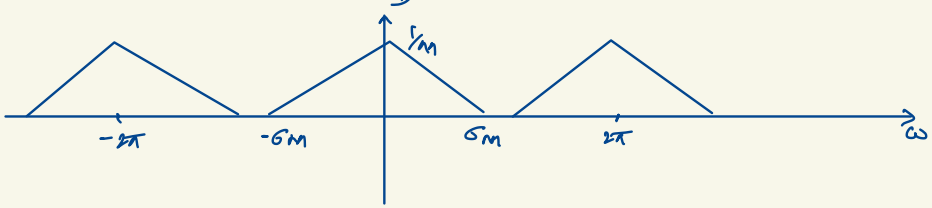
Recovering $x[n]$ from $y_D[n]$

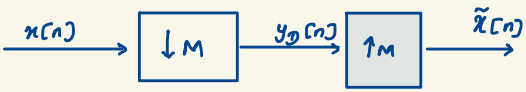



Assumption : $\sigma < \frac{\pi}{M}$



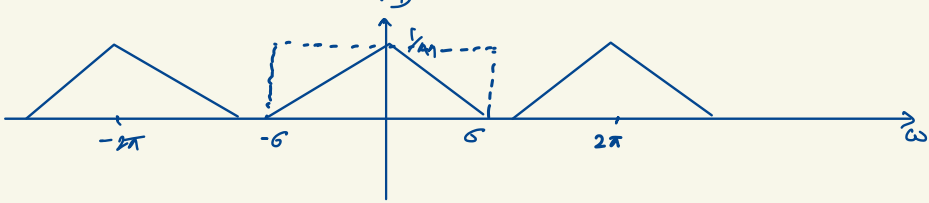

$Y_D(\omega) =$



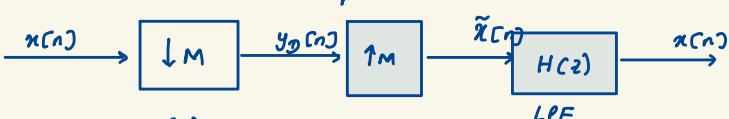



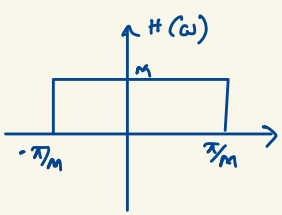

first do expansion.

$Y_D(\omega) =$

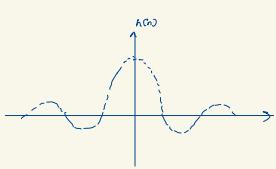



Now we need a lowpass filter to avoid the images.

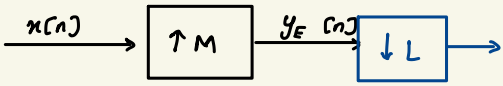



$$h[n] = \frac{\omega_1}{\pi} \cdot \frac{\text{sinc}(\omega_1 n)}{\omega_1 n}$$

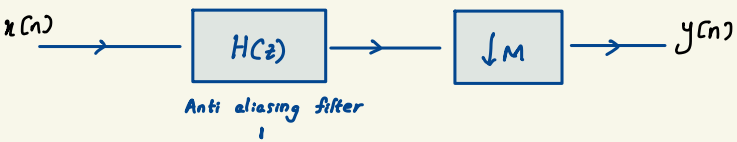


Recovering $x[n]$ from $y_E[n]$

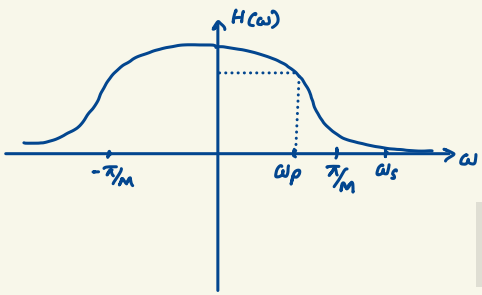



Just use a decimation filling gaps with zeros.

## Decimation Filter



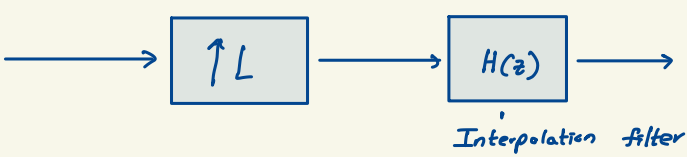

This does a low pass filter, so the image will be blurred when zoomed in, but no aliasing when zoomed out. ∴ Avoid future aliasing.



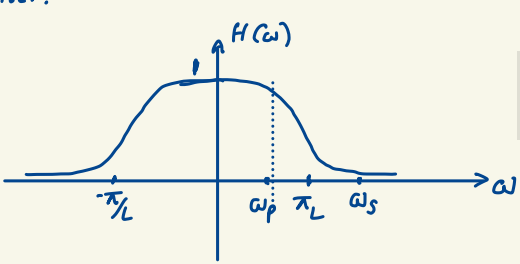

- LPF - blurrs the signal.

$$y(n) = \sum_{k=-\infty}^{\infty} x[k] \cdot h[nM-k]$$

## Interpolation Filter

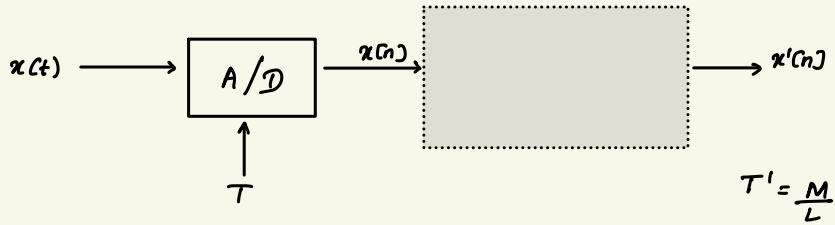



This shows after expansion and avoids imaging. ∴ Anti-imaging filter.

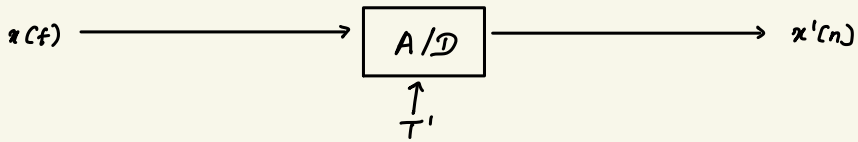



$$y(n) = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-kL]$$

# Fractional Rate Conversion



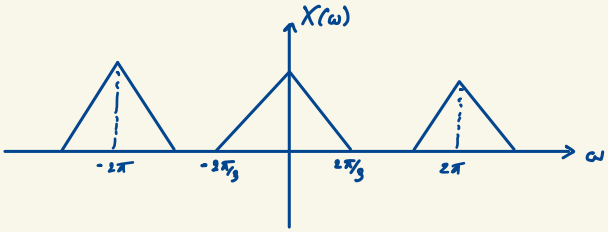

$$T' = \frac{M}{L} T \qquad M, L \in \mathbb{N}$$

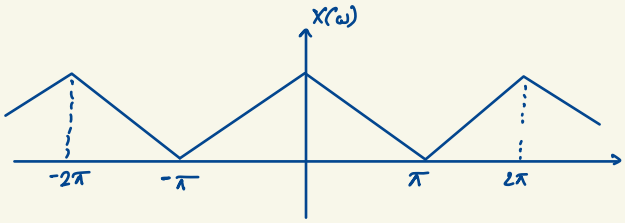


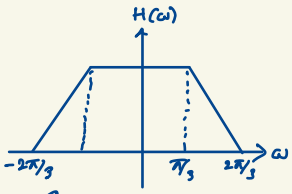

What can we put in the blue box so that boath systems acts identical.

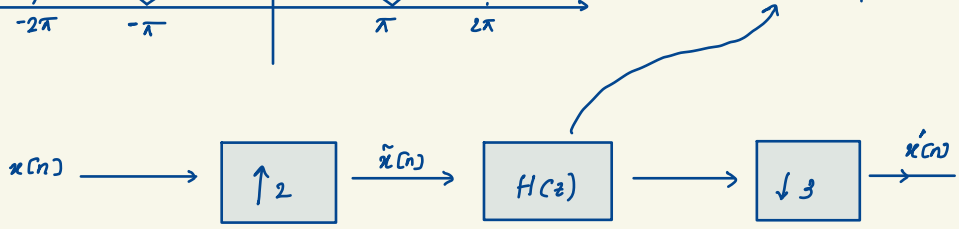
Ex: $M = 3$, $L = 2$. Similar to trying to change the frequency (sampling rate) of an audio in a CD in the digital domain itself.

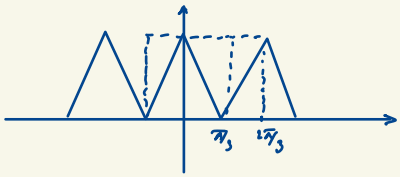
$\sigma = 2\pi/3$



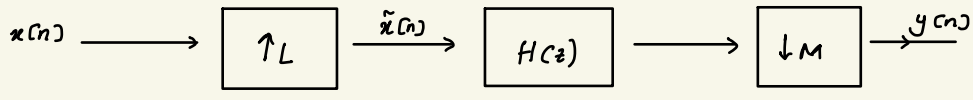



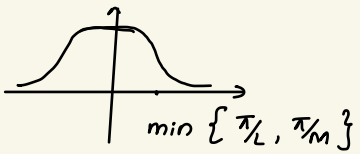






- In general, fractional rate conversion filter is



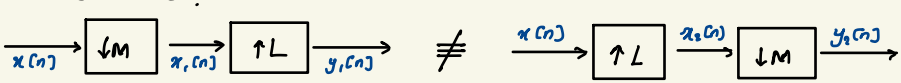




$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[Mn - kL]$$

- Note - Decimation and Expansion are not Time Invariant. But they are Linear

## Interchanging Decimation and Expansion

- In general ↑L and ↓M cannot be interchanged. When can we do this?



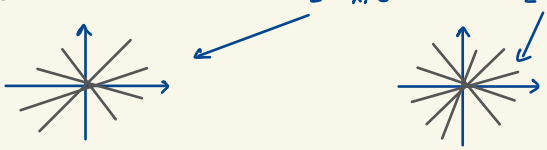

Only when $\gcd(M, L) = 1$

Proof: Done for homework. Use the hint

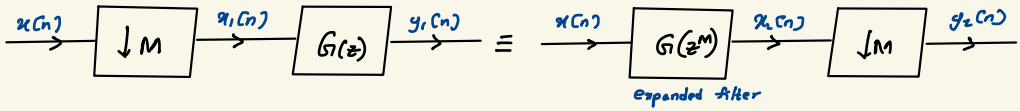
$$Y_1(z) = \frac{1}{M} \sum_{k=0}^{M-1} X\left(W_M^k \cdot z^{\frac{1}{M}}\right)$$

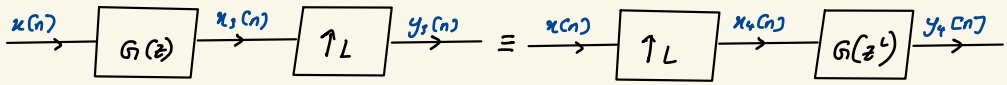
$$Y_2(z) = \frac{1}{M} \sum_{k=0}^{M-1} X\left(W_M^{kL} \cdot z^{\frac{1}{M}}\right)$$

The way to show this is set $\{W_M^k\}$ and $\{W_M^{kL}\}$ should be same.



# The Noble Identities of Multirate DSP

1) 


2) 


Proof: $Y_2(z) = \frac{1}{M} \sum_{K=0}^{M-1} X_2\left(W_M^K z^{1/M}\right)$

$= \frac{1}{M} \sum_{K=0}^{M-1} \left(X(z) \cdot G(z^M)\right)\Big|_{z \leftarrow W_M^K \cdot z^{1/M}}$

$= \left[\frac{1}{M} \sum_{K=0}^{M-1} X\left(W_M^K \cdot z^{1/M}\right)\right] \cdot G\left(\underbrace{W_M^{KM}}_{1} \cdot z\right)$

$X_1(z)$ $\qquad$ $G(z)$

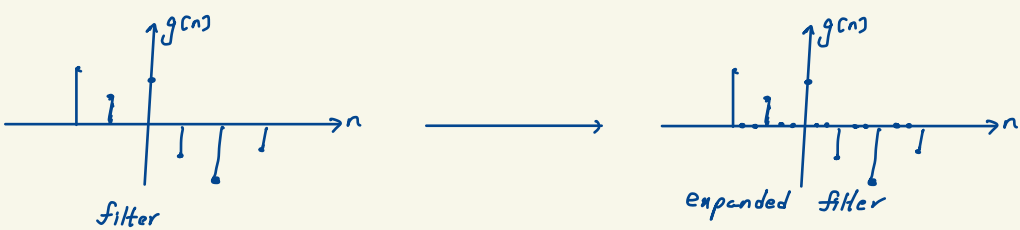
$= Y_1(z)$

$Y_4(z) = X_4(z) \cdot G(z^L) = X(z^L) \cdot G(z^L)$

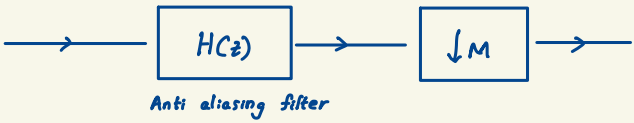
$= \underbrace{X(z) \cdot G(z)}_{X_3(z)}\Big|_{z \leftarrow z^L}$

$= X_3(z^L)$

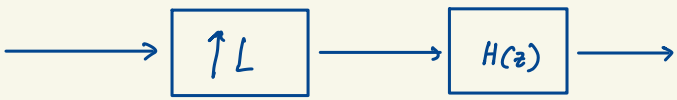
$= Y_3(z)$



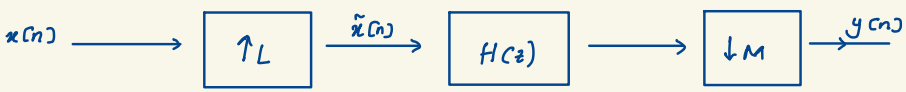
## Decimation Filter




## Interpolation Filter



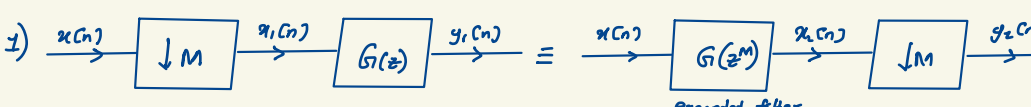

## Fractional Rate Conversion

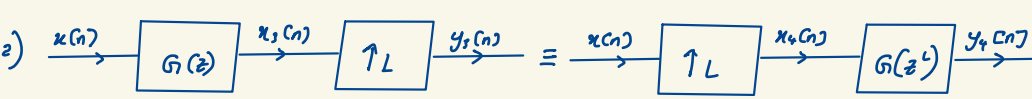



- Decimation and Expansion are Linear, but not TI
-


  Only when $\gcd(M,L) = 1$

## Noble Identities

1) 

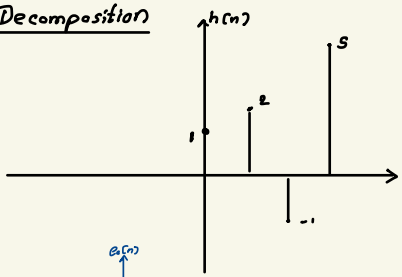

2) 

# Polyphase Decomposition

## Multivariate Example for Polyphase Decomposition

$$H(z) = 1 + 2\cdot z^{-1} - z^{-2} + 5z^{-3}$$

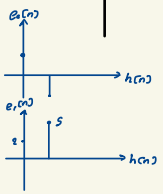
$$H(z) = \underbrace{(1 - z^{-2})}_{\text{even powers}} + z^{-1}\underbrace{(2 + 5z^{-2})}_{\text{even powers}}$$




Assume -

$E_0(z) = 1 - z^{-1}$ These are simple filters

$E_1(z) = 2 + 5z^{-2}$




In general for any $H(z)$, we can write,

$$H(z) = \sum_{n=-\infty}^{\infty} h[2n]\cdot z^{-2n} + z^{-1}\sum_{n=\infty}^{\infty} h[2n+1]\cdot z^{-2n}$$

$$\Rightarrow E_0(z) = \sum_{n=-\infty}^{\infty} h[2n]\cdot z^{-n} \qquad E_1(z) = \sum_{n=\infty}^{\infty} h[2n+1]\cdot z^{-n}$$

$$\Rightarrow \boxed{H(z) = E_0(z^2) + z^{-1}E_1(z^2)}$$

## Polyphase Decomposition for IIR Systems

$$h[n] = \alpha^n u[n] \qquad |\alpha| < 1$$

$$\Rightarrow H(z) = \frac{1}{1-\alpha z^{-1}} = 1 + \alpha z^{-1} + \alpha^2 z^{-2} + \alpha^3 z^{-3} + \dots$$

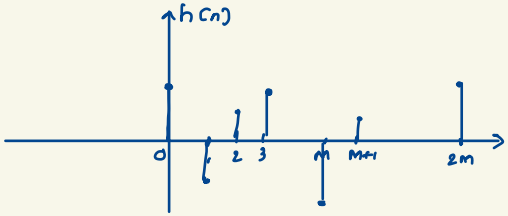
$$= \underbrace{(1 + \alpha^2 z^{-2} + \alpha^4 z^{-4} + \dots)}_{E_0(z) = \frac{1}{1-\alpha^2 z^{-2}}} + \alpha z^{-1}\underbrace{(\alpha^2 z^{-2} + \alpha^4 z^{-4} + \dots)}_{E_1(z) = \frac{\alpha}{1-\alpha^2 z^{-2}}}$$

$$\Rightarrow H(z) = E_0(z^2) + z^{-1}E_1(z^2)$$

$$\Rightarrow \frac{1}{1-\alpha z^{-1}} = \frac{1}{1-\alpha^2 z^{-2}} + z^{-1}\frac{\alpha}{1-\alpha^2 z^{-2}} \qquad 1 + u + u^2 + \dots = \frac{1}{1-u}$$

$$= \frac{1 + \alpha\cdot z^{-1}}{(1-\alpha^2 z^{-2})} = \frac{1}{1-\alpha z^{-1}}$$ which turned to be true.

## Polyphase Decomposition for M Components




Now points are divided to M different components.

$$H(z) = \sum_{n=-\infty}^{\infty} h[Mn]\cdot z^{-nM} + z^{-1}\sum_{n=-\infty}^{\infty} h[Mn+1]\cdot z^{-nM} + \dots$$

$$\dots + z^{-(M-1)\cdot M}\sum_{n=-\infty}^{\infty} h[nM+M]\cdot z^{-nM}$$

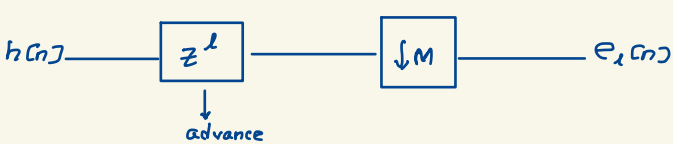
$$\Rightarrow \text{Hence} \quad \boxed{E_l(z) = \sum_{n=-\infty}^{\infty} h[nM+l]\cdot z^{-n} \qquad l = 0, 1, \dots, M-1}$$

$$\Rightarrow \quad H(z) = E_0(z^M) + z^{-1}\cdot E_1(z^M) + \dots + z^{-(M-1)}\cdot E_{M-1}(z^{M-1})$$

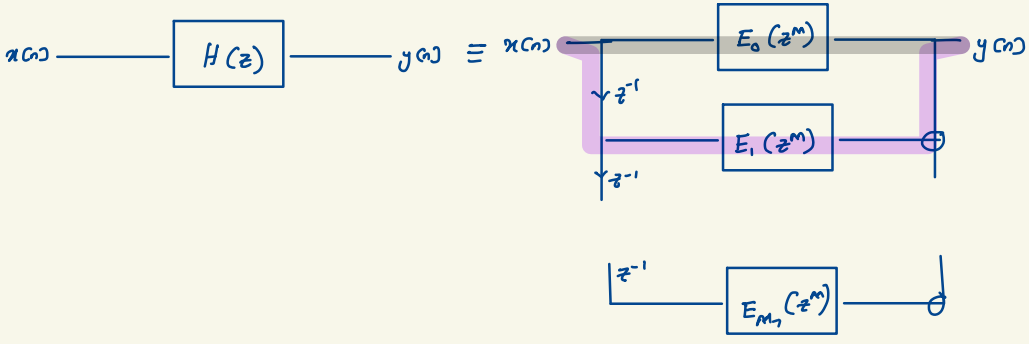
$$\Rightarrow \quad \boxed{H(z) = \sum_{l=0}^{M-1} z^{-l} E_l(z^M)}$$ Type I Polyphase Filter

## Polyphase Components in Time Domain

### Type I Polyphase Components




Type I Polyphase Component



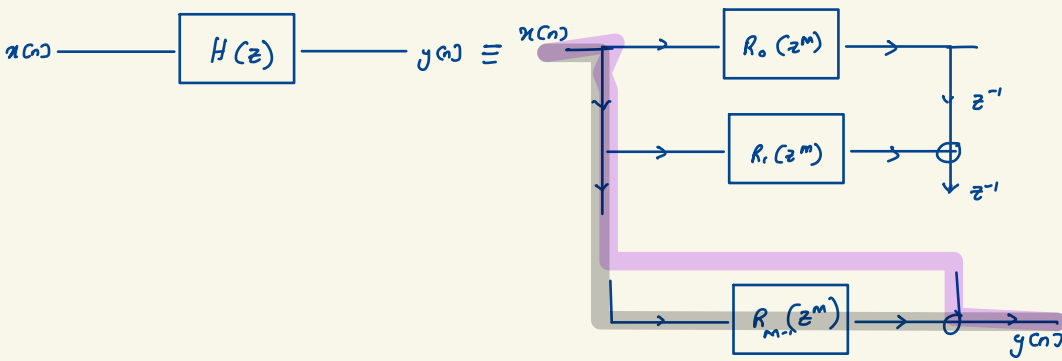
## Type II Polyphase Components

Define $R_l(z) := E_{M-l-1}(z)$ $\quad l = 0, 1, \ldots, M-1$

Switching the order of $l$.

$$\Rightarrow \quad H(z) = \sum_{l=0}^{M-1} z^{-(M-l-1)} \cdot R_l(z)$$

Type I Polyphase Component




$R_l(z)$ is a Type II Polyphase Filter.

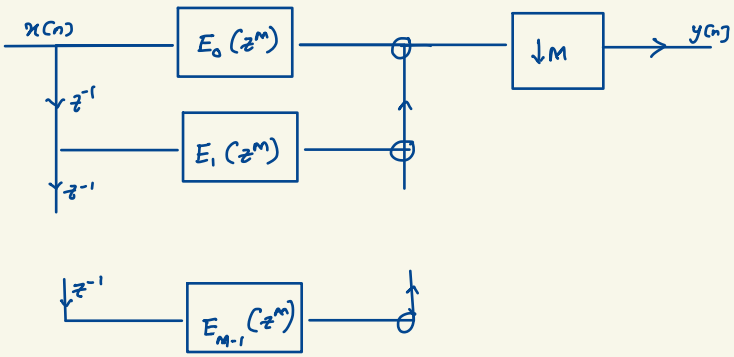
## Computational Complexity of a Decimation Filter (FIR)

- If $h(n)$ has a length $N$. Then we need $N$ multiplications and $N-1$ additions per unit time.

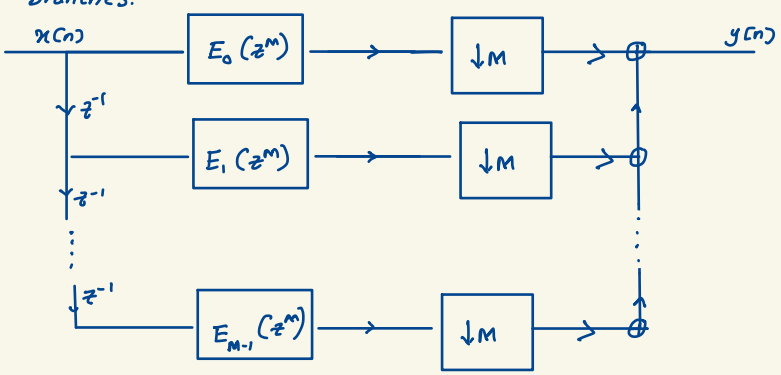
$$\tilde{x}(n) = \sum_{n=0}^{N-1} h(k) \cdot x(n-k)$$

- This block is not efficient since $H(z)$ works in high frequency and $\downarrow M$ kills it. How to make it efficient using polyphase decomposition.

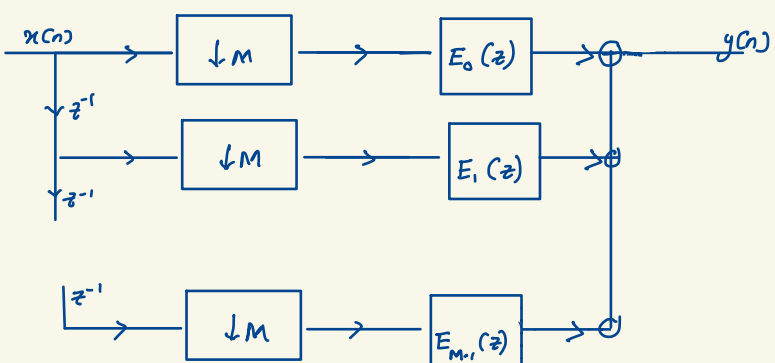
- Replace $H(z)$ by its polyphase components.




- Since this is LTI, we can put the decimation in the branches.




- By using noble identity (1) we can interchange $E_i(z^M)$, $\downarrow M$.

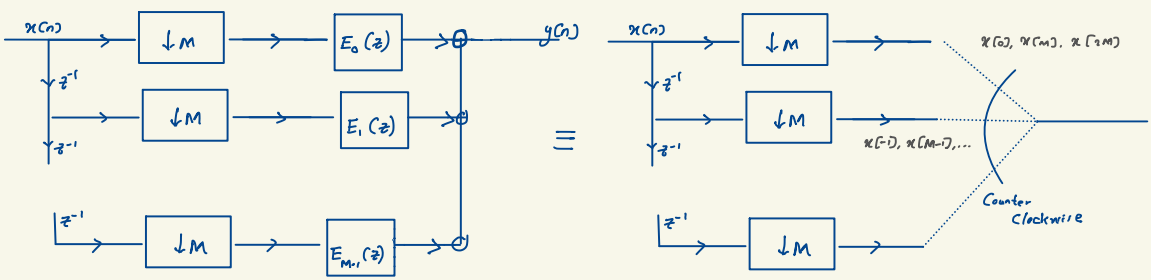



Now things happen at a low rate saving computational complexity.

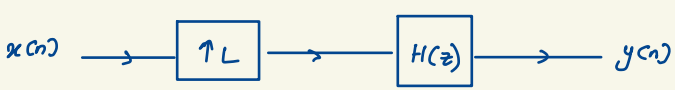
MPUT and APUT (Multiplications Per Unit Time/ Additions Per Unit Time) can be reduced by factor of M

- If $H(z)$ has length $N$, each of $E_\ell(z)$ has length $N/M$. ?
- $N/M$ multiplication per branch. $N$ multiplications in total per $M$ unit of time.
- $N/M$ multiplication per unit time.
- $\frac{N-1}{M}$ multiplications + $M-1$ additions per unit time.

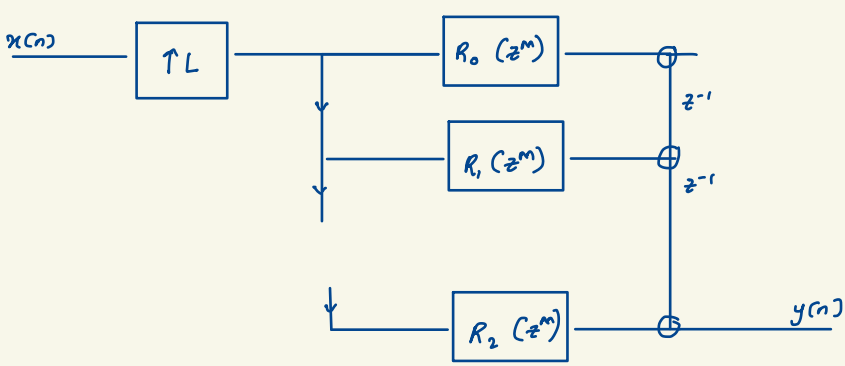
## Multiplexing Implementation of Delay Line in a Decimation Filter.



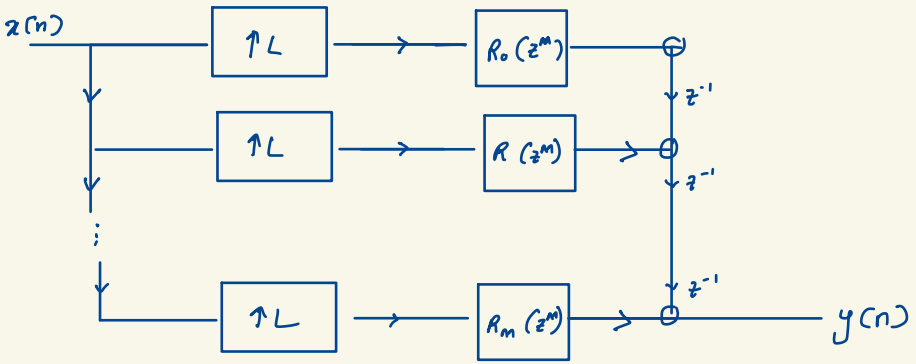

## Computational Complexity of a Interpolation Filter (FIR)



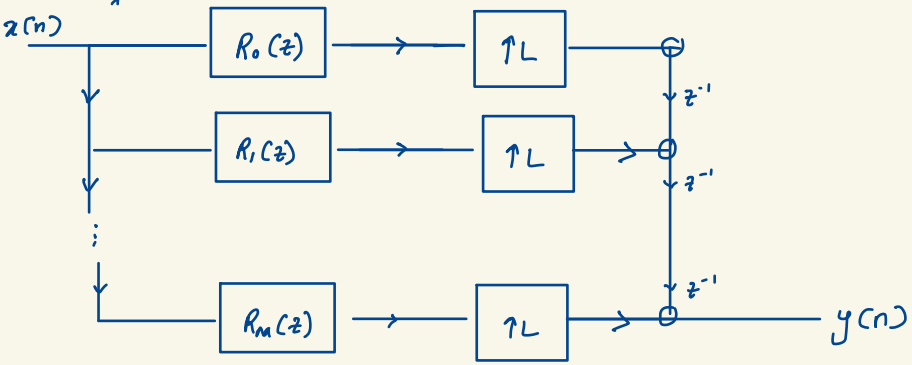

- Replace $H(z)$ by its polyphase components of Type II



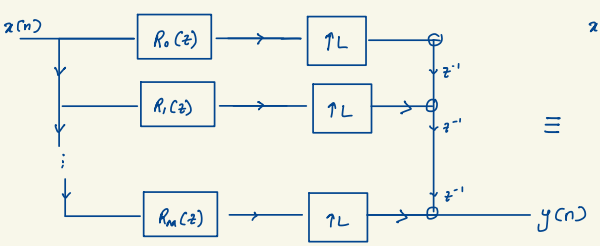
- Since the system is LTI we can put expansions in the branch.




- By using the noble identity II we can interchange the ↑L and $R_i(z^M)$




## Multiplexing Implementation of Delay Line in a Interpolation Filter.

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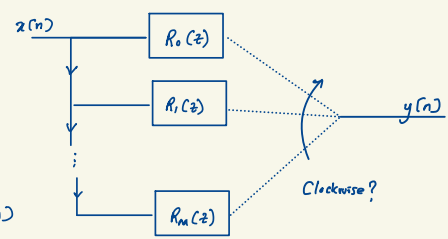
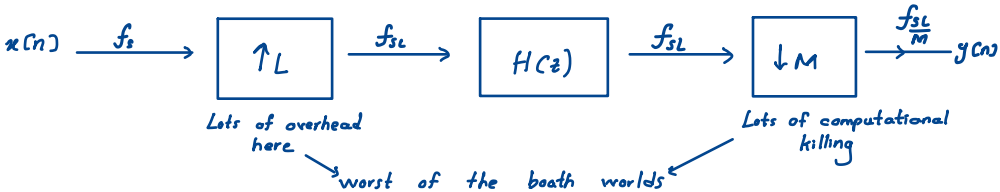
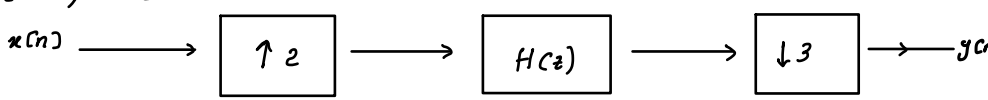


# Computational Complexity of a Functional Rate Convergence Filter



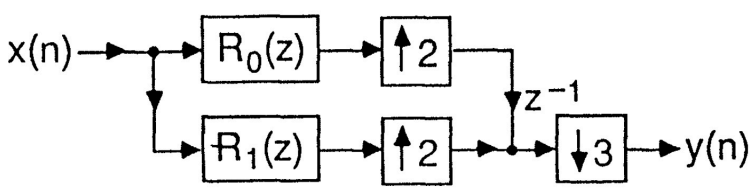

- Type I or II alone cannot bring the computational bottleneck to lowest possible rate of $f_{s/M}$

Ex: $L=2$, $M=3$

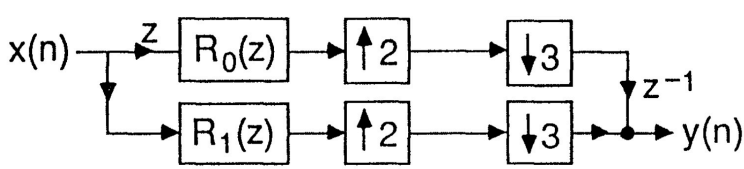



Let's use Type II Polyphase decomposition with $L=2$ on $H(z)$

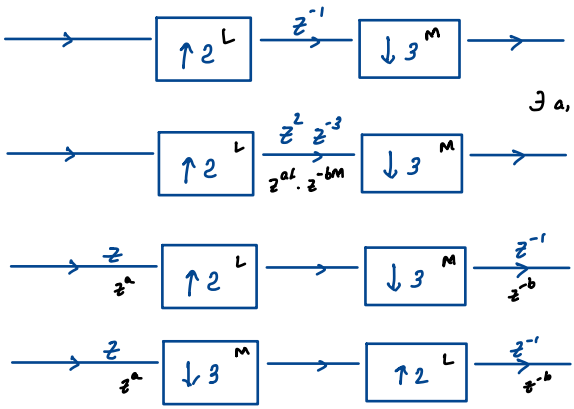
∴ It has 2 components $R_0$ and $R_1$.

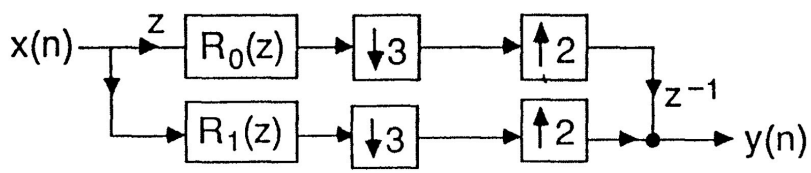


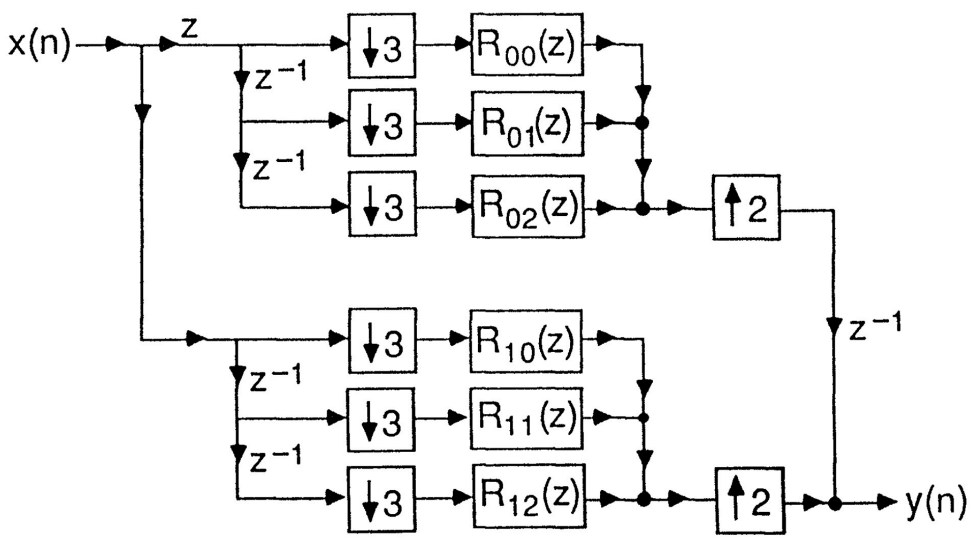

$$R_0(z) = R_{00}(z^3) + z^{-1}R_{01}(z^3) + z^{-2}R_{02}(z^3),$$
$$R_1(z) = R_{10}(z^3) + z^{-1}R_{11}(z^3) + z^{-2}R_{12}(z^3),$$




We can write $z^{-1} = z^{-3} \cdot z^2$. Hence we can interchange ↑2 and ↓3



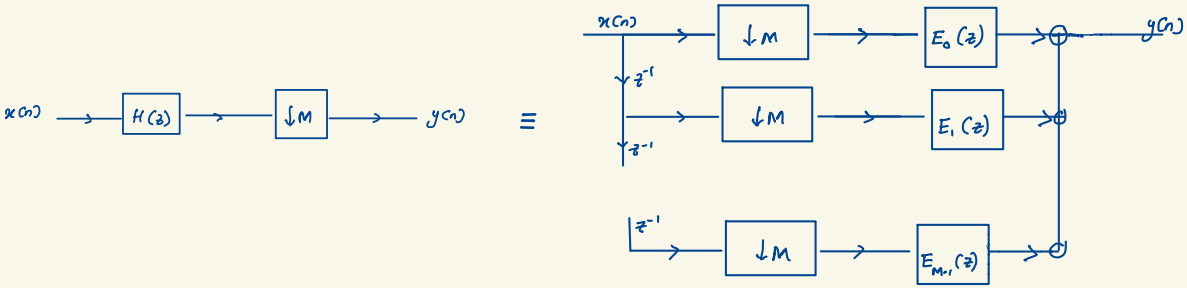




Bezout's Identity.

If $\gcd(M,L) = 1 \iff \exists\, n_0, n_1 \in \mathbb{Z} \quad n_0 M - n_1 L = 1$

# Subband Coding / Multi-stage Implementations

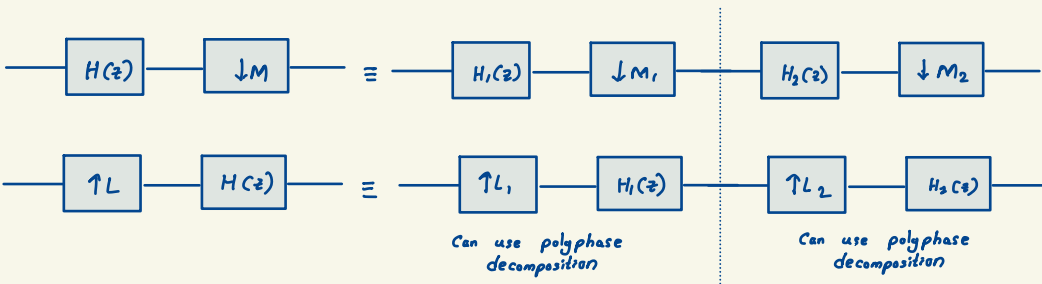



- The main trade off in polyphose decomposition is computational savings vs sensitivity to synchronization accross M/L branches.
- To avoid using large M/L in the polyphase decomposition. Let's consider a factorization

$$M = M_1 M_2$$
$$L = L_1 L_2$$

$M_1, M_2, L_1, L_2 \in \mathbb{N}^+$

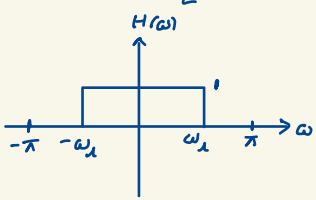
The goal is to find a multistage decomposition



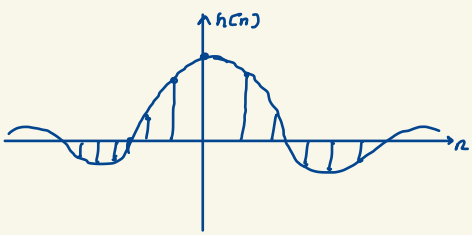

- To implement this 'multi-stage' decomposition, we need classic filter design ideas.
- Reducing M/L reduces the branches and probabilidy of a $l^0$ failure.

# Filter Design Problem

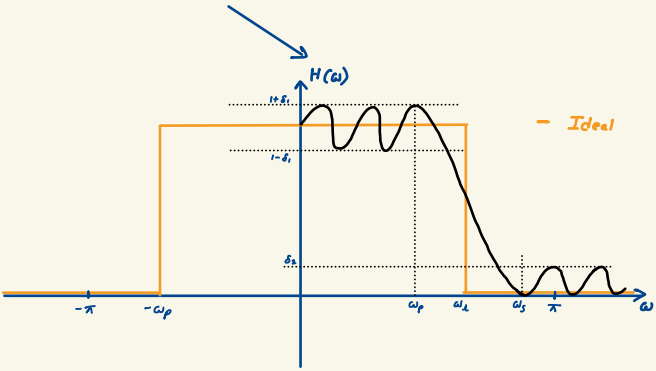
Ideal LPF ⟶ non-ideal LPF with control over it's **error** and **implementation capacity**.




$$h[n] = \begin{cases} \omega_1/\pi & n = 0 \\ \frac{\sin \omega_1 n}{\pi n} & n \neq 0 \end{cases}$$




Very difficult to implement since **non Causal** and **IIR**




$\omega_p$ - pass band frequency

$\omega_s$ - stop band frequency

$\delta_1$ - passband error total

$\delta_2$ - stopband error total

$\omega_p \leq \omega_1 \leq \omega_s$

$\Delta\omega = \omega_s - \omega_p$ : Transition bandwidth

- By admitting these **errors** we can create a filter that is **Causal** and **FIR**.

- Research few decades ago was to discover FIR filters like this,

**Problem**: FIR filter design
Given $\omega_1, \omega_p, \delta_1, \delta_2$ find an FIR filter with the smallest length $N$.

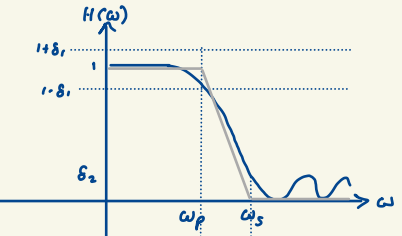
**Solution**: Kaiser-window filter design
Parks-McClellan filter design

In general: $N \sim \frac{g(\delta_1, \delta_2)}{\Delta\omega}$

Generally $N \gg 1$. ($N \sim 150$ to approximate a Sync function)

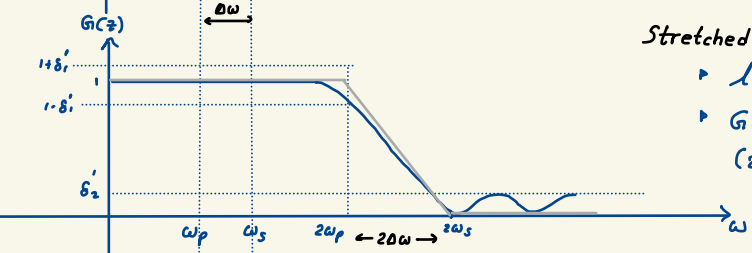
# Interpolated FIR Approach

- Suppose that we have LPF with length $N$, that satisfies all parameters $(\omega_s, \omega_p, \delta_1, \delta_2)$



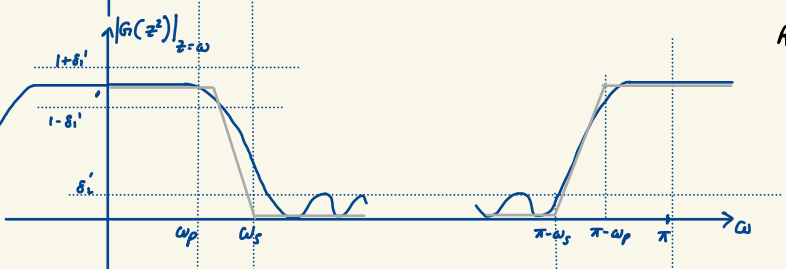

**Desired Narrowband Response $H(\omega)$**

- length $N$
- Generally $\omega_p$ and $\omega_s$ are closer... $\therefore \Delta\omega$ is low. $\therefore N \sim g(\delta_1, \delta_2) \Rightarrow N$ is high.




**Stretched Filter $G(z)$**

- length $N/2$
- $G(z)$ be a FIR LPF filter with $(2\omega_p, 2\omega_s, \delta_1, \delta_2)$




**Response of $G(z^2)$**

- Length $N/2$
- $G(z^2)$ is where signal is Expanded by $\uparrow L=2$
- This shrinks frequency domain by 2. But create images with period $2\pi/L = 2\pi/2 = \pi$

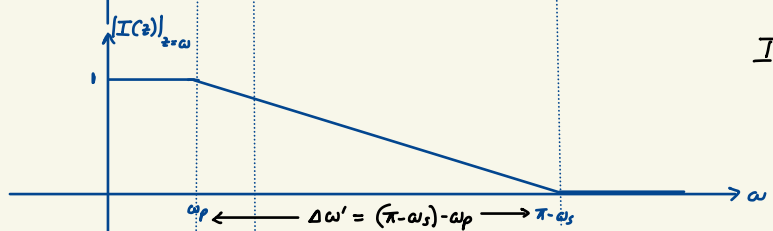
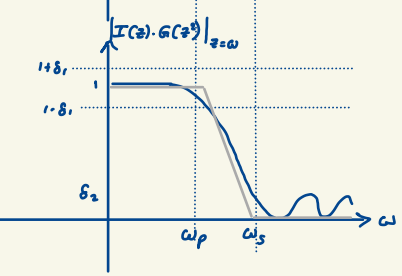


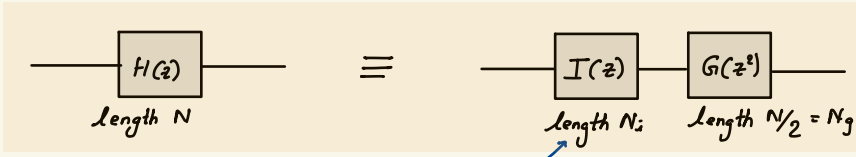

**Image Suppresion Filter $I(\omega)$**

- Idea is to kill odd copies
- Filter is wide because $\Delta\omega'$ is large.



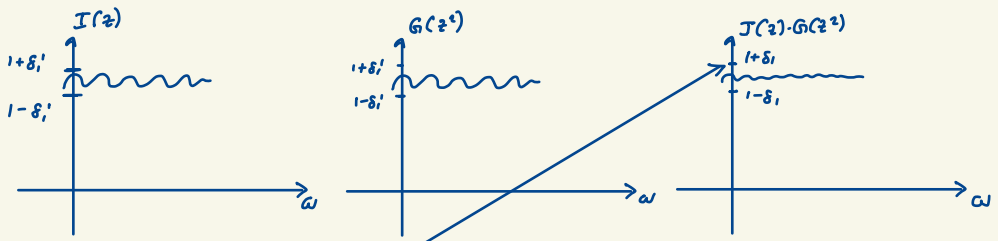

$$H(z) = I(z) \cdot G(z^2)$$

$\delta_1' = \delta_1/2 \quad \delta_2' = \delta_2$




$N_I$ is small since $\pi - \omega_s - \omega_p$ is large.

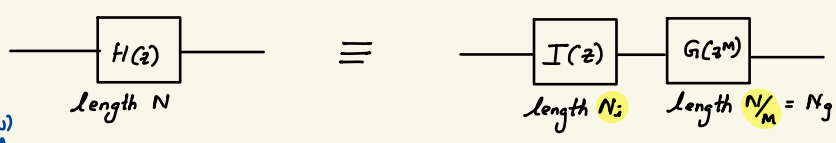
- Adjusting ripple size -
  - $G(z^2) \cdot I(z)$ can have larger ripples. .. Take peak passband ripples of $G(z)$, $I(z)$ as $\delta_1/2 \Rightarrow$ peak passband of overall signall will be $\delta_1$.
  - For peak stop band of $G(z)$, $I(z)$, $\delta_2$ is sufficient.

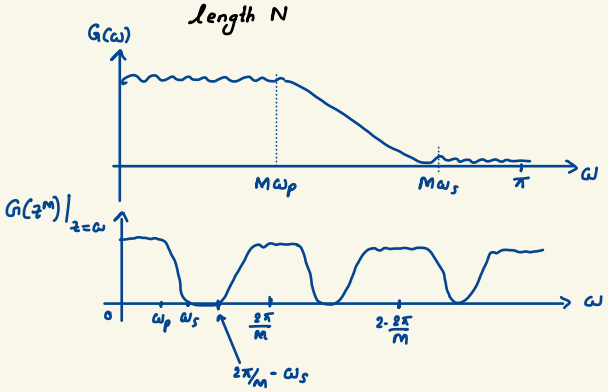



$(1+\delta_1')\cdot(1+\delta_1') = 1 + 2\delta_1' + \delta_1'^2 \simeq 1+\delta_2 \Rightarrow \delta_2 \sim \delta_1'/2$

Hence $N_G > N/2$ to accomadate this.

## G(z) with the Specs $(M\omega_s, M\omega_p, \delta_1, \delta_2)$




$$\Delta\omega' = \frac{2\pi}{M} - \omega_s - \omega_p$$

As $M$ grows, $\Delta\omega' = \frac{2\pi}{M} - \omega_s - \omega_p$ gets smaller, which makes $N_i$ get too larger.

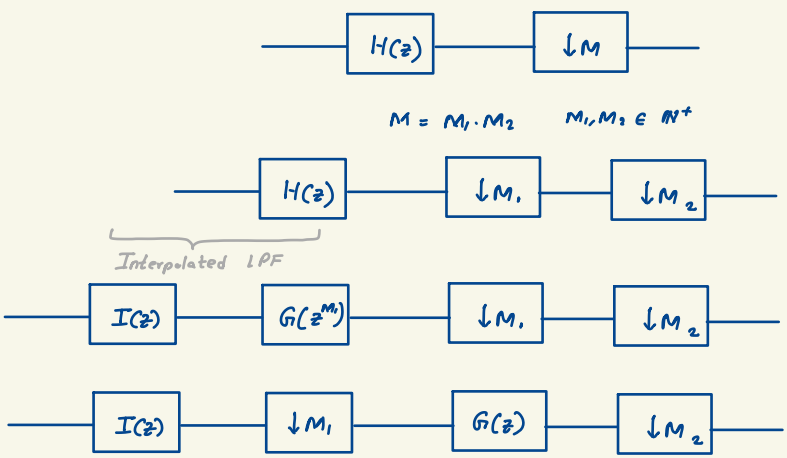
Optimization: What is the optimal $M$ to minimize $\left(\frac{N}{M} + N_i\right)$

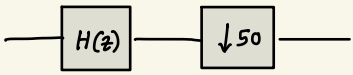
A LPF can be converted HPF or BPF by taking $h[n]$ of LPF and modifying it.

- $h[n] \cdot (-1)^n \longrightarrow$ HPF
- $h[n] \cdot \cos(\omega - \omega_s) \longrightarrow$ Band PF

hence only LPFs are studied.

Goal: Multi-stage design of a Decimation Filter



Ex: 


$H(z)$ is a LPF with $f_p = 70Hz$, $f_s = 80Hz$, $f_0 = 8kHz$, $\delta_1 = 0.01$, $\delta_2 = 0.001$

- Direct design

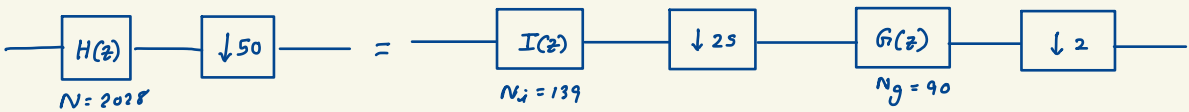
Period $= 2\pi/800 \longrightarrow N = 2028$

- Interpolated FIR design.

$$\underset{M}{50} = \underset{M_1}{25} \times \underset{M_2}{2}$$

Define $G(z)$ with 25 times higher frequencies. $\therefore$ ( $25f_p$, $25f_s$, $\delta_1$, $\delta_2/2$) to accomodate ripples. $\therefore$ Period $25 \times \frac{2\pi}{800} \longrightarrow N_g = 90$ ... from MatLab

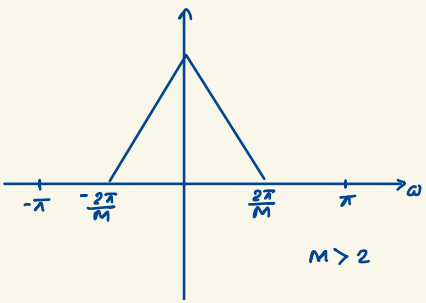
$I(z)$ is designed with $\Delta\omega = 2\pi/M_1 - \omega_s - \omega_p = \frac{17}{800} 2\pi$. and $\delta_1 = 0.005$, $\delta_2 = 0.001$. $\longrightarrow N_i = 139$



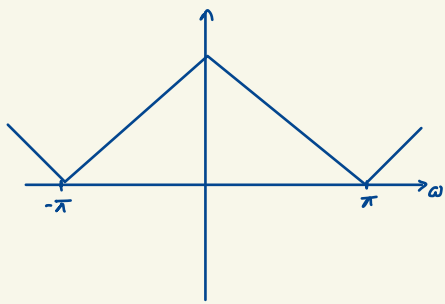

$N = 2028$ $\quad$ $N_i = 139$ $\quad$ $N_g = 90$

- Motivation of the need to work with band limited signals.

**Band limited Signal in DTFT Domain**



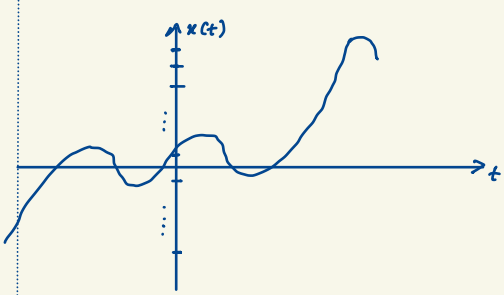

**Non-Band limited Signal in DTFT Domain**




- Signal sampled exactly at the Niquist rate. ∴ Contain in all $\omega$ range.
- If we decimate this create aliases.

Quantization ⟶ Quantizing the Dynamic Range

Decimation

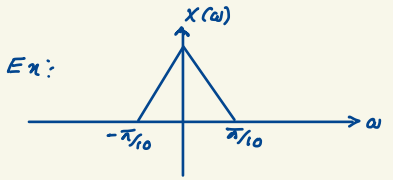



- If it is quantized at N levels, we need $\log_2 N$ bits.
- Sometimes the error can be too much. $|Error| \leq A/2N$

- Trade-off is between how fast the sampling rate vs how low the error in quantising.
- Apart from that quantization add aliasing.

**Imp:-** If the DTFT of $x(n)$ is supported on $(\omega_0, \omega_0 + \frac{2\pi}{M}]$ then it is possible to decimate $x[n]$ by a factor of $M$ without any aliasing.

Ex:

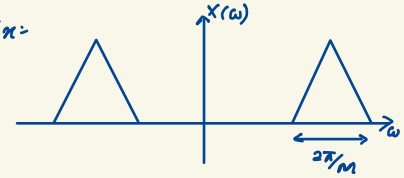



Since $X(\omega)$ is from $(-\pi/10, \pi/10)$, set $\omega_0 = -\pi/10$. $\therefore$ the possible supported range is $\left(-\pi/10, -\pi/10 + \frac{2\pi}{M}\right]$

$\Rightarrow -\pi/10 + 2\pi/M = \pi/10 \Rightarrow M = 10$ is supported

When decimated by $M=10$, this signal does not have aliases.

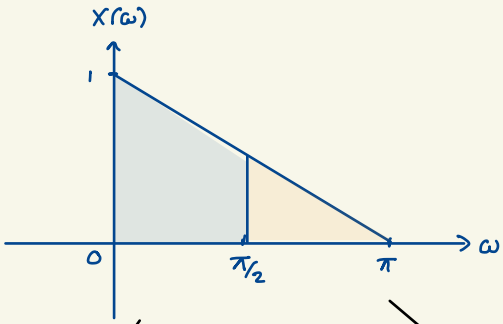
Ex:-




Although the bandwidth can be very high like here effective bandwidth is $2\pi/M$

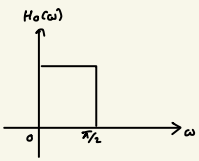
$\therefore$ The images created may not intersect.

# Sub-band Coding

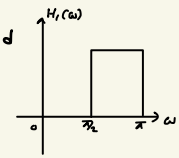


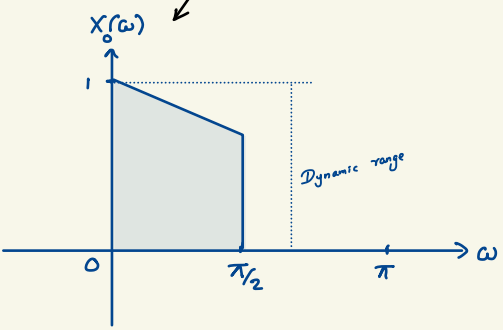

This is a non band limited signal. The goal is to seperate it so the parts can be decimated.

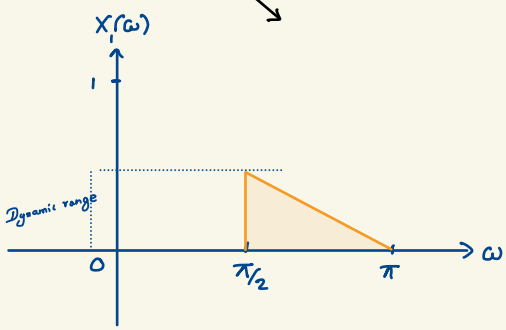
LPF filtered x(t)

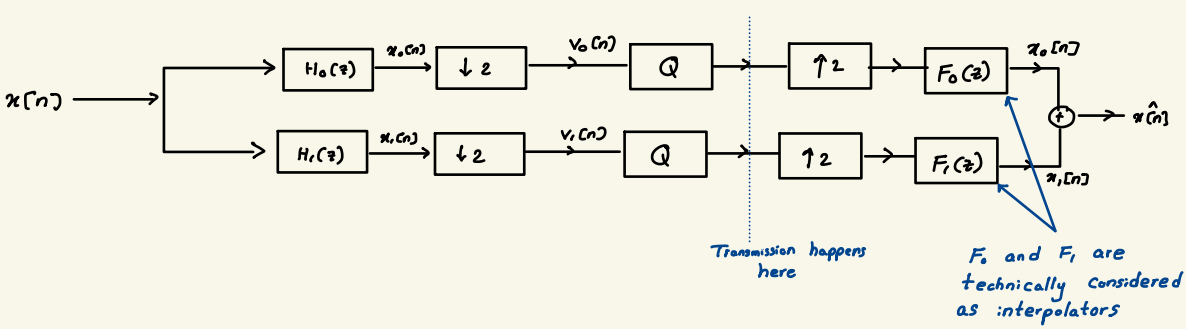



HPF filtered x(t)



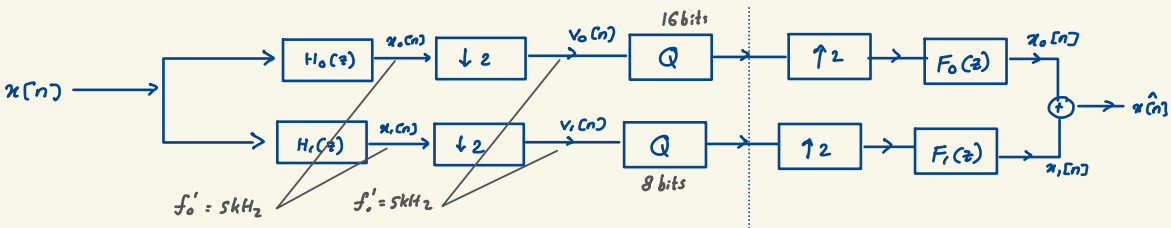

- $x_0[n]$ has higher dynamic range than $x_1[n]$.
- So we can use more bits for quantising $v_0[n]$.

Ex: $f_0 = 10\ kHz$ sampling rate and 16 bits/sample quantization.

$\therefore$ If the signal is transmitted directly, $\therefore$ transmission rate is

$= 10\ kHz \frac{Samples}{s} \times 16 \frac{bits}{Sample} = 160\ kb/s$ //

$\therefore$ If the signal is decomposed using LPF and HPF each part has half the effective bandwidth $\rightarrow f_0' = 5kHz$



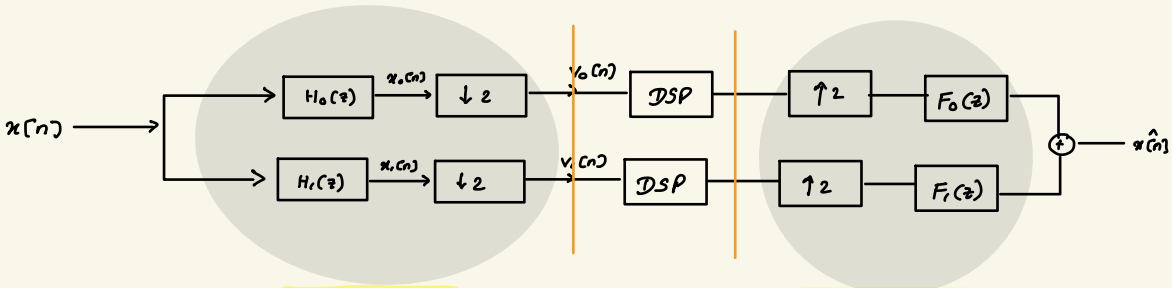

$x_0[n]$ has higher dynamic range than $x_1[n]$. So we can use more bits for quantising $v_0[n]$. $\therefore$ Transmission rate,

$= \left(16 \frac{bits}{sample} \times 5\ kHz \frac{Samples}{s}\right) + \left(8 \times 5kHz\right) = 120\ kb/s$ //

- This can be done with any operations other than quantization.
  $\therefore$ These are called 2-channel filter banks.

# PPV-5

# 2-Channel Filter Banks



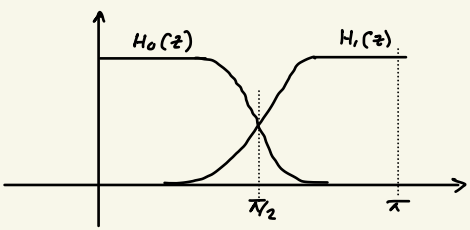

**Analysis Bank**

$H_0(z)$, $H_1(z)$ are Analysis filters.

**Synthesis Bank**

$F_0(z)$ and $F_1(z)$ are synthesis filters

- Assume that DSP unit is an identity for now.
- Typical choices of $H_0(z)$ and $H_1(z)$

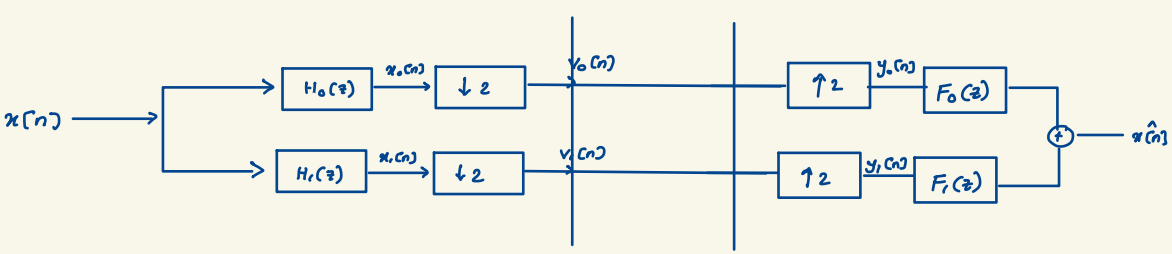



**Applications**

Subband coding/processing of non-band limited signals.

## Requirements for $x[n] = \hat{x}[n]$ "Identity Requirement"

Can $\hat{x}[n]$ be equal to $x[n]$ even when $H_1$ and $H_2$ are non ideal filters. (when DSP unit is identity).



$e^{i\theta} = \cos\theta + i\sin\theta$

$W_M = e^{-i2\pi/M} = e^{-i\pi}$

from Decimation logic

$$V_0[n] = X_0[2n]$$

$$V_0(z) = \frac{1}{M}\sum_{k=0}^{M-1} X_0\left(W_M^k z^{1/M}\right) = \frac{1}{2}\sum_{k=0}^{1} X_0\left(e^{-j\pi k} \cdot z^{1/2}\right)$$

$$V_0(z) = \frac{1}{2} X_0\left(z^{1/2}\right) + \frac{1}{2} X_0\left(-z^{1/2}\right)$$

from filtering

$$X_0(z) = H_0(z) \cdot X(z)$$

from Expansion logic

$$Y_0(z) = X_0(z^2)$$

$$\therefore \hat{X_0}(z) = Y_0(z) \cdot F_0(z) = V_0(z)^2 F_0(z)$$

$$= \left[\frac{1}{2} X_0\left(z^{1/2}\right) + \frac{1}{2} X_0\left(-z^{1/2}\right)\right]^2 \cdot F_0(z)$$

$$= \frac{1}{2} H_0(z) \cdot F_0(z) \cdot X(z) + \frac{1}{2} H_0(-z) \cdot F_0(z) \cdot X(-z)$$

$$\therefore \hat{X_1}(z) = Y_1(z) \cdot F_1(z) = V_1(z)^2 F_1(z)$$

$$= \left[\frac{1}{2} X_1\left(z^{1/2}\right) + \frac{1}{2} X_1\left(-z^{1/2}\right)\right]^2 \cdot F_1(z)$$

$$= \frac{1}{2} H_1(z) \cdot F_1(z) \cdot X(z) + \frac{1}{2} H_1(-z) \cdot F_1(z) \cdot X(-z)$$

Good parts where $X(z)$ is only scaled

$$\therefore \hat{X}(z) = \left[\frac{1}{2} H_0(z) \cdot F_0(z) + \frac{1}{2} H_1(z) \cdot F_1(z)\right] X(z) +$$

$T(z)$ = transfer of the filter bank

$$\left[\frac{1}{2} H_0(-z) \cdot F_0(z) + \frac{1}{2} H_1(-z) \cdot F_1(z)\right] X(-z)$$

$A(z)$ = aliasing of filter bank

| | |
|---|---|
| Transfer | $T(z) = \frac{1}{2} H_0(z) \cdot F_0(z) + \frac{1}{2} H_1(z) \cdot F_1(z)$ |
| Aliasing | $A(z) = \frac{1}{2} H_0(-z)\, F_0(z) + \frac{1}{2} H_1(-z)\, F_1(z)$ |

Ideally - $\begin{cases} T(z) = 1 \\ A(z) = 0 \end{cases} \Longrightarrow \hat{x}[n] = x[n]$

- Can we design the system (reciever with $F_0(z)$ and $F_1(z)$) such that above conditions hold.

## I) For No Aliasing

$\Rightarrow A(z) = 0$

$\Rightarrow \frac{1}{2} H_0(-z) \cdot F_0(z) + \frac{1}{2} H_1(-z) \cdot F_1(z) = 0$

$\Rightarrow$ A sufficient condition for above to be satisfied is,

$$F_0(z) = H_1(-z)$$
$$F_1(z) = -H_0(-z)$$

... z can be negated by flipping the sign of $h_1[n]$ at odd places to get $f_0[n]$

- This is easy to be satisfy at the reciever side. Just get the analysis filters and flip to get synthesis filters.

## II) For no Transfer / Amplitude Shift

- Assume we pick $F_1$ and $F_2$ in above case for no aliasing.

$$T(z) = \frac{1}{2}\left[ H_0(z) \cdot F_0(z) + H_1(z) \cdot F_1(z) \right]$$

$$= \frac{1}{2}\left[ \underbrace{H_0(z) \cdot H_1(-z)}_{W(z)} - \underbrace{H_1(z) \cdot H_0(z)}_{W(-z)} \right]$$

$$= \frac{1}{2}\left[ W(z) - W(-z) \right]$$

$$= \because \text{Even powered terms go away.} \because \text{Only odd powers of } z \text{ remain.}$$

$$= z^{-1} S(z^2)$$

- Although requirement is $T(z) = 1$, it is impossible to achieve since $z^{-1} \cdot S(z^2)$ can't be 1 since $S(z^2)$ polynomial can not cancel $z^{-1}$ term.

**Conjugate Symmetry Property**

if $x[n]$ is real valued $X(\omega)$ is Symmetric

Proof: $X(\omega) = \sum x[n] \cdot e^{-j\omega n}$

Assume $X^*(\omega)$ is complex conjugate of $X(\omega)$

$X^*(\omega) = \sum x[n]^* (e^{-j\omega n})^*$

$= \sum x[n] \cdot e^{j\omega n}$ .... $x[n] \in \mathbb{R}$

$= X(-\omega)$

$\therefore |X(\omega)|$ is Symmetric.

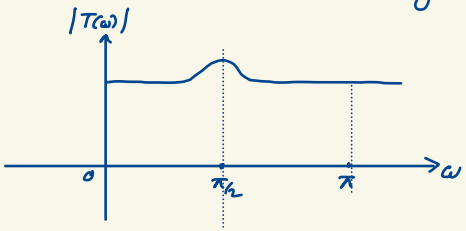
## Symmetry Around $\pi/2$

$\Rightarrow T(\omega) = e^{-j\omega} \cdot S(2\omega)$ — Most of the time $S(\omega)$ has period $2\pi$.

$\Rightarrow T(\omega) = |S(2\omega)|$

$\Rightarrow T(\omega)$ has a period of $\pi$

- If $t[n]$ is real valued it will be symmetric around $\pi/2$.



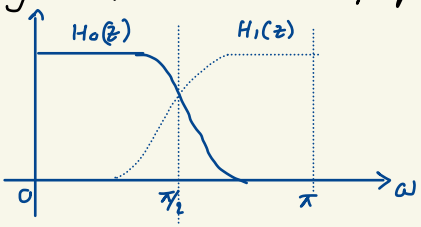

- Symmetricity results in better subband properties. ∴ Most basic way to achieve this is flipping $H_0$ to get $H_1$ ?

## Trade offs

1) Aliasing - $A(z)$
2) Amplitude Distortion - $|T(\omega)|$
3) Phase Distortion - $\angle T(\omega)$
4) Subband Properties

# Quadrature Mirror Filters (QMF) / Mother Filter

- Start by satisfying ④, better subband properties.




$H_1$ is selected to be the Quadrature mirror of $H_0$.

$H_0$ and $H_1$ are QMF

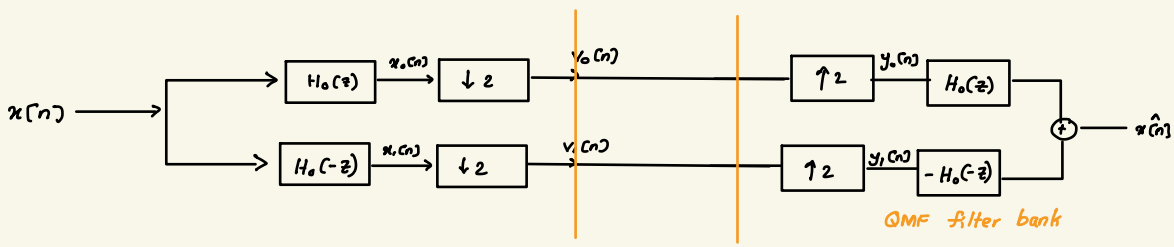
$H_0(z)$ - Designed using the specs $(\omega_p, \omega_s, \delta_1, \delta_2)$ that is suitable.

$H_1(z) = H_0(-z)$ ⟶ ∴ In the $\omega$ domain $|H_1(\omega)| = |H_0(\pi - \omega)|$

- Then let's satisfy ①: $A(z) = 0$

$$F_0(z) = H_1(-z) = H_0(z)$$

$$F_1(z) = -H_0(-z)$$




$$T(z) = \frac{1}{2}\left[H_0(z)H_1(-z) - H_0(-z)\cdot H_1(z)\right]$$

$$\Rightarrow T(z) = \frac{1}{2}\left[H_0^2(z) - H_0(-z)^2\right] \quad \cdots H_1(z) = H_0(-z)$$

Represent analysis and synthesis banks in terms of it's polyphase components,

$$H_0(z) = E_0(z^2) + z^{-1}E_1(z^2) \quad \cdots\cdots \text{Type I polyphase}$$

This can be done even if $H_0$ is FIR/IIR

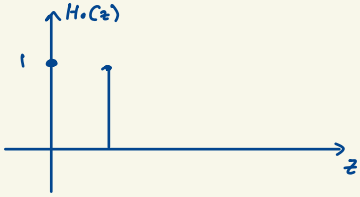
$$\Rightarrow T(z) = \frac{1}{2}\left[E_0(z^2)^2 + z^{-2}\cdot E_1(z^2)^2 + 2\cdot E_0(z^2)\cdot z^{-1}\cdot E_1(z^2) - z^{-2}\cdot E_1(z^2)^2 - E_0(z^2)^2 - 2\cdot E_0(z^2)\cdot E_1(z^2)\cdot(-z)^{-1}\right]$$

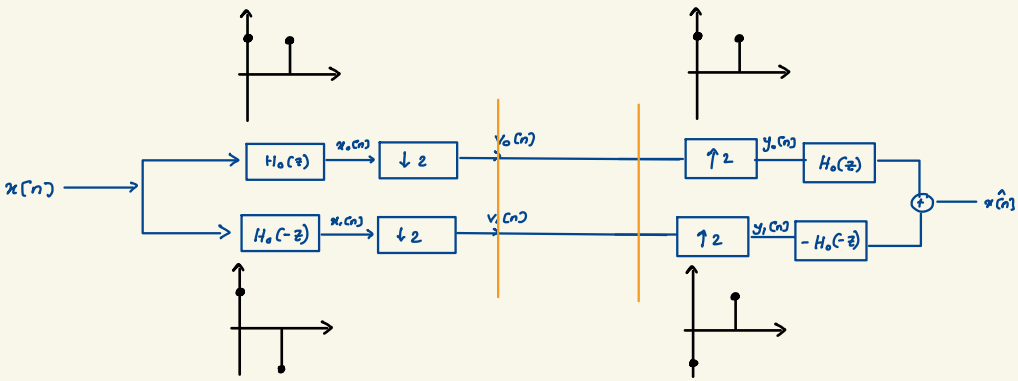
$$= E_0(z^2)\cdot E_1(z^2)\left(z^{-1} - (-z)^{-1}\right)$$

$$\boxed{T(z) = 2\cdot z^{-1}\cdot E_0(z^2)\cdot E_1(z^2)}$$

ideally we want this to be 1. But its not possible.

Ex: $H_0(z) = 1 + z^{-1}$ most basic LPF



## Requirements for $\hat{x}(z) = C \cdot z^{-n_0} \cdot X(z)$ "Perfect Reconstruction"/PR

- $\hat{x}(z) = x(z)$ is "Identity requirement". But this is too stringent as per previous lecture.

  ↓ relax

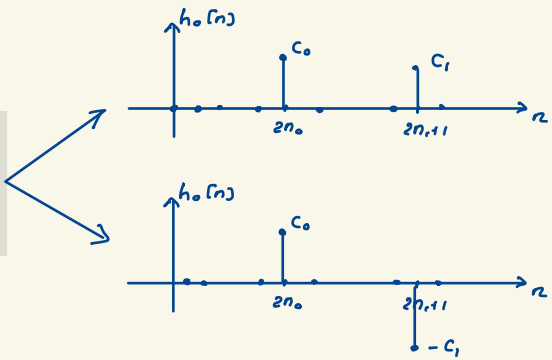
- $\hat{x}(z) = C \cdot z^{-n_0} \cdot X(z)$ ie $T(z) = C \cdot z^{-n_0}$, "Perfect Reconstruction" requirement.

- If $H_0(z)$ is FIR then $E_0(z)$, $E_1(z)$, $T(z)$ are FIR as well.

- $2 \cdot z^{-1} \cdot E_0(z^2) \cdot E_1(z^2) = C \cdot z^{-n_0} \iff E_0(z)$ and $E_1(z)$ are delays.

$$E_0(z) = C_0 \, z^{-n_0}$$
$$E_1(z) = C_1 \, z^{-n_1}$$

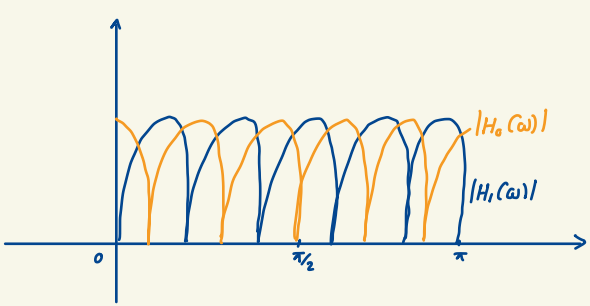
$$\therefore \quad H_0(z) = C_0 \cdot z^{-2n_0} + C_1 \cdot z^{-2n_1-1}$$
$$H_1(z) = C_0 \cdot z^{-2n_0} - C_1 \cdot z^{-2n_1-1}$$




- This works whether or not $H_0(z)$ is LP - Linear Phase. * ?
  To have above form filters can not have sharp cutoff and stop band.

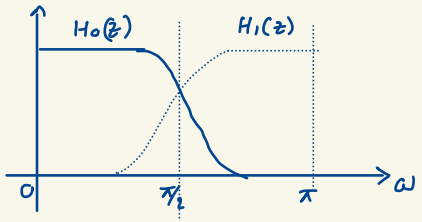
- $T(z)$ becomes delay if $E_1(z) = 1/E_0(z)$. But $E_1, E_2$ are boath not FIR.

- These are one of the worst possible filters.




No of lobes $= 2(n_1 - n_0) + 1$

- If we do 'perfect reconstruction' ⟹ We get poor Subband properties. ④
  we get no 'phase' and 'amplitude' distortion. ② ③




- If we do 'Identity Reconstruction' using QMF we get 'good sub-band' properties ④ ⟹ We get phase / amplitude distortion ②, ③

$$T(z) = \frac{1}{2}\left[ H_0^2(-z) - H_0(-z)^2 \right]$$

* Read about LP, Linear Phase

# Eliminating Phase Distortion with FIR

- Satisfying ①, ③ and ④. Dropping ②.
- All FIR and Linear Phase filters can be designed

Claim: $H_0(z)$ is LP $\Rightarrow$ $T(z)$ is LP $\Rightarrow$ No phase distortion $\Rightarrow$ Pick $h_0$ to be FIR and LP

**Proof:** If $h_0$ is a FIR filter of order $N$ (length $N+1$) $\Rightarrow$ $H_0(z) = \sum_{n=0}^{N} h_0(n) \cdot z^{-n}$

If $h_0[n]$ are real valued coefficients and is Linear Phase (LP) $\Rightarrow$ $h_0(n) = h_0(N \pm n)$

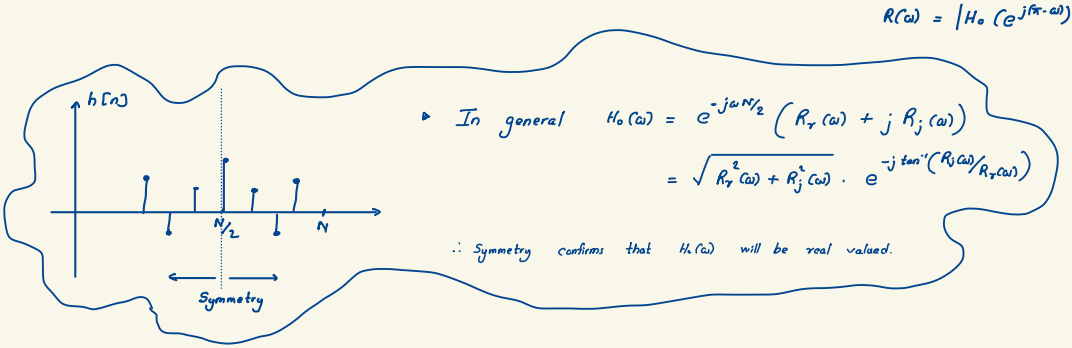
If $h_0$ is lowpass $\Rightarrow$ $h_0(n) = h_0(N-n)$ — Type I FIR LP filters of length $N+1$

From Symmetry and LP $\Rightarrow$ $H_0(\omega) = e^{-j\omega N/2} \cdot R(\omega)$ *

Phase

$R(\omega) \in \mathbb{R}\ \forall \omega$

$R(\omega) = |H_0(e^{j(\pi-\omega)})|$




- In general $H_0(\omega) = e^{-j\omega N/2}\left(R_r(\omega) + j R_j(\omega)\right)$

$= \sqrt{R_r^2(\omega) + R_j^2(\omega)} \cdot e^{-j \tan^{-1}\left(R_j(\omega)/R_r(\omega)\right)}$

$\therefore$ Symmetry confirms that $H_0(\omega)$ will be real valued.

$$\therefore T(\omega) = \tfrac{1}{2}\left[H_0(\omega)^2 - H_1^2(\omega)\right]$$

$$= \tfrac{1}{2} e^{-j\omega N}\left[R^2(\omega) - (-1)^N \cdot R^2(\pi-\omega)\right] \quad \because H_1(\omega) = H_0(\pi-\omega)$$

$$T(e^{j\omega}) = \tfrac{1}{2} e^{-j\omega N}\left[R^2(e^{j\omega}) - (-1)^N \cdot R^2(e^{j(\pi-\omega)})\right]$$

$$T(e^{j\omega}) = \frac{1}{2} e^{-j\omega N} \left[ |H_0(e^{j\omega})|^2 - (-1)^N |H_0(e^{j(\pi-\omega)})|^2 \right]$$

$(-1)^N \in \mathbb{R} + \omega$

$\therefore$ $T(\omega) / T(e^{j\omega})$ is also in the form of 'Linear Phase' filter. $\therefore$ No phase distortion. ■

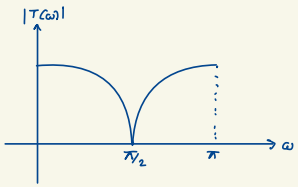
## ①, ②, ④ is Achieved. How To Minimize Amplitude Distortion ③

- When $N$ is even and $\omega = \pi/2$ $\quad N = 2n$

$$RHS = \frac{1}{2} e^{-j\pi/2 N} \left[ R^2(\pi/2) - (-1)^{2n} \cdot R^2(\pi - \pi/2) \right]$$

$$= \frac{1}{2} e^{jN\pi/2} \left[ R^2(\pi/2) - R^2(\pi/2) \right]$$

$$= 0$$




for even $n$, severe amplitude distortion (notch) occurs at $\pi/2$. $N \in$ Odd

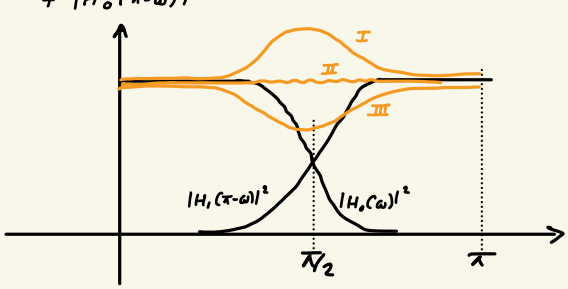
$$T(e^{j\omega}) = \frac{1}{2} e^{-j\omega N} \left[ |H_0(e^{j\omega})|^2 + |H_0(e^{j(\pi-\omega)})|^2 \right]$$

$$|T(\omega)| = \frac{1}{2} \left[ |H_0(\omega)|^2 + |H_0(\pi-\omega)|^2 \right]$$

$|T(\omega)| = \frac{1}{2} [R^2(\omega) + R^2(\pi-\omega)]$

- If $H_0(z)$ is a LPF, $R(\omega) \approx 1$ for $\omega \ll \pi/2$ and $|R(\omega)| \approx 0$ for $\omega \simeq \pi/2$. Main difficulty in handling $|T(\omega)|$ comes in the transition band near $\pi/2$.

$|H_0(\omega)|^2 + |H_0(\pi-\omega)|^2$



- ∴ Goal is to get $\left(|H_0(\omega)|^2 + |H_0(\pi-\omega)|^2 \simeq 1\right)$

- By trial and error or search / optimization we can design FIR LP $H_0(z)$ such that case I and III are always avoided.

$H_0(\omega) = e^{-j\omega N/2} \cdot R(\omega)$

→ When N is even then it's a pure delay.

→ When N is odd, it is equivalent to going to time domain, delay it and sample again.

# Eliminating Amplitude Distortion with IIR

- Satisfying ①, ③ and ②. Dropping ④.
- We have to achieve I. Hence need to force $T(\omega)$ to be bandpass and $T(\omega) = 1$
- Since $T(z) = 2 \cdot z^{-1} E_0(z^2) \cdot E_1(z^2)$ A satisfactory condition would be

  $\Rightarrow$ $T(z) = z^{-1}$ which is a time delay

  $\Rightarrow$ $E_0(z) \cdot E_1(z) = 1/2$

  $\Rightarrow$ $E_1(z) = \dfrac{1}{2 E_0(z)}$

  $\Rightarrow$ We cannot have both $E_0$ and $E_1$ as FIR
- We can pick $E_0(z)$ and $E_1(z)$ to be all-pass filters but IIR

Ex:- All-pass IIR filter $G(z) = \dfrac{z^{-1} - a^*}{1 - a z^{-1}}$ $a \in \mathbb{C}$

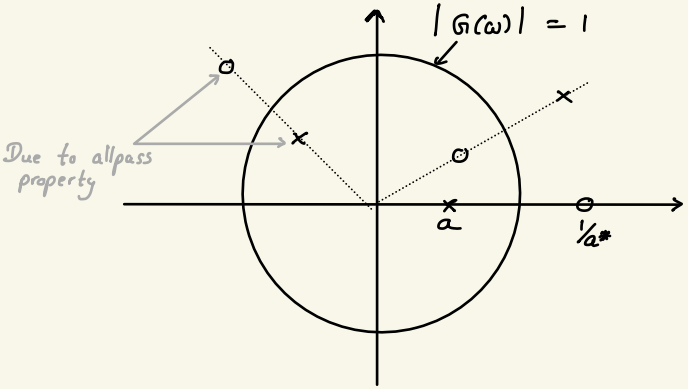
We have poles at $z = a$ and zeros at $z = 1/a^*$

$$G(\omega) = \frac{e^{-j\omega} - a^*}{1 - a \cdot e^{-j\omega}} = e^{-j\omega}\left(\frac{1 - a^* e^{j\omega}}{1 - a \cdot e^{-j\omega}}\right) = e^{-j\omega}\frac{z_0}{z_0^*}$$

$T(z) = 2 \cdot z^{-1} G_0(z^2) \cdot G_1(z^2)$. From polyphase decomposition,

$$\left.\begin{aligned} E_0(z) &= \frac{G_0(z)}{\sqrt{2}} \\ E_1(z) &= \frac{G_1(z)}{\sqrt{2}} \end{aligned}\right\} \text{ Both should be bandpass.}$$

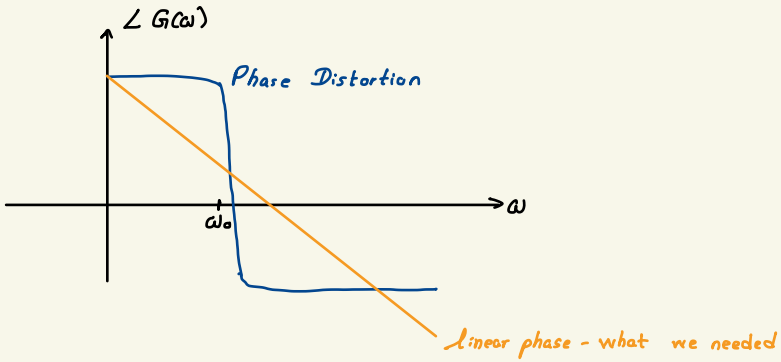
$$|G(\omega)| = \frac{\sqrt{1 + (a^*)^2}}{\sqrt{1 + (-a)^2}} = 1$$




Closer the poles and zeros to unit circle, the phase distortion gets worse.

## ①, ②, ③ is Achieved. How To Minimize Phase Distortion ④

- All pass systems have bad phase response.




- Use Group Delay to shows how much phase distortion we have.

$$\text{Group Delay} = -\frac{d}{d\omega} \angle G(\omega)$$

(∵ What is phase delay in $e^{-j\omega N/2}$ ?)

# Efficient Implementation of QMF Filter Banks

- $H_0(z)$ is the mother filter, all other is derived from this, such that

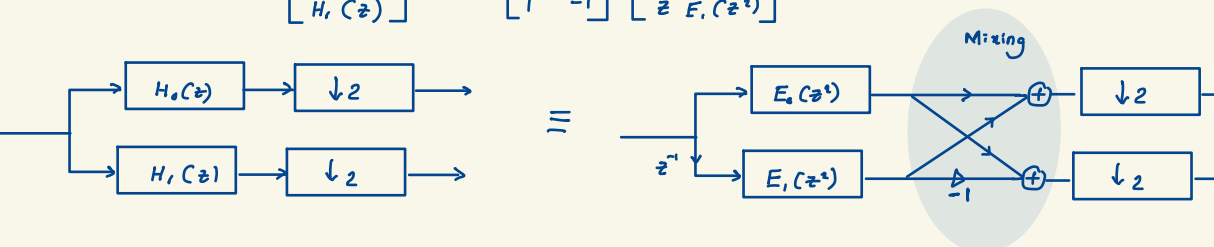
$$H_1(z) = H_0(-z)$$
$$F_0(z) = H_0(z)$$
$$F_1(z) = -H_0(-z)$$

- Then $H_0$ is written using Type I polyphase components.

$$H_0(z) = E_0(z^2) + z^{-1} E_1(z^2)$$

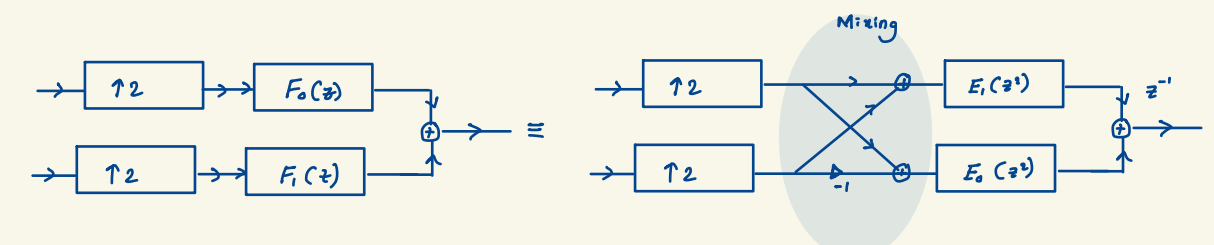
- ∴ Analysis Bank can be written as,

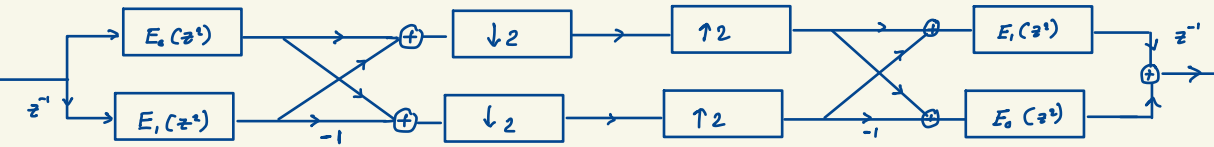
$$\begin{bmatrix} H_0(z) \\ H_1(z) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} E_0(z^2) \\ z^{-1} E_1(z^2) \end{bmatrix}$$



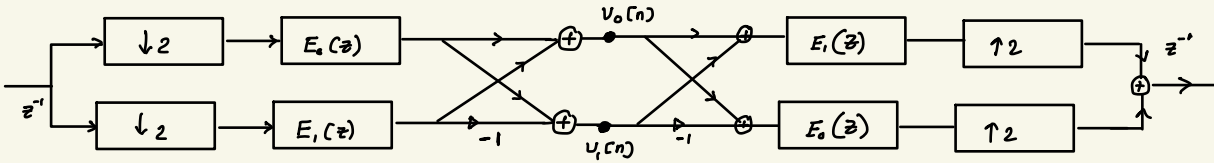

- ∴ Synthesis Bank can be written as,

$$\begin{bmatrix} F_0(z) & F_1(z) \end{bmatrix} = \begin{bmatrix} z^{-1} E_1(z^2) & E_0(z^2) \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$



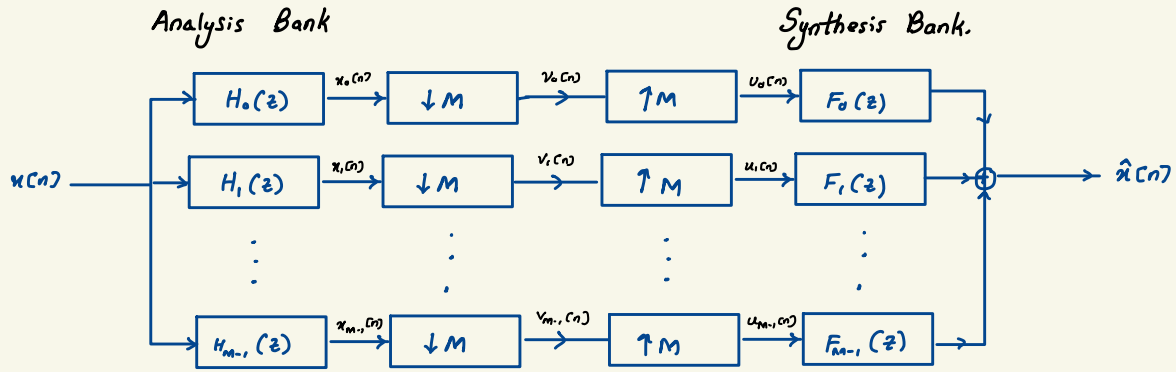


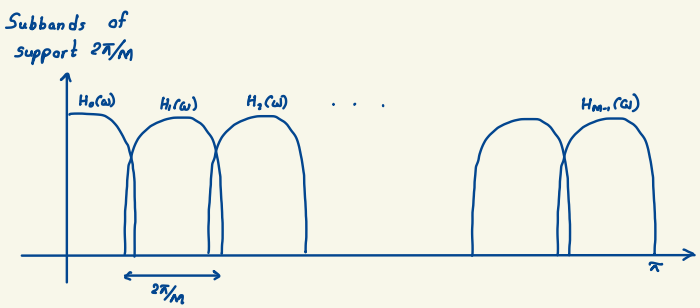

- Shuffling components due to Linear properties and using Noble Identities




- Due to this computational complexity becomes half.

# M-channel Maximally Decimated FB




## How is $\hat{X}(z)$ Related to $X(z)$ Expression of the Reconstructed Signal

$$\hat{X}(z) = \sum_{l=0}^{M-1} A_l(z) \cdot X(W^l z)$$

where $$A_l(z) = \frac{1}{M} \sum_{k=0}^{M-1} \left( H_k(W^l z) \cdot F_k(z) \right) \qquad l = 0, 1, \ldots, M-1 \qquad W = e^{-j 2\pi/M}$$

Proof: From the previous diagram

$$X_k(z) = H_k(z) \cdot X(z) \quad \cdots\cdots\cdots ①$$

$$V_k(z) = \frac{1}{M} \sum_{l=0}^{M-1} X_k(W^l z^{1/M}) \quad \cdots ②$$

$$U_k(z) = V_k(z^M) \quad \cdots\cdots\cdots\cdots ③$$

$$\hat{X}(z) = \sum_{K=0}^{M-1} U_K(z) \cdot F_K(z) \quad \cdots\cdots \textcircled{4}$$

$$\textcircled{0},\textcircled{2} \Rightarrow V_K(z) = \frac{1}{M} \sum_{l=0}^{M-1} H_K\left(W^l z^{1/M}\right) X\left(W^l z^{1/M}\right)$$

$$\textcircled{3} \Rightarrow U_K(z) = \frac{1}{M} \sum_{l=0}^{M-1} H_K\left(W^l z\right) X\left(W^l z\right) \quad \cdots\cdots$$ Substitute $z \to z^M$ in previous line

$$\textcircled{4} \Rightarrow \hat{X}(z) = \sum_{K=0}^{M-1} \left( \frac{1}{M} \sum_{l=0}^{M-1} H_K\left(W^l z\right) X\left(W^l z\right) \right) \cdot F_K(z)$$

$$= \sum_{l=0}^{M-1} \left[ \underbrace{\left( \frac{1}{M} \sum_{K=0}^{M-1} H_K\left(W^l z\right) \cdot F_K(z) \right)}_{A_l(z)} \cdot X\left(W^l z\right) \right]$$

$$= \sum_{l=0}^{M-1} A_l(z) \cdot X\left(W^l z\right)$$

■

(Conside $X(W^l z)$ for $z = e^{j\omega} \rightarrow X(W^l e^{j\omega}) = X(e^{-j2\pi l/m} \cdot e^{j\omega}) = X(e^{j(\omega - 2\pi l/m)})$
$\therefore \hat{X}(e^{j\omega})$ is a linear combination of $X(e^{j\omega})$ and its M-1 shifts.)

## To be Aliasing Free ①

- Ideally we want $A_l(z) = 0$, $l \neq 0$
- $\hat{X}(z) = A_0(z) \cdot X(z)$.
- $A_0(z)$ has the same interpretation as $T(z)$ in the 2-channel case.

# The Alias Component (AC) Matrix

$$\begin{bmatrix} A_0(z) \\ A_1(z) \\ \\ A_{M-1}(z) \end{bmatrix} = \frac{1}{M} \begin{bmatrix} H_0(z) & H_1(z) & \cdots & H_{M-1}(z) \\ H_2(wz) & H_1(wz) & \cdots & H_{M-1}(wz) \\ \vdots & & & \\ H_{M-1}(w^{M-1}z) & H_1(w^{M-1}z) & \cdots & H_{M-1}(w^{M-1}z) \end{bmatrix} \begin{bmatrix} F_0(z) \\ F_1(z) \\ \\ F_{M-1}(z) \end{bmatrix}$$

$$\underline{a}(z) = \mathcal{H}(z) \underline{f}(z)$$

"alias component matrix" — "Transposed Synthesis Bank"

Each column contains all $i^{th}$ aliases of filters.

- To be aliasing free $A_i(z) = 0 \quad \forall i \neq 0$ and $A_0(z) = T(z)$

$$\Rightarrow \underline{a}(z) = \begin{bmatrix} T(z) \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \underline{t}(z)$$

$$\sum_{i=0}^{M-1} H_i(z) \cdot F_i(z) = T(z)$$

$$\sum_{i=0}^{M-1} H_i(w^l z) \cdot F_i(z) = 0 \qquad 0 < l \leq M-1$$

$$\Rightarrow \frac{1}{M} \mathcal{H}(z) \cdot \underline{f}(z) = \underline{t}(z)$$

- Hence if Analysis filters are given (ie $H_i(z)$ $i = 0, 1, \ldots, M-1$) and target transfer $T(z)$ is desired (ex: $T(z) = z^{-k}$) then Synthesis filters are:

$$\underline{f}(z) = M \mathcal{H}^{-1}(z) \cdot \underline{t}(z)$$

If overall effect of an analysis filter is a delay → called perfect reconstruction filter

- Hence if you know what you can tolerate in terms of harmful effects, convert it to $\underline{t}(z)$. If a delay is ok, set it to $z^{-k}$.

- Ex: $H(z) = \begin{bmatrix} z^{-1} & 1-z^{-1} \\ 2+3z^{-1} & -z^{-1} \end{bmatrix} \Rightarrow H(z)^{-1} = \frac{1}{-z^{-2}-2-3z^{-1}} \begin{bmatrix} -z^{-1} & 1-z^{-1} \\ 2+3z^{-1} & z^{-1} \end{bmatrix}$
  - This Denominator can give a pole that's outside the unit circle $\Rightarrow$ Can be unstable
  - Can be FIR
  - Hence this former method can give impractical filters.

- If $H_k(z)$'s are FIR and stable filters, there is no guarantee that the elements of $\underline{f}(z)$ (i.e $M\,\mathcal{H}^{-1}(z)\cdot\underline{t}(z)$) are FIR and stable.

# Polyphase Representation of $\underline{h}(z)$

- ∴ Without trying to design $\underline{f}$ at once, let's take Polyphase components and design them seperately.

$$\underline{h}(z) = \begin{bmatrix} H_0(z) \\ H_1(z) \\ \vdots \\ H_{M-1}(z) \end{bmatrix}$$

- From Type I Polyphase Components of Analysis bank $H_i$ $\longrightarrow$ $H_k(z) = \sum_{l=0}^{M-1} z^{-l} E_{k,l}(z^M)$

$$\underbrace{\begin{bmatrix} H_0(z) \\ H_1(z) \\ \vdots \\ H_{M-1}(z) \end{bmatrix}}_{\underline{h}(z)} = \underbrace{\begin{bmatrix} E_{00}(z^M) & E_{01}(z^M) & \cdots & E_{0,M-1}(z^M) \\ E_{10}(z^M) & E_{11}(z^M) & \cdots & E_{1,M-1}(z^M) \\ \vdots & & & \\ E_{M-1,0}(z^M) & E_{M-1,1}(z^M) & \cdots & E_{M-1,M-1}(z^M) \end{bmatrix}}_{E(z^M)} \underbrace{\begin{bmatrix} 1 \\ z^{-1} \\ \vdots \\ z^{-(M-1)} \end{bmatrix}}_{\underline{e}(z)}$$

"Delay Chain"

$$\underline{h}(z) = E(z^M) \cdot \underline{e}(z)$$

- From Type II Polyphase Components of Synthesis Bank $F_i$ $\longrightarrow$ $F_k(z) = \sum_{l=0}^{M-1} z^{-(M-1-l)} \cdot R_{l,k}(z)$

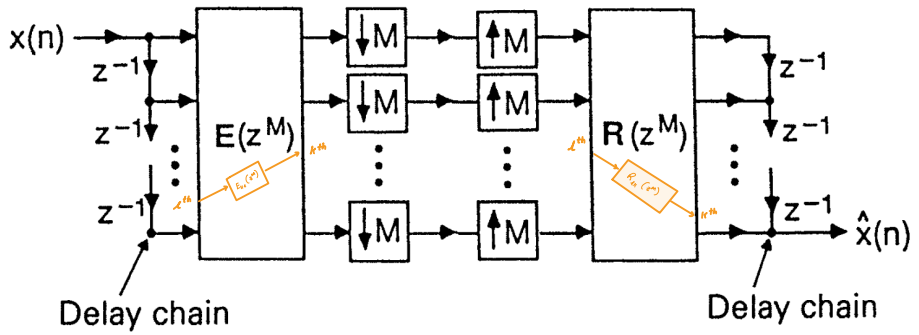
$$\underbrace{\begin{bmatrix} F_0(z) & F_1(z) & \cdots & F_{M-1}(z) \end{bmatrix}}_{\underline{f}^T(z)} = \underbrace{\begin{bmatrix} z^{-M+1} & z^{-M+2} & \cdots & z^{-1} & 1 \end{bmatrix}}_{\underline{e}_B^T(z)} \begin{bmatrix} R_{00}(z^M) & R_{10}(z^M) & \cdots & R_{M-1,0}(z^M) \\ R_{01}(z^M) & R_{11}(z^M) & \cdots & R_{M-1,1}(z^M) \\ \vdots & & & \\ R_{0,M-1}(z^M) & R_{1,M-1}(z^M) & \cdots & R_{M-1,M-1}(z^M) \end{bmatrix}$$

Backward

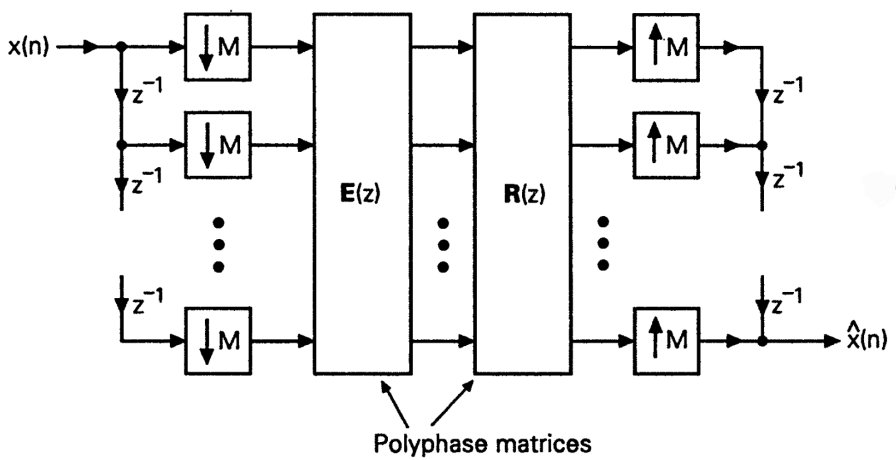
$R(z^M)$

$$\underline{f}^T(z) = \underline{e}_B^T(z) \cdot R(z^M)$$

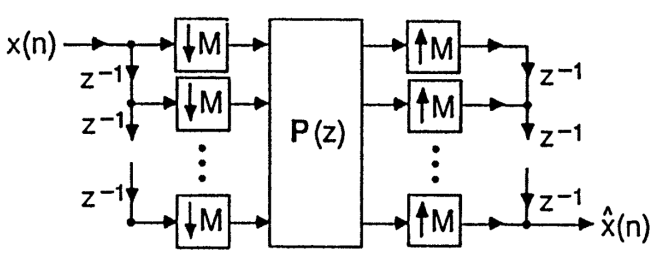
- Hence polyphase representation of M-channel QMF bank




- By using noble identities we can rearrange them so that filter coefficients E(z), R(z) works at lowest rate.




- We combine the matices $P(z) = E(z) \cdot R(z)$



- from polyphase components $\underline{h}(z) = E(z^M) \cdot e(z)$

$$\Rightarrow H^T(z) = [h(z) \quad h(zw) \quad \dots \quad h(z^{M-1}w)]$$

$$= E(z^M) \cdot [e(z) \quad e(wz) \quad \dots \quad e(w^{M-1}z)] \quad \cdots\cdots (1)$$

from definition of Delay chain

$$\Rightarrow e(zw^k) = \begin{bmatrix} 1 \\ z^{-1}w^{-k} \\ \vdots \\ z^{-(M-1)} \cdot w^{-(M-1)\cdot k} \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} 1 & & & & \\ & z^{-1} & & & \\ & & z^{-2} & & \\ & & & \ddots & \\ & & & & z^{-(M-1)} \end{bmatrix}}_{D(z)} \begin{bmatrix} 1 \\ w^{-k} \\ \vdots \\ w^{-(M-1)k} \end{bmatrix} \quad \cdots\cdots (2)$$

From (1) and (2)

$$H(z) = W^+ D(z) \cdot E^T(z^M)$$

## Perfect Reconstruction System

- How should $P(z)$ look like so that entire Filter Bank can achieve perfect reconstruction,

- System has a perfect reconstruction $\Longleftrightarrow$ $R(z)\,E(z) = P(z) = C z^{-m_0} I$ (form exists)

$\Longleftrightarrow$ $T(z) = C z^{-(Mm_0 + M-1)}$

Proof? Pg 237

$\therefore$ If $n_0 = Mm_0 + M-1$

$\hat{x}(n) = C x(n-n_0)$

- We can design the $E_{kl}(z)$ such that $\det(E(z)) = k z^{-n}$. Earlier we could not do this. Now we have lot more degree of freedom. If $E$ is known, $R$ can be found,

$$R(z) = C \cdot z^{-m_0} \cdot E^{-1}(z)$$

- For a FIR "perfect reconstruction" system, we need the above criteria with the fact that $F_i(z)$ s are FIR.

$$\Rightarrow \underline{f}^T(z) = \underline{e}_0^T(z) \cdot R(z) = \underline{e}_0^T(z) \cdot \left[ c \cdot z^{-m_0} \cdot E^{-1}(z) \right]$$

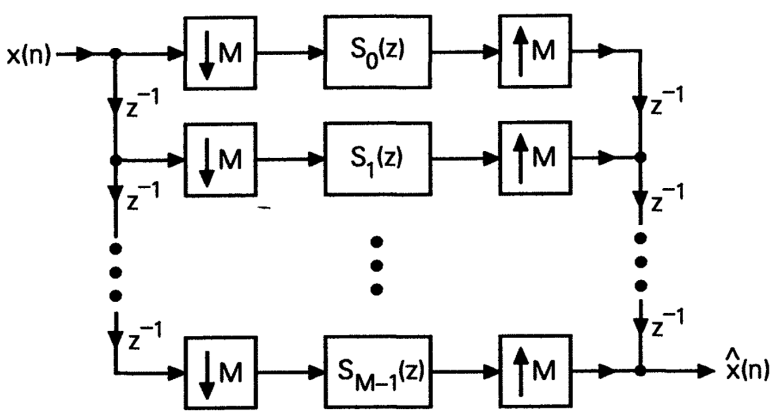
$\Rightarrow$ Problem occurs when $E^{-1}(z)$ has a denominator that gives poles $\neq 0$

$$\therefore \text{PR system that is FIR} \iff \det E(z) = \alpha z^{-k} \quad \alpha \neq 0 \quad k \in \mathbb{Z}$$

## Alias Free Banks

- $P(z) = S(z) \cdot I = \begin{bmatrix} S(z) & \cdots & \cdots & \cdots & - \\ \vdots & & & & \\ \vdots & & S(z) & & \\ \vdots & & & \ddots & \vdots \\ \vdots & & & & S(z) \end{bmatrix} I$

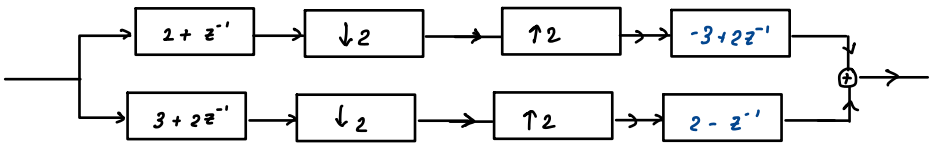
- Above is equivalent to having no mixing in branches.




Exercise :- $T(z) = z^{-(M-1)} \cdot S(z^M) \Rightarrow \hat{X}(z) = z^{-(M-1)} \cdot S(z^M) \cdot X(z)$

* 5.7.2 page 246

Ex:




$E(z) = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \Rightarrow E^{-1}(z) = \frac{1}{2\times 2 - 3\times 1} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

$\therefore$ Setting $R(z) = E^{-1}(z) = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

**Example 5.6.2.**

Let

$$\mathbf{E}(z) = \begin{bmatrix} 1+z^{-1} & 1-z^{-1} \\ 1-z^{-1} & 1+z^{-1} \end{bmatrix}, \tag{5.6.16}$$

which is FIR. Notice that the determinant of this matrix is a delay, as required by (5.6.10). We choose $\mathbf{R}(z)$ to satisfy (5.6.6), that is,

$$\mathbf{R}(z) = cz^{-m_0}\mathbf{E}^{-1}(z) = \frac{cz^{-m_0}}{4} \begin{bmatrix} 1+z & 1-z \\ 1-z & 1+z \end{bmatrix}, \tag{5.6.17}$$

so that the perfect reconstruction condition holds. Choosing $c = 4$ and $m_0 = 1$, this becomes

$$\mathbf{R}(z) = \begin{bmatrix} 1+z^{-1} & -1+z^{-1} \\ -1+z^{-1} & 1+z^{-1} \end{bmatrix}. \tag{5.6.18}$$

The only purpose of $m_0$ has been to avoid the positive powers of $z$ (non causal terms). The analysis and synthesis filters corresponding to the above $\mathbf{E}(z)$ and $\mathbf{R}(z)$ are

$$\begin{aligned} H_0(z) &= 1 + z^{-1} + z^{-2} - z^{-3}, \quad H_1(z) = 1 + z^{-1} - z^{-2} + z^{-3}, \\ F_0(z) &= -1 + z^{-1} + z^{-2} + z^{-3}, \quad F_1(z) = 1 - z^{-1} + z^{-2} + z^{-3}. \end{aligned} \tag{5.6.19}$$

# HAY-2

# Stationary Processes and Models

## Linear Estimations

- Classic DSP : $x[n]$ is deterministic.
  Advanced DSP : $x[n]$ is random variable (r.v / rv)

- A generic discrete stochastic process is a sequence of rv's $\{u[n]\}_{n=-\infty}^{\infty}$. not to be confused with unit step function

- To fully characterise $\{u[n]\}_{n=-\infty}^{\infty}$ we need to know the joint density
  $$p(u[n_1], u[n_2], \dots, u[n_k])$$
  for any $n_1, n_2, \dots, n_k \in \mathbb{N}$ and any $k \in \mathbb{N}^+$

- In general this is infeasible. ⟶ Instead we only focus on 1st and 2nd moments of the $\{u[n]\}_{n=-\infty}^{\infty}$

### The 1st Moment

$$m[n] := E\{u[n]\} \quad \forall n \in \mathbb{Z}$$

$$= \int p(u[n]) \cdot u[n] \cdot du[n] = \int x \, p_n(x) \, dx$$

$p_n$ : density of $u[n]$

This is shown as a Lebesgue integration in 620.

### The 2nd Moment : Auto-correlation

$$r(n, n-k) := E\{u[n] \cdot u^*[n-k]\} \quad \forall n, k \in \mathbb{Z}$$

: Auto-Covariance

$$c(n, n-k) := E\{(u[n] - m[n])(u^*[n-k] - m^*[n-k])\}$$

- This is still too much to track. Hence we define WSS.

# Wide Sense Stationary (WSS)

If, $\{u_n\}_{n=-\infty}^{\infty}$

(1) $E[u(n)] = m(n) = m$ (independant of $n$)

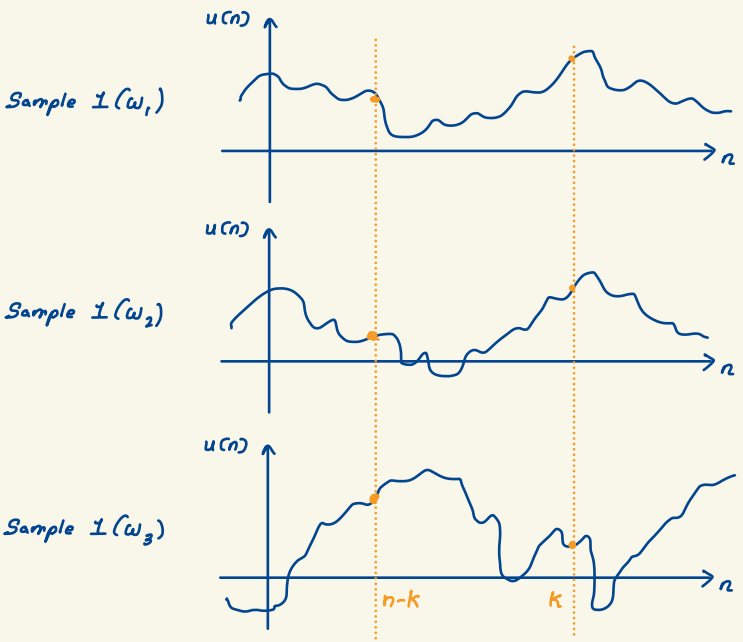
(2) $E[u(n)\cdot u^*(n-k)] = r(n, n-k) = r[k]$ $\forall\ n,k \in \mathbb{Z}$ (Auto correlation sequence)

$c(n, n-k) = c[k]$ $\forall\ n,k \in \mathbb{Z}$ (Auto covariance sequence)

a WSS process can be characterised by $m$ and $r[k]$ $k \in \mathbb{Z}$.

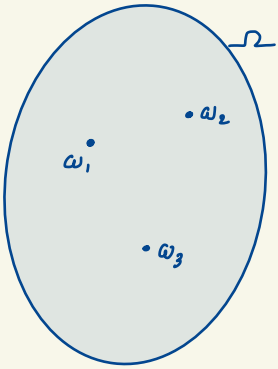
$$c[k] = E\left[(u(n)-m)(u[n-k]^* - m^*)\right] = r[k] - |m|^2$$

- An example - $\omega_1, \omega_2, \ldots, \omega_n$ are the temperature curves of a room. Each day we have a new realization of this rv.




$\{u_\omega[n]\}_{n=-\infty}^{\infty}$

pick $\omega$ - a member of a porbability space



## Strict Sense Stationary (SSS)

$P(u[n])$ is Independant of $n$

$P(u[n], u[n-k])$ is only a function of $k$.

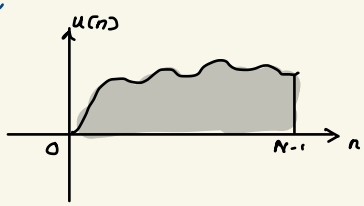
$\therefore$ WSS $\Rightarrow$ SSS

$\not\Leftarrow$

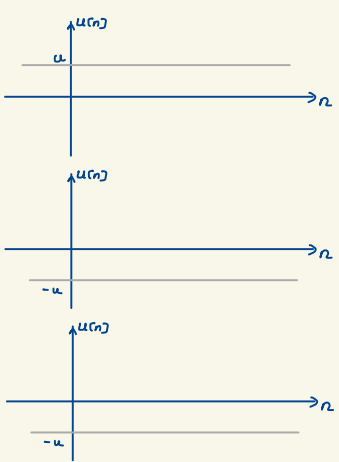
## Statistical Averaging vs Sample Averaging

- Statistical averaging $\longrightarrow$ The process mentioned earlier with covariances.
- Sample averaging of a process is,

$$\hat{m}[n] = \frac{1}{N} \sum_{n=0}^{N-1} u[n]$$




- As $N \to \infty$ would $\lim_{K \to \infty} \hat{m}(N) = m$? . Ingeneral no. Consider the following example for a counter point.
- Ex: $u[n] = u \quad \forall n \in \mathbb{N}$. where $u$ is a zero-mean r.v.




$m = E[u] < 0$

# Mean Ergodic Theorem

- The challange in quantifying $m$ is that we need to know $p(u[n])$ for some $n$, to compute: $\int u[n]\, p(u[n]) \cdot du[n]$
- Instead we set a condition s.t. $\hat{m}(n)$ converges to $m$ in the "Mean Square Error" sense,

$$\lim_{N\to\infty} E\left[|\hat{m}(N) - m|^2\right] = 0 \quad \Rightarrow \quad \text{Called "Mean Ergodic Process"}$$

- Ergodicity is assumed in most of the scientific processes where something is observed over a long time and averaged.

$$E\left[|\hat{m}(N) - m|^2\right] = E\left[\left|\frac{1}{N}\sum_{n=0}^{N-1} u[n] - m\right|^2\right]$$

$$= E\left[\left|\frac{1}{N}\sum_{n=0}^{N-1} (u[n]-m)\right|^2\right]$$

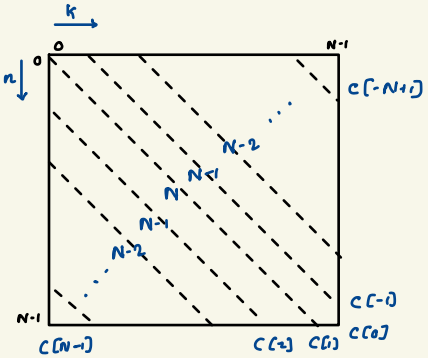
$$= E\left[\frac{1}{N^2}\left\{\sum_{n=0}^{N-1}(u[n]-m)\right\}\left\{\sum_{n=0}^{N-1}(u^*[k]-m^*)\right\}\right]$$

$$= \frac{1}{N^2}\sum_{k=0}^{N-1}\sum_{n=0}^{N-1} \underbrace{E\left[(u[n]-m)(u^*[n]-m)\right]}_{C[n-k]}$$

$$= \frac{1}{N^2}\sum_{k=0}^{N-1}\sum_{n=0}^{N-1} C[n-k]$$

$$= \frac{1}{N^2}\sum_{l=-N+1}^{N-1} (N-|l|)\, C[l]$$

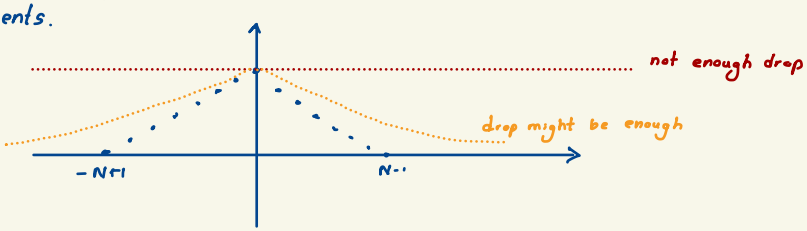
$$= \frac{1}{N}\sum_{l=-N+1}^{N-1}\left(1-\frac{|l|}{N}\right)\cdot C[l]$$




Mean Ergodicity:

$$\lim_{N\to\infty} \frac{1}{N}\sum_{l=-N+1}^{N-1}\left(1-\frac{|l|}{N}\right)\cdot C[l] = 0$$

- This generally means when auto covariance drops (sufficiently fastly) when the events considered are spread in time, then it can be "mean ergodic". ∴ Practically also this is assumed in scientific experiments.




Sample Auto-correlation

$$\hat{r}(N,k) := \frac{1}{N} \sum_{n=0}^{N-K+1} u[n] \cdot u^*[n-k]$$

Correlation Ergodicity

$$\lim_{N\to\infty} E\left[ \left| \hat{r}(N,k) - r(k) \right|^2 \right] = 0 \quad \forall k$$

- Mean ergodicity is not enough, we need correlation ergodicity ?

# Correlation Matrix

- We consider a consecutive sequence of $\{u(n)\}_{n=-\infty}^{\infty}$ of length $M$ ending at time $n$:

$$\ldots, u[n-M+1], u[n-M+2], \ldots, u[n-1], u[n], u[n+1], .$$

$$\underline{u}[n] = \begin{bmatrix} u[n] \\ u[n-1] \\ u[n-2] \\ \vdots \\ u[n-M+1] \end{bmatrix} \in \mathbb{C}^M$$

Correlation Matrix of order M $= R = R_M = E\left[\underline{u}[n] \cdot \underline{u}^H[n]\right]$

$$\begin{bmatrix} r[0] & r[1] & \cdots & r[M-1] \\ r[-1] & r[0] & \cdots & r[M-2] \\ \vdots & & & \\ r[-M+1] & \cdots & r[-1] & r[0] \end{bmatrix}$$

Proof:-

$$\underline{u}[n] \cdot \underline{u}^H[n] = \begin{bmatrix} u[n] \\ \vdots \\ u[n-M+1] \end{bmatrix} \begin{bmatrix} u^*[n] & \cdots & u^*[n-M+1] \end{bmatrix}$$

$$= \begin{bmatrix} u[n]\cdot u^*[n] & u[n]\cdot u^*[n-1] & \cdots & u[n]\cdot u^*[n-M+1] \\ u[n-1]\cdot u^*[n] & u[n-1]\cdot u^*[n-1] & \cdots & u[n-1]\cdot u^*[n-M+1] \\ \vdots & & & \\ u[n-M+1]\, u^*[n] & u[n-M+1]\cdot u^*[n-1] & & u[n-M+1]^*\, u[n-M+1] \end{bmatrix}$$

$$R = E\left[\underline{u}[n]\cdot\underline{u}^H[n]\right] = \begin{bmatrix} E[u[n]\cdot u^*[n]] & E[u[n]\cdot u^*[n-1]] & \cdots & E[u[n]\cdot u^*[n-M+1]] \\ E[u[n-1]\cdot u^*[n]] & E[u[n-1]\cdot u^*[n-1]] & \cdots & E[u[n-1]\cdot u^*[n-M+1]] \\ \vdots & & & \\ E[u[n-M+1]\, u^*[n]] & E[u[n-M+1]^*\cdot u[n-1]] & \cdots & E[u[n-M+1]^*\, u[n-M+1]] \end{bmatrix}$$

$$= \begin{bmatrix} r[0] & r[1] & \cdots & r[M-1] \\ r[-1] & r[0] & \cdots & r[M-2] \\ \vdots & & & \\ r[-M+1] & \cdots & r[-1] & r[0] \end{bmatrix}$$

... If the system is WSS, autocorrelation $r(n,n-k) = r(k) = E[u[n]\cdot u^*[n-k]]$

■

## Properties of Correlation Matrix - $R_M$

of a WSS Process

1) $R_M$ is a Toeplitz matrix when $\underline{u}[n]$ is WSS.
i.e. all entries along the diagonals are equal.

- For real valued data $r(-k) = r(k)^* = r(k)$. Hence correlation matrix is symmetric.

Hermissian operation

$$(\ )^H = [(\ )^*]^T$$

2) $R_M$ is Hermitian $\Rightarrow$ $r[i-j] = r^*[j-i]$

$$R_M^H = R_M$$

Proof:

$$r[k] = E[u[n] \cdot u^*[n-k]]$$

$$r^*[k] = E[u^*[n]\, u[n-k]]$$

$$r^*[-k] = E[u^*[n] \cdot u[n+k]] = E[u[n+k] \cdot u^*[n]] = r[k] \quad \forall k \in \mathbb{Z}$$

If wss

$$r^*[0] = r[0] \Rightarrow r[0] \in \mathbb{R}.$$

3) $R_M$ is non-negative definite (i.e. positive semidefinite) $\Rightarrow$ $\forall\ \underline{x} \in \mathbb{C}^M$ $\underline{x}^H R_M \underline{x} \geq 0$

And almost always positive definite.

Proof: Claim: $\underline{x}^H R_M \underline{x} \in \mathbb{R}$

and $\Rightarrow$ Correlation matrix is almost always non-singular. Has an inverse.

$$(\underline{x}^H R_M \underline{x})^* = (\underline{x}^H R_M \underline{x})^H \quad \cdots\cdots$$

Since $\underline{x}^H R_M \underline{x}$ is a scalar, geting a transpose is not required

$$= \underline{x}^H R_M^H (\underline{x}^H)^H$$

$$= \underline{x}^H R_M x \quad \cdots\cdots \text{ Property (2)}$$

$$\underline{x}^H R_M \underline{x} = \underline{x}^H E\{u[n] \cdot \underline{u}^H[n]\} \cdot \underline{x}$$

$$= E\{\underline{x}^H u[n] \cdot \underline{u}^H[n] \cdot \underline{x}\}$$

$$= E\{\underbrace{\underline{x}^H u[n]}_{z} \cdot \underbrace{(\underline{x}^H \cdot \underline{u}[n])^*}_{z^*}\} \quad \cdots\cdots (\underline{x}^H)^* = \underline{x}$$

$$= E\{z \cdot z^*\}$$

$$= E\{|z|^2\} \geq 0$$

4) Eigenvalues of $R_M$ are $\mathbb{R}$

Proof: If $\underline{v}$, $\lambda$ are eigenvector, eigenvalue pairs,

$$\Rightarrow R_M \cdot \underline{v} = \lambda \underline{v}$$

$$\Rightarrow \underline{v}^H R_M \underline{v} = \lambda \underbrace{(\underline{v}^H v)}_{I} \quad \cdots\cdots$$

multiply by $\underline{v}^H$

$\lambda$ scalar

Why?

$$\Rightarrow \quad \lambda = \underline{v}^H R_M \underline{v}$$

$$\lambda^* = \lambda^H = \left(\underline{v}^H R_M \underline{v}\right)^H = \underline{v}^H \cdot R_M^H \cdot (\underline{v}^H)^H = \underline{v}^H R_M \cdot \underline{v} = \lambda \quad \Longrightarrow \quad \lambda \in \mathbb{R}$$

5) Backward ordering property.

$$\underline{u}[n] = \begin{bmatrix} u[n] \\ u[n-1] \\ \vdots \\ u[n-M+1] \end{bmatrix} \longleftrightarrow R_M \qquad \underline{u}^B[n] = \begin{bmatrix} u[n-M+1] \\ u[n-M+2] \\ \vdots \\ u[n] \end{bmatrix} \longleftrightarrow R_M^T$$

6) $R_M$ admits recursive construction

$$\underline{u}[n] \longleftrightarrow R_M$$

$$\begin{bmatrix} u[n+1] \\ \underline{u}[n] \end{bmatrix} \longleftrightarrow R_{M+1}$$

$$R_{M+1} = \left[\begin{array}{c|c} r[0] & \underline{r}^H \\ \hline \underline{r} & R_M \end{array}\right] = \left[\begin{array}{c|c} R_M & \underline{r}^{B*} \\ \hline \underline{r}^{BT} & r_0 \end{array}\right]$$

Don't get confused with $\underline{r}$ (vector) and $r$ (scalar).

$$\underline{r} = \begin{matrix} r(-1) \\ r(-2) \\ \vdots \\ r(-M) \end{matrix}$$

Proof: From definition of $R_M$

$$R_{M+1} = E\left\{ \begin{bmatrix} u[n+1] \\ \underline{u}[n] \end{bmatrix} \begin{bmatrix} u^*[n+1] & \underline{u}^H[n] \end{bmatrix} \right\}$$

$$R_{M+1} = \left[\begin{array}{c|ccccc} r[0] & r[1] & r[2] & \cdots & r[M-1] & r[M] \\ \hline r[-1] & r[0] & r[1] & \cdots & r[M-2] & r[M-1] \\ \vdots & \vdots & & & \vdots & \vdots \\ r[-M+1] & r[-M+2] & r[-M+3] & \cdots & r[0] & r[1] \\ r[-M] & r[-M+1] & r[-M+2] & \cdots & r[2] & r[0] \end{array}\right] \begin{matrix} \\ \\ \\ \\ R_M \end{matrix}$$

$\underline{r}$

Let $\underline{r} = \begin{bmatrix} r(-1) \\ r(-2) \\ \vdots \\ r[-M] \end{bmatrix} = \begin{bmatrix} r^*[1] \\ r^*[2] \\ \vdots \\ r^*[M] \end{bmatrix} \Longrightarrow R_M = \begin{bmatrix} r[0] & \underline{r}^H \\ \underline{r} & R_M \end{bmatrix}$

Similarly we can divide $R_m$ as follows,

$$R_{M+1} = \begin{bmatrix} r[0] & r[1] & r[2] & \cdots & r[M-1] & r[M] \\ r[-1] & r[0] & r[1] & \cdots & r[M-2] & r[M-1] \\ \vdots & \vdots & & & \vdots & \vdots \\ r[-M+1] & r[-M+2] & r[-M+3] & \cdots & r[0] & r[1] \\ r[-M] & r[-M+1] & r[-M+2] & \cdots & r[2] & r[0] \end{bmatrix}$$

($R_m$ : top-left block; $(\underline{r}^B)^*$ : top-right column; $(\underline{r}^B)^T$ : bottom-left row)

$$\Rightarrow R_{M+1} = \begin{bmatrix} R_m & (\underline{r}^B)^* \\ (\underline{r}^B)^T & r_0 \end{bmatrix}$$

## White Noise Process

Definition - A WSS process is called 'White Noise' if,

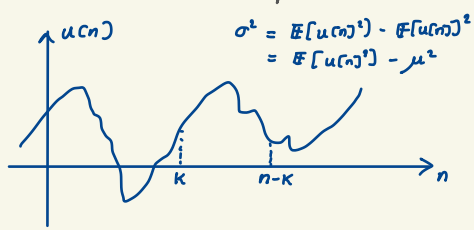
$$c[k] = \sigma^2 \delta[k] = \begin{cases} \sigma^2 & k = 0 \\ 0 & o.w \end{cases} \quad \text{for some } \sigma^2 > 0$$

Ex: Show that $u[n] \overset{iid}{\sim} N(\mu, \sigma^2)$ is a white noise process

Answer: $r[k] = E\{u[n] \cdot u[n-k]\}$

$$= \begin{cases} E\{u[n] \cdot u[n-k]\} & k \neq 0 \\ E\{u[n]^2\} & k = 0 \end{cases}$$

$$= \begin{cases} \mu^2 & k \neq 0 \\ \mu^2 + \sigma^2 & k = 0 \end{cases}$$




$\sigma^2 = E[u[n]^2] - E[u[n]]^2$
$= E[u[n]^2] - \mu^2$

$$\Rightarrow c[k] = r[k] - \mu^2 = \begin{cases} 0 & k \neq 0 \\ \sigma^2 & k = 0 \end{cases} \Rightarrow u[n] \text{ is a white noise}$$

Ex: Complex Harmonic Process

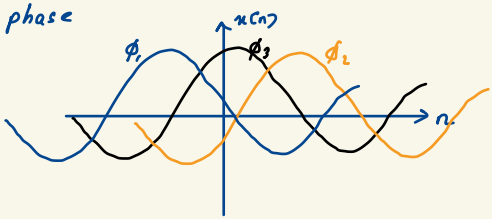
$$x[n] = A \cdot e^{j(2\pi f_0 n + \phi)}$$

Show that $x[n]$ is W.S.S and find its mean and auto correlation.

$A > 0$ - real constant
$f_0$ a real constant
$\phi \sim$ uniform $[-\pi, \pi)$

Answer $\quad Re\{x[n]\} = A \cdot \cos(2\pi f_0 n + \phi)$

Consider $\phi_1, \phi_2, \phi_3$ realizations $\therefore$ All are sign waves, with only difference of starting phase




$$E\{x[n]\} = \int_{-\pi}^{\pi} \frac{1}{2\pi} A \cdot e^{j(2\pi f_0 n + \phi)} d\phi = 0 \quad \cdots \quad f_\phi(\phi) = \frac{1}{2\pi} \quad -\pi < \phi < \pi$$

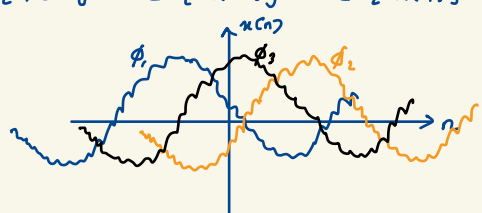
$$E\{x[n] \cdot x^*[n-k]\} = E\left\{ A\, e^{j(2\pi f_0 n + \phi)} \cdot A\, e^{-j(2\pi f_0 (n-k) + \phi)} \right\}$$

$$= A^2 e^{j2\pi f_0 k} \cdot$$

$$= r[k].$$

Ex: $Y[n] = X[n] + u[n]$ Combination from previous 2 examples. Assume $X[n]$ and $u[n]$ are independant. Show $Y[n]$ is wss.

$$E\{Y[n]\} = E\{x[n]\} + E\{u[n]\} = 0$$




$$E\{Y[n] \cdot Y^*[n-k]\} = E\left\{ \left(A \cdot e^{j2\pi f_0 n + \phi} + u[n]\right) \cdot \left(A \cdot e^{-j(2\pi f_0 (n-k) \cdot + \phi} + u[n]\right) \right\}$$

$$= A^2 \cdot e^{j2\pi f_0 k} + \sigma^2 \cdot \delta[k] = r[k] \Rightarrow \text{White noise}$$

**Ex:** Find $R_M$ for $u[n]$, $x[n]$, $y[n]$

**Answer:** ▶ For $u[n] \overset{iid}{\sim} N(0, \sigma^2) \Rightarrow \gamma_u[k] = \sigma^2 \delta[k]$

$$R_M = \begin{bmatrix} \gamma[0] & \gamma[1] & \cdots & \gamma[M-1] \\ \gamma[-1] & \gamma[0] & \cdots & \\ \vdots & & & \\ \gamma[-M+1] & \cdots & & \gamma[0] \end{bmatrix} = \begin{bmatrix} \sigma^2 & \cdots & - \\ - & \sigma^2 & \ddots \\ & \ddots & \sigma^2 \end{bmatrix} = \sigma^2 I$$

▶ For $x[n] = A \cdot e^{j2\pi f_0 n + \phi} \Rightarrow \gamma_x[k] = A^2 \cdot e^{j2\pi f_0 k}$

$$R_M = A^2 \begin{bmatrix} 1 & e^{j2\pi f_0} & e^{j2\pi 2f_0} & \cdots & e^{j2\pi(M-1)f_0} \\ e^{-j2\pi f_0} & 1 & \ddots & \ddots & \vdots \\ e^{-j2\pi 2f_0} & \ddots & \ddots & \ddots & e^{j2\pi 2f_0} \\ \vdots & \ddots & \ddots & 1 & e^{j2\pi f_0} \\ e^{-j2\pi(M-1)f_0} & \cdots & e^{-j2\pi 2f_0} & e^{-j2\pi f_0} & 1 \end{bmatrix}$$

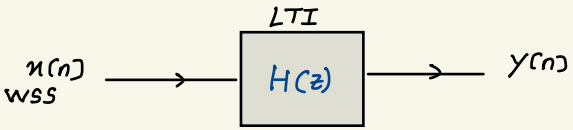
$$R_M = A^2 \begin{bmatrix} 1 \\ e^{-j2\pi f_0} \\ e^{-j2\pi 2f_0} \\ \vdots \\ e^{-j2\pi(M-1)f_0} \end{bmatrix} \begin{bmatrix} 1 & e^{j2\pi f_0} & e^{j2\pi 2f_0} & \cdots & e^{j2\pi(M-1)f_0} \end{bmatrix}$$

If $\underline{e}(f_0) = \begin{bmatrix} 1 \\ e^{-j2\pi f_0} \\ e^{-j2\pi 2f_0} \\ \vdots \\ e^{-j2\pi(M-1)f_0} \end{bmatrix} \Rightarrow R_M = A^2 \underline{e}(f_0) \cdot \underline{e}^H(f_0)$

Delay line

▶ $\therefore$ for $y[n] \longrightarrow R_M = A^2 \underline{e}(f_0) \cdot \underline{e}^H(f_0) + \sigma^2 I$

# WSS Process and LTI Systems




Is $y(n)$ WSS. If so find it's mean and variance.

- We know from WSS property $\longrightarrow$ $E[x(n)] = m_x$

  $E[x(n)\, x^*[n-k]] = r_x[k] \quad \forall k$

- Since $y(n)$ is the output of system $\longrightarrow$ $y(n) = \sum_{k=-\infty}^{\infty} x[n-k] \cdot h[k]$

$$\therefore E\{y(n)\} = E\left\{\sum_{k=-\infty}^{\infty} x[n-k]\, h[k]\right\}$$

$$= \sum_{k=-\infty}^{\infty} E\{x[n-k]\} \cdot h[k] \quad \cdots\cdots$$ Assuming BIBO we can take $\Sigma$ out due to absolute summable

$$= m_x \sum_{k=-\infty}^{\infty} h[k] \quad \cdots\cdots\cdots \quad E\{x[n-k]\} = m$$

$$\boxed{m_y = m_x \cdot H(0)} \quad \cdots\cdots\cdots \quad \sum_{k=-\infty}^{\infty} h[k] = H(0)$$

- First compute cross correlation of $x(n)$ and $y(n)$

$$E\{y[n+k] \cdot x^*[n]\} = E\left\{\left(\sum_{l=-\infty}^{\infty} x[n+k-l] \cdot h[l]\right) \cdot x^*[n]\right\}$$

$$= \sum_{l=-\infty}^{\infty} h[l] \cdot \underbrace{E\{x[n+k-l] \cdot x^*[n]\}}_{r_x''[k-l]} \quad \cdots$$ BIBO assumption

$$= r_x[k] * h[k]$$

- Then what we actually need is,

$$E\{y[n+k]\cdot y^*[n]\} = E\left\{y[n+k]\cdot\left(\sum_{l=-\infty}^{\infty} x^*[n-k]\cdot h^*[l]\right)\right\}$$

$$= \sum_{l=-\infty}^{\infty} h^*[l]\cdot E\{y[n+k]\cdot x^*[n-l]\}$$

$$= r_x[k+l] * h[k+l]$$

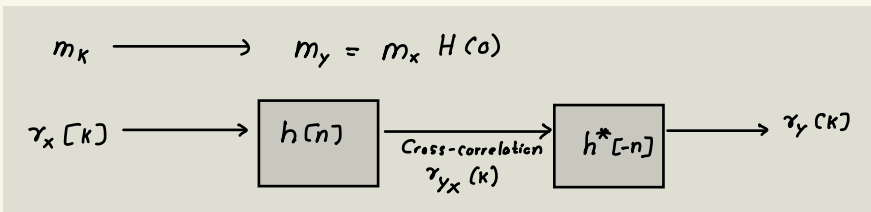
from previous cross correlation

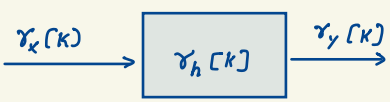
Substitute $l' \to -l$

$$= \sum_{l'=-\infty}^{\infty} h^*[-l']\cdot E\{r_x[k-l'] * h[k-l']\}$$

$$\boxed{r_y[k] = r_x[k] * h[k] * h^*[-k]}$$

..... determint auto correlation of $h(n)$




$h[k] * h^*[-k]$

deterministic auto-corelation of $h[k]$

# Power Spectral Density (PSD)

- PSD of WSS process is,

$$P_x(\omega) := \mathrm{DTFT}\{\gamma_x[k]\} = \sum_{k=-\infty}^{\infty} \gamma_k[k] \cdot e^{-j\omega k}$$

$$\gamma_x[k] = \frac{1}{2\pi} \int_0^{2\pi} P_x(\omega) \cdot e^{j\omega k} \, d\omega$$

$\gamma_x[k]$ autocorrelation of $x$ for lag $k$.

- We can define the $z$-Transform of $\gamma_x[k]$

$$P_x(z) := \sum_{k=-\infty}^{\infty} \gamma_x[k] \cdot z^{-k}$$

## Properties of PSD

1) Although $\gamma_x[k]$ can be complex valued, due to it's conjugate symmetry. $P_x(\omega)$ is always real valued.

$$\gamma_x[k] = \gamma_x^*[k] \quad \forall k \implies P_x(\omega) = P_x^*(\omega)$$

$$P_x(z) = P_x^*\left(\frac{1}{z^*}\right)$$

2) The power of a WSS process $E[|x[n]|^2]$ is propotional to the area under $P_x(\omega)$.

$$E[|x[n]|^2] = \frac{1}{2\pi} \int_0^{2\pi} P_x(\omega) \cdot d\omega$$

Proof: $E[|x[n]|^2] = \gamma_x[0] = \frac{1}{2\pi} \int_0^{2\pi} P_x(\omega) \cdot d\omega$

3) If $\gamma_x[k]$ is also real valued then PSD is symmetric around zero.

$$P_x(\omega) = P_x(-\omega)$$

$$P_x(z) = P_x^*(z^*)$$

4) WSS and LTI systems in the DTFT / z-Domain turns out to be,

$$P_y(z) = P_x(z) \cdot H(z) \cdot H^*\left(\frac{1}{z^*}\right)$$

$$P_y(\omega) = P_x(\omega) \cdot H(\omega) \cdot H^*(\omega) = P_x(\omega) \cdot |H_x(\omega)|^2$$

Proof: Claim:- $h[k] \xleftrightarrow{z} H(z)$ then $h^*[-k] \xleftrightarrow{z} H^*\left(1/z^*\right)$

Proof :- $$A(z) = \sum_{k=-\infty}^{\infty} h^*[-k] \cdot z^{-k}$$

$$A^*(z) = \sum_{k=-\infty}^{\infty} h[-k] \cdot (z^*)^{-k}$$

$$= \sum_{k'=-\infty}^{\infty} h[k'] \cdot (z^*)^{k'} \quad \cdots \text{ Substitute } -k \to k'$$

$$= \sum_{k'=-\infty}^{\infty} h[k'] \cdot \left(\frac{1}{z^*}\right)^{-k'}$$ Put $k' \to k$

$$= \sum_{k=-\infty}^{\infty} h[k] \cdot \left(\frac{1}{z^*}\right)^{-k} = H\left(z = 1/z^*\right) = H\left(1/z^*\right)$$

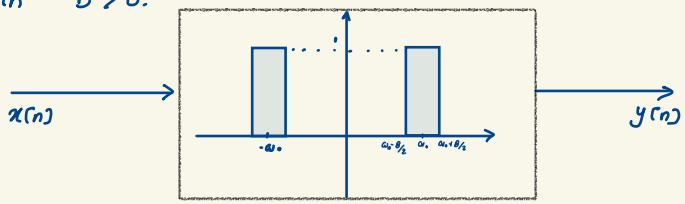
$$\Rightarrow \left(A^*(z)\right)^* = H^*\left(\frac{1}{z^*}\right) \Rightarrow A(z) = H^*\left(1/z^*\right)$$ ∎

5) If $h[n]$ is real valued $\Longrightarrow P_y(z) = P_x(z) \cdot H(z) \cdot H^*(z^*)$

6) The PSD $P_x(\omega)$ is non-negative, $P_x(\omega) \geq 0 \quad \forall \omega$

# What is PSD ?

- $H(\omega)$ is an ideal bandpass filter (BPF) at $\omega_0$ with symmetric bandwidth $B > 0$.




- Let's compute the power of $y[n]$ when $B \ll 1$ :

$$E[|y[n]|^2] = r_y[0] = \frac{1}{2\pi}\int_0^{2\pi} P_y(\omega)\, d\omega$$

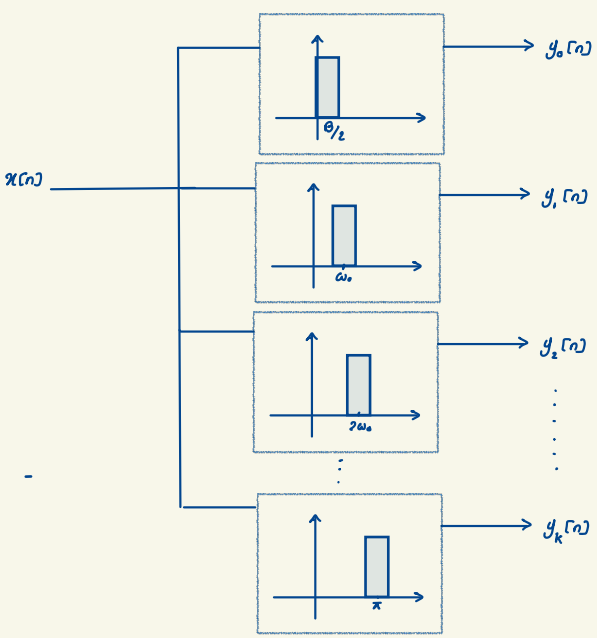
$$= \frac{1}{2\pi}\int_0^{2\pi} P_x(\omega)\cdot |H(\omega)|^2\, d\omega$$

$$= \frac{1}{2\pi}\int_{\omega_0 - B/2}^{\omega_0 + B/2} P_x(\omega)\cdot d\omega \quad \cdots\cdots$$

For simplicity consider the one sided BPF with range $\omega_0 - B/2$, $\omega_0 + B/2$

$$= \frac{B}{2\pi} P_x(\omega_0) + O(B^2) \quad \cdots\cdots$$

from Mean Value Theorem and Taylor Expansion



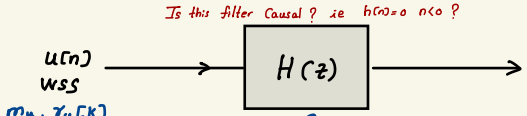
Auto-regressive Moving Average (ARMA) Modeling

# Stochastic Models

1) Autoregressive models
2) Moving average models
3) Mixed autoregressive and moving average models

Suppose $H(z)$ is a rational transfer function

Is this filter Causal ? ie $h[n]=0$ $n<0$ ?




$m_x = H(0) \cdot m_u$

$r_x[k] = r_u[k] * h[k] * \tilde{h}[-k]$

If we observe $x[n]$ for a finite horizon, can we find $H(z)$ that could have produced it based on the input $u[n]$ ?

- Almost all LTI system can be approximated arbitarily well with a rational transfer function. (Yule 1927)

$$H(z) = \frac{\sum_{l=0}^{q} b[l] \cdot z^{-l}}{\sum_{l=0}^{p} a[l] \cdot z^{-l}}$$

- Yule's logic - Time series $x[n]$ consisting of highly correlated observations may be generated by applying a series of statistically independant shocks to a Linear filter

Present value of model output $a[0] \cdot x[n]$ + linear combination of past values of model output $\sum_{l=1}^{p} a[l]\, x[n-l]$ = linear combination of present and past values of model output $\sum_{l=0}^{q} b[l] \cdot u[n-l]$

Equivalent to having a constant coefficient difference equation.

$$a[0] \cdot x[n] = -\sum_{l=1}^{p} a[l] \cdot x[n-l] + \sum_{l=0}^{q} b[l] \cdot u[n-l]$$

" Auto-regression on $x[n]$ " " moving average of $u[n]$ "

$$\Rightarrow H(z) = \frac{X(z)}{U(z)}$$

ARMA (p, q)

p - degree of denominator

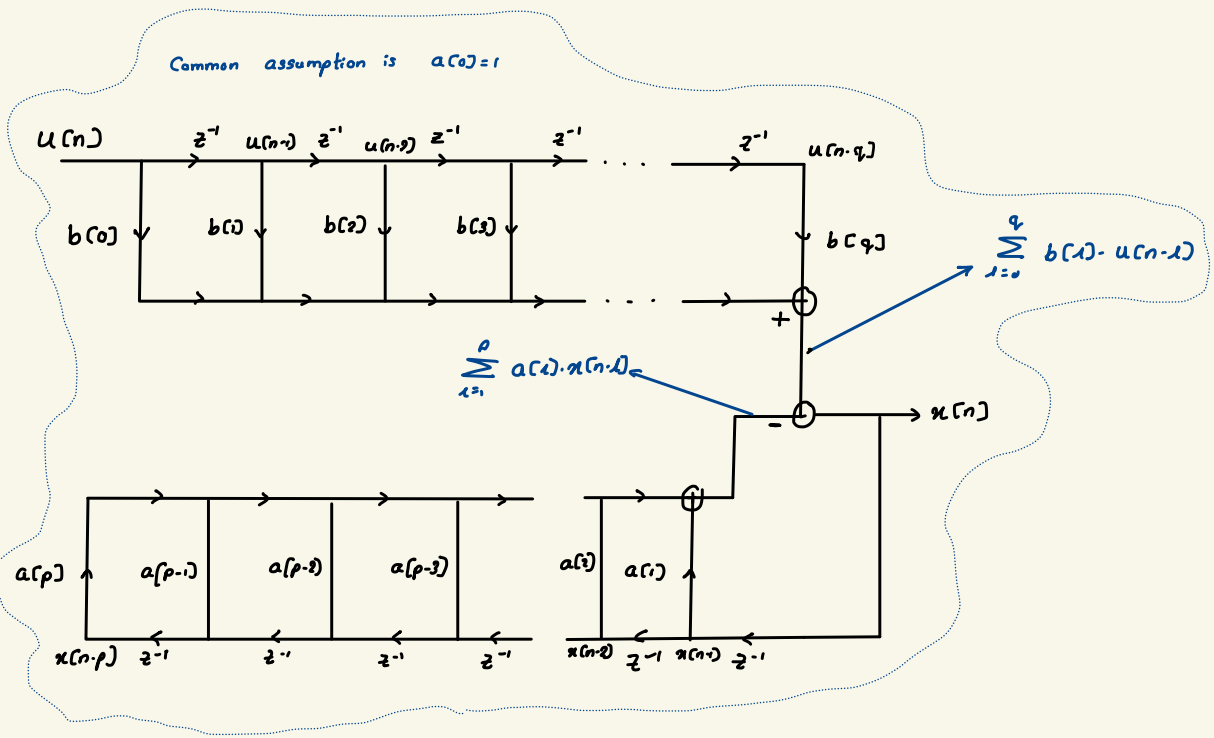
q - degree of numerator

- The common assumption is that $u[n]$ is zero mean and white with variance $\sigma^2$, then

$$r_u[k] = \sigma^2 \delta[k]$$

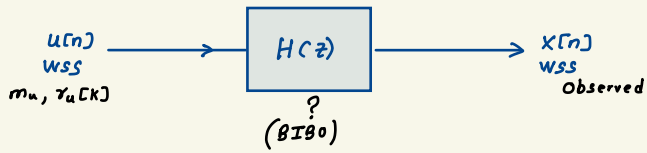
$$P_u(\omega) = \sigma^2$$

$$H(z) = \frac{B(z)}{A(z)} \longrightarrow P_x(\omega) = \sigma^2 |H(\omega)|^2 = \sigma^2 \frac{|B(\omega)|^2}{|A(\omega)|^2}$$



# Motivation for ARMA Models

1) Suppose $H(z)$ is a rational transfer function




$$m_X = H(0) \cdot m_u$$
$$r_X[k] = r_u[k] * h[k] * h^*[-k]$$
$$P_X(\omega) = P_u(\omega)\,|H(\omega)|^2$$

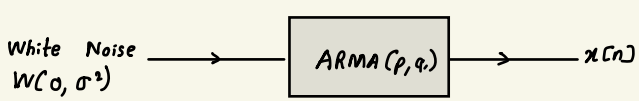
If we observe $X[n]$ for a finite horizon, can we find $H(z)$ that could have produced it based on the input $u[n]$?

2) Allmost all LTI systems can be approximated by an ARMA $(p,q)$ for large enough $p$ and $q$

$$H(z) = \frac{\sum_{l=0}^{q} b[l] \cdot z^{-l}}{\sum_{l=0}^{p} a[l] \cdot z^{-l}}$$

$$a[0] \cdot X[n] = -\sum_{l=1}^{p} a[l] \cdot x[n-l] + \sum_{l=0}^{q} b[l] \cdot u[n-l]$$

3) ARMA can be a useful generative model for WSS processes.



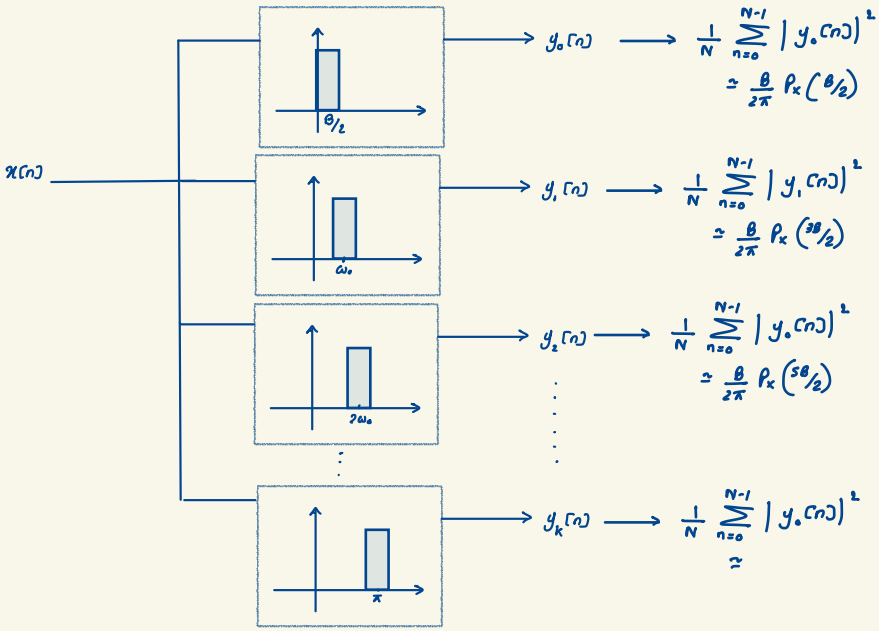

$$P_X(\omega) = P_u(\omega) \cdot |H(\omega)|^2 = \sigma^2 |H(\omega)|^2$$

$P_X(\omega)$: measure in the lab; $\sigma^2 |H(\omega)|^2$: model

Sometimes $P_x(\omega)$ is easy to measure. (eg:- RF signals)

Sometimes $r_x[k]$ is easy to estimate. (eg:- Correlation ergodic process and large $N$)

$$\hat{\gamma}_x[k] = \frac{1}{N}\sum_{n=k}^{N-1} x[n]\cdot x^*[n-k] \xrightarrow[N\to\infty]{\text{quadratic mean}} \gamma_x[k] = E\left[x[n]\cdot x^*[n-k]\right]$$




**Quadratic Mean Convergence**

$$\lim_{N\to\infty} E\left\{\left|\hat{\gamma}_x[k]-\gamma_x[k]\right|^2\right\} = 0 \iff \hat{\gamma}_x[k] \xrightarrow{q.m.} \gamma_x[k]$$

# Yule - Walker Equations

→ System of linear equations to solve AR coefficients

- Let's assume that $\hat{r}_x[k]$ is quite accurate and so that $r_x[k]$ is known.
  By observing the output values for N samples

- Find $a[l]$, $b[l]$ given $r_x[k]$ for $l = 1, 2, \ldots, p$ $k = \pm 1, \pm 2, \ldots$
  Assume $a[0] = b[0] = 1$

$$P_x(\omega) = P_u(\omega) \cdot |H(\omega)|^2 = \sigma^2 |H(\omega)|^2$$

measure in the lab — model

$$P_x(z) = P_u(z) \cdot H(z) \cdot H^*\left(\frac{1}{z^*}\right)$$

Where $P_x$ is the Power Spectral density of $x_n$

Hence.

$$P_x(\omega) = DTFT\left\{r_x[k]\right\}$$

$$\therefore \quad H(z) = \frac{1 + b[1] \cdot z^{-1} + \ldots + b[q] \cdot z^{-q}}{1 + a[1] \cdot z^{-1} + \ldots + a[p] \cdot z^{-p}} = \frac{B(z)}{A(z)}$$

$$P_x(z) = \underbrace{\sigma^2}_{\substack{z \text{ transform of} \\ r_u[k] = \sigma^2 \delta[k]}} \cdot \underbrace{H(z)}_{\substack{z\text{-transform} \\ \text{of} \\ h[k]}} \cdot \underbrace{H^*\left(\frac{1}{z^*}\right)}_{z\text{-transform of } h^*[-k]}$$

Yule said this $u[n]$ can be Gaussian

$$P_x(z) = \sigma^2 \cdot \frac{B(z)}{A(z)} \cdot H^*\left(\frac{1}{z^*}\right)$$

$$\Rightarrow \underbrace{A(z) \cdot P_x(z)}_{\substack{\text{unknown} \\ \text{known}}} = \underbrace{\sigma^2}_{\text{Known}} \cdot \underbrace{B(z)}_{\text{unknown}} \cdot \underbrace{H^*\left(\frac{1}{z^*}\right)}_{\text{Unknown}}$$

$h[k] \xleftrightarrow{z} H(z)$ then $h^*[-k] \xleftrightarrow{z} H^*\left(\frac{1}{z^*}\right)$

$$\Rightarrow \sum_{l=0}^{p} a[l]\, r_x[k-l] = \sigma^2 \sum_{l=0}^{q} b[l] \cdot h^*[l-k]$$

.. Product in z domain is Convolution in time domain

$$\Rightarrow \underbrace{a[0]}_{1} \cdot r_x[k] = -\sum_{l=1}^{p} a[l] \cdot r_x[k-l] + \sigma^2 \sum_{l=0}^{q} b[l] \cdot h^*[l-k]$$

From causality of $h^*$ filter

$$\Rightarrow \begin{cases} r_x[k] = -\sum_{l=1}^{p} a[l] \cdot r_x[k-l] + \sigma^2 \sum_{l=0}^{q-k} h^*[l] \cdot b[l+k] & k = 0, 1, 2, \ldots, q \\ r_x[k] = -\sum_{l=1}^{p} a[l] \cdot r_x[k-l] & k \geq q+1 \end{cases}$$

- known
- unknown

found from $\sum_{l=0}^{p} a[l] \cdot r_x[-l]$

∴ Yule- walke equations are

$$\boxed{\begin{aligned} r_x[k] &= -\sum_{l=1}^{p} a[l] \cdot r_x[k-l] + \sigma^2 \sum_{l=0}^{q-k} h^*[l] \cdot b[l+k] \qquad k = 0, 1, 2, \ldots, q \\ r_x[k] &= -\sum_{l=1}^{p} a[l] \cdot r_x[k-l] \qquad k \geq q+1 \end{aligned}}$$

- Since $h^*$ terms are there, these equations are highly non linear.
- But we have unlimited equations with only finite unknowns.

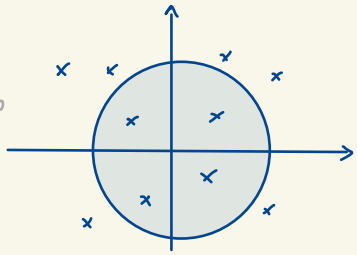
Back To

# Stochastic Models

Past values of the input u(n) are not used

## Autoregressive Models / AR(p) Models
## All-pole Models / ARMA(p,0)

$$x[n] = -\sum_{l=1}^{p} a[l]\cdot x[n-l] + \underbrace{b[0]\cdot u[n]}_{\text{like an error term}}$$



If $b[0]=1$ for simplicity $\Rightarrow$

$$H(z) = \frac{1}{\sum_{l=0}^{p} a[l]\cdot z^{-l}}$$

Since this is an all pole model, $a[0]=b[0]=1 \quad b[l]=0 \;\forall\, l\neq 0$

$\therefore B(z) = 1$

$A(z) = 1 + a[1]z^{-1} + \dots + a[p]\cdot z^{-p}$

$$\therefore H(z) = \frac{B(z)}{A(z)} = \frac{1}{1 + a[1]z^{-1} + \dots + a[p]\cdot z^{-p}}$$

$$h[n] = \text{Inverse } z\text{ Transform}\left\{\frac{1}{1 + a[1]z^{-1} + \dots + a[p]\cdot z^{-p}}\right\}$$

. . . Initial Value Theorem for Causal signals

$$h[0] = \lim_{z\to\infty} H(z) = \lim_{z\to\infty} \frac{1}{1 + a[1]\cdot z^{-1} + \dots + a[p]\cdot z^{-p}} = 1 \quad \cdots (*)$$

$\left(h[0] = \lim_{n\to\infty} H(z) \quad H(z) = \sum h[n]\cdot z^{-n}\right)$

Q: Given $r_x[k]$ (found by estimating using $n$ samples) $k = 0, \pm1, \pm2, \dots$ estimate $a[l]$, $l = 1, \dots, p$

Since $q=0$, Yule Walker equation become,

$$\begin{cases} r_x[0] = -\sum_{l=1}^{p} a[l]\cdot r_x[-l] + \sigma^2 \sum_{l=0}^{0} h^*[l]\cdot b[l] & k=0 \\ r_x[k] = -\sum_{l=1}^{p} a[l]\cdot r_x[k-l] & k \geq 1 \end{cases}$$

$$\begin{cases} r_x[0] = -\sum_{l=1}^{p} a[l]\cdot r_x[-l] + \sigma^2 \underbrace{h^*[0]}_{1\text{ from }(*)}\underbrace{b[0]}_{1} & k=0 \\ r_x[k] = -\sum_{l=1}^{p} a[l]\cdot r_x[k-l] & k \geq 1 \end{cases}$$

Let's pick Yule Walker equations for $k = 1, 2, 3, \dots, p$. (Why do we pick first $p$? $\longrightarrow$ It's easier to estimate correlations due to small lags. $\therefore$ Easier to estimate them)

Actual Yule Walker Equations for the AR processes

$$\begin{bmatrix} r_x[1] \\ r_x[2] \\ \vdots \\ r_x[p] \end{bmatrix} = - \begin{bmatrix} r_x[0] & r_x[-1] & r_x[-2] & \cdots & r_x[-p+1] \\ r_x[1] & r_x[0] & r_x[-1] & \cdots & r_x[-p+2] \\ & & & & \\ r_x[p-1] & r_x[p-2] & r_x[p-3] & \cdots & r_x[0] \end{bmatrix} \begin{bmatrix} a[1] \\ a[2] \\ \vdots \\ a[p] \end{bmatrix}$$

$$\underline{r} \qquad R_p^T \qquad \underline{a}$$

$$\hat{\underline{r}} = -\hat{R}_p^T \, \hat{\underline{a}}$$

$$\hat{\underline{a}} = -\left(\hat{R}_p^T\right)^{-1} \hat{\underline{r}}$$

Claim :-

$$x[n] = -\sum_{l=1}^{p} a[l] \cdot x[n-l] + b[0] \cdot u[n]$$

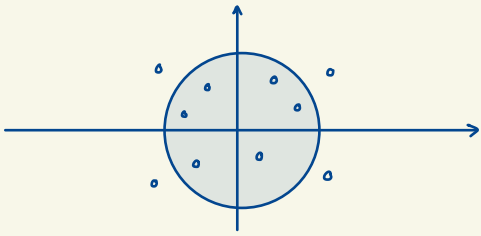
then, $x[n]$ is assymptotically stationary $\iff$ All poles of the AR process lies inside the unit circle

Past values of the output $x[n]$ are not used.

## Moving Average Models / MA(q) Models
## All-zero Models / ARMA(0, q)

$$x[n] = \sum_{l=0}^{q} b[l] \cdot u[n-l]$$

$$H(z) = \sum_{l=0}^{q} b[l] \, z^{-l}$$



Since this is a all-zero model, $a[0] = b[0] = 1$ $\quad a[l] = 0 \;\; \forall l \neq 0$

$$B(z) = 1 + b[1] \cdot z^{-1} + \ldots + b[q] \cdot z^{-q}$$

$$A(z) = 1$$

$H(z) = \frac{B(z)}{A(z)} = B(z)$

$\therefore$ Zeroes are at $B(z) = H(z) = 0$

Q: Given $r_x[k]$ $\quad k = 0, \pm 1, \pm 2, \ldots$ estimate $b[l]$, $l = 1, \ldots, q$

$$\begin{cases} r_x[k] = -\sum_{l=1}^{p} a[l] \cdot r_x[k-l] + \sigma^2 \sum_{l=0}^{q-k} h^*[l] \cdot b[l+k] & 0 \leq k \leq q \\ r_x[k] = -\sum_{l=1}^{p} a[l] \cdot r_x[k-l] = 0 & q+1 \leq k \end{cases}$$

(a[l] = 0 for MA)

- In case I, $h[l] = b[l]$ hence equations become following. ($q+1$ quadratic equations for $q$ unknowns $b[1], \ldots, b[q]$)

$$\begin{cases} r_x[k] = \sigma^2 \sum_{l=0}^{q-k} b^*[l]\cdot b[l+k] & 0 \le k \le q \\ r_x[k] = 0 & q+1 \le k \end{cases}$$

- Since this is not a nice solution as in AR models, AR models are widely used than MA models.

  AR models has Yule Walker equations.

Ex:- Consider ARMA(1,1) process. Write it as a AR($\infty$) process.

$$H(z) = \frac{1+b_1 z^{-1}}{1+a_1 z^{-1}}$$

$$H_{AR}(z) = \frac{1}{1+c_1 z^{-1}+\dots}$$

$$\therefore H(z) = H_{AR}(z) \Rightarrow \frac{1}{H(z)} = \frac{1}{H_{AR}(z)}$$

$$\Rightarrow \frac{1+a_1 z^{-1}}{1+b_1 z^{-1}} = 1 + c_1 z^{-1} + c_2 z^{-2} + c_3 z^{-3}$$

$$\Rightarrow 1+a_1 z^{-1} = \left[1 + z^{-1}(c_1+b_1) + z^{-2}(c_2+b_1 c_1) + z^{-3}(c_3+b_1 c_2)+\dots\right]$$

$$\Rightarrow 1 + a_1 z^{-1} = 1 + (c_1+b_1) z^{-1} + \sum_{i=2}^{\infty} z^{-i}(c_i + b_1 c_{i-1})$$

$$\Rightarrow c_0 = 1$$

$$a_1 = c_1 + b_1 \Rightarrow c_1 = a_1 - b_1$$

$$0 = c_k + c_{k-1} b_1 \Rightarrow c_k = c_{k-1}(-b_1)$$

$$\Rightarrow c_i = (a_1-b_1)(-b)^{i-1}$$

$$\Rightarrow c_i = \begin{cases} (a_1-b_1)\cdot(-b)^{i-1} & i \neq 0 \\ 1 & i = 0 \end{cases}$$

Ex:- Write it as a MA process, $H_{MA}(z) = 1 + d_1 z^{-1} + \dots$

$$\therefore H(z) = H_{MA}(z) \Rightarrow \frac{1+b_1 z^{-1}}{1+a_1 z^{-1}} = 1 + d_1 z^{-1} + d_2 z^{-2} + \dots$$

$$\Rightarrow 1 + b_1 z^{-1} = 1 + z^{-1}(a_1+d_1) + z^{-2}(d_2 + a_1 d_1) + \dots$$

$$0 = d_k + a_1 d_{k-1}$$

$$d_k = d_{k-1}(-a_1)$$

$$\Rightarrow d_i = \begin{cases} 1 & i = 0 \\ (b_1-a_1)(-a_1)^{k-1} & i \neq 0 \end{cases}$$

# Asymptotically Stationary Processess

Consider $x[0], x[-1], \ldots, x[-p+1]$. $\therefore$ breaking down the process,

$$\Rightarrow \quad H(z) = \frac{1}{\sum_{k=0}^{p} a_k \cdot z^{-k}} = \sum_{k=1}^{p} \frac{A_k}{1 - p_k \cdot z^{-1}}$$

$$\Rightarrow \quad h[n] = \sum_{k=1}^{p} A_k \cdot p_k^{n} + \sum_{k=1}^{p} \underbrace{c_k \cdot \gamma_k^{n}} \cdot \cos(\omega_k n + \phi_k) \quad *$$

$|p_i| < 1$ $\qquad$ $p_i = \gamma_i e^{\pm j\omega_i}$

# Wold Decomposition Theorem

Any WSS process can be decomposed as $x[n] = u[n] + s[n]$ where,

1) $u[n]$ and $s[n]$ are uncorrelated : $E[u[n] \cdot s^*[m]] = 0 \quad \forall m,n$

2) $u[n]$ is an MA($\infty$) WSS process:

$$u[n] = \sum_{l=0}^{\infty} b[l] \cdot v[n-l] \quad \text{s.t } v[n] \text{ is a white noise and } \sum_{l=0}^{\infty} |b[l]|^2 < \infty \text{ and } b[0] = 1$$

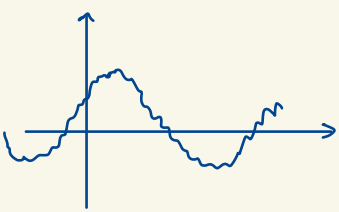
3) $s[n]$ is predictable (deterministic) process :

$$s[n] = -\sum_{l=1}^{\infty} a[l] \cdot s[n-l]$$

Ex: $y[n] = s[n] + u[n]$

where $u[n] \sim W(0, \sigma^2)$ $\quad s[n] = A \cdot e^{j(2\pi f_0 n + \varphi)}$

$$r_y[k] = A^2 \cdot e^{j 2\pi f_0 k} + \sigma^2 \delta[k]$$



Claim: $s[n]$ can be exactly expressed in terms of $s[n-1]$ and $s[n-2]$, Since

$$s[n] = \alpha \cdot s[n-1] + \beta \cdot s[n-2]$$

## Special Case

When $H(z)/B(z)$ All zero filter is minimum phase $\Rightarrow$ All zeros shold lie inside unit circle

$\Rightarrow u[n] = \sum_{l=0}^{\infty} b[l] \cdot v[n-l]$ $\Rightarrow$ $H(z)$ can be replaced with a all pole filter with $a[n] = b[n]^*$

$\Rightarrow U(z) = B(z) \cdot V(z)$

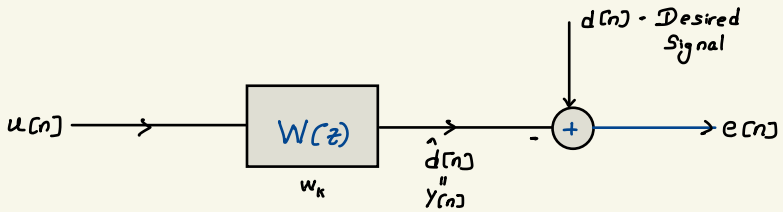
$\Rightarrow H(z) = \dfrac{U(z)}{V(z)} = \dfrac{B(z)}{A(z)} = B(z) = \sum_{n=0}^{\infty} b[l] \cdot z^{-l}$

# HAY - 5

Three problems we want to solve under Linear Prediction

1) Filtering - Wiener Filters
2) Forward Linear Prediction - FLP
3) Backward Linear Prediction - BLP

# Wiener Filters




The goal is to design the signal d[n]. We might know some statistical probabilities of d[n]. How can we use this to design the filter so e[n] is small.

1) Filtering - We should not depend on the future values of d.
∴ Causal

$$x[n] = d[n] + v[n]$$
$$x[n],\ x[n-1],\ \ldots,\ x[n-k+1]$$

2) Smoothing - Non-causal

$$x[n] = d[n] + v[n]$$

Random BS? by TA when Babadi was away.

3) Prediction

$$d[n] = x[n+i] \qquad i \geq 1$$

4) Deconvolution

$$u[n] = d[n] * g[n] + v[n].$$

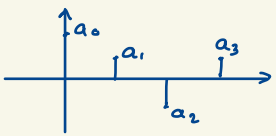
- Assumptions on W(z)
  1) linear
  2) Filter operates in discrete time
- Always go for FIR filters due to stability and easiness in implementation
  IIR filter involves boath feedforward and feedback. ∴ can be unstable eventhough it is computationally efficient.
- The higher the M, more accurate the prediction. But u(n-M+1) will be very less correlated to d(n).

## Goal

- Use $u[n], u[n-1], \ldots, u[n-M+1]$ $M$ observations in the past and the present to predict $d[n]$. We need to use underline statistical dependancies between $u(n)$ and $d(n)$. $\underline{u}[n] = \begin{bmatrix} u[n] \\ u[n-1] \\ \vdots \\ u[n-M+1] \end{bmatrix}$ - observations

## Assumption

- The system is Causal and LTI, FIR filter with coeffecients $a_0, a_1, \ldots, a_{M-1}$




$$y[n] = \hat{d}[n] = \sum_{k=0}^{M-1} w_k^* \cdot u[n-k]$$

$$\underline{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_{M-1} \end{bmatrix}$$

filter coefficients

In the lecture filter coeff. was taken as $[a_0, \ldots, a_{M-1}]^T$ but conjugation was dropped for some reason. effectively $\underline{w} = \underline{a}^*$

## Cost Function (J)

Mean square error (MSE) criteria.

$$\Rightarrow \{a_0, a_1, \ldots, a_{M-1}\} = \underset{a_0, \ldots, a_{M-1}}{\operatorname{argmin}} \; E\{|e[n]|^2\} \cdots E\{e[n] \cdot e^*[n]\}$$

$$= \underset{a_0, \ldots, a_{M-1}}{\operatorname{argmin}} \; E[|y[n] - \hat{d}[n]|^2]$$

$$= \underset{a_0, \ldots, a_{M-1}}{\operatorname{argmin}} \; E\left[\left|y[n] - \sum_{k=0}^{M-1} w_k^* \cdot u[n-k]\right|^2\right]$$

$$= \min J_w$$

### Summary.

1) $W(z) = \sum_{k=0}^{M-1} w_k \cdot z^{-k}$

2) $\hat{d}[n] = y[n] = \sum_{k=0}^{M-1} w_k^* \cdot u[n-k] = \underline{w}^H \cdot \underline{u}[n] \ldots\ldots = \underline{a}^T \underline{u}[n]$ in lecture

Why – book says due to complexness of the $\underline{u}[n]$ baseband.

3) $e[n] = d[n] - \sum_{k=0}^{M-1} w_k^* \cdot u[n-k] = d[n] - y[n]$

4) $J = E[e[n] \cdot e^*[n]] \ldots\ldots ①$

# Principle of Orthogonality

If cost function $J$ attains the minimum value at $e = e_o[n]$

$e_o[n] \perp$ each input at $n$ $\quad u[n-k]$ $\quad k = 0, 1, 2, .., M-1$

$e_o[n] \perp \underline{u}[n]$

$\mathbb{E}\left[ u[n-k] \cdot e_o^*[n] \right] = 0$

Proof: Assume we have infinite samples. $\therefore u[n], u[n-1], \ldots$ i.e $M \to \infty$.

Then

$$y[n] = \sum_{k=0}^{\infty} w_k^* \, u[n-k] \qquad n = 0, 1, \ldots$$

Assume $w_k = a_k + j b_k$

$$\frac{\partial e[n]}{\partial a_k} = \frac{\partial}{\partial a_k} \left[ d[n] - \sum_{k=0}^{\infty} (a_k + j b_k)^* u[n-k] \right] = -u[n-k]$$

$$\frac{\partial e[n]}{\partial b_k} = +j \, u[n-k]$$

$$\frac{\partial e^*[n]}{\partial a_k} = \frac{\partial}{\partial a_k} \left[ d^*[n] - \sum_{k=0}^{\infty} (a_k + j b_k) \, u^*[n-k] \right] = -a_k u^*[n-k]$$

$$\frac{\partial e^*[n]}{\partial b_k} = -j \, u^*[n-k]$$

Hence by differentiating $J$, in terms of $w_k$ we get

$$\Rightarrow \nabla_k J = \frac{\partial}{\partial w_k} J = \frac{\partial}{\partial a_k} J + j \frac{\partial}{\partial b_k} J$$

differentiating real and imaginary parts seperately.

From ① $\Rightarrow$ $= \frac{\partial}{\partial w_k} \mathbb{E}[e[n] e^*[n]] = \frac{\partial}{\partial a_k} \mathbb{E}[e[n] e^*[n]] + j \frac{\partial}{\partial b_k} \mathbb{E}[e[n] e^*[n]]$

Hence at the optimum point $\nabla_k J = 0$

$$\Rightarrow \frac{\partial J}{\partial a_k} + j \frac{\partial J}{\partial b_k} = 0$$

$$\Rightarrow \frac{\partial}{\partial a_k} \mathbb{E}[e[n] e^*[n]] + j \frac{\partial}{\partial b_k} \mathbb{E}[e[n] e^*[n]] = 0$$

By using previous 4 derivations,

$$\Rightarrow \mathbb{E}\left[ \frac{\partial e(n)}{\partial a_k} \cdot e^*(n) + \frac{\partial e^*(n)}{\partial a_k} e(n) + j \frac{\partial e(n)}{\partial b_k} \cdot e^*(n) + j \frac{\partial e^*(n)}{\partial b_k} \cdot e(n) \right] = 0$$

$$\Rightarrow \mathbb{E}\left[ -u(n-k) e^*(n) - a_k u^*(n-k) e(n) + j \left( +j\, u(n-k) e^*(n) - j\, u^*(n-k) e(n) \right) \right] = 0$$

$$\Rightarrow \quad \mathbb{E}\left[ -2\, u(n-k) \cdot e^*(n) \right] = 0$$

$$\Rightarrow \quad \mathbb{E}\left[ u(n-k) \cdot e^*(n) \right] = 0$$

Corollary :- At the optimum mean square estimation

Estimation error $\perp$ Estimate of the desired response

$$e_o(n) \perp y_o(n) \qquad i.e \quad \mathbb{E}\left[ y(n)_o \cdot e_o^*(n) \right] = 0$$

For $k = 1, 2, 3, \ldots, M-1$

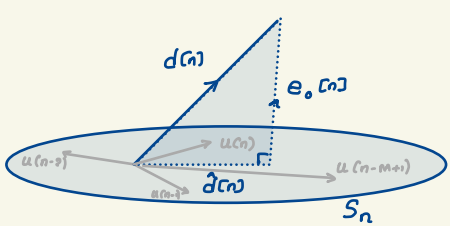
$$\Rightarrow \frac{\partial J}{\partial a_k^*} = \frac{\partial}{\partial a_k^*} \mathbb{E}\left[ e(n) \cdot e^*(n) \right]$$

$$= \mathbb{E}\left[ e(n) \cdot \frac{\partial}{\partial a_k^*} \left[ d^*(n) - \sum_{j=0}^{M-1} w_j^* \cdot x^*(n-j) \right] \right]$$

$$= \mathbb{E}\left[ e(n) \cdot \left( -x^*(n-k) \right) \right] = 0$$

$$\Rightarrow \quad \mathbb{E}\left[ e_o(n) \cdot x^*(n-k) \right] = 0$$

$$\Rightarrow \quad \mathbb{E}\left[ e_o(n) \cdot \hat{d}_o(n) \right] = 0$$




$S_n$ = set of all linear combinations of $u(n), \ldots, u(n-M+1)$

$$y_o(n) = \hat{d}[n|S_n]$$

# Normal Equations / Wiener-Hopf Equations

$$\underline{p} = E\left[\underline{u}[n] \cdot d^*[n]\right] = \begin{bmatrix} E[u[n] \cdot d^*[n]] \\ E[u[n-1]\, d^*[n]] \\ \vdots \\ E[u[n-M+1] \cdot d^*[n]] \end{bmatrix} = \begin{bmatrix} r_{du}[0] \\ r_{du}[1] \\ \vdots \\ r_{du}[M-1] \end{bmatrix} \qquad \underline{u}[n] = \begin{bmatrix} u[n] \\ u[n-1] \\ \vdots \\ u[n-M+1] \end{bmatrix}$$

Cross correlation between input and desired signal (u, d)

$$R = \begin{bmatrix} r[0] & r[1] & \cdots & r[M-1] \\ r[-1] & r[0] & \cdots & r[M-2] \\ \vdots & & & \\ r[-M+1] & \cdots & \cdots & r[0] \end{bmatrix} \qquad \underline{w}_o = \begin{bmatrix} w_{o1} \\ w_{o2} \\ \vdots \\ w_{oM-1} \end{bmatrix}$$

$R$: Autocorrelation of filter inputs $u[n]$

$\underline{w}_o$: filter coefficients at optimal MSE prediction

$$\boxed{R\,\underline{w}_o = \underline{p}} \quad \cdots \quad R\,\underline{a}_o^* = \underline{p} \text{ in lecture}$$

## Proof I

$R^T \underline{w}_o^* = \underline{p}^*$ ?

$$\begin{aligned} J &= E[e[n] \cdot e^*[n]] \\ &= E\left[\left(d[n] - \sum_{k=0}^{M-1} w_k^* u[n-k]\right)\left(d^*[n] - \sum_{k=0}^{M-1} w_k \cdot u^*[n-k]\right)\right] \\ &= E[|d[n]|^2] - E\left[d[n] \cdot \sum_{k=0}^{M-1} w_k\, u^*[n-k]\right] - E\left[d^*[n] \sum_{k=0}^{M-1} w_k^* \cdot u[n-k]\right] \\ &\quad + E\left[\sum_{k=0}^{M-1}\sum_{l=0}^{M-1} a_k^* \cdot a_l \cdot u[n-k] \cdot u^*[n-l]\right] \\ &= E[|d[n]|^2] - \sum_{k=0}^{M-1} w_k\, E[d[n] \cdot u^*[n-k]] - \sum_{k=0}^{M-1} w_k^*\, E[d^*[n] \cdot u[n-k]] \\ &\quad + \sum_{k=0}^{M-1}\sum_{l=0}^{M-1} a_k^* \cdot a_l\, E[u[n-k] \cdot u^*[n-l]] \\ &= E[|d[n]|^2] - \underline{w}^T \cdot \underline{p}^* - (\underline{w}^*)^T \underline{p} + (\underline{w}^*)^T R\, \underline{w} \\ &= E[|d[n]|^2] - \underline{w}^T \cdot \underline{p}^* - \underline{w}^H \underline{p} + \underline{w}^H R\, \underline{w} \quad \cdots (*) \end{aligned}$$

$$\begin{aligned} (R \cdot \underline{w} - \underline{p})^H \cdot R^{-1} \cdot (R\underline{w} - \underline{p}) &= (\underline{w}^H R^H - \underline{p}^H)(\underline{w} - R^{-1}\underline{p}) \\ &= \underline{w}^H R^H \underline{w} - \underline{p}^H \underline{w} - \underline{w}^H \underline{R^H R^{-1}} \underline{p} + \underline{p}^H R^{-1} \underline{p} \\ &= \underline{w}^H R^H \underline{w} - \underline{p}^* \underline{w}^T - \underline{w}^H \underline{p} \underline{I} + \underline{p}^H R^{-1} \underline{p} \end{aligned}$$

Not sure

From the above expressions (*)

$$J = \underbrace{E[|d[n]|^2] - \underline{p}^H R^{-1} \underline{p}}_{J_{min}} + (R\underline{w} - \underline{p})^H R^{-1} (R\underline{w} - \underline{p})$$

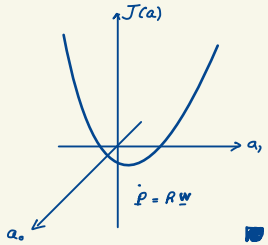
$\therefore$ Minimum can only be $J_{min}$, since latter part is a perfect square

$\therefore$ At the minimum point,

$\Rightarrow R\underline{w}_o - \underline{p} = 0$ ... Normal equation

$\Rightarrow R\underline{w}_o = \underline{p}$

$\Rightarrow \underline{w}_o = R^{-1}\underline{p}$




## Proof II

▶ From the orthogonality principle, the minimum error occurs at,

$$\mathbb{E}\left[ e[n] \cdot u^*[n-k] \right] = 0 \qquad k = 0, \ldots, M-1$$

By substition for $e[n]$, we get M set of equations,

$$\Rightarrow \mathbb{E}\left[ \left( d[n] - \sum_{l=0}^{M-1} w_{ol}^* \, u[n-l] \right) \cdot u^*[n-k] \right] = 0$$

$$\Rightarrow \mathbb{E}\left[ d[n] \cdot u^*[n-k] \right] = \mathbb{E}\left[ \sum_{l=0}^{M-1} w_{ol}^* \, u[n-l] \cdot u^*[n-k] \right]$$

$$\Rightarrow \mathbb{E}\left[ d^*[n] \, u[n-k] \right] = \sum_{l=0}^{M-1} w_{ol} \cdot \mathbb{E}\left[ u[n-l]^* \, u[n-k] \right]$$

$$\Rightarrow k\text{-}l^{th} \text{ element of vector } \underline{p} \text{ ie } p(-k) = \sum_{l=0}^{M-1} w_{ol} \cdot r(l-k)$$

$$\Rightarrow \underline{p} = R\underline{w}_o$$

$\Rightarrow \mathbb{E}\left[ d[n] \cdot u^*[n-k] \right] = \mathbb{E}\left[ \sum_{l=0}^{M-1} w_{ol}^* \, u[n-l] \cdot u^*[n-k] \right]$

$kl$ element of $R_m$; $kl$ element of $R_m^T$; $\sum_l \left(R_m^T\right)_{kl} \cdot w_{ol}^*$

$\Rightarrow \left(A\underline{w}^*\right)_k = \sum_l A_{kl} \cdot w_{ol}^*$

$\Rightarrow \left(\underline{p}^*\right)_k = \left(R_m^T \underline{w}^*\right)_k$

▶ We can convert the above to matrix form

$\Rightarrow R_m^T \underline{a} = \underline{p}^* \qquad R_m \underline{a}^* = \underline{p}$

# Error Performance Surface

Since $\underline{w}_o = \underline{a}_o^*$ in lecture

$$\text{Var}\left(\hat{d}[n]\right) = \underline{p}^H \underline{w}_o = \underline{w}_o^H \underline{p} = \underline{a}_o^T \underline{p}$$

**Proof 1** Since at the minimum point $R^{-1}\underline{p} = \underline{w}_o$

$$\therefore J_{min} = E\left[|d[n]|^2\right] - \underline{p}^H R^{-1} \underline{p} = E\left[|d[n]|^2\right] - \underline{p}^H \underline{w}_o \quad \cdots (1)$$

Since $d[n] = e_o[n] + \hat{d}[n] \Rightarrow \sigma_d^2 = J_{min} + \text{var}\left(\hat{d}[n]\right)$

$$\Rightarrow J_{min} = \sigma_d^2 - \text{var}\left(\hat{d}[n]\right) \quad \cdots (2)$$

$$(1), (2) \Rightarrow \text{Var}\left(\hat{d}[n]\right) = \underline{p}^H \underline{w}_o$$

---

**Proof 2 (Babadi's)**

$$\hat{d}[n] = \left(\underline{w}^*\right)^T \cdot \underline{u}[n] = \underline{w}^H \cdot \underline{u}[n] = \sum_{k=0}^{M-1} w_k^* \cdot u[n-k]$$

Assuming $m_u = 0$

$$\text{Var}\left(\hat{d}[n]\right) = E\left[\hat{d}[n] \cdot \hat{d}^*[n]\right] = E\left[\underline{w}^H \underline{u}[n] \cdot \underline{u}^H[n] \cdot \underline{w}\right]$$

$$= \underline{w}^H \cdot \underbrace{E\left[\underline{u}[n] \cdot \underline{u}^H[n]\right]}_{R} \cdot \underline{w}$$

$$= \underline{w}^H R \underline{w}$$

$$= \underline{w}_o^H \underline{p} \quad \ldots \quad R\underline{w}_o = \underline{p}$$

$$= \underline{p}^H \underline{w}_o \quad \ldots \text{ since a scalar}$$

■

# Minimum Mean Square Error

$$\therefore \quad J_{min} = \sigma_d^2 - \underline{p}^H \underline{w}_o$$

In lecture $\left(\sigma_d^2 - \underline{p}^H \underline{a}_o^*\right)$ $\left(\sigma_d^2 - \underline{a}_o^T \underline{p}\right)$

$\sigma_d^2$ : Error in absence of any observation.

If we try to predict $d[n]$ by some un-related signal $u[n]$, then $\underline{p}$ - correlation will be very low and we will have just $\sigma_d^2$

Proof:-

Assume zero mean in $u(n)$ input signal and the $\hat{d}[n] = y[n]$ estimated Signal

$$\Rightarrow \quad d[n] = e[n] - \hat{d}[n]$$

$$\Rightarrow \text{Var}(d[n]) = E\left[|d[n]|^2\right]$$

$$= E\left[(e[n] + \hat{d}[n])(e^*[n] + \hat{d}^*[n])\right]$$

$$= E\left[|e[n]|^2\right] + E\left[|\hat{d}[n]|^2\right] + E\left[e[n] \cdot d^*[n]\right] + E\left[e^*[n] \cdot d[n]\right]$$

$E[|e[n]|^2]$ = $J_{min}$ ; $E[|\hat{d}[n]|^2]$ = Var $[\hat{d}(n)]$ ; $E[e[n] \cdot d^*[n]] = 0$ from principle of orthogonality ; $E[e^*[n] \cdot d[n]] = 0$

$$\Rightarrow \quad J_{min} = \underbrace{\text{var}(d[n])}_{\sigma_d^2} - \text{Var}(\hat{d}[n])$$

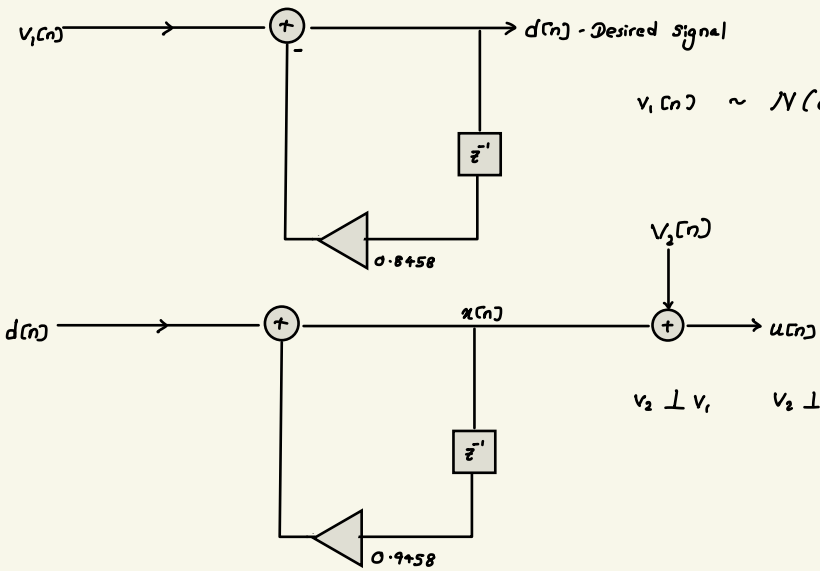
$\therefore$ From (1)

$$\Rightarrow \quad J_{min} = \sigma_d^2 - \underline{p}^H \underline{w}_o$$

■

# Numerical Example

## Example




$v_1[n] \sim \mathcal{N}(0, \sigma_1^2)$ where $\sigma_1^2 = 0.27$

$v_2 \perp v_1$ $\quad v_2 \perp x$ $\quad \sigma_2^2 = 0.1$

(a) What is $\{x[n]\}$

$$d[n] = -0.8458\, d[n-1] + v_1[n] \implies H_1(z) = \frac{1}{1 + 0.8458\, z^{-1}}$$

$$x[n] = 0.9458 \cdot x[n-1] + d[n] \implies H_2(z) = \frac{1}{1 - 0.9458 \cdot z^{-1}}$$

$$\therefore H(z) = H_1(z) \cdot H_2(z) = \frac{1}{1 - 0.1\, z^{-1} - 0.8\, z^{-2}}$$

$\Rightarrow$ $x[n]$ is an AR(2) process.

$\Rightarrow$ $x[n] - \underbrace{0.1}_{a_1}\, x[n-1] - \underbrace{0.8}_{a_2}\, x[n-2] = v_1[n]$

(b) What is the correlation matrix of $u[n]$ ?

$$u[n] = x[n] + v_2[n] \implies R_u = R_x + R_{v_2} \quad \cdots\cdots (1)$$

$$R_u = E[\underline{u}[n] \cdot \underline{u}^H[n]] = \begin{bmatrix} r_u(0) & r_u(1) \\ r_u^*(1) & r_u(0) \end{bmatrix} \quad \cdots\cdots (2)$$

By using the Yule Walker equation for AR process in part @,

$$\Rightarrow \begin{bmatrix} r_x(0) & r_x(1) \\ r_x(-1) & r_x(0) \end{bmatrix} \rightarrow R\,\underline{w} = \underline{r} \leftarrow \begin{pmatrix} r_x(1) \\ r_x(2) \end{pmatrix}$$

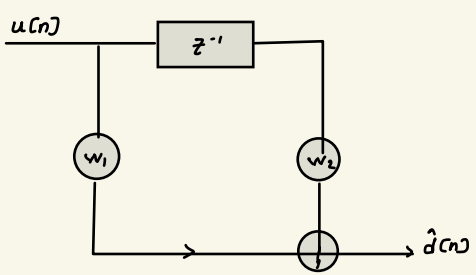
$$\Rightarrow \quad r_x(0) = \left(\frac{1+a_2}{1-a_2}\right) \frac{\sigma_i^2}{(1+a_2)^2 - a_1^2} = 1$$

$$r_x(1) = \frac{-a_1}{1+a_2}\, r_x(0) = 0.5$$

$$\Rightarrow \quad R_{v_2} = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}$$

$$\therefore \quad R_u = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} + \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} = \begin{bmatrix} 1.1 & 0.5 \\ 0.5 & 1.1 \end{bmatrix}$$

(c)




What are $w_1$, $w_2$, $J_{min}$.

$$\Rightarrow \quad \underline{p} = E\left[ d^*[n] \begin{pmatrix} u[n] \\ u[n-1] \end{pmatrix} \right] \quad \ldots\ldots (1)$$

$$\Rightarrow \quad E[\, d^*[n]\, u[n]\,] = E[\, (x[n] - 0.9458 \cdot x[n-1])(x[n] + v_2[n])\,]$$

$$= r_x(0) - 0.9458 \cdot r_x(-1)$$

$$= 1 - 0.9458 \times 0.5$$

$$= 0.5271$$

$$\Rightarrow \quad E[d^*(n) \cdot u(n-1)] = r_x(1) - 0.9458 \cdot r_x(0) = -0.4458$$

$$\therefore \quad \underline{w}_o = R^{-1}\underline{p} = \begin{bmatrix} 0.8360 \\ -0.7853 \end{bmatrix} //$$

from ① $\quad \underline{p} = \begin{bmatrix} 0.5271 \\ -0.4458 \end{bmatrix} //$

ⓓ What is $J_{min}$ ?

$$J_{min} = E[|d(n)|^2] - \underline{w}_o^T \underline{p} = 0.1579 //$$

This can be found using $d(n) = -0.8458\, d(n-1) + v_1(n)$

- Why do we need to emphasize on complex values ?

When we are sending a high frequency component, the infomation is contained in boath amplitude and phase.

$$s(n) = \underbrace{A(n)}_{\text{Info}} \cdot \cos(2\pi f_c n + \underbrace{\varphi(n)}_{\text{Info}})$$

$$\therefore u(n) = \underbrace{u_I(n)}_{\text{Inface}} + j\underbrace{u_Q(n)}_{\text{Quadrature}}$$

∴ To figure out Quadrature port, we need complex values

## Differentiability of a Complex Function $f(x)$

For a $f: \mathbb{R} \to \mathbb{C}$ function to be differentiable on a region $D$,

1) $f$ should be continuos.

2) Chauchy Riemann Condition -

$$f(z) = u(x,y) + j \cdot v(x,y)$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

Ex: Consider $f(z) = |z|^2$ here $z \cdot z^* = (x^2+y^2) + ix$ . ∴ This does not comply with cauchy riemann condition.

## Theorem

If $f(z, z^*)$ is a real valued function, then $\nabla_{\underline{z}^*} f(z, z^*)$ is the vector pointing in the direction of maximum change.

Corollary: Stationary points of $f(z, z^*)$ are the solutions to $\nabla_{z^*} f(z, z^*) = 0$

Ex: Consider $J = E[|d(n)|^2] - \underline{a}^H \underline{p}^* - \underline{p}^T \underline{a} + \underline{a}^H R \underline{a}$

| | $\underline{a}^H \underline{z}$ | $\underline{z}^H \underline{a}$ | $\underline{z}^H A \underline{z}$ |
|---|---|---|---|
| $\nabla_{\underline{z}}$ | $\underline{a}^*$ | 0 | $A^T \underline{z}^* = (A\underline{z})^*$ |
| $\nabla_{\underline{z}^*}$ | 0 | $\underline{a}$ | $A\underline{z}$ |

∴ From the table,

$$\nabla_{\underline{a}}^* J = -\underline{p}^* + R^T \underline{a}$$

where $R^T \underline{a} = \underline{p} \Rightarrow R\underline{a}^* = \underline{p}$

equation for lowest point

This agrees with what we saw before.

# HAY - 6

# Linear Predictions

## 1) Forward Linear Prediction (FLP)

- We have observed $u[n-1], u[n-2], \ldots, u[n-M]$. Can we predict $u[n]$ based on a linear combination of them?

- This is same as in filtering problem, but $d[n] = u[n]$ and observations are $u[n-1], \ldots, u[n-M]$

- Goal: Find $c_1, c_2, \ldots, c_M$ such that

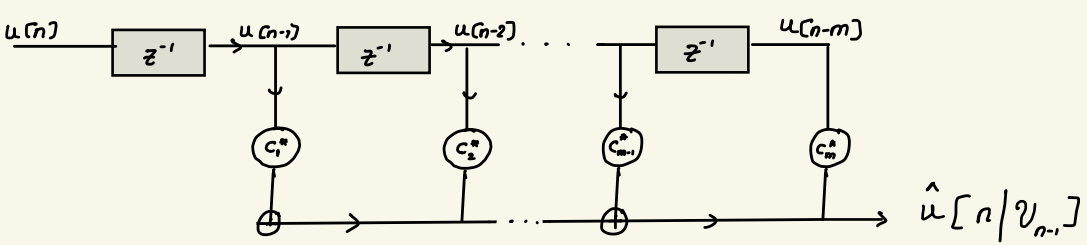
$$\hat{u}(n|\mathcal{U}_{n-1}) \simeq \sum_{k=1}^{M} c_k^* \cdot u[n-k] = \underline{c}^H \underline{u}[n-1]$$

in book filter coefficients was $\underline{w}_{f,k}$ i.e. $\underline{c} = \underline{w}_f$

is the optimal prediction of $u[n]$ in the RMSE sense.

$$\underline{u}[n-1] = \begin{bmatrix} u[n-1] \\ u[n-2] \\ \vdots \\ u[n-M] \end{bmatrix} \qquad \underline{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_M \end{bmatrix}$$

- The role of $\underline{a}_i^*$'s are played by $\underline{c}_i$'s here. Not exactly. some confusion is with conjugation.




$$\hat{u}[n|\mathcal{U}_{n-1}] = \sum_{k=1}^{M} c_k^* \cdot u[n-k] = \underline{c}^H \cdot \underline{u}[n-1]$$

$\therefore$ The prediction error $= f_M[n] := u[n] - \hat{u}[n/\mathcal{U}_{n-1}]$

$$f_M[n] = u[n] - \sum_{k=1}^{M} c_k^* \, u[n-k]$$

In Wiener filters this was $J_{min}$ .. $J_{min} = \sigma_d^2 - \underline{a}^T \underline{p} = \sigma_d^2 - \underline{w}^H \underline{p}$

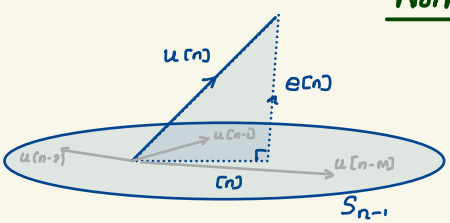
$\therefore$ Prediction error variance $= P_M := Var(f_M[n]) = E\{(f_M[n])^2\}$ ..... Since tap inputs are assumed to be zero mean

(i.e. Minimum Mean Square Error)

(i.e. Forward Prediction Error Power)

## Normal Equations for FLP

(Wiener Hopf)




- From the Wiener Filters we had $\underline{c}$'s instead of $\underline{a}^*$s.

$\Rightarrow R_M \underline{a}^* = \underline{p}$ . ($\underline{w}$ in the book)

Replace $\underline{a}^*$ from $\underline{c}$ and $\underline{p}$ from $\underline{r}$. dont't be confused with old $\underline{p}$ - cross correlation and new $\underline{p}$ - FP error power $\Rightarrow$

$$\boxed{R_M \underline{c} = \underline{r}}$$ Normal Equations for FLP

- To use the same logic we used in Wiener filters, we need to figure out correlation matrix of input $\underline{u}[n-1]$, and cross correlation of input $\underline{u}[n-1]$ and output $u[n]$.

### I) Correlation matrix of the tap input

$$R = E[\underline{u}[n-1] \cdot \underline{u}^H[n-1]] = E[\underline{u}[n] \cdot \underline{u}[n]^H] \quad \cdots \text{ } u \text{ is wss}$$

$$R = \begin{bmatrix} r(0) & r(1) & \cdots & r(M-1) \\ r(-1) & r(0) & \cdots & r(M-2) \\ \vdots & & & \\ r(-M+1) & \cdots & \cdots & r(0) \end{bmatrix}$$

.. $r(k)$ - autocorrelation of $u$ for lag $k$.

Revice: $R_M$ is a Toeplitz matrix when $\underline{u}[n]$ is wss.

## II) Cross-correlation vector

$$\underline{r} = E[\underline{u}^*[n] \cdot \underline{u}[n-1]] = \begin{bmatrix} r(-1) \\ r(-2) \\ \vdots \\ r(-M) \end{bmatrix} \Rightarrow \underline{r} = \begin{bmatrix} r(-1) \\ r(-2) \\ \vdots \\ r(-M) \end{bmatrix}$$

in wiener filters this was $\underline{p}$

## III) Variance of $u[n]$

Since $u[n]$ is assumed to be zero mean $\Rightarrow$

$$\text{Var}[u[n]] = E[|u[n]|^2] = E[u[n] \cdot u^*[n]] = r(0)$$

$$\text{Var}[u[n]] = r(0) \quad \cdots \quad \sigma_d^2 \text{ in Wiener filters}$$

- From the Wiener filter derivation for $J_{min}$, which is $\rho_M$ now, we can get

$$J_{min} = \sigma_d^2 - \underline{p}^H \cdot \underline{w}_o$$

$$\boxed{\rho_M = r[0] - \underline{r}^H \cdot \underline{c} = r[0] - \underline{c}^H \cdot \underline{r}}$$

## Relationship of Linear Prediction (Wiener Hopf Equation) and Autoregressive Modeling (Yule Walker Equations)

$$f_M[n] = u[n] - \sum_{k=1}^{M} c_k^* u[n-k]$$

$$= \sum_{k=0}^{M} a_{M,k}^* u[n-k] \quad \text{where} \quad a_{M,k} = \begin{cases} 1 & k = 0 \\ -c_k^* & k \neq 0 \end{cases}$$

This is similar to autoregressive (AR) special case in Yule Walker equations. So we can use Yule Walker.

Yule Walker:

$$\begin{bmatrix} r(0) & r(-1) & \cdots & r(-p+1) \\ r(1) & \cdots & & \\ \vdots & & & \\ & & & r(0) \end{bmatrix} \begin{bmatrix} a(1) \\ \vdots \\ a(p) \end{bmatrix} = \begin{bmatrix} -r(1) \\ \vdots \\ -r(p) \end{bmatrix}$$

# Augmented Normal Equations (Wiener Hopf)

$$R_M \underline{c} = \underline{r} \implies \underline{r} \cdot 1 - R_M \underline{c} = 0$$

$$P_M = r[0] - \underline{r}^H \underline{c} \implies r[0] \cdot 1 - \underline{r}^H \underline{c} = P_M$$

$$\begin{bmatrix} r[0] & \underline{r}^H \\ \underline{r} & R_M \end{bmatrix} \begin{bmatrix} 1 \\ -\underline{c} \end{bmatrix} = \begin{bmatrix} P_M \\ \underline{0} \end{bmatrix} \implies R_{M+1} \, \underline{a}_M = \begin{bmatrix} P_M \\ \underline{0}_M \end{bmatrix}$$

Augmented Normal Equations

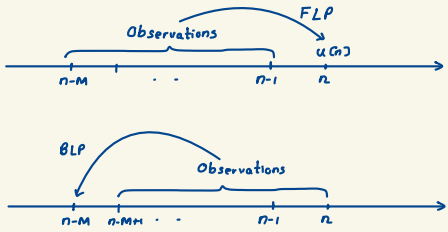
$R_{M+1}$ $\underline{a}_M$

$\underline{a}_M \in \mathbb{C}^{M+1}$

$R_{M+1} \in \mathbb{C}^{(M+1) \times (M+1)}$

# 2) Backward Linear Prediction

- Given observations $u[n], u[n-1], \ldots, u[n-M+1]$ find a Linear prediction of $u[n-M] = d[n]$




- Assume that filter coefficients are $\underline{g}$ and input $\underline{u}[n]$ is

$$\underline{g} = \begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_n \end{bmatrix} \qquad \underline{u}[n] = \begin{bmatrix} u[n] \\ u[n-1] \\ \vdots \\ u[n-M+1] \end{bmatrix}$$

$$\hat{u}[n-M \mid \mathcal{U}_n] = \sum_{k=1}^{M} g_k^* u[n+1-k] = \underline{g}^H \underline{u}[n]$$

Backward prediction error $= b_M[n] = u[n-M] - \hat{u}[n-M/\mathcal{U}_n]$

Minimum mean square Prediction Error $= P_M = \text{Var}\{b_M[n]\} = E[|b_M[n]|^2]$

## Normal Equations for BLP

For Wiener Filter we had $R_M \underline{a}^* = \underline{p}$. We replace $\underline{p} \to \underline{r}$ and $\underline{a}^* \to \underline{g}^*$. Hence,

$$R_M \underline{g} = \underline{r}^{B*}$$

## I) Correlation matrix of the tap input

$$R = E[\underline{u}[n] \cdot \underline{u}[n]^H] = \begin{bmatrix} r(0) & r(1) & \cdots & r(M-1) \\ r(-1) & r(0) & \cdots & r(M-2) \\ \vdots & & & \\ r(-M+1) & \cdots & \cdots & r(0) \end{bmatrix}$$

Revice
$R_M$ is a Toeplitz matrix when $\underline{u}[n]$ is WSS.

$u$ is WSS
$r(k)$ - autocorrelation of $u$ for lag $k$.

## II) Cross-correlation vector

$$\underline{r} = \mathbb{E}\left[\underline{u}^*[n-M]\cdot u[n]\right] = \begin{bmatrix} u^*[n-M]\,u[n] \\ u^*[n-M]\,u[n-1] \\ \vdots \\ u^*[n-M]\,u[n-M+1] \end{bmatrix} = \begin{bmatrix} r(M) \\ r(M-1) \\ \vdots \\ r(1) \end{bmatrix} \Rightarrow \text{new } \underline{r} = \underline{r}^{B^*}$$

In the FLP, we had $\underline{r}$ - cross correlation as $\underline{r} = \begin{bmatrix} r(-1) \\ r(-2) \\ \vdots \\ r(-M) \end{bmatrix}$ i.e. $\underline{p}$ in Wiener filters

but in the BLP, we can see $\underline{r}$ is reverse ordered and conjugated since $r(-k) = r(k)^*$

## III) Variance of $u[n]$

Since $u[n]$ is assumed to be zero mean $\Rightarrow$ $\mathrm{Var}[u[n-M]] = \mathbb{E}[|u[n-M]|^2]$

$= \mathbb{E}[u[n-M]\cdot u^*[n-M]]$

$= r(0)$ ... wss

$\mathrm{Var}[u[n]] = r(0)$ .... $\sigma_d^2$ in Wiener filters

- From the Wiener filter derivation for $J_{min}$, which is $P_M$ now, we can get

Wiener $\underline{p}$ → FLP $\underline{r}$ → BLP $\underline{r}^{B^*}$

$$J_{min} = \sigma_d^2 - \underline{p}^H\cdot\underline{w}_o$$

$$P_M = r[0] - \left(\underline{r}^{B^*}\right)^H\underline{g} = r[0] - \underline{r}^{B^T}\underline{g}$$

$$\boxed{P_M = r[0] - \underline{r}^{B^T}\underline{g}}$$

## Connection Between BLP and FLP

$$\boxed{\underline{g}^{B^*} = \underline{c} \qquad \underline{g} = \underline{c}^{B^*}}$$

$$R_M\,\underline{g} = \underline{r}^{B^*} \Rightarrow (R_M\,\underline{g})^* = \underline{r}^B \Rightarrow R_M^*\,\underline{g}^* = \underline{r}^B$$

$$\Rightarrow R_M^T\cdot\underline{g}^* = \underline{r}^B \qquad R_M \text{ is Hermission}$$

$$\Rightarrow R_M\cdot\underline{g}^{B^*} = \underline{r}$$

Backward Prediction Error Power = Forward Prediction Error Power

Compare this with FLP prediction $R_M\,\underline{c} = \underline{r}$

# Augmented Normal Equations for BLP

- From the BLP known factors

$$\text{MMSE} \longrightarrow r[0] - \underbrace{g^H \underline{r}^{B^*}}_{=\underline{r}^{B^T} g} =: P_M \implies r[0] - \underline{r}^{B^*} g = P_M$$

$$\text{Normal Equations} \longrightarrow R_M g^{B^*} = \underline{r} \implies R_M g^{B^*} = \underline{r}$$

$$\implies R_M g = \underline{r}^{B^*}$$

$$\implies r^{B^*} - R_M g = \underline{0}$$

$$\therefore \underbrace{\left[\begin{array}{c|c} R_M & \underline{r}^{B^*} \\ \hline \underline{r}^{B^T} & r[0] \end{array}\right]}_{R_{M+1}} \underbrace{\begin{bmatrix} -g \\ 1 \end{bmatrix}}_{\underline{a}_M^{B^*}} = \begin{bmatrix} \underline{0} \\ P_M \end{bmatrix}$$

where

$$\underline{a}_M = \begin{bmatrix} 1 \\ -\underline{c} \end{bmatrix} \implies \underline{a}_M^{B^*} = \begin{bmatrix} -\underline{c}^{B^*} \\ 1 \end{bmatrix} = \begin{bmatrix} -g \\ 1 \end{bmatrix}$$

| M-tap Linear Predictors | Wiener Filters | FLP | BLP |
|---|---|---|---|
| Filter Coefficients | $\underline{a} = \underline{w}^*$ | $\underline{c}^*$ | $\underline{g}^*$ |
| Input | $\underline{u}[n]$ | $\underline{u}[n-1]$ | $\underline{u}[n]$ |
| Desired response | $d[n]$ | $u[n]$ | $u[n-M]$ |
| Estimated signal | $\underline{a}^T \underline{u}[n]$ | $\underline{c}^H \underline{u}[n-1]$ | $\underline{g}^H \underline{u}[n]$ |
| Estimated Error | $e[n] = d[n] - \underline{a}^T \underline{u}[n]$ | $f_M[n] = u[n] - \underline{c}^H \underline{u}[n-1]$ | $b_M[n] = u[n-M] - \underline{g}^H \underline{u}[n]$ |
| Correlation Matrix | $R_M$ | $R_M$ | $R_M$ |
| Cross-correlation Vector | $\underline{p}$ | $\underline{r}$ | $\underline{r}^{B*}$ |
| MMSE | $\sigma_d^2 - \underline{a}^T \underline{p} =: J_{min}$ | $r[0] - \underline{c}^H \underline{r} =: P_M$ | $r[0] - \underline{g}^H \underline{r}^{B*} =: P_M$ |
| Normal Equations | $R_M \underline{a}^* = \underline{p}$ | $R_M \underline{c} = \underline{r}$ | $R_M \underline{g}^{B*} = \underline{r}$ |
| Augmented Normal Equations | - | $R_{M+1} \underline{a}_M = \begin{bmatrix} P_M \\ \underline{0} \end{bmatrix}$ | $R_{M+1} \underline{a}_M^{B*} = \begin{bmatrix} \underline{0} \\ P_M \end{bmatrix}$ |

In book $\underline{a} = \underline{w}^*$

$$\underline{a}_M = \begin{bmatrix} 1 \\ -\underline{c} \end{bmatrix} \qquad \underline{a}_M^{B*} = \begin{bmatrix} -\underline{c}^{B*} \\ 1 \end{bmatrix} = \begin{bmatrix} -\underline{g} \\ 1 \end{bmatrix}$$

# 3) Levinson - Durbin Recurssion

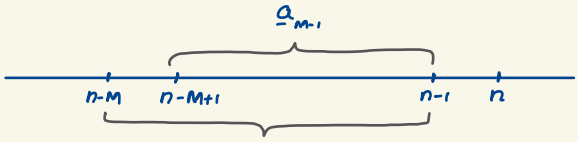
Use Topleitz property of $R_M$ to solve Winer Hopf equation iteratively

- Solving normal equations in brute force requires inverting $R_M$ that needs $O(M^3)$ multiplications.
- Let's use time symmetry to solve the normal equations with only $\frac{M(M+1)}{2}$ multiplications.
- Idea - Recurssive Solution Suppose that we know the FLP solution with $(m-1)$ taps $\underline{a}_{M-1}$ and MMSE $P_{M-1}$.
- Assume filter coefficients for FLP to predict $u[n]$ from $u[n-1], .. u[n-M]$ are $\underline{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_M \end{bmatrix}$ At the $M^{th}$ iteration of Levinson Durbin we know them to be $\underline{c} = \begin{bmatrix} c_{M,1} \\ c_{M,2} \\ \vdots \\ c_{M,M} \end{bmatrix}$, hence $\underline{a}_M$ in extended equations become,

$$\underline{a}_M = \begin{bmatrix} 1 \\ -\underline{c} \end{bmatrix} = \begin{bmatrix} 1 \\ -c_{M-1,1} \\ -c_{M-1,2} \\ \vdots \\ -c_{M-1,M-1} \end{bmatrix} \cdots = \begin{bmatrix} a_{M-1,0} \\ a_{M-1,1} \\ a_{M-1,2} \\ \vdots \\ a_{M-1,M-1} \end{bmatrix} \quad P_{M-1}$$

why negatives. Wasn't it conjugates?

- Goal : How to modify $\underline{a}_{M-1}$ to obtain $\underline{a}_M$ ?



$$\underline{a}_m = \begin{bmatrix} \underline{a}_{m-1} \\ 0 \end{bmatrix} + \Gamma_m \begin{bmatrix} 0 \\ \underline{a}_{m-1}^{B*} \end{bmatrix}$$

$$\Gamma_m = \frac{-\Delta_{m-1}}{\rho_{m-1}} = -\frac{\underline{r}_m^{B^T}\,\underline{a}_{m-1}}{\rho_{m-1}}$$

$$\rho_m = \rho_{m-1}\left(1 - |\Gamma_m|^2\right)$$

Levinson Durbin Recursion

Proof: Assume that the $m$ taps at time $m$ can be approximated using $(m-1)$ taps at time $m-1$ by using some error vector $\underline{\tilde{\delta}}_m$ and $\Gamma_m = a_{m,m} \neq 0$

$$\underline{a}_m = \begin{bmatrix} \underline{a}_{m-1,0} \\ \underline{a}_{m-1,1} \\ \vdots \\ \underline{a}_{m-1,m-1} \\ 0 \end{bmatrix} + \Gamma_m \begin{bmatrix} 0 \\ \tilde{\delta}_{m,1} \\ \tilde{\delta}_{m,2} \\ \vdots \\ 1 \end{bmatrix}$$

From Augmented Normal Equations for FLP, if filter coefficients for

$$R_{m+1}\,\underline{a}_m = \begin{bmatrix} \rho_m \\ \underline{0} \end{bmatrix}$$

$$R_{m+1}\begin{bmatrix} \underline{a}_{m-1} \\ 0 \end{bmatrix} + R_{m+1}\,\Gamma_m \begin{bmatrix} 0 \\ \tilde{\delta}_m \end{bmatrix} = \begin{bmatrix} \rho_m \\ \underline{0} \end{bmatrix} \quad \cdots \text{ by above approximation}$$

$$\left[\begin{array}{c|c} R_m & \underline{r}_m^{B*} \\ \hline \underline{r}_m^{B^T} & r(0) \end{array}\right] \begin{bmatrix} \underline{a}_{m-1} \\ 0 \end{bmatrix} + \Gamma_m \begin{bmatrix} r(0) & \underline{r}^H \\ \underline{r} & R_m \end{bmatrix} \begin{bmatrix} 0 \\ \tilde{\delta}_m \end{bmatrix} = \begin{bmatrix} \rho_m \\ \underline{0} \end{bmatrix}$$

$$\begin{bmatrix} R_m\,\underline{a}_{m-1} \\ \underline{r}_m^{B^T}\,\underline{a}_{m-1} \end{bmatrix} + \Gamma_m \begin{bmatrix} \underline{r}^H \tilde{\underline{\delta}}_m \\ R_m\,\tilde{\underline{\delta}}_m \end{bmatrix} = \begin{bmatrix} \rho_m \\ \underline{0}_m \end{bmatrix} \quad \cdots \Delta_{m-1} = \underline{r}_m^{B^T} \cdot \underline{a}_{m-1}$$

$\underline{r}_m^{B^T}\,\underline{a}_{m-1} =: \Delta_{m-1}$

$$\begin{bmatrix} R_m & \underline{a}_{m-1} \\ \underline{0}_{m-1} & \\ \Delta_{m-1} & \end{bmatrix} + \Gamma_m \begin{bmatrix} \underline{r}^H \underline{\tilde{\delta}}_m \\ R_m \underline{\tilde{\delta}}_m \end{bmatrix} = \begin{bmatrix} P_m \\ \underline{0}_{m-1} \\ 0 \end{bmatrix}$$

By considering only coloured raws

$$R_m \cdot \underline{\tilde{\delta}} = \begin{bmatrix} \underline{a}_{m-1} \\ -\Delta_{m-1} \\ \hline \Gamma_m \end{bmatrix} \quad \cdots (1)$$

But we also know that from extended normal equations at M-1

$$R_m \cdot \underline{a}^{B*}_{m-1} = \begin{bmatrix} \underline{0}_{m-1} \\ P_{m-1} \end{bmatrix} \quad \cdots (2)$$

From (1) and (2), Since $\tilde{\delta}_{m,m} = 1$ and $a^{B*}_{m-1} = 1$ the ratio of between vectors (1) and (2) are 1,

$$\Rightarrow \tilde{\delta}_m = \underline{a}^{B*}_{m-1}$$

$$\Rightarrow \Gamma_m = \frac{-\Delta_{m-1}}{P_{m-1}}$$

■

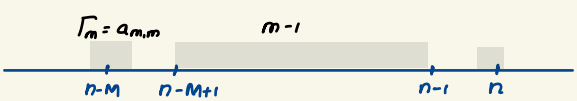
## Initialization

For Levinson Derbin recurssion, we can have the intialization as,

$$a_{m,m-1} = 0$$

$$P_0 = r[0]$$

$$\Delta_0 = \left[r[-1]\right]^T [1] = r[-1]$$



## Algorithm

For $m = 1, 2, \ldots, M$

① $\Gamma_m = \dfrac{-\Delta_{m-1}}{\rho_{m-1}}$

② For $k = 1, \ldots, m$

$$a_{m,k} = a_{m-1,k} + \Gamma_m \, a^*_{m-1,m-k}$$

end

③ $\Delta_m = \underline{r}^{B^T}_{m+1} \, \underline{a}_m$

$\rho_m = \rho_{m-1} \left(1 - |\Gamma_m|^2\right)$

output $\underline{a}_M$

end

## Complexity

No of multiplications $= 1 + 2 + \ldots + M = \sum_{m=1}^{M} m = \dfrac{M(M+1)}{2}$ → $+2M$

# Properties of Prediction Error

## Forward Prediction Error

$$f_m[n] = u[n] - \underline{c}^H \underline{u}[n-1] = \sum_{k=0}^{m} a_{m,k}^* \, u[n-k]$$

where $a_{m,k} = \begin{cases} 1 & k=0 \\ -c_k & k \neq 0 \end{cases}$

(0) $\text{Var}[f_m[n]] = E[|f_m[n]|^2] = \rho_m$

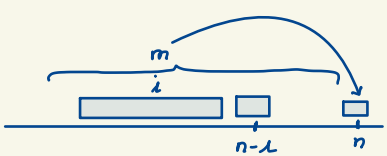
(1) $E[f_m[n] \cdot u^*[n-k]] = 0 \qquad 1 \leq k \leq m$ — from principle of orthogonality

(2) $E[f_m[n] \cdot f_i^*[n]] = \rho_m \qquad i \leq m$

$f_i^*[n] = \left(u[n] - \sum_{k=0}^{i} a_{i,k} \, u[n-k]\right)^*$

(3) $E[f_m[n] \, f_i[n-l]] = 0 \qquad 1 \leq l \leq m-i \qquad i < m$

Whatever predictions within the m taps are orthogonal




(4) $E[f_m[n+m] \, f_i^*[n+i]] = \begin{cases} \rho_m & i=m \\ 0 & \text{o.w} \end{cases}$

# HAY - 7

# Kalman Filters

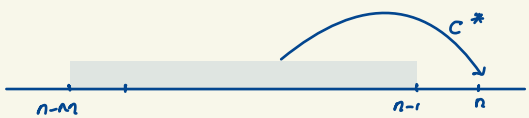
- Use the notation $y[n]$ instead of $u[n]$. Suppose that we observe $y[1], y[2], \ldots, y[n]$

$$\underbrace{\alpha[n]}_{\text{Innovation Process}} := f_{n-1}[n] = y[n] - \hat{y}[n \mid Y_{n-1}]$$

$$Y_{n-1} := \{y[1], \ldots, y[n-1]\}$$

FLP ⟶

- Goal: Predict $u[n]$ given $u[n-1], \ldots, u[n-M]$

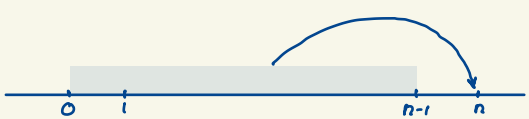



$$\hat{x}[n \mid Y_{n-1}] = \sum_{k=1}^{M} c_k^* \cdot u[n-k] \quad \text{where} \quad R_M \cdot \underline{c} = \underline{r}$$

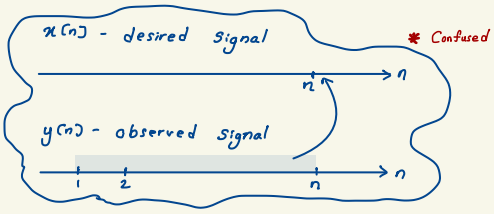
- This $c = R_M^{-1} \cdot \underline{r}$ can be effectively solved using Levinson Durbin recurssion.

Filtering / Kalman ⟶

- Goal - Estimate $y[n]$ given observations $y[1], \ldots, y[n-1]$




- We don't use a fixed tap of length M. At any given time n, we use all n-1 time point observations.




- If we try to do one step prediction as in FLP, $\hat{y}[n]$ prediction becomes a linear combination of $y[1], \ldots, y[n-1]$

$$\therefore \hat{y}[n \mid Y_{n-1}] = \text{linear combination of } y[1], \ldots, y[n-1]$$

$$\therefore \underset{\text{Error}}{\text{Forward Prediction}} = f_{n-1}[n] = y[n] - \hat{y}[n \mid Y_{n-1}]$$

$f_{n-1}[n] \equiv \alpha[n]$ — Innovation Process

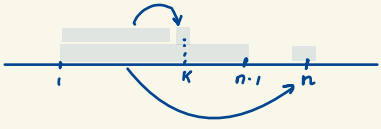
## Properties of $\alpha[n]$ in FLP Sense

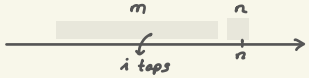
1) Principle of Orthogonality

$$E[\alpha[n] \cdot y^*[k]] = 0 \qquad 1 \leq k \leq n-1$$

2) $E[\alpha[n] \cdot \alpha^*[k]] = 0 \qquad 1 \leq k \leq n-1$ ···· From property ④ in FLP

$$E\{f_m[n] \cdot f_i[n-1]\} = 0 \qquad 1 \leq i \leq m-1$$




3) Span $\{y[1], y[2], \ldots, y[n]\}$ = Span $\{\alpha[1], \alpha[2], \ldots, \alpha[n]\}$

Proof: From the definition of the innovation process,

$$\alpha[1] = f_0[1] = y[1]$$

$$\alpha[2] = f_1[2] = y[2] - \underbrace{c_{1,1}\, y[1]}_{\hat{y}[2 \mid Y_1]} = y[2] - \hat{y}[2 \mid Y_1]$$

$$\alpha[3] = f_2[3] = y[3] - \underbrace{c_{2,1}\, y[2] - c_{2,2}\, y[1]}_{\hat{y}[3 \mid Y_2]} = y[3] - \hat{y}[3 \mid Y_2]$$

$$\vdots \qquad\qquad \vdots \qquad\qquad \vdots$$

$$\alpha[n] = f_{n-1}[n] = y[n] - \underbrace{c_{n,1}\, y[n-1] - c_{n,2}\, y[n-2] - \ldots - c_{n,n}\, y[1]}_{\hat{y}[n \mid Y_{n-1}]} = y[n] - \hat{y}[n \mid Y_{n-1}]$$

$$\begin{bmatrix} \alpha[1] \\ \alpha[2] \\ \vdots \\ \alpha[n] \end{bmatrix} = \begin{bmatrix} 1 & & & & \\ -c_{1,1} & 1 & & \phi & \\ -c_{2,2} & -c_{2,1} & 1 & & \\ -c_{3,3} & -c_{3,2} & -c_{3,3} & 1 & \\ \vdots & & & & \\ -c_{n-1,n-1} & \cdots & & -c_{n-1,1} & 1 \end{bmatrix} \begin{bmatrix} y[1] \\ y[2] \\ \vdots \\ y[n] \end{bmatrix}$$

$\det = 1$

Since the determinant of a triangle matrix is a product of the diagonal

$\Rightarrow \quad \underline{\alpha}(n) = X \cdot \underline{y}(n)$ where $\det(X) = 1$. $\therefore$ $\underline{y}(n)$ is another representation of $\underline{\alpha}(n)$. Hence they have the same span. ■

## Innovation Process and Grand Schmidt Normalization

In the above proof, each $\alpha[i]$ step is the difference of $y[i]$ from $\hat{y}[i \mid Y_{j..}]$ which is like removing all the linear combinations of previous '$y$' values from current proces. (As in Grand Schmidt process)

$\therefore$ This is effectively converting $y[i]$'s to another set of basis $\alpha[i]$'s that are orthogonal.

$\therefore$ $\alpha[i]$'s can be independantly used to solve for each '$n$' Directions in the span. $\therefore$ Coupled simultaneous equation are becomming de-coupled.

## How $\alpha[i]$'s Reduce Complexity

| Goal - Want to predict $x[n]$ using $y[1], y[2], \ldots, y[n]$ | Goal - Want to predict $x[n]$ using $\alpha[1], \alpha[2], \ldots \alpha[n]$ |
|---|---|
| $R_n = \begin{bmatrix} r_y[0] & r_y[1] & \cdots & r_y[n-1] \\ r_y[-1] & r_y[0] & \cdots & r_y[n-2] \\ \vdots & & & \\ r_y[-n+1] & r_y[-n+2] & \cdots & r_y[0] \end{bmatrix}$ | $R_n = \begin{bmatrix} E[\alpha[1]\alpha^*[1]] & 0 & \cdots & \phi \\ 0 & E[\alpha[2]\cdot\alpha^*[2]] & & \vdots \\ \vdots & \phi & & \\ & & & E[\alpha[n]\cdot\alpha^*[n]] \end{bmatrix}$ |
| $\underline{P} = E\left[x^*[n] \begin{bmatrix} y[1] \\ y[2] \\ \vdots \\ y[n] \end{bmatrix}\right]$ | $\underline{P} = \begin{bmatrix} E[x^*[n]\cdot\alpha[1]] \\ E[x^*[n]\cdot\alpha[2]] \\ \vdots \\ E[x^*[n]\cdot\alpha[n]] \end{bmatrix}$ |

$$\hat{x}[n|Y_n] = \sum_{k=0}^{n-1} a_k \cdot y[n-k]$$

$$\Rightarrow R_n \underline{a}^* = \underline{\rho}$$

At best requires $O(n^2)$ multiplications.

$$\hat{x}[n|Y_n] = \sum_{k=1}^{n} b_k \cdot \alpha[k]$$

$$\Rightarrow R_n \underline{b}^* = \underline{\rho}$$

$k = 1, 2, \ldots, n$

$$\Rightarrow b_k^* = \frac{E\left[x^*[n] \cdot \alpha[k]\right]}{E\left[\alpha[k]\ \alpha^*[k]\right]}$$

At best requires $O(n^2)$ multiplications.

## How having a model for $x[n]$ Reduces Complexity

we can bring $O(n)$ to $O(1)$ with a model

Ex: Assume the model for $x[n]$ is

$$x[n] = X + \varepsilon[n]$$

$(x - x[n-1])$

where $X$ is independant of $n$ and $\varepsilon[n]$ are noise that is zero mean.

$$\Rightarrow \hat{x}[n|Y_n] = \underbrace{\sum_{k=1}^{n-1} b_k \cdot \alpha[k]}_{} + b_n \alpha[n]$$

$$\Rightarrow = \hat{x}[n-1|Y_{n-1}] + b_n \cdot \alpha[n]$$

where all the $b_k$'s for $k = 1, .., n-1$ are know from previous step

$\therefore$ Assumption for $n$'th is

$$b_n = \frac{E\left[x^*[n] \cdot \alpha[k]\right]}{E\left[\alpha[k] \cdot \alpha^*[k]\right]} = \frac{E\left[x^*[n-1] \cdot \alpha[k]\right]}{E\left[\alpha[n] \cdot \alpha^*[k]\right]}$$

$$\Rightarrow \hat{x}[n|Y_n] = \underbrace{\hat{x}[n-1|Y_{n-1}]}_{\text{known from previous step}} + \underbrace{\frac{E\left[x^*[n-1] \cdot \alpha[k]\right]}{E\left[\alpha[n] \cdot \alpha^*[k]\right]}}_{x^*[n] = x^*[n-1] + \left(\varepsilon[n] \cdot \varepsilon[n-1]\right)} \cdot \alpha[n]$$

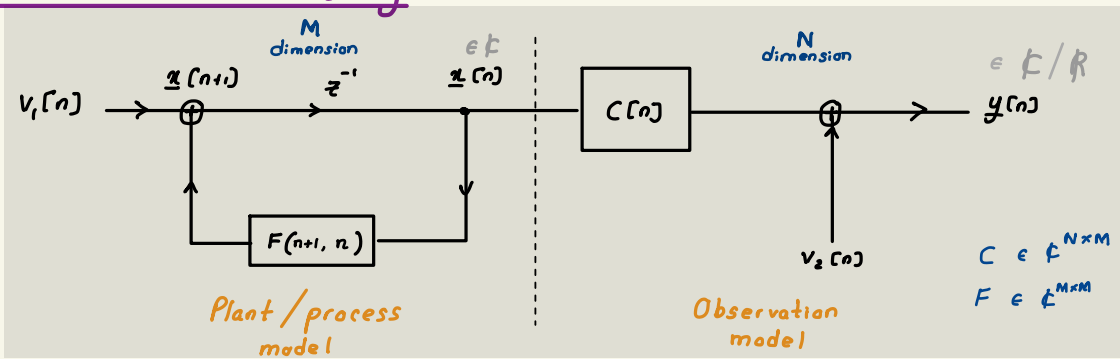
which is independant of $\alpha[n]$

$\therefore$ 2 Multiplications

$\Rightarrow$ Only $O(1)$ complexity is needed

- Think as if this is a diffusion process.
- $v_1$ is the starting noise, $v_2$ is the noise added iteratively.

# The Kalman Filter Setting

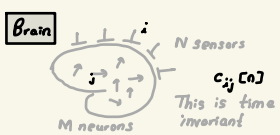


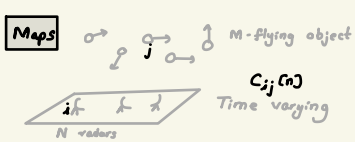

These all are at time n.
∴ We talk about spatial dimensions.

$C[n]$ - Projection matrix. Project M to N dimensions.
$v_2$ - Noise in the Iterative process
$v_1$ - Driving Input

Goal: Linear Prediction of m dimensional X with an N dimensional basis.

- Ex: Google Maps
  ECG and Neurons - Trying to predict $10^5$ neurons from 100 scalp sensors




Always F will be given which will include lot of physics and observing pilot data.

$$\underline{x}[n+1] = \begin{bmatrix} \Delta v_x & 0 & 0 \\ 0 & \Delta v_y & 0 \\ 0 & 0 & \Delta v_z \end{bmatrix} \underline{x}[n] + \underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}}_{\text{Change of speed}}$$

- Due to $F(n)$ this is no longer WSS

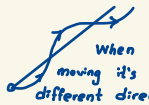
$$\underline{x}[n+1] = F[n+1, n] \cdot \underline{x}[n] + \underline{v}_1[n] \quad \text{... process equation} \qquad (F \in \mathbb{C}^{M \times M})$$

$$\underline{y}[n] = C[n] \cdot \underline{x}[n] + \underline{v}_2[n] \quad \text{.. measurement equation} \qquad (\underline{y} \in \mathbb{C}^N,\ C \in \mathbb{C}^{N \times M})$$

- Typically $N \ll\ll M$.

$$E\left[\underline{v}_1[n]\,\underline{v}_1^H[k]\right] = \begin{cases} Q_1[n] & n=k \\ 0 & n\neq k \end{cases}$$

variance



When a plane is moving it's deviations in different directions might be correlated. $Q_1$ models that

$$E\left[\underline{v}_2[n]\,\underline{v}_2^H[k]\right] = \begin{cases} Q_2[n] & n=k \\ 0 & n\neq k \end{cases}$$

$Q_2$ models correlation matrix over 2 sensors/radars.

$$E\left[\underline{v}_1[n]\,\underline{v}_2^H[k]\right] = 0 \qquad \forall\, n,k$$

Radar has nothing to do with planes

Sensors has nothing to do with Brain

Innovation Process: $\underline{\alpha}[n] = \underline{y}[n] - \hat{\underline{y}}[n \mid Y_{n-1}]$

## Properties of Innovation Process

1) $E\left[\underline{\alpha}[n]\cdot \underline{y}^H[n]\right] = \underline{0}$ $\qquad 1 \leq k \leq n-1$

2) $E\left[\alpha[n]\cdot \underline{\alpha}^H[n]\right] = \underline{a}$ $\qquad 1 \leq k \leq n-1$

3) $\text{Span}\{\underline{y}[1], \ldots, \underline{y}[n]\} = \text{Span}\{\alpha[1], \ldots, \alpha[n]\}$

4) $E\left[\underline{x}[k]\cdot \underline{v}_2^H[n]\right] = 0$ $\qquad \forall\, k,n$

5) $E\left[\underline{y}[k]\cdot \underline{v}_2^H[n]\right] = 0$ $\qquad 1 \leq k \leq n-1$

6) $E\left[\underline{y}[k]\cdot \underline{v}_1^H[n]\right] = 0$ $\qquad 1 \leq k \leq n-1$

## 2) Prediction Problem

Minimise variance of the error

**Goal :** Estimate $\underline{x}[n+1]$ given $y[1], y[2], \ldots, y[n]$ In MMSE Sense.

$$\hat{\underline{x}}[n+1/Y_n] = \sum_{k=1}^{n} B'[n] \cdot \underline{y}[k]$$

From the principle of orthogonality, at the best MMSE estimate $\hat{x}[n+1|Y_n]$ should be orthogonal to all data samples

$$\Rightarrow \underline{x}[n+1] - \hat{\underline{x}}[n+1/Y_n] \perp \underline{y}[k] \quad * \; 1 \leq k \leq n$$

$$\Rightarrow \mathbb{E}\left\{\left(\underline{x}[n+1] - \hat{\underline{x}}[n+1/Y_n]\right) \underline{y}^H[k]\right\} = 0$$

Rather than doing the prediction of the output at n+1 (ie $\underline{x}[n+1]$) it is easier to predict sensors at n+1 (ie $y[n+1]$). This can be used later to find $x[n+1]$

To find — $\hat{\underline{x}}[n+1/Y_n]$

Since span $\{y[1], \ldots, y[n]\}$ = span $\{\alpha[1], \ldots, \alpha[n]\}$ where $\alpha$ is the innovation process. ($\alpha[n] = y[n] - \hat{y}[n/Y_{n-1}]$), It is easier work over the "Innovation Process" due to orthogonality of them.

Minimise variance of the error

**New Goal :** Estimate $\underline{x}[n+1]$ given $y[1], y[2], \ldots, y[n]$ In MMSE Sense.

$$\hat{\underline{x}}[n+1/Y_n] = \sum_{k=1}^{n} B[n] \cdot \underline{\alpha}[k] \quad (1)$$

Vector $\in \mathbb{C}^M$ Matrix $\in \mathbb{C}^{M \times N}$ Vector $\in \mathbb{C}^N$

Similar to FLP equation, $\underbrace{\hat{u}[n+1/U_n]}_{\text{Scalar}} = \sum_{k=0}^{m} \underbrace{b_k}_{\text{Scalar}} \cdot \underbrace{u[n-k]}_{\text{Scalar}}$

- Forward Prediction Error of $\underline{x}[n]$ = $\underline{\varepsilon}[n,n-1] = \underline{x}[n] - \hat{\underline{x}}[n|Y_{n-1}]$ (2)

- Correlation Matrix of the Innovation Process $\alpha$ = $R(n) = E[\underline{\alpha}[n]\,\underline{\alpha}^H[n]]$ (3)

- Correlation Matrix of the Prediction State Error (MxM) = $K(n,n-1) = E[\varepsilon(n,n-1)\cdot\varepsilon^H(n,n-1)]$ (4)

$$\underline{\alpha}[n] = C[n]\cdot\underline{\varepsilon}[n,n-1] + \underline{v}_2[n] \quad (5)$$

Proof: From measurement equation, $\Rightarrow y[n] = C[n]\,\underline{x}[n] + \underline{v}_2[n]$ ..... (1)

Condition measurement on $Y_{n-1}$ $\Rightarrow \hat{y}[n|Y_{n-1}] = C[n]\cdot\hat{\underline{x}}[n|Y_{n-1}] + \underline{v}_2[n|Y_{n-1}]$

$v_2$ is noise, it can not be predicted using $Y_{n-1}$ $\Rightarrow \hat{y}[n|Y_{n-1}] = C[n]\cdot\hat{\underline{x}}[n|Y_{n-1}]$ .... (3)

From (1), (2)

$$\underline{\alpha}[n] = y[n] - \hat{y}[n|Y_{n-1}]$$

$$\Rightarrow \underline{\alpha}[n] = C[n]\cdot\Big(\underbrace{\underline{x}[n] - \hat{\underline{x}}[n|Y_{n-1}]}_{\underline{\varepsilon}[n,n-1]}\Big) + \underline{v}_2[n]$$

$$\Rightarrow \underline{\alpha}[n] = C[n]\cdot\underline{\varepsilon}[n,n-1] + \underline{v}_2[n]$$

■

$$R[n] = C[n]\cdot K[n,n-1]\cdot C^H[n] + Q_2[n] \quad (6)$$

Proof: If $\underline{y} = A\underline{x} \Rightarrow Cov(\underline{y}) = A\cdot Cov(\underline{x})\cdot A^H$. Apply this to (5). Since $\varepsilon \perp v_2$ we get the above expression

■

$$\boxed{E\left[\varepsilon[n+1,n]\cdot \underline{\alpha}^H[m]\right] = \underline{0}} \qquad m=1,2,\dots,n \qquad (7)$$

Proof: According to principle of orthogonality, the predicted state error vector is orthogonal to innovation process. Since error in predicting $x[n]$ using $Y_{n-1}$ hasn't seen the innovations. ∎

## Estimation of the State Using Innovation Process

$$\boxed{B[m] = E\left[x[n+1]\cdot \underline{\alpha}^H[m]\right]\cdot R_m^{-1}} \qquad m=1,2,\dots,n \qquad (8)$$

Proof: From (2) and (1)

$$\Rightarrow E\left[\left(x[n+1]-\hat{x}[n+1\,|\,Y_n]\right)\underline{\alpha}^H[m]\right] = \underline{0}$$

$$\Rightarrow E\left[\left(x[n+1]-\sum_{k=1}^{n} B[k]\,\underline{\alpha}[k]\right)\underline{\alpha}^H[m]\right] = \underline{0} \qquad \cdots \text{ from (1)}$$

$$\Rightarrow E\left[x[n+1]\cdot\underline{\alpha}^H[m]\right] = \sum_{k=1}^{n} B[k]\cdot E\left[\underline{\alpha}[k]\,\underline{\alpha}^H[k]\right] = B[m]\cdot \underbrace{E\left[\underline{\alpha}[m]\cdot\underline{\alpha}^H[m]\right]}_{R[m] \text{ from (3)}}$$

$$\Rightarrow B[m] = E\left[x[n+1]\cdot\underline{\alpha}^H[m]\right]\cdot R_m^{-1}$$

∎

- If we rephrased (1) as an optimization problem to find $B[i]$'s such that we will get minimum MMSE error it becomes

$$\Rightarrow \min_{B[k]\text{'s}} E\left\{\left\|\underline{x}[n+1]-\hat{\underline{x}}[n+1\,|\,Y_n]\right\|^2\right\}$$

$$\Rightarrow \min_{B[k]\text{'s}} E\left\{\left\|\underline{x}[n+1]-\sum_{k=1}^{n} B[n]\cdot\underline{\alpha}[k]\right\|^2\right\} \qquad \cdots \text{ from (1)}$$

If we partial differentiate $\frac{\partial H}{\partial B[m]} = 0$ then we will get (8) again.

- The $i^{th}$ equation only has $B[i]$'s since we used a set of $\alpha[i]$'s which are orthogonal.

- What are these $B[m]$'s. We want to predict high dimensional ($M$) $\hat{x}[n+1|Y_n]$ vector using low dimensional $\alpha[i]$'s. After figuring out these projections $B$ over innovation process, convert them to $C$s in observation model.

$$\begin{bmatrix} \\ M \\ \\ \end{bmatrix}\begin{bmatrix} \hat{x} \end{bmatrix} = \begin{bmatrix} \\ M \\ \\ \end{bmatrix}\begin{bmatrix} N \\ B[1] \\ \\ \end{bmatrix}\begin{bmatrix} \\ N \\ \\ \end{bmatrix}\begin{bmatrix} \alpha[1] \end{bmatrix} + \begin{bmatrix} \\ M \\ \\ \end{bmatrix}\begin{bmatrix} N \\ B[2] \\ \\ \end{bmatrix}\begin{bmatrix} \\ N \\ \\ \end{bmatrix}\begin{bmatrix} \alpha[2] \end{bmatrix} + \cdots + \begin{bmatrix} \\ M \\ \\ \end{bmatrix}\begin{bmatrix} N \\ B[n] \\ \\ \end{bmatrix}\begin{bmatrix} \\ N \\ \\ \end{bmatrix}\begin{bmatrix} \alpha[n] \end{bmatrix}$$

$$\underline{\alpha}[n] = \underline{y}[n] - \hat{\underline{y}}[n|Y_{n-1}]$$

$$\hat{\underline{y}}[n|Y_{n-1}] = C[n] \cdot \hat{x}[n|Y_{n-1}]$$

$$\boxed{\underline{\alpha}[n] = \underline{y}[n] - C[n]\,\hat{x}[n|Y_{n-1}]} \quad \cdots (*)$$

## How Can Plant / Process Model Helps Reduce Complexity

$$\boxed{\hat{\underline{x}}[n+1|Y_n] = F[n+1,n] \cdot \hat{\underline{x}}[n|Y_{n-1}] + G[n] \cdot \underline{\alpha}[n]} \quad (9)$$

Kalman Equation

Proof: From (1)

$$\Rightarrow \hat{\underline{x}}[n+1|Y_n] = \sum_{k=1}^{n} B[n] \cdot \underline{\alpha}[k]$$

$$\Rightarrow = \sum_{k=1}^{n} \left( E\left[ \underline{x}[n+1] \cdot \underline{\alpha}^H[k] \right] \cdot R_k^{-1} \right) \cdot \underline{\alpha}[k] \quad \cdots \text{from } (8)$$

$$\Rightarrow = \sum_{k=1}^{n-1} \left( E\left[ \left( F[n+1,n] \cdot \underline{x}[n] + \underline{v}_1[n] \right) \underline{\alpha}^H[k] \right] \cdot R_k^{-1} \right) \cdot \underline{\alpha}[k] \quad \cdots \text{process equation}$$

($\underline{v}_1[n] \perp \underline{\alpha}[k]$, $k \le n$)

$$+ E\left[ \underline{x}[n+1] \cdot \underline{\alpha}^H[n] \right] R_n^{-1} \cdot \underline{\alpha}[n]$$

$$\Rightarrow = F[n+1,n] \sum_{k=1}^{n-1} \left( E\left[ \underbrace{\underline{x}[n] \cdot \underline{\alpha}^H[k]}_{B[k]} \right] \cdot R_k^{-1} \right) \cdot \underline{\alpha}[k]$$

$$+ E\left[ \underline{x}[n+1] \cdot \underline{\alpha}^H[n] \right] R_n^{-1} \cdot \underline{\alpha}[n]$$

$$\Rightarrow = F[n+1,n] \underbrace{\sum_{k=1}^{n-1} B[k]\,\underline{\alpha}[k]}_{\hat{\underline{x}}[n|Y_{n-1}]} + E\left[ \underline{x}[n+1] \cdot \underline{\alpha}^H[n] \right] R_n^{-1} \cdot \underline{\alpha}[n] \quad \text{from } (8)$$

$$\Rightarrow = F[n+1,n] \cdot \hat{\underline{x}}[n|Y_{n-1}] + \underbrace{E\left[ \underline{x}[n+1] \underline{\alpha}^H[n] \right] R_n^{-1}}_{G[n]} \underline{\alpha}[n] \quad \text{from } (1)$$

$G[n]$: Kalman Gain

$$\Rightarrow = F[n+1,n] \cdot \hat{\underline{x}}[n|Y_{n-1}] + G[n] \cdot \underline{\alpha}[n]$$

■

- Now calculating the $\hat{\underline{x}}[n+1|y_n]$ is a recurssive process that keeps updating what was known in the previous step. ∴ This is now $O(1)$ complexity.

## Kalman Gain

Definition - $$G[n] = \mathbb{E}\left[\underline{x}[n+1]\,\underline{\alpha}^H[n]\right] \cdot R^{-1}[n] \quad (10)$$

$$G[n] = F[n+1,n]\,K[n,n-1]\,C^H[n]\left(C^H[n]\,K[n,n-1]\,C^H[n] + Q_2[n]\right)^{-1} \quad (11)$$

Proof: $\mathbb{E}\left[\underline{x}[n+1]\,\underline{\alpha}^H[n]\right]$ = $\mathbb{E}\left[\left(\underbrace{F[n+1,n]\cdot\underline{x}[n] + \underline{v}_1[n]}_{\perp \text{ orthogonal}}\right)\underline{\alpha}^H[n]\right]$ .... process equation

$= \mathbb{E}\left[F[n+1,n]\cdot\underline{x}[n]\,\underline{\alpha}^H[n]\right]$

$= F[n+1,n]\cdot\mathbb{E}\left[\underline{x}[n]\,\underline{\alpha}^H[n]\right]$

$= F[n+1,n]\cdot\mathbb{E}\left[\underline{x}[n]\left(\overbrace{C[n]\cdot\underline{\varepsilon}[n,n-1] + \underline{v}_2[n]}^{\alpha[n]}\right)^H\right]$ ... from (5)

$= F[n+1,n]\cdot\mathbb{E}\left[\underline{x}[n]\left(\underline{\varepsilon}[n,n-1]^H C[n]^H + \underline{v}_2^H[n]\right)\right]$ (Orthogonal)

$= F[n+1,n]\cdot\mathbb{E}\left[\underline{x}[n]\cdot\underline{\varepsilon}[n,n-1]^H\right]\cdot C[n]^H$

$= F[n+1,n]\cdot\mathbb{E}\left[\left(\underbrace{\left(\underline{x}[n] - \hat{\underline{x}}[n|y_{n-1}]\right)}_{\varepsilon[n,n-1] \text{ from } (2)} + \underbrace{\hat{\underline{x}}[n|y_{n-1}]}_{\text{Orthogonal}}\right)\underline{\varepsilon}[n,n-1]^H\right]\cdot C[n]^H$

$= F[n+1,n]\cdot\mathbb{E}\left[\underbrace{\underline{\varepsilon}[n,n-1]\cdot\underline{\varepsilon}[n,n-1]^H}_{K[n,n-1] \text{ from } (4)}\right]\cdot C[n]^H$

$= F[n+1,n]\cdot K[n,n-1]\cdot C[n]^H$

Substituting this to the definition of Kalman Gain,

$G[n] = \underbrace{\mathbb{E}\left[\underline{x}[n+1]\,\underline{\alpha}^H[n]\right]\cdot R^{-1}[n]}$

$= \overbrace{F[n+1,n]\,K[n,n-1]\,C^H[n]\,R^{-1}[n]}$ Substitute $R[n]$ from (5)

$$= F[n+1,n]\,K[n,n-1]\,C^H[n]\left(C^H[n]\,K[n,n-1]\,C^H[n] + Q_2[n]\right)^{-1}$$

How to find this: Use Riccati Equations

# Riccati Equations

This is a method to find $k[n,n-1]$ recursively. ie relate $k[n+1,n]$ to $k[n,n-1]$

$$K[n+1,n] = F[n+1,n]\left(K[n,n-1] - F^{-1}[n+1,n]\,G[n]\,C[n]\,K[n,n-1]\right)F^H[n+1,n] + Q_1[n]$$

Discrete Algebraic Riccati Equation (DARE)

Proof: From forward error of $\underline{x}[n]$

$$\underline{\varepsilon}[n+1,n] = \underline{x}[n+1] - \underline{\hat{x}}[n+1\,|\,Y_n]$$

$$= \left(\overbrace{F[n+1,n]\cdot\underline{x}[n] + \underline{v}_1[n]}^{\text{Process Equation}}\right) - \left(\overbrace{F[n+1,n]\cdot\underline{\hat{x}}[n\,|\,Y_{n-1}] + G[n]\cdot\underline{\alpha}[n]}^{\text{Kalman Equation (11)}}\right)$$

$$= F[n+1,n]\cdot\left(\underbrace{\underline{x}[n] - \underline{\hat{x}}[n\,|\,Y_{n-1}]}_{\underline{\varepsilon}[n,n-1]\text{ from (3)}}\right) + \underline{v}_1[n] - G[n]\cdot\underline{\alpha}[n]$$

$$= F[n+1,n]\cdot\underline{\varepsilon}[n,n-1] + \underline{v}_1[n] - G[n]\cdot\left(\underbrace{C[n]\cdot\underline{\varepsilon}[n,n-1] + \underline{v}_2[n]}_{\alpha[n]\text{ from (5)}}\right)$$

$$= \left(F[n+1,n] - G[n]\cdot C[n]\right)\cdot\underline{\varepsilon}[n,n-1] + \underline{v}_1[n] - G[n]\cdot\underline{v}_2[n]$$

If $\underline{y} = A\underline{x} \Rightarrow \text{Cov}(\underline{y}) = A\cdot\text{Cov}(\underline{x})\cdot A^H$. Apply this to above. Since $\varepsilon \perp v_1 \perp v_2$ we get

$$\Rightarrow \text{Cov}\left(\underline{\varepsilon}[n+1,n]\right) = \left(F[n+1,n] - G[n]\cdot C[n]\right)\cdot\text{Cov}\left(\underline{\varepsilon}[n,n-1]\right)\left(F[n+1,n] - G[n]\cdot C[n]\right)^H + \text{Cov}\left(\underline{v}_1[n]\right) + G[n]\cdot\text{Cov}\left(\underline{v}_2[n]\right)\cdot G[n]^H$$

(Why positive)

$$\Rightarrow \overbrace{K[n+1,n]}^{\text{from (4)}} = \left(F[n+1,n] - G[n]\cdot C[n]\right)\cdot\overbrace{K[n,n-1]}^{\text{from (4)}}\cdot\left(F[n+1,n] - G[n]\cdot C[n]\right)^H + \text{Cov}\left(\underline{v}_1[n]\right) + G[n]\cdot\text{Cov}\left(\underline{v}_2[n]\right)\cdot G[n]^H$$

$$= \left(F[n+1,n] - G[n]\cdot C[n]\right)\cdot K[n,n-1]\cdot\left(F[n+1,n]^H - C[n]^H G[n]^H\right) + \text{Cov}\left(\underline{v}_1[n]\right) + G[n]\cdot\text{Cov}\left(\underline{v}_2[n]\right)\cdot G[n]^H$$

$\text{Cov}(\underline{v}_1[n]) = Q_1[n]$, $\text{Cov}(\underline{v}_2[n]) = Q_2[n]$

$$= F[n+1,n]\cdot k[n,n-1]\cdot F[n+1,n]^H - F[n+1,n]\cdot k[n,n-1]\cdot C[n]^H\cdot G[n]^H$$
$$- G[n]\cdot C[n]\cdot k[n,n-1]\cdot F[n+1,n]^H + G[n]\cdot C[n]\cdot k[n,n-1]\cdot C[n]^H G[n]^H$$
$$+ Q_1[n] + G[n]\cdot Q_2[n]\cdot G[n]^H$$

$$= F[n+1,n]\cdot k[n,n-1]\cdot F[n+1,n]^H - F[n+1,n]\cdot k[n,n-1]\cdot C[n]^H\cdot G[n]^H$$
$$- G[n]\cdot C[n]\cdot k[n,n-1]\cdot F[n+1,n]^H + Q_1[n]$$
$$+ G[n]\cdot \underbrace{\left(C[n]\,k[n,n-1]\,C[n]^H + Q_2[n]\right)}_{=R[n] \text{ from } (6)}\cdot G[n]^H$$

$$= F[n+1,n]\cdot k[n,n-1]\cdot F[n+1,n]^H - F[n+1,n]\cdot k[n,n-1]\cdot C[n]^H\cdot G[n]^H$$
$$- G[n]\cdot C[n]\cdot k[n,n-1]\cdot F[n+1,n]^H + Q_1[n] + G[n]\cdot R[n]\cdot G[n]^H$$

It can be shown that $F[n+1,n]\cdot k[n,n-1]\cdot C[n]^H\cdot G[n]^H = G[n]\cdot R[n]\cdot G[n]^H$

$$K[n+1,n] = F[n+1,n]\cdot k[n,n-1]\cdot F[n+1,n]^H - G[n]\cdot C[n]\cdot k[n,n-1]\cdot F[n+1,n]^H + Q_1[n]$$

$$= F[n+1,n]\cdot k[n,n-1]\cdot F[n+1,n]^H - \underbrace{F[n+1,n]\cdot F[n+1,n]^{-1}}_{I} G[n]\cdot C[n]\cdot k[n,n-1]\cdot F[n+1,n]^H + Q_1[n]$$

$$= F[n+1,n]\cdot\Big(k[n,n-1] - F[n+1,n]^{-1} G[n]\cdot C[n]\cdot k[n,n-1]\Big) F[n+1,n]^H + Q_1[n]$$

$K[n]$ ⊛⊛

$$= F[n+1,n]\,K[n]\cdot F[n+1,n]^H + Q_1[n] \quad \cdots ⊛⊛⊛$$

DARE

.. We know $G[n]$ from (10) in terms of $K[n-1,n]$ and previous step.

# Initialization of Kalman Recursion

Initialize $G[0]$, $\alpha[0]$ from prior guesses

$$G[n] = F[n+1, n]\, K[n, n-1]\, C^H[n] \left( C^H[n]\, K[n, n-1]\, C^H[n] + Q_2[n] \right)^{-1} \quad (11)$$

$$\alpha[n] = \underline{y}[n] - C[n]\, \hat{\underline{x}}[n \mid Y_{n-1}] \quad (*)$$

$$\hat{\underline{x}}[n+1 \mid Y_n] = F[n+1, n] \cdot \hat{\underline{x}}[n \mid Y_{n-1}] + G[n] \cdot \underline{\alpha}[n] \quad (9)$$ Kalman Equation

$$K[n] = k[n, n-1] \cdot - F[n+1, n]^{-1} G[n] \cdot C[n] \cdot k[n, n-1] \quad (*)(7)$$

$$K[n+1, n] = F[n+1, n]\, K[n] \cdot F[n+1, n]^H + Q_1[n] \quad (*)(*)(*)$$ DARE

# KAY - 4

and Kay Book Chapter 2 - 1988
"Spectral Estimation"

# Classical Spectral Estimation

## Non-Parametric Estimation

- Classical DSP assumes that we have $x[n]$ $n \in \mathbb{Z}$
- In modern, practical DSP $x[n]$ is only avaible for $n = 1, 2, \ldots, N-1$
- This results in problems in various parts of DSP. One such area is PSD. It is used in many cases since it has nice properties due to Fourier Transform.

$$\text{Power Spectral Density} = \text{PSD} = \sum_{k=-\infty}^{\infty} r_x[k] e^{-j\omega k}$$

- Since now we don't have $(-\infty, \infty)$ range,

  1) Estimate $r_x[k]$ using $x[n]$ $n = 1, 2, \ldots, N-1$
  2) Truncate the sum over $(-N, N)$

- Goal of this part of course,

  Goal : Design "good" estimators of $P(f)$ based on a finite set of observations $x[n]$, $n = 1, 2, .., N-1$ (under assumption X is WSS)

## Good Estimator

1) Assymptotically Unbiased :

$$\lim_{N \to \infty} E[\hat{P}(f)] = P(f)$$

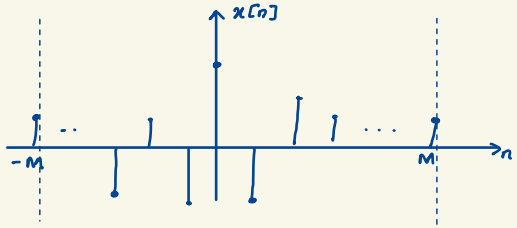
2) Assymptotically consistancy :

$$\lim_{n \to \infty} E\left[\left|\hat{P}(f) - E[\hat{P}(f)]\right|^2\right] = 0$$

$$\lim_{n \to \infty} \text{Var}(\hat{P}(f)) = 0$$

# 1) The Periodigram Estimator (PER)

- Suppose we have observed $x[n]$ for $N = 2M+1$ samples symmetrically around $n=0$.




Periodigram Estimator

$$\hat{P}(f) = \frac{1}{2M+1} \left| \sum_{n=-M}^{M} x[n] \cdot e^{-j2\pi fn} \right|^2$$

DTFT

$$f = \frac{\omega}{2\pi}$$ normalized to match with results from physics

- ∴ If we assume the signal is from 0 to N sample.

$$\hat{P}_{PER}(f) = \frac{1}{N} \left| \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi fn} \right|$$

- Ex: $\hat{P}_{PER}(f) = \sum_{K=-(N-1)}^{N-1} \hat{\gamma}_x[k] \cdot e^{-j2\pi fn}$ where

$$\hat{\gamma}_x[k] = \frac{1}{N} \sum_{n=0}^{N-1-k} x^*[n] \cdot x[n+k]$$

} Correlogram Estimators

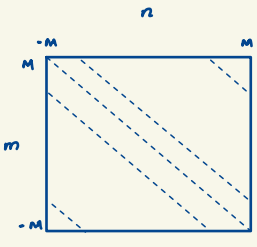
## Assumptotical Unbiasness of PER

- $P(f)$ or true PSD is a deterministic function of $f$. But $\hat{P}(f)$ is a rv. To be a good estimator it should be at least unbiased. (ie $E[\hat{P}(f)] = P(f)$)

$$E\{\hat{P}(f)\} = \frac{1}{2M+1} E\left\{\left|\sum_{n=-M}^{M} x[n]\cdot e^{-j2\pi f n}\right|^2\right\}$$

$$= \frac{1}{2M+1} E\left\{\left(\sum_{n=-M}^{M} x[n]\cdot e^{-j2\pi f n}\right)\left(\sum_{m=-M}^{M} x^*[m]\cdot e^{+j2\pi f m}\right)\right\}$$

$$= \frac{1}{2M+1} \sum_{n=-M}^{M}\sum_{m=-M}^{M} \underbrace{E[x[n]\cdot x^*[m]]}_{r_x[n-m]}\cdot e^{-j2\pi f(n-m)}$$




$$= \frac{1}{2M+1}\sum_{n=-M}^{M}\sum_{m=-M}^{M} r_x[n-m]\cdot e^{-j2\pi f(n-m)}$$

$$= \frac{1}{2M+1}\sum_{K=-2M}^{2M}(2M+1-|k|)\cdot r_x[k]\, e^{-j2\pi f k}$$

$$= \sum_{K=-2M}^{2M}\left(1-\frac{|k|}{2M+1}\right)\underbrace{r_x[k]\cdot e^{-j2\pi f k}}$$

For this $E[\hat{P}(f)] = P(f)$ we need 2 requirements

- Summation should be infinite instead of $-2M$ to $2M$
- $1-\frac{|k|}{2M+1} = 1$

Not a good estimator.

- Let's check if $E[\hat{P}(f)] = P(f)$ at least when $M \to \infty$

$$\lim_{M\to\infty} E\{\hat{P}(f)\} = \lim_{M\to\infty}\left[\sum_{K=-2M}^{2M}\left(1-\frac{|k|}{2M+1}\right) r_x[k]\cdot e^{-j2\pi f k}\right]$$

$$= \sum_{K=-\infty}^{\infty} r_x[k]\cdot e^{-j2\pi f k} - \lim_{M\to\infty}\left[\sum_{K=-2M}^{2M}\frac{|k|\cdot r_x[k]\cdot e^{-j2\pi f k}}{2M+1}\right]$$

Square Integrability

If $\sum_{K=-2M}^{2M}|k|\cdot r_x[k]\cdot e^{-j2\pi f k}$ grows slower than $M$

$$= \sum_{K=-\infty}^{\infty} r_x[k]\, e^{-j2\pi f k}$$

$$= P(f)$$

- $\therefore$ If $\sum_{K=-2M}^{2M}|k|\,|r_x[k]|$ grows slower than $M$ $\longrightarrow$ Then this estimator becomes "Assymptotically Unbiased"

# Assymptotically Consistancy of PER

- The bias of the auto-correlation is $\longrightarrow$ Bias $= \frac{-k}{N} r_x[k]$

Proof :- $\hat{r}_x[k] = \frac{1}{N} \sum_{n=0}^{N-1-k} x^*[n] \cdot x[n+k]$

$$E\{\hat{r}_x[k]\} = \frac{1}{N} \sum_{n=0}^{N-1} E[x^*[n] \cdot x[n+k]]$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} r_x[k]$$

$$= \frac{N-k}{N} r_x[k]$$

$$\therefore \text{Bias} = E\{\hat{r}_x[k]\} - r_x[k] = \frac{-k}{N} r_x[k]$$

- When $k$ is higher the bias /error is high.

- $\text{Var}(\hat{\rho}_{PER}(f))$ what happens to this at $N \to \infty$

Theorem: $\text{Var}[\hat{\rho}_{PER}(f_k)] = \begin{cases} \rho^2(f_k) & f_k \neq 0, N/2 \\ 2\rho^2(f_k) & f_k = 0, N/2 \end{cases}$ $\quad f_k = \frac{k}{N}$, $k = 0, 1, \ldots, N/2$

$$\propto \rho^2(f) \neq 0$$

$\therefore$ Var $\not\to 0$ when $N \to \infty$. $\therefore$ This is not assymptotically consistant.

# Discussion 12

Show that the periodigram estimator is an inconsistant estimator of PSD.

Since we are under the assumption that $x[n]$ is a white noise process in PSE,

$$P_{xx}(f) = \sigma_x^2$$

ie we assume variance is constant over time. $\therefore$ Pick $f = 0$

$\therefore$ From Periodigram equation,

$$\hat{P}(f) = \frac{1}{2M+1} \left| \sum_{n=-M}^{M} x[n] \cdot e^{-j2\pi f n} \right|^2$$

from hint $\Rightarrow$

$$\hat{P}_{PER}(f=0) = \frac{1}{N} \left| \sum_{n=0}^{N-1} x[n] \right|^2$$

$$= \frac{1}{N} \sigma_x^2 N \left( \sum_{n=0}^{N-1} \frac{x[n]}{\sigma_x \sqrt{N}} \right)^2$$

$$= \sigma_x^2 \left( \sum_{n=0}^{N-1} \frac{x[n]}{\sigma_x \sqrt{N}} \right)^2 \quad \ldots . \textcircled{1}$$

- What is the distribution of $\hat{P}_{PER}(0)$. Use the following hint $Z = \sum_{n=0}^{N-1} \frac{x[n]}{\sigma_x \sqrt{N}}$ and $S = \sum_{n=0}^{N-1} x[n]$, find the distribution of $Z$ and $S$

Since we have assumed $x[n]$ is normal $\Rightarrow$ $x[n] \sim \mathcal{N}(0, \sigma_x^2)$
Since summation of gaussian distributions are gaussian

$$\mathbb{E}[S] = \sum \mathbb{E}[x[n]] = 0$$
$$\text{Var}[S] = \sum \text{Var}[x[n]] = N \sigma_x^2$$

$$\Rightarrow \quad S = \mathcal{N}(0, N\sigma_x^2)$$

$$Z = \frac{S}{\sigma_x \sqrt{N}} \Rightarrow E[Z] = \frac{E[S]}{\sigma_x \sqrt{N}} = 0$$

$$\Rightarrow Z = N(0,1) \quad \cdots (2)$$

$$Var[Z] = \frac{Var(S)}{\sigma_x^2 N} = \frac{N\sigma_x^2}{\sigma_x^2 N} = 1$$

Now we can combine (1) and (2)

$$\Rightarrow \hat{\rho} = \sigma_x^2 Z^2$$

$$\Rightarrow Z^2 = \hat{\rho} / \sigma_x^2 \quad \cdots (3)$$

- By definition when $Z$ is a normal $Z^2$ has a chi-squared distribution with 1 degree of freedom.

$$\therefore Z^2 \sim \chi_1^2$$

$$\Rightarrow E[Z^2] = 1 \qquad Var(Z^2) = 2$$

**Chi-Squared Distribution - $\chi_N$**

$$Y \sim \chi_N^2$$

$$\chi_N^2 = \sum_{i=1}^{N} Z_i^2 \quad \text{where} \quad Z_i \sim N(0,1)$$

$N$ - number of independant squared standard normal variables being added up.

$$E[Y] = N \qquad Var[Y] = 2N$$

From (3) $\Rightarrow E[\hat{\rho}/\sigma_x^2] = 1 \qquad Var[\hat{\rho}/\sigma_x^2] = 2$

$$\Rightarrow E[\hat{\rho}] = \sigma_x^2 \qquad Var[\hat{\rho}] = 2\sigma_x^4$$

$\therefore$ When $N$ goes to infinity variance does not shrink. $\therefore$ It's an inconsistant estimator of PSD.

■

# 2) Averaged Periodigram Estimator (AVPER)

- How doing same experiment / estimation repeatedly helps to make an estimator "Assymptotically Constant".

- Suppose we have K different sets of "uncorrelated" Periodigram Estimators.

$$\hat{\rho}_{PER}^{(1)}(f),\ \hat{\rho}_{PER}^{(2)}(f),\ \ldots,\ \hat{\rho}_{PER}^{(K)}(f)$$

$$\hat{\rho}_{AVPER}(f) = \frac{1}{K}\sum_{K=1}^{K}\hat{\rho}_{PER}(f)$$

**Assymptotically Unbiased**

$$\lim_{N\to\infty} E\left[\hat{\rho}_{AVPER}(f)\right] = \rho(f)$$

$$Var\left[\hat{\rho}(f)\right] = \begin{cases} \frac{1}{K}\rho^2(f_K) & f_K = 0, N/2 \\ \frac{2}{K}\rho^2(f_K) & \text{otherwise} \end{cases}$$

$$\lim_{N\to\infty} Var\left[\hat{\rho}_{AVPER}(f)\right] \propto 1/K\ \rho^2(f)$$ - small variance for large K

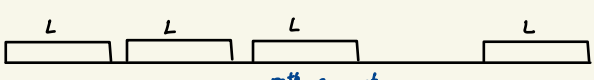
**Assymptotically Consistant**

## How to Attain Uncorrelated Requirement

We can't practically do the experiments at completely different dates so $x[0], \ldots, x[N-1]$ becomes uncorrelated

$$N = KL$$

Do the experiment at once. Divide the desired data $x[0], \ldots, x[N-1]$ into K consecutive segments of length L.




$$x^{(m)}[n] = x[n + mL] \qquad n = 0, 1, \ldots, L-1$$

Assume $x^{(i)}[n]$ are uncorrelated

$$\hat{\rho}_{PER}^{(m)} = \frac{1}{L}\left|\sum_{n=0}^{L-1} x^{(m)}[n]\cdot e^{-j2\pi fn}\right|^2$$

$$\hat{\rho}_{AVERAGE}(f) = \frac{1}{K}\sum_{k=1}^{K}\hat{\rho}_{PER}^{(k)}(f)$$

## Bias - Variance Trade off

$L\uparrow \longrightarrow$ Bias $\downarrow \longrightarrow K\downarrow \longrightarrow$ Variance $\uparrow$

$L\downarrow \longrightarrow$ Bias $\uparrow \longrightarrow K\uparrow \longrightarrow$ Variance $\downarrow$

## Discussion 12

Show that the averaged periodigram estimator is a consistant estimator of PSD.

$$\hat{\rho}_{AVPER}(0) = \frac{1}{N}\sum_{m=1}^{N-1}\hat{\rho}_{PER}^{(m)}(0) = \frac{1}{N}\sum_{m=0}^{N-1}x^2[m]$$

Since $x[n]$ is assumed to be real, white and Gaussian $x(n)$ is $N(0,\sigma_x^2)$.

$$\Rightarrow \sigma_x^2 = Var(X[n]) = E[x^2[n]] - \underbrace{E[x[n]]^2}_{0} = E[x^2[n]]$$

Since $x[m] = \sigma_x z$ where $Z \sim N(0,1)$

$$\Rightarrow x^2[m] = \sigma_x^2 z^2 \Rightarrow z^2 = x^2[m]/\sigma_x^2 \sim \chi_1^2 \Rightarrow Var\left(\frac{x^2[m]}{\sigma_x^2}\right) = 2 \Rightarrow Var(x^2[m]) = 2\sigma_x^4$$

$$\therefore E\left[\hat{\rho}_{AVPER}(0)\right] = E\left[\frac{1}{N}\sum_{m=0}^{N-1}x^2[m]\right] = \frac{1}{N}\sum_{m=0}^{N-1}\underbrace{E[x^2[m]]}_{\sigma_x^2}$$

$$= \frac{1}{N}\cdot N\sigma_x^2$$

$$= \sigma_x^2 \quad .$$

$$\therefore \text{Var}\left[\hat{\rho}_{AVPER}(0)\right] = \text{Var}\left[\frac{1}{N}\sum_{m=0}^{N-1} x^2[m]\right] = \frac{1}{N^2}\sum_{m=0}^{N-1} \text{Var}\left[x^2[m]\right]$$

$$= \frac{1}{N^2} \cdot N \ \text{Var}\left[x^2[m]\right]$$

$$= \frac{1}{N} \cdot \underbrace{\text{Var}\left[x^2[m]\right]}$$

$$= \frac{2\sigma_x^4}{N}$$

$\therefore$ As $N \to \infty$ $\text{Var}\left(\hat{\rho}_{AVPER}(0)\right)$ goes to 0 $\therefore$ This is a consistent estimator.

# 3) Correlogram Estimators

These methods first estimate the autocorrelation $r[k] = \hat{r}[k]$ and take the fourier transform.

$$\text{Power Spectral Density} = PSD = \sum_{k=-\infty}^{\infty} r_x[k] e^{-j\omega k}$$

$$\simeq \sum_{k=-N+1}^{N-1} \hat{r}_x[k] \cdot e^{-j\omega k}$$

$$\hat{\rho}_{PER}(f) = \sum_{k=-(N-1)}^{N-1} \hat{r}_x[k] \cdot e^{-j2\pi fn}$$

$$\hat{r}_x[k] = \frac{1}{N} \sum_{n=0}^{N-1-k} x^*[n] \cdot x[n+k]$$

The bias of the auto-correlation is $\longrightarrow$ Bias $= \frac{-k}{N} r_x[k]$

Proof :- $\hat{r}_x[k] = \frac{1}{N} \sum_{n=0}^{N-1-k} x^*[n] \cdot x[n+k]$

$$E\{\hat{r}_x[k]\} = \frac{1}{N} \sum_{n=0}^{N-1} E[x^*[n] \cdot x[n+k]]$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} r_x[k]$$

$$= \frac{N-k}{N} r_x[k]$$

$\therefore$ Bias $= E\{\hat{r}_x[k]\} - r_x[k] = \frac{-k}{N} r_x[k]$

- When $k$ is higher the bias / error is high.

## Homework 6 - Problem 1 / Problem 2

**1.** Recall that:

$$P(f) = \sum_{k=-\infty}^{\infty} r_x[k] e^{-j2\pi fk}$$

A *correlogram* estimate of the PSD is obtained by replacing $r_x[k]$ by an estimate $\widehat{r}_x[k]$ and replacing the infinite sum by a finite sum:

$$\widehat{P}_{\mathsf{CORR}}(f) = \sum_{k=-L}^{L} \widehat{r}_x[k] e^{-j2\pi fk}$$

where $L$ is an arbitrary positive integer.

(a) Show that $\widehat{r}_x^{(u)}[k] := \frac{1}{N-k} \sum_{n=0}^{N-k-1} x[n+k]x^*[n]$, for $0 \le k \le N-1$ is an unbiased estimator of $r_x[k]$, i.e., $\mathbb{E}\left\{\widehat{r}_x^{(u)}[k]\right\} = r_x[k]$.

(b) Let $\widehat{P}_{\mathsf{CORR}}^{(1)}(f)$ be a correlogram where we use $\widehat{r}_x^{(u)}[k]$ as the estimator $\widehat{r}_x[k]$. Show that:

$$\mathbb{E}\left\{\widehat{P}_{\mathsf{CORR}}^{(1)}(f)\right\} = P(f) * W(f),$$

where

$$W(f) := \frac{\sin(\pi f(2L+1))}{\sin(\pi f)}.$$

**2.** Here, we construct another correlogram estimate of the PSD.

(a) Show that $\widehat{r}_x^{(b)}[k] := \frac{1}{N} \sum_{n=0}^{N-k-1} x[n+k]x^*[n]$, for $0 \le k \le N-1$ is a biased estimator of $r_x[k]$, such that

$$\mathbb{E}\left\{\widehat{r}_x^{(b)}[k]\right\} = \left(1 - \frac{|k|}{N}\right) r_x[k].$$

(b) Let $\widehat{P}_{\mathsf{CORR}}^{(2)}(f)$ be a correlogram where we use $\widehat{r}_x^{(b)}[k]$ as the estimator $\widehat{r}_x[k]$. Show that:

$$\mathbb{E}\left\{\widehat{P}_{\mathsf{CORR}}^{(2)}(f)\right\} = P(f) * W(f) * |T(f)|^2,$$

where $W(f)$ is the same as in Problem 1 and

$$T(f) := \frac{e^{-j\pi f(N-1)}}{\sqrt{N}} \frac{\sin(\pi fN)}{\sin(\pi f)}.$$

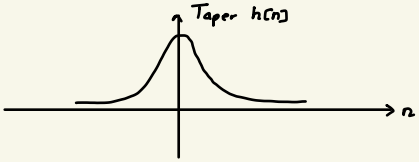
(c) Show that when $L = N-1$, $\widehat{P}_{\mathsf{CORR}}^{(2)}(f)$ can be expressed as

$$\widehat{P}_{\mathsf{CORR}}^{(2)}(f) = \mathbf{e}_N^H(f) \widetilde{\mathbf{R}}_N \mathbf{e}_N(f)$$

Tapering is a solution to global bias in periodograms, from side lobe curse.

# 4) Tappered Estimate ($\hat{\rho}_{TAPERED}(f)$)

$$\hat{\rho}_{TAPERED}(f) = \frac{1}{N}\left|\sum_{n=0}^{N-1} h[n]\cdot x[n]\cdot e^{-j2\pi fn}\right|^2$$



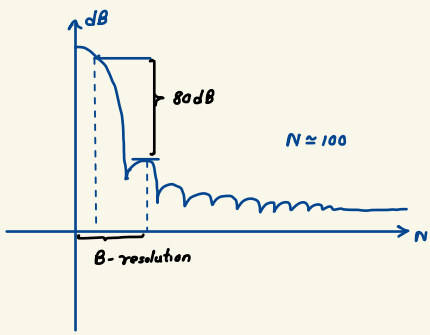

- When we have an ideal taper (which is a Dirac delta) original signal is recovered.

$$\mathbb{E}\left[\hat{\rho}_{TAPERED}(f)\right] = |H(f)|^2 * P(f)$$

- $$\hat{\rho}_{TAPERED}(f) = \frac{1}{N}\left|\sum_{n=0}^{N-1} h[n]\cdot x[n]\cdot e^{-j2\pi fn}\right|^2$$

$$= \frac{1}{N}\sum_{n=0}^{N-1}\sum_{m=0}^{N-1} h[n]\cdot h^*[m]\cdot x[n]\cdot x^*[m]\cdot e^{-j2\pi f(n-m)}$$

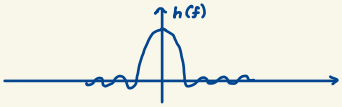
$h_N(n) \xleftrightarrow{FT} H_N(f)$




- Depending on the tapering function, - $h[n]$ we can have different types of tappered estimates.

Ex:- $H_R(f) = \frac{\sin(N\pi f)}{N\cdot\sin(\pi f)}$ $\Longrightarrow$ Then $\lim_{N\to\infty}\mathbb{E}[\hat{\rho}(f)] = \rho(f)$

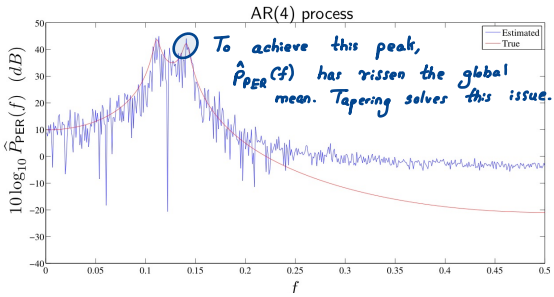
$$\mathbb{E}[\hat{\rho}_{PER}(f)] = |H_R(f)|^2 * P(f)$$



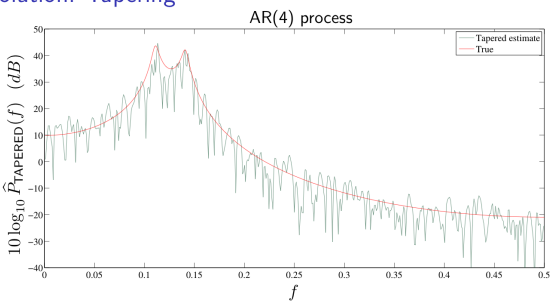
## The Periodogram

The estimate of $P_X(f)$ is given by:

$$\widehat{P}_{\text{PER}}(f) = \sum_{k=-(N-1)}^{N-1} \widehat{r}_x[k] e^{-j2\pi fk} = \frac{1}{N} \left| \sum_{k=0}^{N-1} x[n] e^{-j2\pi fn} \right|^2$$




## Solution: Tapering




The global bias is dramatically improved!

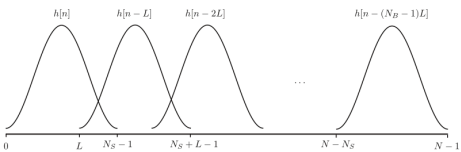
# 5) Average Tappered Estimate ($\hat{P}_{AVTAPERED}(f)$)

$$\hat{P}_{AVTAPERED}(f) = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{N}\left|\sum_{n=0}^{N-1} h^{(k)}[n]\cdot x[n]\cdot e^{-j2\pi fn}\right|^2$$

## Ex: Welch Overlapping Segment Averaging

### Homework 6 - Problem 3

**3.** The Welch periodogram estimate is obtained as follows: the observed data $x[0], x[1], \cdots, x[N-1]$ is divided into $N_B$ overlapping segments of length $N_S$ each with a shift parameter $L$. Then, each segment of tapered with a window $h[n]$ of length $N_S$:




Suppose that $0 < L \le N_S$ and $(N_B-1)L = N - N_S$, as shown in the figure. Let $\widehat{P}_\ell(f)$ be the tapered periodogram estimate of the data of length $N_S$ starting at index $\ell$:

$$\widehat{P}_\ell(f) := \left|\sum_{n=0}^{N_S-1} h[n]x[n+\ell]e^{-j2\pi fn}\right|^2, \qquad \ell = 0, 1, \cdots, N-N_S.$$

The Welch spectral estimate is formed by averaging the $N_B$ tapered estimates given by:

$$\widehat{P}_{\text{Welch}}(f) := \frac{1}{N_B}\sum_{b=0}^{N_B-1}\widehat{P}_{bL}(f).$$

(a) Show that:

$$\mathbb{E}\left\{\widehat{P}_{\text{Welch}}(f)\right\} = P(f) * |H(f)|^2,$$

where $H(f)$ is the DTFT of $h[n]$.

(b) Assuming that $x[n]$ is real-valued and Gaussian, we have the identity:

$$\mathbb{E}\{x[k]x[\ell]x[m]x[n]\} = r_x[k-\ell]r_x[m-n] + r_x[k-m]r_x[\ell-n] + r_x[k-n]r_x[m-\ell].$$

Using this identity, show that:

$$\mathbb{E}\{\widehat{P}^2_{\text{Welch}}(f)\} = \frac{1}{N_B^2}\sum_{b=0}^{N_B-1}\sum_{c=0}^{N_B-1}\iint P(f')P(f'')\Big[|H(f-f')|^2|H(f-f'')|^2 + e^{j2\pi(f'-f'')(b-c)L}H(f-f')H^*(f+f'')H(f+f')H^*(f-f'') + e^{j2\pi(f'-f'')(b-c)L}|H(f-f')|^2|H(f-f'')|^2\Big]df'df''.$$

(c) Using the results of parts (a) and (b) show that:

$$\text{Var}\{\widehat{P}_{\text{Welch}}(f)\} = \frac{1}{N_B^2}\sum_{b=0}^{N_B-1}\sum_{c=0}^{N_B-1}\iint P(f')P(f'')e^{j2\pi(f'+f'')(b-c)L}H(f-f')H^*(f+f'')H(f+f')H^*(f-f'')df'df'' + \frac{1}{N_B^2}\sum_{b=0}^{N_B-1}\sum_{c=0}^{N_B-1}\left|\int P(f')e^{j2\pi f'(b-c)L}|H(f-f')|^2df'\right|^2.$$

(d) Assuming that that the taper $h[n]$ has high side-lobe suppression with a bandwidth of $B$, and that $P(f)$ is relatively constant around $f$, show that for $f > B$, the first term in the expression of the variance is negligible and the variance can be expressed as:

$$\text{Var}\{\widehat{P}_{\text{Welch}}(f)\} \approx \frac{P(f)^2}{N_B}\sum_{k=-N_B+1}^{N_B-1}\left(1-\frac{|k|}{N_B}\right)\left|\int e^{j2\pi f'kL}|H(f-f')|^2df'\right|^2 = \frac{P(f)^2}{N_B}\sum_{k=-N_B+1}^{N_B-1}\left(1-\frac{|k|}{N_B}\right)\left|\sum_{n=0}^{N_S-1}h[n]h[n+kL]\right|^2$$

(e) Discuss the bias-variance trade-off of the Welch estimate as compared to the tapered periodogram estimate.

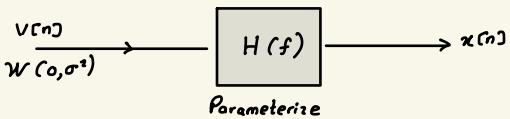
# KAY - 6

and Kay Book Chapter 2 - 1988
"Spectral Estimation"

# Autoregressive Spectral Estimation

## Parametric Estimation

- So far we saw 4 different Non parametric estimators. Now we assume parametric resolution on $x[n]$




$$\text{PSD of } x[n] = P_x(f) = \sigma^2 \left| H(f) \right|^2$$

- We assume $H(f)$ is a fractional transfer function. ie

$$H(z) = \frac{\sum_{k=0}^{q} b_k z^{-k}}{\sum_{k=0}^{p} a_k z^{-k}} = ARMA(p,q)$$

- If we have estimates for $a_k$'s and $b_k$'s, (assume $\hat{a}_k$, $\hat{b}_k$) we can directly form an estimate of $P(f)$.

$$\hat{P}(f) = \frac{\sigma^2 \left| \sum_{k=0}^{q} \hat{b}_k \cdot e^{-j2\pi fn} \right|^2}{\left| \sum_{k=0}^{p} \hat{a}_k \cdot e^{-j2\pi fn} \right|^2}$$

Goal : Estimate the parameters of the model $\hat{a}_k$, $\hat{b}_k$ from using $x[n]$, $n = 0, 1, \ldots, N-1$

Assumption: Given $x[n]$, $n = 0, 1, \ldots, N-1$ suppose that we have good estimates of $r_x[k]$ given by $\hat{r}_x[k]$ for $k = 0, 1, \ldots, p$ $(p \ll N)$. One such estimate is

$$\hat{r}_x[k] = \frac{1}{N} \sum_{n=0}^{N-1-k} x^*[n] \cdot x[n+k] \qquad k = 0, 1, \ldots, p$$

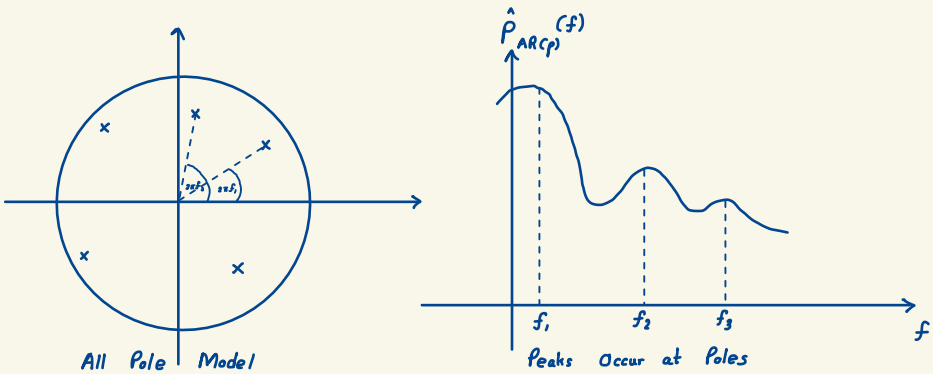
- called Correlogram

# 1) AR(P) Spectral Estimators $\left(\hat{P}_{AR(P)}(f)\right)$

- Ideal when there are peaks / resonance values in the signal. (Ex: EEG)

$$H(z) = \frac{1}{A(z)}$$

$$\hat{P}_{AR(P)}(f) = \frac{\sigma^2}{\left| \sum_{k=0}^{P} \hat{a}_k \cdot e^{-j2\pi fn} \right|^2}$$




- How to estimate $\sigma^2$ and $\hat{a}_k$ s $\Longrightarrow$ Use Yule Walker Equations

$$\hat{r}_x[0] = -\sum_{l=1}^{P} a[l] \cdot \hat{r}_x[-l] + \sigma^2$$

$$\hat{r}_x[k] = -\sum_{l=1}^{P} a[l] \cdot \hat{r}_x[k-l] \quad \cdots \quad k = 1, 2, \ldots, P \quad \cdots \quad R_P \underline{a} = -\underline{r}$$

# Yule - Walker Equations - Recap

- Let's assume that $\hat{r}_x[k]$ is quite accurate and so that $r_x[k]$ is known.

- Find $a[l]$, $b[l]$ given $r_x[k]$ for $l = 1, 2, \ldots, p$ $\quad k = \pm 1, \pm 2, \ldots$

  Assume $a[0] = b[0] = 1$

$$\therefore \quad H(z) = \frac{1 + b[1]\cdot z^{-1} + \ldots + b[q]\cdot z^{-q}}{1 + a[1]\cdot z^{-1} + \ldots + a[p]\cdot z^{-p}} = \frac{B(z)}{A(z)}$$

$$P_x(z) = \underbrace{\sigma^2}_{\substack{z \text{ transform of} \\ r_x[k] = \sigma^2\delta[k]}} \cdot \underbrace{H(z)}_{\substack{z\text{-transform} \\ \text{of} \\ h[k]}} \cdot \underbrace{H^*\left(\frac{1}{z^*}\right)}_{z\text{-transform of } h^*[-k]}$$

$$P_x(z) = \sigma^2 \cdot \frac{B(z)}{A(z)} \cdot H^*\left(\frac{1}{z^*}\right)$$

$$\Rightarrow \quad \underbrace{A(z)}_{\text{unknown}} \cdot \underbrace{P_x(z)}_{\text{known}} = \underbrace{\sigma^2}_{\text{known}} \cdot \underbrace{B(z)}_{\text{unknown}} \cdot \underbrace{H^*\left(\frac{1}{z^*}\right)}_{\text{Unknown}}$$

$$\Rightarrow \quad \sum_{l=0}^{p} a[l]\, r_x[k-l] = \sigma^2 \sum_{l=0}^{q} b[l]\cdot h^*[l-k] \quad \left(\text{Since } h^*[l-k] \text{ is causal and substituting } l-k \longrightarrow l' \;..\; \text{Product in } z \text{ domain is Convolution in time domain}\right)$$

$$\Rightarrow \begin{cases} r_x[k] = -\sum_{l=1}^{p} a[l]\cdot r_x[k-l] + \sigma^2 \sum_{l=0}^{q-k} h^*[l]\cdot b[l+k] & k = 0, 1, 2, \ldots, q \\ r_x[k] = -\sum_{l=1}^{p} a[l]\cdot r_x[k-l] & k \geq q+1 \end{cases}$$

$\therefore$ Yule- walke equations are

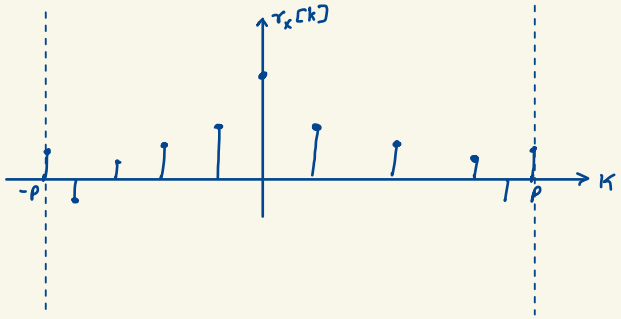
$$r_x[k] = -\sum_{l=1}^{p} a[l]\cdot r_x[k-l] + \sigma^2 \sum_{l=0}^{q-k} h^*[l]\cdot b[l+k] \qquad k = 0, 1, 2, \ldots, q$$

$$r_x[k] = -\sum_{l=1}^{p} a[l]\cdot r_x[k-l] \qquad k \geq q+1$$

- Since $h^*$ terms are there, these equations are highly non linear.
- But we have unlimited equations with only finite unknowns.

# 2) Maximum Entropy Spectral Estimator (MESE)

- Suppose we have reliable estimators $r_x[k] = 1, 2, \ldots, p$. Consider the set of all WSS processes that have the first $p$ elements of $r_x[k]$ given by $\hat{r}_x[k]$




- Assume we want to fill the outside $[-p, p]$ values for $r_x[k]$. To be fair and to not assume things that are not given, we have to be careful to introduce least amount of structure possible while filling. And the system should remain WSS. How to do this?

  ↳ Yule Walker Equations!

## Maximal Entropy Principle

Q:- Cross entropy between $P(f)$ (PSD of $x[n]$) and $1$ (PSD of $W(0,1)$) should be maximum to achieve least structure. ie.

$$\max_{P(f)} \int_{-1/2}^{1/2} \ln P(f)\, df$$

such that $\int_{-1/2}^{1/2} P(f)\, e^{j2\pi fn}\, df = \hat{r}_x[k] \qquad k = 0, 1, \ldots, p$

A- Answer to above optimization problem is Yule Walker equations where

$$r_x[k] = -\sum_{l=1}^{p} a[l] \cdot r_x[k-l] \qquad k \geq q+1$$

$a[l]$ are solutions to $R_p \underline{a} = \underline{r}$

**Proof - HW 7 - Problem 1**

@ PSD of $x[n]$ = $P(f) = \sum_{k=-\infty}^{\infty} r_x[k]\, e^{-j2\pi fk}$ ..... ⓪

$$\Rightarrow \frac{\partial}{\partial r_x[k]} = e^{-j2\pi fk} \quad \cdots\cdots ①$$

- The constraint is $\int_{-1/2}^{1/2} P(f) \cdot e^{2\pi f l}\, df = \hat{r}_x[l] \qquad |l| = 0, 1, \ldots, p$

$$\Rightarrow \underbrace{\hat{r}_x[l] - \int_{-1/2}^{1/2} P(f) \cdot e^{2\pi f l}\, df}_{g(l)} = 0 \quad \cdots\cdots ②$$

- The maximization criteria $\rightarrow$ $f(l) = \int_{-1/2}^{1/2} \ln P(f)\, df$ ..... ③

- By using the Lagrangian Multiplier as $\lambda_l$'s

$$\Rightarrow L = f(l) + \sum_{l=-p}^{p} \lambda_l g(l)$$

$$\Rightarrow L = \int_{-1/2}^{1/2} \ln P(f)\, df + \sum_{l=-p}^{p} \lambda_l \left( \hat{r}_x[l] - \int_{-1/2}^{1/2} P(f) \cdot e^{2\pi f l}\, df \right) \quad \cdots ② \text{ and } ③$$

$$\Rightarrow \frac{\partial L}{\partial r_x[k]} = \int_{-1/2}^{1/2} \frac{1}{P(f)} \cdot \underbrace{\frac{\partial P(f)}{\partial r_x[k]}}_{\text{from } ①} df + \sum_{l=-p}^{p} \lambda_l \left( \underbrace{\frac{\partial \hat{r}_x[l]}{\partial r_x[k]}}_{\substack{\| \\ 0 \\ \text{Since } \hat{r}_x[l] \text{ is} \\ \text{Constant}}} - \int_{-1/2}^{1/2} \underbrace{\frac{\partial P(f)}{\partial r_x[k]}}_{\text{from } ①} e^{j2\pi fl}\, df \right)$$

$$\Rightarrow \frac{\partial L}{\partial r_x[k]} = \int_{-1/2}^{1/2} \frac{1}{P(f)}\, e^{-j2\pi fk}\, df - \sum_{l=-p}^{p} \lambda_l \int_{-1/2}^{1/2} e^{-j2\pi f(k-l)}\, df$$

$$\Rightarrow \frac{\partial L}{\partial r_x[k]} = \int_{-1/2}^{1/2} \frac{1}{P(f)}\, e^{-j2\pi fk}\, df - \sum_{l=-p}^{p} \lambda_l\, \delta[k-l] \qquad \delta[k-l] \rightarrow \begin{cases} 1 & k=l \\ 0 & o.w \end{cases}$$

$\therefore$ when $|k| > p$ the constraint term (second one) becomes zero.

$$\Rightarrow \frac{\partial L}{\partial r_x[k]} = \begin{cases} \int_{-1/2}^{1/2} \frac{1}{P(f)}\, e^{-j2\pi fk}\, df - \lambda_k & |k| \leq p \\ \int_{-1/2}^{1/2} \frac{1}{P(f)}\, e^{-j2\pi fk}\, df & o.w \end{cases}$$

(b) When $k = 0, 1, \ldots, p$ $\quad \frac{\partial L}{\partial r_x[k]} = \int_{-1/2}^{1/2} \frac{1}{P(f)} e^{-j2\pi fk} df - \lambda_k$

$\therefore$ By setting the partial derivative to zero,

$$\Rightarrow \int_{-1/2}^{1/2} \frac{1}{P(f)} e^{-j2\pi fk} df - \lambda_k = 0$$

$$\Rightarrow \int_{-1/2}^{1/2} \frac{1}{P(f)} e^{-j2\pi fk} df = \lambda_k$$

$\Rightarrow \frac{1}{P(f)}$ and $\lambda_k$ is a DTFT pair for $0 \le |k| \le p$

$$\Rightarrow \sum_{k=-\infty}^{\infty} \lambda_k e^{j2\pi fk} = \frac{1}{P(f)}$$

(c) By factorizing $P(f)$ real-valued non-negative function,

$$\sum_{k=-p}^{p} \lambda_k e^{j2\pi fk} = \frac{1}{\sigma^2} \left( \sum_{l=0}^{p} a_l e^{j2\pi fl} \right) \left( \sum_{l=0}^{p} a_l^* e^{-j2\pi fl} \right)$$

From part (b)

$$P(f) = \frac{1}{\sum_{k=-\infty}^{\infty} \lambda_k e^{j2\pi fk}} = \frac{\sigma^2}{\left( \sum_{l=0}^{p} a_l e^{j2\pi fl} \right) \left( \sum_{l=0}^{p} a_l^* e^{-j2\pi fl} \right)}$$

$$= \frac{\sigma^2}{\left| \sum_{l=0}^{p} a_l e^{j2\pi fl} \right|^2} \quad \cdots\cdots (4)$$

Taking the inverse DTFT version of (a),

$$\Rightarrow r_x[l] = \int_{-1/2}^{1/2} P(f) e^{j2\pi fl} df$$

$$\Rightarrow r_x[k-l] = \int_{-1/2}^{1/2} \left( \frac{\sigma^2}{\left| \sum_{l=0}^{p} a_l e^{j2\pi fl} \right|^2} \right) \cdot e^{j2\pi f(k-l)} df \quad \cdots \text{from (4)}$$

$$\Rightarrow r_x[k-l] = \sigma^2 \int_{-1/2}^{1/2} \frac{e^{j2\pi f(k-l)}}{\left|\sum_{l=0}^{p} a_l e^{j2\pi f l}\right|^2}\, df \quad \cdots\cdots (5)$$

$$\therefore \sum_{l=0}^{p} a_l^* r_x[k-l] = \sum_{l=0}^{p} a_l^* \cdot \sigma^2 \int_{-1/2}^{1/2} \frac{e^{j2\pi f(k-l)}}{\left|\sum_{l=0}^{p} a_l e^{j2\pi f l}\right|^2}\, df \quad \cdots \text{ from } (5)$$

$$= \sigma^2 \int_{-1/2}^{1/2} \frac{\sum_{l=0}^{p} a_l^* e^{j2\pi f(k-l)}}{\left|\sum_{l=0}^{p} a_l e^{j2\pi f l}\right|^2}\, df$$

$$= \sigma^2 \int_{-1/2}^{1/2} \frac{e^{j2\pi f k} \cancel{\sum_{l=0}^{p} a_l^* e^{-j2\pi f l}}}{\left(\sum_{l=0}^{p} a_l e^{j2\pi f l}\right)\cancel{\left(\sum_{l=0}^{p} a_l^* e^{-j2\pi f l}\right)}}\, df$$

$$= \sigma^2 \int_{-1/2}^{1/2} \frac{e^{j2\pi f k}}{\sum_{l=0}^{p} a_l e^{j2\pi f l}}\, df$$

(d) Consider, $H(z) = \dfrac{1}{\sum_{l=0}^{p} a_l z^{l}}$ where $a_0 = 1$. Since $z$ has only positive powers, this is a non-causal filter.

$$\Rightarrow h[k] = 0 \quad \forall \; k > 0 \quad \cdots\cdots (6)$$

Consider the inverse DTFT of $H(z)$, $z = e^{j2\pi f}$

$$h[k] = \int_{-1/2}^{1/2} \frac{1}{\sum_{l=0}^{p} a_l z^{l}} e^{j2\pi f k}\, df$$

$$= \int_{-1/2}^{1/2} \frac{e^{j2\pi f k}}{\sum_{l=0}^{p} a_l e^{j2\pi f l}}\, df$$

From (6) when $k>0 \Rightarrow h[k] = 0 \Rightarrow \int_{-1/2}^{1/2} \frac{e^{j2\pi f k}}{\sum_{l=0}^{p} a_l e^{j2\pi f l}}\, df = 0$

when $k=0 \Rightarrow h[0] = \int_{-1/2}^{1/2} \frac{e^{j2\pi f 0}}{\sum_{l=0}^{p} a_l e^{j2\pi f l}}\, df = \lim_{z \to 0} H(z) = \frac{1}{a_0} = 1$

$$\therefore \int_{-1/2}^{1/2} \frac{e^{j2\pi f 0}}{\sum_{l=0}^{p} a_l e^{j2\pi f l}}\, df = \begin{cases} 1 & k = 0 \\ 0 & \text{o.w} \end{cases}$$

ⓔ From part ⓒ we have,

$$\Rightarrow \quad \sigma^2 \int_{-1/2}^{1/2} \frac{e^{j2\pi f k}}{\sum_{l=0}^{p} a_l e^{j2\pi f l}} \, df = \sum_{l=0}^{p} a_l^* \, r_x[k-l]$$

$$= a_0^* \, r_x[k-0] + \sum_{l=1}^{p} a_l^* \, r_x[k-l]$$

$$= r_x[k] + \sum_{l=1}^{p} a_l^* \, r_x[k-l]$$

$$\Rightarrow \quad r_x[k] = \sigma^2 \int_{-1/2}^{1/2} \frac{e^{j2\pi f k}}{\sum_{l=0}^{p} a_l e^{j2\pi f l}} \, df - \sum_{l=1}^{p} a_l^* \, r_x[k-l]$$

from ⓓ

$$\Rightarrow \quad r_x[k] = \begin{cases} \sigma^2 \cdot (1) - \sum_{l=1}^{p} a_l^* \, r_x[0-l] & k=0 \\ \sigma^2(0) - \sum_{l=1}^{p} a_l^* \, r_x[k-l] & \text{o.w} \end{cases}$$

$$= \begin{cases} \sigma^2 - \sum_{l=1}^{p} a_l^* \, r_x[-l] & k=0 \\ \sum_{l=1}^{p} a_l^* \, r_x[k-l] & \text{o.w} \end{cases}$$

$$r_x[0] = -\sum_{l=1}^{p} a_l^* \, r_x[-l] - \sigma^2$$

$$r_x[k] = -\sum_{l=1}^{p} a_l^* \, r_x[k-l] \qquad k \geq 1$$

These are the Yule Walker equations for AR(p) process with autocorrelation estimates for the first $p$ lags

$$r_x[l] = \hat{r}_x[l] \qquad |l| = 0, 1, \ldots, p$$

$\therefore$ MESE corresponds to the AR(p) process.

# 3) Least Square Estimator

$$\min_{\{a_k\}} \sum_{n=p}^{N-1} \left( x[n] - \sum_{k=1}^{p} a_k \, x[n-k] \right)^2$$

Least Square Estimator is given by,

$$\hat{a}_p^{LS} = -\left(x^T x\right)^{-1} x^T x_{p:N-1}$$

$$\underbrace{\begin{bmatrix} x[p] \\ x[p+1] \\ \vdots \\ x[N-1] \end{bmatrix}}_{x_{p:N-1}} = -\underbrace{\begin{bmatrix} x[p-1] & x[p-2] & \cdots & x[0] \\ x[p] & x[p-1] & \cdots & x[1] \\ \vdots & \vdots & & \vdots \\ x[N-2] & x[N-3] & \cdots & x[N-p] \end{bmatrix}}_{x} \underbrace{\begin{bmatrix} a_{p,1} \\ a_{p,2} \\ \vdots \\ a_{p,p} \end{bmatrix}}_{\underline{a}_p} + \begin{bmatrix} w[p] \\ w[p+1] \\ \vdots \\ w[N-1] \end{bmatrix}$$

Claim : HW 7 - Problem 2

Yule Walker estimation $\hat{\underline{a}}_p^{YW}$ for an AR(p) process is equivalent to Least square estimate $\hat{\underline{a}}_p^{LS}$ applied to a properly zero paded data.

Proof: Let's rewrite the above matrix equation by using the properly zero padded data by adding 'p' zeros to the beginning and 'p' zeros to the end.

$$y[n] = \begin{cases} 0 & -p \leq n < 0 \\ x[n] & 0 \leq n \leq N-1 \\ 0 & N < n \leq N+p-1 \end{cases}$$

$$\underline{y}_{0:N+p-1} = -\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ y[0] & 0 & \cdots & 0 & 0 \\ y[1] & y[2] & \cdots & 0 & 0 \\ \vdots & & & & \\ y[p-2] & y[p-3] & \cdots & y[0] & 0 \\ y[p-1] & y[p-2] & \cdots & y[1] & y[0] \\ \vdots & & & & \\ y[N-2] & y[N-3] & \cdots & y[N-p] & y[N-p-1] \\ y[N-1] & y[N-2] & \cdots & y[N-p+1] & y[N-p] \\ 0 & y[N-1] & \cdots & y[N-p+2] & y[N-p+1] \\ \vdots & & & & \\ 0 & 0 & \cdots & y[N-1] & y[N-2] \\ 0 & 0 & \cdots & 0 & y[N-1] \end{bmatrix}}_{Y} \underbrace{\begin{bmatrix} a_{p,1} \\ a_{p,2} \\ \vdots \\ a_{p,p} \end{bmatrix}}_{\underline{a}_p} + \begin{bmatrix} w'[0] \\ w'[1] \\ \vdots \\ w'[p-1] \\ w[p] \\ \vdots \\ w[N-1] \\ w'[N] \\ w'[N+1] \\ \vdots \\ w'[N+p-2] \\ w'[N+p-1] \end{bmatrix}$$

$\therefore$ Innovation is equal to actual $w[n]$ for $n = p, p+1, \ldots, N-1$

▶ Hence by using $Y$ we have,

$$\Rightarrow \frac{1}{N} Y^T Y = \frac{1}{N} \begin{bmatrix} 0 & y[0] & y[1] & \cdots & y[N-2] & y[N-1] & 0 & \cdots & 0 \\ 0 & 0 & y[0] & \cdots & y[N-3] & y[N-2] & y[N-1] & \cdots & 0 \\ \vdots & \vdots & \vdots & & & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & y[N-p+1] & y[N-p] & y[N-p+1] & \cdots & y[N-1] \end{bmatrix} \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ y[0] & 0 & \cdots & 0 & 0 \\ y[1] & y[2] & \cdots & 0 & 0 \\ \vdots & & & & \\ y[p-2] & y[p-3] & \cdots & y[0] & 0 \\ y[p-1] & y[p-2] & \cdots & y[1] & y[0] \\ \vdots & & & & \\ y[N-2] & y[N-3] & \cdots & y[N-p] & y[N-p-1] \\ y[N-1] & y[N-2] & \cdots & y[N-p+1] & y[N-p] \\ g & y[N-1] & \cdots & y[N-p+2] & y[N-p+1] \\ \vdots & & & & \\ 0 & 0 & \cdots & y[N-1] & y[N-2] \\ 0 & 0 & \cdots & 0 & y[N-1] \end{bmatrix}$$

By using the fact

$$\hat{r}_x[k] = \sum_{n=0}^{N-k-1} y[n+k] \cdot y[n] = \frac{1}{N} \sum_{n=0}^{N-k-1} x[n+k] \cdot x[n] \qquad 0 \leq k \leq N-1$$

$$\Rightarrow \frac{1}{N} Y^T Y = \begin{bmatrix} \hat{r}_x[0] & \hat{r}_x[1] & \cdots & \hat{r}_x[p-1] \\ \hat{r}_x[1] & \hat{r}_x[0] & \cdots & \hat{r}_x[p-2] \\ \vdots & & & \vdots \\ \hat{r}_x[p-1] & \hat{r}_x[p-2] & \cdots & \hat{r}_x[0] \end{bmatrix} = \hat{R}_p \quad \cdots (2)$$

- Similarly by using $y_{0:N+p-1}$ we have

$$\frac{1}{N} Y^T y_{0:N+p-1} = \frac{1}{N} \begin{bmatrix} 0 & y[0] & y[1] & \cdots & y[N-2] & y[N-1] & 0 & \cdots & 0 \\ 0 & 0 & y[0] & \cdots & y[N-3] & y[N-2] & y[N-1] & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & y[N-p+1] & y[N-p] & y[N-p+1] & \cdots & y[N-1] \end{bmatrix} \begin{bmatrix} y[0] \\ y[1] \\ \vdots \\ y[p-1] \\ y[p] \\ \vdots \\ y[N-1] \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \hat{\gamma}_x[1] \\ \hat{\gamma}_x[2] \\ \vdots \\ \hat{\gamma}_x[p] \end{bmatrix}$$

$$= \underline{\hat{\gamma}}_p \quad \cdots\cdots (3)$$

∴ Least Square Estimator in terms of $y(n)$'s according to (1) is,

$$\underline{\hat{a}}_p^{LS} = -(Y^T Y)^{-1} Y^T y_{0:N+p-1}$$

$$= -\left(\frac{1}{N} Y^T Y\right)^{-1} \left(\frac{1}{N} Y^T y_{0:N+p-1}\right)$$

$$= -\hat{R}_p^{-1} \hat{\gamma}_p \quad \cdots\cdots\cdots \text{from } (2), (3)$$

$$= \underline{\hat{a}}_p^{YW}$$

# KAY - 13

# 13. Sinusuidal Parameter Estimation

## 13 6 Frequency Estimation

- So far we talked about spectral estimation.

- Goal :-
  Design "good" estimators of $P(f)$ (Power Spectral Density (PSD)) based on a finite set of observations $x[n]$, $n = 1, 2, .., N-1$ (under assumption X is WSS)
  Estimate should be Assymptotically Unbiased and Assymptotically consistant.

- Hidden Assumption - $P(f)$ exists. But there are useful signals that needed to be analysed where $P(f) \to \infty$. These connot be estimated using Spectral Estimation methods.

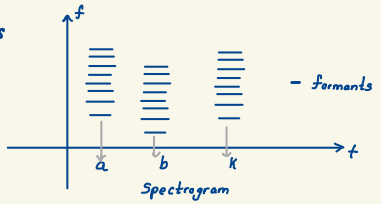
### Harmonic Processes

Harmonic Processes of form

$$x[n] = \sum_{i=1}^{p} A_i e^{j(2\pi f_i n + \phi_i)} + w(n)$$

$$\phi_i \sim \text{Unif}[0, 2\pi) \quad \text{and iid} \qquad w[n] \sim \mathcal{N}(0, \sigma_w^2)$$

- Harmonic Processes are useful models in many fields.
  - Sonar / Radar - $f_i$ - Doppler frequency that is related to the velocity. $p$ - no. of flying objects.
  - Speech Processing - $p$ - no. of formants ($C = 10$)



- $\hat{P}(f)$ won't give $E(\hat{P}(f)) = P(f)$ in this example.
- Can we estimate $f_i$'s and $A_i$'s without using the parametric / non-parametric spectral estimation methods.
- Use frequency estimation opposed to Spectral Estimation.

**New Goal :-** Finding the $f_i$'s and $A_i$'s of a Harmonic Process observed in the presence of noise $w[n]$, that is assumed to be white ($w[n] \sim W(0, \sigma_w^2)$).

$$x[n] = \sum_{i=1}^{p} A_i \, e^{j(2\pi f_i n + \phi_i)} + w[n]$$

- From HW4 Problem 2

$$r_x[k] = \sum_{i=1}^{p} A_i^2 \, e^{j2\pi f_i k} + \sigma_w^2 \, \delta[k] \qquad A_i \in \mathbb{R}$$

$$P(f_i) = A_i^2 \cdot \delta(f - f_i) \longrightarrow \text{Notational Convention}$$

Given $M$ lags, $i = 1, 2, \ldots, p$

$$R_x = \underbrace{\sum_{i=1}^{p} A_i^2 \, \underline{e}_i \, \underline{e}_i^H}_{R_S} + \underbrace{\sigma_w^2 I}_{R_w} \Rightarrow \boxed{R_x = R_S + R_w}$$

$R_S$: Signal AR Matrix

$R_w$: Noise AR Matrix

where $\underline{e}_i = \begin{bmatrix} 1 \\ e^{-j2\pi f_i} \\ e^{-j4\pi f_i} \\ \vdots \\ e^{-j2(M-1)\pi f_i} \end{bmatrix}$

Assumption $M > p$

## HW 4 - Problem 2

Consider a process $x[n] = \sum_{i=1}^{N} A_i e^{j(\omega_i n + \phi_i)}$, where $A_i$'s are a given set of real-valued amplitudes, $\omega_i \in [0, 2\pi]$ are a given set of frequencies, and $\phi_i \sim \mathsf{unif}[0, 2\pi)$ are i.i.d. Show that the correlation matrix of $M$ subsequent samples of $x[n]$ can be expressed as:

$$\mathbf{R}_M = \sum_{i=1}^{N} A_i^2 \mathbf{e}_i \mathbf{e}_i^H,$$

where $\mathbf{e}_i := \begin{bmatrix} 1 & e^{-j\omega_i} & e^{-j2\omega_i} & \cdots & e^{-j(M-1)\omega_i} \end{bmatrix}^\top$. What is the rank of the matrix $\mathbf{R}_M$?

▶ Assume the $i^{th}$ term is $u_i[n] = A_i e^{j(\omega_i n + \phi_i)}$

$$m_i[n] = E\{u_i[n]\} = \frac{1}{2\pi} \int_{-\pi}^{\pi} A_i \cdot e^{j(\omega_i n + \phi_i)} \, d\phi_i = \frac{A_i e^{j\omega_i n}}{2\pi} \int_{0}^{2\pi} e^{j\phi_i} \, d\phi_i = 0$$

$$r_i[k] = E\{u_i[n] \cdot u_i^*[n-k]\} = E\left\{ A_i \cdot e^{j(\omega_i n + \phi_i)} \cdot A_i e^{-j(\omega_i (n-k) + \phi_i)} \right\}$$

$$= A_i^2 \cdot e^{j\omega_i k}$$

Since $\phi_i$'s are taken from iid autocorrelation of $x[n]$:

$$r_x[k] = \sum_{i=1}^{N} r_i[k] = \sum_{i=1}^{N} A_i^2 e^{j\omega_i k} \quad \ldots\ldots (1)$$

$$R_M = E\left[ \underline{x}[n] \cdot \underline{x}^H[n] \right]$$

Consider the $(k,l)^{th}$ term in $R_M$ matrix. One term $(n-k)$ comes from $\underline{x}$ and $(n-m)$ comes from $\underline{x}^H$

$$\therefore (R_M)_{k,l} = E\{x[n-k] \cdot x^*[n-l]\} = r_x[l-k] = \sum_{i=1}^{N} A_i^2 e^{j\omega_i (l-k)}$$

$$= \sum_{i=1}^{N} A_i^2 \cdot e^{j\omega_i l} \cdot e^{-j\omega_i k}$$

$$= \sum_{i=1}^{N} A_i^2 (e_i)_l (e_i)_k^*$$

$$\Rightarrow R_M = \sum_{i=1}^{N} A_i^2 \cdot e_i \cdot e_i^H$$

# Signal Subspace

- We consider the $R_S$ and it's structure,

$$R_S = \sum_{i=1}^{p} A_i^2 \, \underline{e}_i \, \underline{e}_i^H$$

$$= \underbrace{\begin{bmatrix} \underline{e}_1 & \underline{e}_2 & \cdots & \underline{e}_p \end{bmatrix}}_{S} \underbrace{\begin{bmatrix} A_1^2 & & & \\ & A_2^2 & & \\ & & \ddots & \\ & & & A_p^2 \end{bmatrix}}_{\mathcal{D}} \underbrace{\begin{bmatrix} \underline{e}_1^H \\ \underline{e}_2^H \\ \vdots \\ \underline{e}_p^H \end{bmatrix}}_{S^H}$$

$$\Rightarrow R_S = S \mathcal{D} S^H$$

$$\Rightarrow R_S \, \underline{a} = \underbrace{S}_{p \times p} \underbrace{\mathcal{D} S^H}_{p \times M} \underbrace{\underline{a}}_{M \times 1}$$

$$\Rightarrow R_S \, \underline{a} = \sum_{i=1}^{p} \alpha_i \, \underline{e}_i$$

(*) Rank of $R_S$ is $p$, and it's range is spanned by $(\underline{e}_1, \underline{e}_2, \ldots, \underline{e}_p)$. This is the Signal Subspace.

Claim :- Eigenvectors of $R_S$ and $R_x$ are Identical

Proof :- Consider the eigenvalue decomposition of $R_x$. Since $R_x$ is Hermitian it would have $M$ real eigenvalues and $M$ orthonormal eigenvectors. Assume $(\lambda_i, \underline{v}_i)$ $i = 1, 2, \ldots, M$ are eigenvalue, vector pairs of $R_x$

$$\Rightarrow \lambda_i \, \underline{v}_i = R_x \, \underline{v}_i$$

$$\boxed{\underline{v}_i^H \underline{v}_j = \delta_{i,j}}$$

$$\Rightarrow \lambda_i \, \underline{v}_i = (R_S + R_w) \, \underline{v}_i$$

$$\Rightarrow R_S \, \underline{v}_i = \lambda_i \, \underline{v}_i - \underline{R_w \, \underline{v}_i} = \lambda_i v_i - (\sigma_w^2 I) \underline{v}_i \ldots \quad \text{since noise is white}$$

$$\Rightarrow R_S \, \underline{v}_i = (\lambda_i - \sigma_w^2) \cdot \underline{v}_i$$

$\therefore \underline{v}_i$'s are also the eigenvectors of $R_S$ with (**) $\lambda_i - \sigma_w^2$ as eigenvalues. ∎

$\underline{v}_i$ = eigenvectors of $R_x$ and $R_s$ $\quad i = 1, 2, \ldots, M$

Eigenvalues of $R_s = \lambda_i^{(s)} = \begin{cases} \lambda_i - \sigma_w^2 & \text{if } \lambda_i \geq \sigma_w^2 \\ 0 & \text{o.w} \end{cases}$

Proof: $\because$ Assume

Eigenvalues of $R_x \longrightarrow \lambda_1, \lambda_2, \ldots, \lambda_p, \ldots, \lambda_M \qquad M > p$

From (*) $\Rightarrow$ $R_s$ has rank $p$

From (**) $\Rightarrow$ $R_s$ has eigenvalues $\lambda_i - \sigma_w^2 \quad i = 1, 2, \ldots, M$

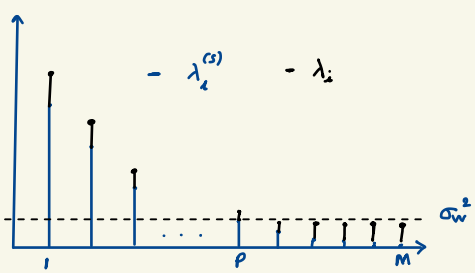
$\Rightarrow$ $p$ of the $\lambda_i$'s are greater than $\sigma_w^2$ and rest is equal to $\sigma_w^2$

$\therefore$ We can get

Eigenvalues of $R_s \longrightarrow \lambda_1 - \sigma_w^2, \lambda_2 - \sigma_w^2, \ldots, \lambda_p - \sigma_w^2, \underbrace{0, 0, \ldots, 0}_{M-p}$

$\longrightarrow \lambda_1^{(s)}, \lambda_2^{(s)}, \ldots, \lambda_p^{(s)}, 0, 0, \ldots, 0$

Eigenvalues of $R_s = \lambda_i^{(s)} = \begin{cases} \lambda_i - \sigma_w^2 & \text{if } \lambda_i \geq \sigma_w^2 \\ 0 & \text{o.w} \end{cases}$




$\therefore$ If $M > p$, then the smallest eigenvalue of $R_x$ is $\sigma_w^2$

$\blacksquare$

$\underline{e}_l^H \underline{v}_i = 0 \qquad \forall \; i = p+1, \ldots, M$
$\qquad\qquad\qquad\qquad\quad l = 1, \ldots, p$

Proof: For $i = p+1, p+2, \ldots, M$ we have

$\longrightarrow R_s \underline{v}_i = \underline{0}$

$\longrightarrow \underbrace{S}_{M \times p} D \underbrace{S^H}_{p \times 1} v_i = \underline{0}$

Since $SD$ has rank $P$ it should not have a linear combination resulting in $\underline{0}$

$$\Rightarrow S^H \underline{v}_i = \underline{0} \quad \forall i$$

$$\Rightarrow \underbrace{\text{Span}\{\underline{e}_1, \underline{e}_2 \ldots \underline{e}_P\}}_{\text{Signal Subspace}} \perp \underbrace{\text{Span}\{\underline{v}_{P+1}, \underline{v}_{P+2}, \ldots, \underline{v}_M\}}_{\text{Noise Subspace}} \quad \cdots (1)$$

Also we have,

$$\text{Span}\{\underline{v}_1, \underline{v}_2, \ldots, \underline{v}_P\} \perp \text{Span}\{\underline{v}_{P+1}, \ldots, \underline{v}_M\} \quad \cdots (2)$$

From (1) and (2)

$$\text{Span}\{\underline{v}_1, \underline{v}_2, \ldots, \underline{v}_P\} \equiv \text{Span}\{\underline{e}_1, \underline{e}_2 \ldots \underline{e}_P\}$$

$$\underline{e}_l^H \underline{v}_i = 0 \quad \forall \; i = P+1, \ldots, M, \quad l = 1, \ldots, P$$

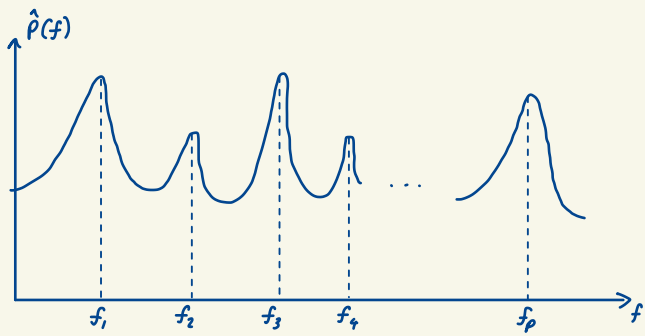
## Frequency Estimation Function

$$\hat{P}(f) = \frac{1}{\sum_{i=P+1}^{M} \alpha_i \left| \underline{e}(f)^H \underline{v}_i \right|^2}$$

DTFT Square of $v_i$ at frequency $-f$

where $\underline{e}(f) = \begin{bmatrix} 1 \\ e^{-j2\pi f_1} \\ e^{-j4\pi f_1} \\ \vdots \\ e^{-j2(M-1)\pi f_1} \end{bmatrix}$ and $\alpha_i$'s are some pre-defined weights.

Depending on algorithm there are different weights



$$\underline{e}^H(f)\,\underline{v}_i = \sum_{n=0}^{M-1} v_i[n]\cdot e^{-j2\pi(-f)n} = \text{DTFT of } \underline{v}_i \text{ as a sequence of frequency } -f$$

$$\Rightarrow \underline{v}_i^H R_x \underline{v}_i = \lambda_i \qquad i = 1,2,\ldots,p$$

$$\Rightarrow \underline{v}_i^H (R_s + R_w)\,\underline{v}_i = \lambda_i$$

$$\Rightarrow \underline{v}_i^H \left(\sum_{l=1}^{p} A_l^2\, \underline{e}_l\, \underline{e}_l^H + \sigma_w^2 I\right)\underline{v}_i = \lambda_i$$

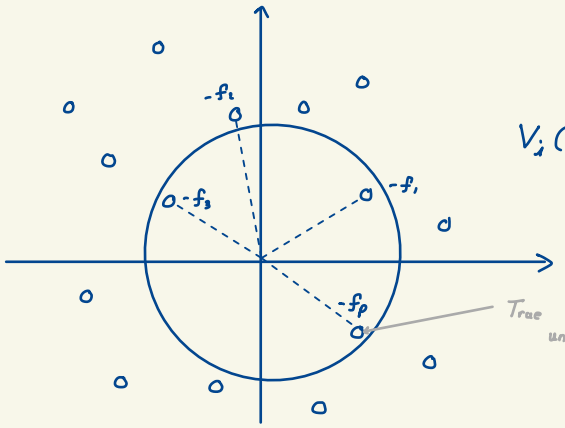
$$\Rightarrow \sum_{l=1}^{p} A_l^2 \left(\underline{v}_i^H \underline{e}_l(f)\right)\left(\underline{e}_l^H(f)\cdot \underline{v}_i\right) = \lambda_i - \sigma_w^2$$

$$\Rightarrow \sum_{l=1}^{p} A_l^2 \underbrace{\left|\, e^H(f_l)\cdot \underline{v}_i \,\right|^2}_{B_{li}} = \lambda_i - \sigma_w^2 \qquad i = 1,2,\ldots,p$$

$$B = \begin{bmatrix} |e^H(f_1)\underline{v}_1|^2 & \cdots & |e^H(f_i)\underline{v}_1|^2 & \cdots & |e^H(f_p)\underline{v}_1|^2 \\ \vdots & & \vdots & & \vdots \\ |e^H(f_1)\underline{v}_j|^2 & \cdots & |e^H(f_i)\underline{v}_j|^2 & \cdots & \\ \vdots & & & & \\ |e^H(f_1)\underline{v}_M|^2 & \cdots & \cdots & \cdots & \end{bmatrix}$$

$$\Rightarrow B \begin{bmatrix} A_1^2 \\ A_2^2 \\ \vdots \\ A_p^2 \end{bmatrix} = \begin{bmatrix} \lambda_1 - \sigma_w^2 \\ \lambda_2 - \sigma_w^2 \\ \vdots \\ \lambda_p - \sigma_w^2 \end{bmatrix}$$

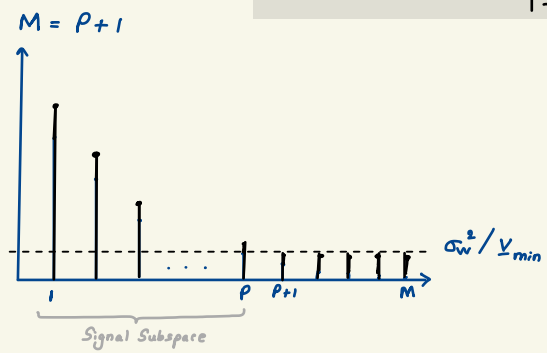
$$\Rightarrow \begin{bmatrix} A_1^2 \\ A_2^2 \\ \vdots \\ A_p^2 \end{bmatrix} = B^{-1} \begin{bmatrix} \lambda_1 - \sigma_w^2 \\ \lambda_2 - \sigma_w^2 \\ \vdots \\ \lambda_p - \sigma_w^2 \end{bmatrix}$$



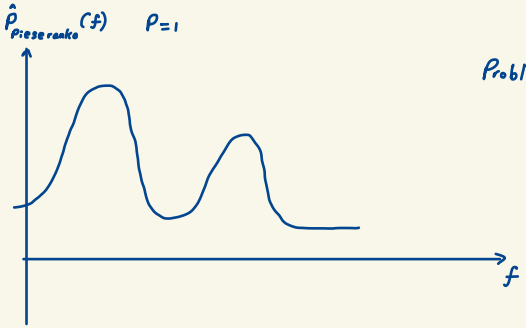

$$V_i(z) = \sum_{n=0}^{M-1} v_i[n] \cdot z^{-n}$$

## 1) The Pisarenko Method

$$\hat{P}_{pisarenko}(f) = \frac{1}{\left| \underline{e}^H(f) \cdot \underline{v}_{min} \right|}$$




$$\lambda_{min} = \sigma_w^2$$
$$\alpha_{p+i} = 1$$




Problem in Pisarenko Method

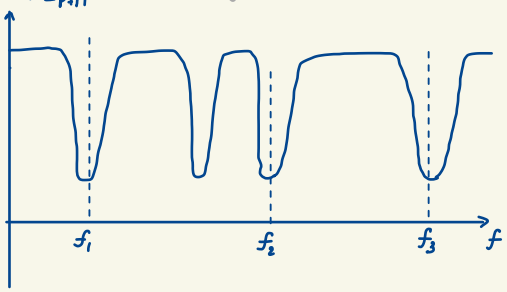
True peaks cannot be identified from false ones. In p=1, this method usually gives 2 peaks.

# 2) Multiple Signal Classification (MUSIC) Algorithm

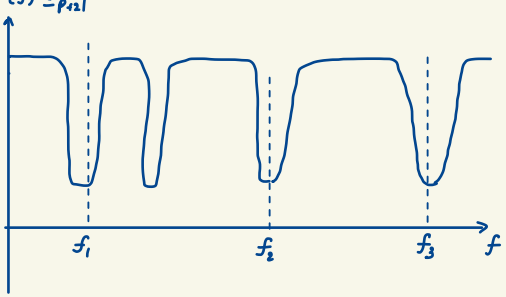
$$\hat{\rho}_{MUSIC}(f) = \frac{1}{\sum_{i=p+1}^{M} |e^H(f) \cdot \underline{v}_i|^2} = \frac{1}{\| e(f)^H \underline{v}_n \|^2}$$

Unlike in Pisarenko, values are first averaged before inverting.

$|e^H(f) \cdot \underline{v}_{p+1}|^2$

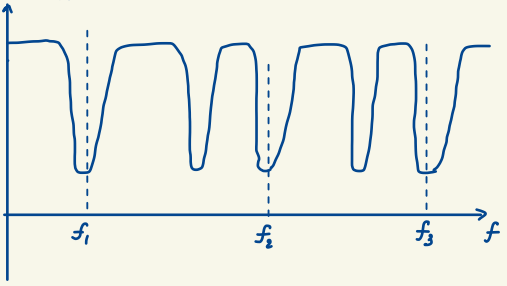



$|e^H(f) \cdot \underline{v}_{p+2}|^2$



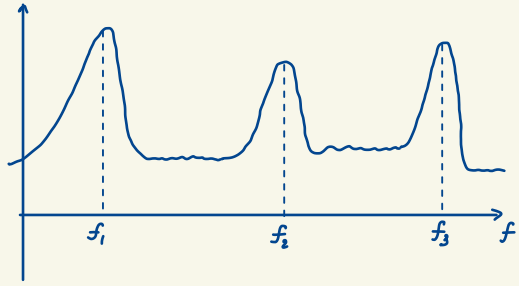

$\vdots$

$|e^H(f) \cdot \underline{v}_M|^2$




$\underline{v}_n = [\, v_{p+1} \; v_{p+2} \; \dots \; v_M \,] \in \mathbb{C}^{M \times (M-p)}$ is the matrix with columns containing all the M-p noise eigenvectors

$\hat{\rho}_{MUSIC}(f)$ $M > p$



HW 7 – Problem 3

Instead of using $\underline{v}_n$ (noise eigenvectors) a different matrix $\underline{V}$ can be used as a reasonable choice to identify the $p$ frequencies of the harmonic process where,

$$\underline{V} = \frac{\underline{v}_n \underline{v}_n^H e_1}{e_1^H \underline{v}_n \underline{v}_n^H e_1} \qquad \hat{P}(f) = \frac{1}{|e(f)^H v|^2} \qquad e_1 = [1\ 0\ 0 \dots 0]^T$$

$\underline{V}$ – lies in noise subspace
first element is 1
has smallest possible norm

Proof: $\underline{V}_n = [\underline{v}_{p+1} \ \underline{v}_{p+2} \dots \underline{v}_M] \Longrightarrow$ Columns of $\underline{V}_n$ contains M-p noise eigenvectors

$\Longrightarrow \underline{V}_n \underline{V}_n^H$ projects any vector onto the subspace spanned by the columns of $V_n$

- Assume $\underline{a} = \underline{V}_n^H \underline{x}$ then $\underline{V} = \underline{V}_n \underline{V}_n^H \underline{x} = \underline{V}_n \underline{a}$

$$= \sum_{i=p+1}^{M} a_{i-p} \underline{v}_i$$

$\Longrightarrow \underline{V}$ is a linear combination of columns of $V_n$

$\Longrightarrow \underline{V}$ lies in the noise subspace for any $\underline{x}$

Assume $\underline{V} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix}$ $\qquad \underline{V} = \underline{V}_n \underline{V}_n^H \underline{x} \quad \dots (1)$

- If $\underline{x}$ is choosen to satisfy following

$\underline{x}^H \underline{V}_n \underline{V}_n^H e_1 = 0 \Longrightarrow \underline{V} e_1 = 0 \Longrightarrow \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} [1\ 0 \dots 0] = 0$

$\Rightarrow v_1 = 1$

$\therefore$ Property (i) and (ii) are satisfied.

$$\|\underline{v}\|^2 = \underline{v}^H \underline{v} = \left(\underline{v}_n \underline{v}_n^H \underline{x}\right)^H \left(\underline{v}_n \underline{v}_n^H \underline{x}\right)$$

$$= \underline{x}^H \underline{v}_n \underbrace{\underline{v}_n^H \underline{v}_n}_{I} \underline{v}_n^H \underline{x} \quad \cdots\cdots \quad \underline{v}_n^H \underline{v}_n = I \text{ since columns are eigenvectors of Hermitian matrix}$$

$$= \underline{x}^H \underline{v}_n \underline{v}_n^H \underline{x}$$

Now the optimization problem is to figure out an $\underline{x}$, which gives the smallest norm. According to part (b) that is,

$$\min_{\underline{x}} \quad \underline{x}^H \underline{v}_n \underline{v}_n^H \underline{x}$$

While satisfying property (i) and (ii). For that required condition according to part @ is

$$\underline{x}^H \underline{v}_n \underline{v}_n^H \underline{e} = 1$$

$\therefore$ The lagrangian is given by,

$$\mathcal{L} = \underline{x}^H \underline{v}_n \underline{v}_n^H \underline{x} + \lambda\left(1 - \underline{x}^H \underline{v}_n \underline{v}_n^H \underline{e}\right)$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \underline{x}^H} = \underline{v}_n \underline{v}_n^H \underline{x} + \lambda\left(-\underline{v}_n \underline{v}_n^H \underline{e}\right)$$

$$\overset{\text{At } \frac{\partial \mathcal{L}}{\partial \underline{x}^H} = 0}{\Rightarrow} \underline{v}_n \underline{v}_n^H \underline{x} + \lambda\left(-\underline{v}_n \underline{v}_n^H \underline{e}\right) = 0$$

$$\Rightarrow \underbrace{\underline{v}_n \underline{v}_n^H \underline{x}}_{\text{from } ①} = \lambda \underline{v}_n \underline{v}_n^H \underline{e}$$

$$\Rightarrow \underline{v} = \lambda \underline{v}_n \underline{v}_n^H \underline{e} \quad \cdots\cdots ②$$

$\therefore$ From constraints $\longrightarrow$ $\underline{x}^H \underline{v}_n \underline{v}_n^H \underline{e}_1 = 1$

$$\Rightarrow \underline{v}^H \underline{e}_1 = 1$$

$$\Rightarrow \left(\lambda \underline{v}_n \underline{v}_n^H \underline{e}_1\right)^H \underline{e}_1 = 1 \quad \cdots\cdots \text{ from } ②$$

$$\Rightarrow \lambda^* = \frac{1}{\underline{e}_1^H \underline{v}_n \underline{v}_n^H \underline{e}_1}$$

$$\hat{P}(f) = \frac{1}{|\underline{e}(f)^H \underline{v}|^2}$$

from part @, (b) and (c) $\underline{v} = \frac{\underline{v}_n \underline{v}_n^H \underline{e}_1}{\underline{e}_1^H \underline{v}_n \underline{v}_n^H \underline{e}_1}$ satisfies all 3 properties.

Since this $\hat{P}(f)$ is inversely propotional to inner product of $e(f)$ and a represented in the noise subspace, it is a reasonable choice.

# THEO - 9

# Sparsity Aware Learning

## Concepts and Theoretical Foundations

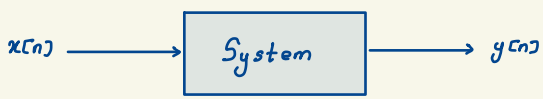
But the first part is from a different chapter

- In classical DSP, sampling can be done at no cost.
  Eg: $x[n]$, $n = 0, 1, \ldots, N-1$.

- In modern DSP, $N$ may not be large enough and it may be quite costly to sample on demand.
  Eg: MRI

## Applications

### 1) System Identification




- Goal - Identify the system using pairs of measurements / inputs $\{x[n], y[n]\}_{n=0}^{N-1}$

- Ex: Consider the system is a human body using MRI applications.
  $x[n]$ - RF pulse
  $y[n]$ - RF response

### 2) DFT

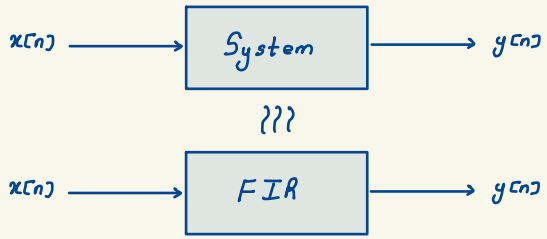
$$x[n] \xrightarrow{DFT} X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi kn}{N}} \quad \ldots \quad N/K = f_K$$

$$X[k] \xrightarrow{IDFT} x[n] = \frac{1}{N} \sum_{n=0}^{N-1} X[k]\, e^{j\frac{2\pi kn}{N}}$$

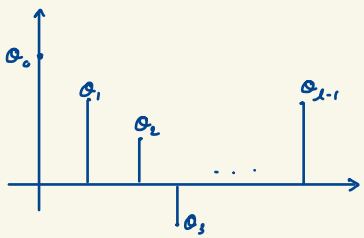
- Generally to find $X[k]$ $\forall$ $k = 0, 1, \ldots, N-1$ we need $x[n]$ $\forall$ $k = 0, 1, \ldots, N-1$

- Goal - Can we still find $X[k]$ $\forall$ $k = 0, 1, \ldots, N-1$ but only using a subset of $x[n]$ for $n \subset \{0, 1, \ldots, N-1\}$

# Linear Models for System Identification




- Assume that $\theta_i$ are the impulse response of estimated FIR filter




$$y[n] = \sum_{i=0}^{L-1} \theta_i \, x[n-i] \qquad n = 0, 1, \ldots, N-1$$

Goal :- Can we estimate a "Good" FIR filter $\{\theta_0, \theta_1, \ldots, \theta_{L-1}\}$ using $\{x[n], y[n]\}_{n=0}^{N-1}$?

Q :- Estimate $\underline{\theta}_0$ such that $\underline{y} = X\underline{\theta}_0 + \underline{e}$

$\underline{y} \in \mathbb{R}^N$
$X \in \mathbb{R}^{N \times L}$
$\underline{\theta} \in \mathbb{R}^L$
$\underline{e} \in \mathbb{R}^N$

- What is meant by "Good" $\Longrightarrow$ Good = Minimum MSE ?

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_{L-1} \end{bmatrix} \in \mathbb{R}^L \qquad \hat{\underline{\theta}} = \begin{bmatrix} \hat{\theta}_0 \\ \hat{\theta}_1 \\ \vdots \\ \hat{\theta}_{L-1} \end{bmatrix} \longrightarrow \text{estimate}$$

$$\underline{\theta}_0 = \begin{bmatrix} \theta_{0,0} \\ \theta_{1,0} \\ \vdots \\ \theta_{L-1,0} \end{bmatrix} \in \mathbb{R}^L \longrightarrow \text{true } \underline{\theta}$$

# Minimum Variance Unbiased Estimator (MVUE)

- If we assume that "Good" estimator should have minimum MSE

$$MSE = MSE(\underline{\hat{\theta}}) = E\left[\|\underline{\hat{\theta}} - \underline{\theta}_0\|^2\right]$$

- $\therefore$ 'Best' $\underline{\theta}$ is the value where $\underset{\underline{\hat{\theta}}}{\text{argmin}}\ MSE(\underline{\hat{\theta}})$

- Consider the special case $\lambda = 1$

$$MSE(\hat{\theta}) = E\left[(\hat{\theta} - \theta_0)^2\right]$$

$$= E\left[\left(\hat{\theta} - E[\hat{\theta}] + E[\hat{\theta}] - \theta_0\right)^2\right]$$

$$= E\left[\left(\hat{\theta} - E[\hat{\theta}]\right)^2\right] + E\left[\underbrace{\left(E[\hat{\theta}] - \theta_0\right)^2}_{\text{Bias of } \hat{\theta}}\right] + 2E\left[\left(\hat{\theta} - E[\hat{\theta}]\right)\left(E[\hat{\theta}] - \theta_0\right)\right]$$

$$= Var(\hat{\theta}) + bias^2(\hat{\theta})$$

- Many approaches focus on starting with estimates that are unbiased ie. $E[\hat{\theta}] = \theta_0$ and then trying to minimize $Var(\hat{\theta})$ hoping that it will result in smallest MSE.

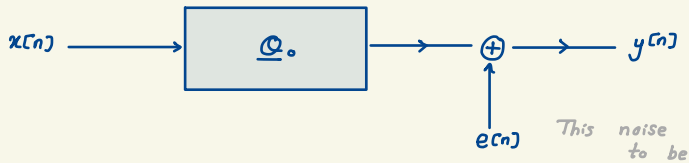
$$MSE(\hat{\theta}) = Var(\hat{\theta})$$

$$\min_{\hat{\theta}} Var(\hat{\theta}) \quad \text{such that} \quad E[\hat{\theta}] = \theta_0$$

impractical optimization problem since to look for minimum MSE we need $\theta_0$ itself. An unknown parameter

# Least Squares (LS) Estimator

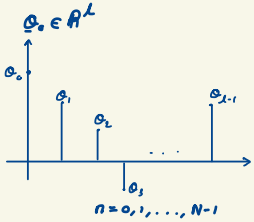
- Back to the system Identification problem




This noise is assumed to be zero mean

$$\underline{\hat{\theta}}_{LS} = \underset{\underline{\theta}}{\operatorname{argmin}} \sum_{n=0}^{N-1} \left( y[n] - \sum_{i=0}^{L-1} \theta_i \, x[n-i] \right)^2$$

$$\underline{\hat{\theta}}^{LS} = \underset{\underline{\theta}}{\operatorname{argmin}} \sum_{n=0}^{N-1} \left( y[n] - \underline{\theta}^T \underline{x}[n] \right)^2$$




$$\underline{x}[n] = \begin{bmatrix} x[n] \\ x[n-1] \\ \vdots \\ x[n-L+1] \end{bmatrix} \qquad \underline{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_{L-1} \end{bmatrix}$$

Since $\underline{x}[0] = \begin{bmatrix} x[0] \\ x[-1] \\ \vdots \\ x[-L+1] \end{bmatrix}$

Let's assume that we know the past values of $x$

- At the optimal point,

$$\Rightarrow \frac{\partial}{\partial \underline{\theta}} \underline{\hat{\theta}}_{LS} = \sum_{n=0}^{N-1} 2 \left( y[n] - \underline{\theta}^T \underline{x}[n] \right) \left( \underline{x}[n] \right) = 0$$

$$\Rightarrow \sum_{n=0}^{N-1} y[n] \cdot \underline{x}[n] = \sum_{n=0}^{N-1} \underline{\theta}^T \underline{x}[n] \cdot \underline{x}[n]$$

$$= \left( \sum_{n=0}^{N-1} \underline{x}[n] \cdot \underline{x}^T[n] \right) \underline{\theta}$$

- By looking at the vector $\underline{y}$ and matrix $\underline{X}$

$$\underline{y} = \begin{bmatrix} y[0] \\ y[1] \\ \vdots \\ y[N-1] \end{bmatrix} \in \mathbb{R}^{N \times 1} \qquad \underline{X} = \begin{bmatrix} \underline{x}[0]^T \\ \underline{x}[1]^T \\ \vdots \\ \underline{x}[N-1]^T \end{bmatrix} \in \mathbb{R}^{N \times L}$$

$$\Rightarrow \quad X^T y = (X^T X)\,\underline{\theta}$$

$$\Rightarrow \quad \boxed{\underline{\hat{\theta}}^{LS} = (X^T X)^{-1} X^T y}$$

## Bias

- Now consider the expectation of this estimator,

$$\mathbb{E}[\underline{\hat{\theta}}^{LS}] = \mathbb{E}[(X^T X)^{-1} X^T y]$$

$$= \mathbb{E}[(X^T X)^{-1} X^T (X\underline{\theta}_0 + \underline{e})] \quad \cdots\cdots \quad \underline{e} = \begin{bmatrix} e[0] \\ e[1] \\ \vdots \\ e[N-1] \end{bmatrix}$$

$$= \mathbb{E}[\underbrace{(X^T X)^{-1} X^T X}_{I}\,\underline{\theta}_0 + (X^T X)^{-1} X^T e)]$$

Signal X and noise e are independant

$$= \underline{\theta}_0 + \mathbb{E}[(X^T X)^{-1} X^T]\,\underbrace{\mathbb{E}[e]}_{0 \text{ from assumption}}$$

$$= \underline{\theta}_0$$

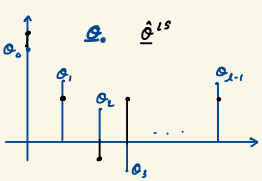
∴ This LS is unbiased but might have high variance. ∴ Solution is to use ridge regression.

## Variance

- In practice the LS solution can have quite a high variance. Let's find the variance of LS assuming $e[n] \sim \mathcal{N}(0, \sigma^2)$

Since out $\underline{\hat{\theta}}^{LS}$ is an estimation for

$$\underline{\hat{\theta}}^{LS} = \underline{\theta}_0 + (X^T X)^{-1} X^T \underline{e} \quad \cdots \text{ why?}$$




$$\Rightarrow \quad \underline{\hat{\theta}}^{LS} - \underline{\theta}_0 = (X^T X)^{-1} X^T \underline{e}$$

$$\Rightarrow \quad \mathrm{Cov}[\underline{\hat{\theta}}^{LS}] = (X^T X)^{-1} X^T \underbrace{\mathrm{Cov}[\underline{e}]}_{\sigma^2 I} \times (X^T X)^{-1}$$

$$= \sigma^2 (X^T X)^{-1}$$

$$\mathrm{Var}[\underline{\hat{\theta}}^{LS}] = \mathbb{E}[\|\underline{\hat{\theta}}^{LS} - \underline{\theta}_0\|^2] = Tr\{\mathrm{Cov}[\underline{\hat{\theta}}^{LS}]\} = \sigma^2 Tr\{(X^T X)^{-1}\}$$

∴ Noise amplification. Poor scaling with eigenvalues of $X^T X$.

If $\lambda_{min}(X^T X) = 0.001 \Rightarrow Tr\{(X^T X)^{-1}\} \geq 1000$

# Lagrange Multiplier Proof to LS Solution (Home Work 8 - Problem 1)

**Problem 1.** Suppose that $\mathbf{X} \in \mathbb{R}^{N \times \ell}$ for $N < \ell$, and that the rank of $\mathbf{X}$ is $N$. Show that the solution to the $\ell_2$-minimization problem:

$$\min_{\boldsymbol{\theta}} \ \frac{1}{2}\|\boldsymbol{\theta}\|_2^2 \qquad \text{s.t.} \qquad \mathbf{y} = \mathbf{X}\boldsymbol{\theta},$$

is given by $\widehat{\boldsymbol{\theta}} := \mathbf{X}^\top \left(\mathbf{X}\mathbf{X}^\top\right)^{-1} \mathbf{y}$.

*Hint: The Lagrange multipliers corresponding to the $N$ constraints in $\mathbf{y} - \mathbf{X}\boldsymbol{\theta} = \mathbf{0}$ are of the form: $\sum_{i=1}^{N} \lambda_i (y_i - (\mathbf{X}\boldsymbol{\theta})_i)$, which can be expressed as $\boldsymbol{\lambda}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})$, where $\boldsymbol{\lambda} := [\lambda_1 \ \lambda_2 \ \cdots \ \lambda_N]^\top$.*

This is a convex optimization problem with an equality constraints. ∴ Use method of Lagrangian multipliers.

$$\text{Objective function} = f(\underline{\theta}) = \frac{1}{2}\|\underline{\theta}\|_2^2 = \frac{1}{2}\underline{\theta}^T\underline{\theta}$$

$$\text{Constraints} = g_i(\underline{\theta}) = y_i - (X\underline{\theta})_i = 0 \quad i = 1, \ldots, N \ldots \text{ i.e } \underline{y} - X\underline{\theta} = 0 \quad (1)$$

$$\Rightarrow \therefore \text{Lagrangian} = \mathcal{L}(\underline{\theta}, \underline{\lambda}) = f(\underline{\theta}) + \sum_{i=1}^{N} \lambda_i g_i(\underline{\theta}) \quad \cdots \lambda_i \text{ are lagrangian multipliers}$$

$$= \frac{1}{2}\underline{\theta}^T\underline{\theta} + \sum_{i=1}^{N} \lambda_i (y_i - (X\underline{\theta})_i)$$

$$= \frac{1}{2}\underline{\theta}^T\underline{\theta} + \underline{\lambda}^T \cdot (\underline{y} - X\underline{\theta})$$

$$= \frac{1}{2}\underline{\theta}^T\underline{\theta} - \underline{\lambda}^T(X\underline{\theta} - \underline{y}) \quad \cdots \underline{\lambda} = [\lambda_1, \ldots \lambda_N]^T$$

$$= \frac{1}{2}\underline{\theta}^T\underline{\theta} - \underline{\lambda}^T X\underline{\theta} + \underline{\lambda}^T\underline{y}$$

$$\Rightarrow \nabla_{\underline{\theta}} \mathcal{L} = \frac{\partial}{\partial \underline{\theta}} \left( \frac{1}{2}\underline{\theta}^T\underline{\theta} - \underline{\lambda}^T X\underline{\theta} + \underline{\lambda}^T\underline{y} \right)$$

($\underline{\lambda}^T\underline{y}$: independant of $\underline{\theta}$; $\frac{\partial}{\partial \underline{\theta}}\left(\frac{1}{2}\underline{\theta}^T\underline{\theta}\right)$ is equivalent to $\frac{d}{dx}\left(\frac{1}{2}x^2\right)$)

$$= \underline{\theta} - X^T\underline{\lambda} + 0 \quad \cdots \text{using rules of vector calculus}$$

At the optimum point $\hat{\underline{\theta}}$, $\nabla_{\underline{\theta}} \mathcal{L}$ becomes '0'.

$$\Rightarrow \quad \hat{\underline{\theta}} - X^T\underline{\lambda} = 0 \implies \hat{\underline{\theta}} = X^T\underline{\lambda} \quad \cdots\cdots (2)$$

Substituting ②, to constraint ①,

$$\Rightarrow \quad \underline{y} - X\hat{\underline{\alpha}} = 0$$
$$\Rightarrow \quad \underline{y} - X X^T \underline{\lambda} = 0$$
$$\Rightarrow \quad \underline{y} = X X^T \underline{\lambda}$$

It is given that $X$ is a $N \times d$ matrix with rank $N$ where $N < d$. $\therefore$ $X$ is a full row rank matrix. $\therefore$ $XX^T$ is positive definite. $\therefore$ inver tible.

$$\Rightarrow \quad (XX^T)^{-1} \underline{y} = \underline{\lambda} \quad \cdots\cdots ③$$

$$②, ③ \Rightarrow \quad \hat{\underline{\alpha}} = X^T (XX^T)^{-1} \underline{y}$$

■

# Shrinkage

- Speciual case $\lambda = 1$. Consider we already have an unbiased estimator of $\theta_0$

$$\hat{\theta}^u : \text{unbiased estimator of } \theta_0$$

- Now let's make this unbiased estimator, biased by force.

$$\hat{\theta}^b = (1+\alpha) \cdot \hat{\theta}^u$$

Claim: There exist a range of $\alpha$'s for which

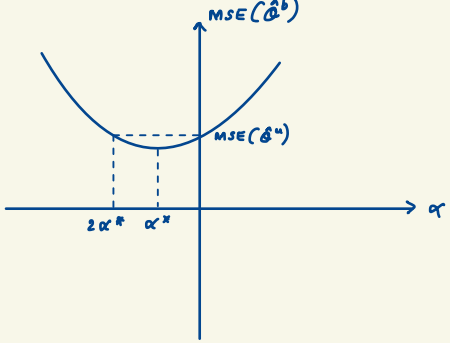
$$MSE(\hat{\theta}^b) < MSE(\hat{\theta}^u)$$

Proof: $\mathbb{E}[\hat{\theta}^b] - \theta_0 = (1+\alpha)\theta_0 - \theta_0 = \alpha\theta_0$ hence this is a biased estimator.

$$\begin{aligned} MSE(\hat{\theta}^b) &= Var(\hat{\theta}^b) + bias^2(\hat{\theta}^b) \\ &= (1+\alpha)^2 Var(\hat{\theta}^u) + \alpha^2\theta_0^2 \\ &= (1+\alpha)^2 MSE(\hat{\theta}^u) + \alpha^2\theta_0^2 \end{aligned}$$

$$\therefore \frac{d}{d\alpha} MSE(\hat{\theta}^b) = 2(1+\alpha) MSE(\hat{\theta}^u) + 2\alpha\theta_0^2 = 0$$

At the local minimum $\longrightarrow$ $\alpha^* = -\dfrac{MSE(\hat{\theta}^u)}{\theta_0^2 + MSE(\hat{\theta}^u)}$




$\therefore$ For $2\alpha^* < \alpha < 0$ we have $MSE(\hat{\theta}^b) < MSE(\hat{\theta}^u)$ ■

- Finding $\alpha^*$ is not practical again, since it also have $\theta_0$ parameters, which are the values we are trying to figure out in the first place. But at least we know shrinkage can improve upone the unbiased estimator.

# Regularization

- Instead a technique called regularization is used.

$$\hat{\underline{\theta}}^{LS} = \underset{\underline{\theta}}{\operatorname{argmin}} \sum_{n=0}^{N-1} \left(y[n] - \underline{\theta}^T \underline{x}[n]\right)^2$$

$$\hat{\underline{\theta}}^{Reg} = \underset{\underline{\theta}}{\operatorname{argmin}} \sum_{n=0}^{N-1} \left(y[n] - \underline{\theta}^T \underline{x}[n]\right)^2 \quad s.t \quad \|\underline{\theta}\|^2 \leq \rho$$

$l_2$ regularized

- But this is not easy to solve since it need quadratic optimization.

# Least Absolute Shrinkage and Selection Operator (LASSO)

## Ridge Regression

- This uses L2 norm for regularization

- To achieve easy computation, the quadratic constraints are removed using this method.

- We can achieve better MSE by using $\lambda$ carefully. But may be biased.

$$\underline{\hat{\theta}}^{Ridge} = \underset{\underline{\theta}}{\operatorname{argmin}} \sum_{n=0}^{N-1} \left(y[n] - \underline{\theta}^T \underline{x}[n]\right)^2 + \lambda \|\underline{\theta}\|_2^2$$

L2 Norm

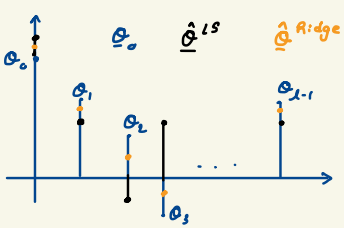
$\lambda$: Ridge parameter, $\lambda \geq 0$

Proof??

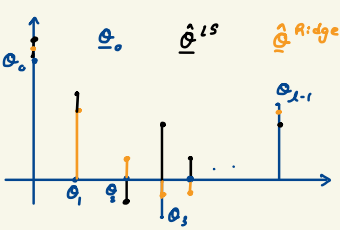
$$\underline{\hat{\theta}}^{Ridge} = \left(X^T X + \lambda I\right)^{-1} X^T y$$

- $\lambda = 0 \rightarrow \underline{\hat{\theta}}^{Ridge} = \underline{\hat{\theta}}^{LS}$
  $\lambda = \infty \rightarrow \underline{\hat{\theta}}^{Ridge} = 0$

$\therefore$ Effectively $\lambda$ is moving the answer between 0 and LS.




- Often times the unknown $\underline{\theta}_0$ is "sparse". That means non-zero elements of $\underline{\theta}_0$ is much smaller than $L$. $\therefore$ The above plot is more likely to be like,




Sparse $\theta_0$

What it means $\underline{\theta}$ to be sparse?

Assume collecting audio signals from a room. There might be are 20 objects in the room that can reflect an audio. But only 2 might be big enough to make an impact.

# Sparse $\underline{\theta}$'s and L1 Norm

$$L_1 \text{ Norm} = \|\underline{\theta}\|_1 = \sum_{i=1}^{L} |\theta_i| \qquad \text{Taxicab norm/ Manhattan norm}$$

$$L_2 \text{ Norm} = \|\underline{\theta}\|_2 = \sqrt{\sum_{i=1}^{L} |\theta_i|^2}$$

$$L_p \text{ Norm} = \|\underline{\theta}\|_p = \left(\sum_{i=1}^{L} |\theta_i|^p\right)^{1/p} \qquad p \geq 1$$

- Let's use the $L_1$ norm, instead of the $L_2$ norm for regularization
- Let's assume $X^TX = I$ for simplicity. ∴ Then closeform solutions of LS, Ridge and L1 becomes.

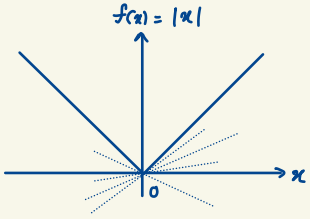
$$\hat{\underline{\theta}}^{LS} = \underbrace{(\underline{X}^T\underline{X})^{-1}}_{I}\underline{X}^T\underline{y} = \underline{X}^T\underline{y}$$

$$\hat{\underline{\theta}}^{Ridge} = \underbrace{(X^TX}_{I} + \lambda I)^{-1}\underbrace{X^Ty}_{\hat{\theta}^{LS}} = \frac{X^Ty}{1+\lambda}$$

$$\hat{\underline{\theta}}^{L1} = \underset{\underline{\theta}}{\text{argmin}} \sum_{n=0}^{N-1}\left(y[n] - \underline{\theta}^T\underline{x}[n]\right)^2 + \underbrace{\lambda\|\underline{\theta}\|_1}_{\text{L1 norm regularization}} = \|\underline{y} - \underline{X}\underline{\theta}\|_2^2 + \lambda\|\theta\|_1$$

- To differentiate $\hat{\underline{\theta}}^{L1}$ we need to modify the differetiation of $\|\underline{\theta}\|_1$ since $f(x) = |x|$ does not have a differentiation at $x=0$. Let's define "Sub derivative" as,

$$\partial|x| = \begin{cases} 1 & x > 0 \\ [-1, 1] & x = 0 \\ -1 & x < 0 \end{cases}$$




∴ Slope at $x=0$ is another variable we have to pick from the set $[-1, 1]$.

revise vector differentiation

$$\Rightarrow \nabla_{\underline{\theta}} = \frac{\partial}{\partial \underline{\theta}} \hat{\underline{\theta}}^{L1} = -2x^T(y - x\underline{\theta}) + \lambda \cdot \partial_{\underline{\theta}} \left( \sum_{i=1}^{J} |\underline{\theta}|_j \right) = 0$$

$$\begin{bmatrix} \partial|\theta|_0 \cdots > \epsilon\,[-1,1] \\ \partial|\theta|_1 \cdots > \epsilon\,[-1,1] \\ \vdots \\ \partial|\theta|_J \cdots > \epsilon\,[-1,1] \end{bmatrix} \quad \therefore J \text{ new choices to pick. Lot of unknowns.}$$

$$\Rightarrow -2x^T y + 2\underbrace{x^T x}_{I}\theta^{L1} + \lambda \cdot \partial_{\underline{\theta}} \|\underline{\theta}^{L1}\|_1 = \underline{0}$$

$$\Rightarrow -\hat{\underline{\theta}}^{LS} + \underline{\theta}^{L1} + \frac{\lambda}{2}\partial_{\underline{\theta}} \|\underline{\theta}^{L1}\|_1 = \underline{0}$$

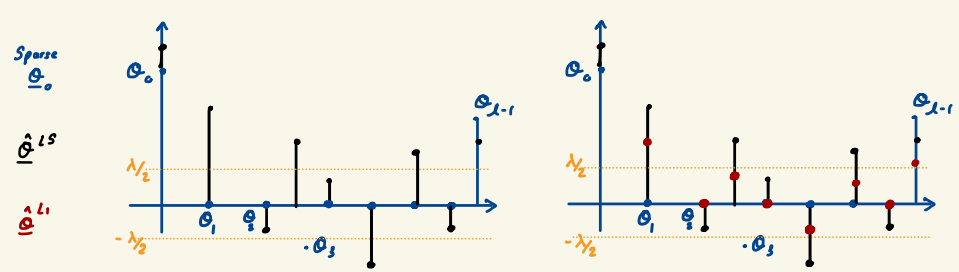
$$\hat{\theta}_j^{L1} > 0 \Rightarrow -\hat{\theta}_j^{LS} + \hat{\theta}_j^{L1} + \frac{\lambda}{2}\underbrace{\partial|\hat{\underline{\theta}}^{L1}|}_{+1} = 0 \Rightarrow \hat{\underline{\theta}}_j^{L1} = \hat{\underline{\theta}}_j^{LS} - \frac{\lambda}{2}$$

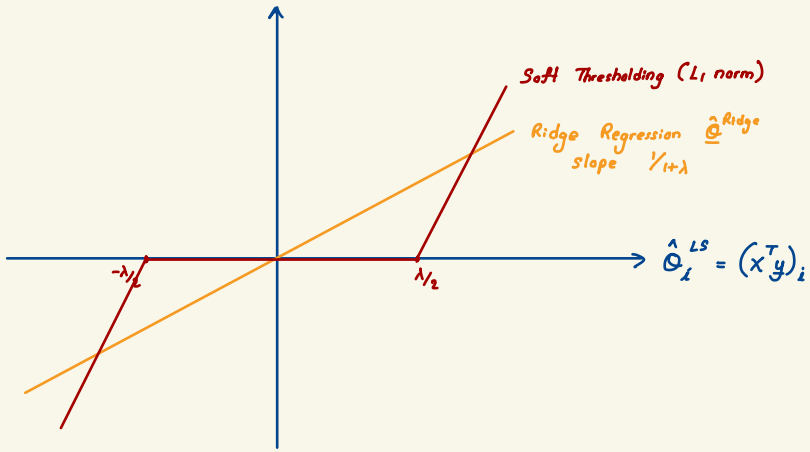
$$\hat{\theta}_j^{L1} < 0 \Rightarrow -\hat{\theta}_j^{LS} + \hat{\theta}_j^{L1} + \frac{\lambda}{2}\underbrace{\partial|\hat{\underline{\theta}}^{L1}|}_{-1} = 0 \Rightarrow \hat{\underline{\theta}}_j^{L1} = \hat{\underline{\theta}}_j^{LS} + \frac{\lambda}{2}$$

$$\hat{\theta}_j^{L1} = 0 \Rightarrow -\hat{\theta}_j^{LS} + 0 + \frac{\lambda}{2}\underbrace{\partial|\hat{\underline{\theta}}^{L1}|}_{\epsilon\,[-1,1]} = 0 \Rightarrow \hat{\underline{\theta}}_j^{LS} \leq \frac{\lambda}{2}$$

$$\Rightarrow \hat{\underline{\theta}}_j^{L1} = \begin{cases} \hat{\underline{\theta}}_j^{LS} - \frac{\lambda}{2} & \hat{\theta}_j^{LS} > \lambda/2 \\ 0 & \hat{\theta}_j^{L1} = 0 \Rightarrow -\lambda/2 \leq \hat{\theta}_j^{LS} \leq \lambda/2 \\ \hat{\underline{\theta}}_j^{LS} + \frac{\lambda}{2} & \hat{\theta}_j^{LS} < -\lambda/2 \end{cases}$$

$$\Rightarrow \hat{\underline{\theta}}_j^{L1} = sgn(\hat{\underline{\theta}}_j^{LS}) \cdot \max\left( |\hat{\underline{\theta}}_j^{LS}| - \lambda/2, 0 \right)$$






$L1$ norm has a killing effect. But ridge regression does not kill stuff, it just diminish the small noises

If $\underline{\theta}$ is sparse we actually need stuff to be killed

- What if $X^T X \neq I$

$N$ - no. of measurements

$L$ - no. of unknowns

**Challenge** $N < L$

If $N < L$, $X^T X$ can never be equal to identity.

But usually $N < L$.

i.e No of observations $[y, \underline{y}]$ < No of parameters

∴ There can be lot of $\underline{\theta}$'s giving these outputs

$$\underline{y} = X \underline{\theta}_0 + \underline{e}$$

$$\begin{bmatrix} \underline{y} \end{bmatrix}_N = \begin{bmatrix} & L & \\ & X & \\ \text{one Input} & & \end{bmatrix} \begin{bmatrix} \underline{\theta}_0 \end{bmatrix}_L + \begin{bmatrix} \underline{e} \end{bmatrix}$$

$\left.\begin{array}{l} \text{Ran}(X) \ll N \\ X^T X \in \mathbb{R}^{L \times L} \end{array}\right\} \Rightarrow X^T X$ can not be an Identity $\Longrightarrow$ What to do?

This problem is solved under the assumption of perfect noise and sparsity of $\underline{\theta}$.

$L_0$ and $L_1$ minimizers result in the same solution.

# In Search of the Sparsest Solution

Q - How can we estimate $l$ unknowns $\underline{\theta}$ from $N$ observations $\underline{y}$ → $\begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$

Sensing Matrix $N \times l$ input matrix

$l \times 1$ parameter vector

$$\underline{y} = X\underline{\theta} \qquad N < l$$

(Noiseless case $\underline{e} = 0$)

- Let

$$\text{null}(X) = \{ \underline{z} \in \mathbb{R}^l : X\underline{z} = 0 \} \quad \cdots (1)$$

Since $N < l$ from previous logic we know that $\underline{y} = X\underline{\theta}$ has lot's of solutions. Let $\Theta$ be set of all of them,

$$\Theta = \{ \underline{\theta} \in \mathbb{R}^l : X\underline{\theta} = \underline{y} \} \quad \cdots (2)$$

If $\underline{\theta}_0$ is a solution to $\underline{y} = X\underline{\theta}$

$$\Rightarrow \quad \underline{y} = X\underline{\theta}_0$$

$$\Rightarrow \quad \underline{y} - \underline{y} = X\underline{\theta}_0 - X\underline{\theta} \qquad \forall \underline{\theta} \in \Theta$$

$$\Rightarrow \quad 0 = X(\underline{\theta}_0 - \underline{\theta})$$

$$\Rightarrow \quad \underline{\theta}_0 - \underline{\theta} \in \text{null}(X)$$

$$\Rightarrow \quad \Theta = \underline{\theta}_0 + \text{null}(X)$$

i.e set of $\Theta$ lie in a translated space of null(X)

- Extra constraints are needed to pick one $\underline{\theta}$ out of $\Theta$,

∴ We need more constraints ⟶ A: Use regularization.

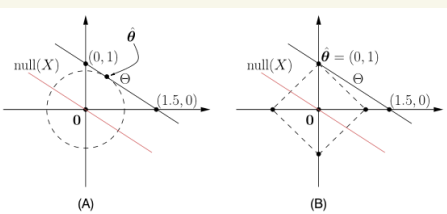



**FIGURE 9.7**

The set of solutions Θ is an affine set (gray line), which is a translation of the null($X$) subspace (red line). (A) The $\ell_2$ norm minimizer: The dotted circle corresponds to the smallest $\ell_2$ ball that intersects the set Θ. As such, the intersection point, $\hat{\boldsymbol{\theta}}$, is the $\ell_2$ norm minimizer of the task in (9.18). Note that the vector $\hat{\boldsymbol{\theta}}$ contains no zero component. (B) The $\ell_1$ norm minimizer: The dotted rhombus corresponds to the smallest $\ell_1$ ball that intersects Θ. Hence, the intersection point, $\hat{\boldsymbol{\theta}}$, is the solution of the constrained $\ell_1$ minimization task of (9.21). Note that the obtained estimate $\hat{\boldsymbol{\theta}} = (0, 1)$ contains a zero.

# $L_2$

1) The Minimum $L_2$ Norm Solution ⟶ $\min_{\underline{\theta}} \|\underline{\theta}\|_2^2$ s.t. $\underline{y} = X\underline{\theta}$ ⟶ Unique solution $\hat{\underline{\theta}} = X^T(X^TX)^{-1}\underline{y}$

(Pick $\underline{\theta}$ such that it has the shortest length)

- Not sparcity preserving. And high contributing $\theta_i$ values won't be identified due to smoothing effect.

Definition:- The $l_0$ (quasi) norm of a vector $\underline{\theta}$ is defined as $\|\underline{\theta}\|_0$ = No. of nonzero elements of $\underline{\theta}$

# $L_0$

2) The minimum $L_0$ quasi norm Solution ⟶ $\min_{\underline{\theta}} \|\underline{\theta}\|_0$ s.t $\underline{y} = X\underline{\theta}$ ⟶ The solution is combinatorial. Need to consider all $2^l - 1$ possible non-zero element locations

- Assume perfect noiseless measurements, and that $\underline{\theta}_0$ is sparse. Then $L_0 \equiv L_1$ and we can simply use $L_1$. And it trivially shows $L_0$ should also be unique when $L_1$ is unique.

# $L_1$

3) The Minimum $L_1$ Norm Solution ⟶ $\min_{\underline{\theta}} \|\underline{\theta}\|_1$ s.t. $\underline{y} = X\underline{\theta}$ ⟶ Q: Solution? A: Use Linear Programming

- This can be solved using linear programming.
- Generaly non convex $L_p$, $P<1$ norms are sparcity promoting. $L_1$ is the closest we can get to achieving sparcity while being convex.

# Geometric Interpretation of $L_1$

**Example 9.2.** Consider a sparse vector parameter $[0, 1]^T$, which we assume to be unknown. We will use one measurement to *sense* it. Based on this single measurement, we will use the $\ell_1$ minimizer of (9.21) to recover its true value. Let us see what happens.

We will consider three different values of the "sensing" (input) vector $\boldsymbol{x}$ in order to obtain the measurement $y = \boldsymbol{x}^T\boldsymbol{\theta}$: (a) $\boldsymbol{x} = [\frac{1}{2}, 1]^T$, (b) $\boldsymbol{x} = [1, 1]^T$, and (c) $\boldsymbol{x} = [2, 1]^T$. The resulting measurement, after sensing $\boldsymbol{\theta}$ by $\boldsymbol{x}$, is $y = 1$ for all three previous cases.

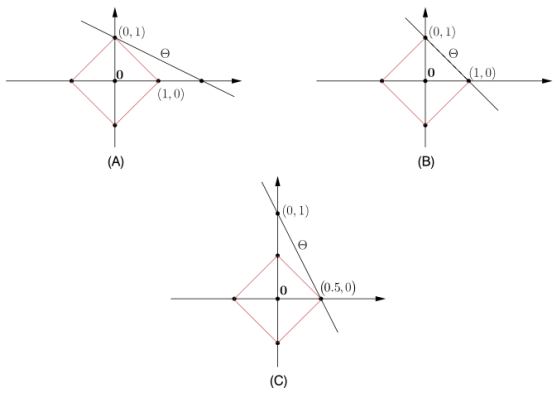



**FIGURE 9.9**

(A) Sensing with $\boldsymbol{x} = [\frac{1}{2}, 1]^T$. (B) Sensing with $\boldsymbol{x} = [1, 1]^T$. (C) Sensing with $\boldsymbol{x} = [2, 1]^T$. The choice of the sensing vector $\boldsymbol{x}$ is crucial to unveiling the true sparse solution (0,1). Only the sensing vector $\boldsymbol{x} = [\frac{1}{2}, 1]^T$ identifies uniquely the desired (0,1).

What are the required constraints on X? *

- ∴ We have to measure X - input / sensing matrix so that the $L_1$ ball touches the $\Theta$ plane [ $\Theta = \underline{\theta}_0 + \text{null}(X)$ ] in a 'gentle' way.
- Normalize the columns of X by dividing from $L_2$ Euclidean norm.

# Uniqueness of $L_0$ Minimizer

- First we need to find conditions under which answer for $L_0$ above gives a unique solution.

**Definition:** Rank - maximum number of linear independant columns.

**Definition:** Spark - minimum number of linear dependant colums or ranks of $X$.

Ex: $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ Column 1 = Column 2 + Column 3 $\therefore$ rank = 2

$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$

rank $(X) = 4$

No column is a scaled another column. $\therefore$ Spark $\neq 2$

Column 6 = Column 2 + Column 3 $\therefore$ Spark = 3

Computing spark $(X)$, requires examining all $2^t$ sub-column sets and checking their linear dependance.

i.e $\underline{z}$ is an element in null$(X)$

**Lemma:** If $X\underline{z} = 0 \implies \|\underline{z}\|_0 \geq \text{Spark}(X)$

**Proof:** Proof by contradiction. Assume $\|\underline{z}\|_0$ is less than spar $(X)$ and show that if $X\underline{z} = 0$, the we can find a set of colums that are LC.

Book gives this in terms of $L_0$ normalizer. But $L_0 \equiv L_1$

Donaho and Eland (2003)

**Theorem:** If the true $\underline{\theta}_0$ has only $S$ non zero elements (out of $d$) and if $S < \frac{1}{2}\,\text{Spark}(X)$, then the solution $\underline{\theta}_0$ to

$$\min_{\underline{\theta}} \|\underline{\theta}\|_0 \quad \text{s.t} \quad X\underline{\theta} = \underline{y}$$

is unique.

**Proof:** Proof by contradiction. Suppose that $\underline{\theta}'$ is another solution that is the sparsest and $\neq \underline{\theta}_0$.

$$\left.\begin{aligned} X\underline{\theta}_0 &= \underline{y} \\ X\underline{\theta}' &= \underline{y} \end{aligned}\right\} \Rightarrow \quad X\underbrace{(\underline{\theta}_0 - \underline{\theta}')}_{\underline{z}} = \underline{0}$$

By applying the previous lemma by taking $\underline{z} = \underline{\theta}_0 - \underline{\theta}'$ we get

$$\Rightarrow \|\underline{z}\|_0 = \|\underline{\theta}_0 - \underline{\theta}'\|_0 \geq \text{Spark}(X)$$

$$\Rightarrow \|\underline{\theta}_0\|_0 + \|\underline{\theta}'\|_0 \geq \|\underline{\theta}_0 - \underline{\theta}'\|_0 \geq \text{Spark}(X) \quad \cdots \text{triangle inequality} \rightarrow \text{works for } L_0 \text{ norm}$$

$$\Rightarrow \|\underline{\theta}_0\|_0 \geq \text{Spark}(X) - \|\underline{\theta}'\|_0$$

$$\geq \text{Spark}(X) - \tfrac{1}{2}\,\text{Spark}(X) \quad \cdots \text{hypothesis/assumption}$$

$$\geq \tfrac{1}{2}\,\text{Spark}(X) > \|\underline{\theta}_0\|$$

\# Since then $\underline{\theta}'$ is not the sparsest

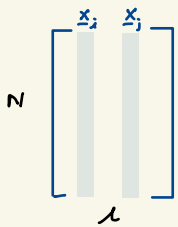
* Constraint on $X$ for 'gentle' touch.

If parameter vector $\underline{\theta}$ is known to have at most $S$ non zero elements, pick input sensing vector $X$, such that $\text{Spark}(X) > 2S$.

# Mutual Coherence

Definition :- $$\mu(X) = \max_{\substack{i,j \\ i \neq j}} \frac{|\underline{x}_i^T \underline{x}_j|}{\|\underline{x}_i\|_2 \|\underline{x}_j\|_2}$$

- Computing $\mu(X)$ requires $\binom{L}{2}$ inspections. This is farely effecient compared to spark which need $2^L$ inspections

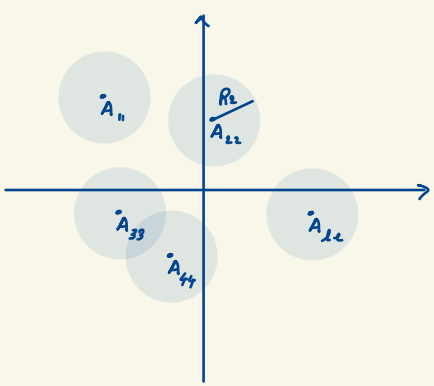



equivalent to cosine between $x_i, x_j$

Theorem :- Gershgorin's Disc (Circle) Theorem

For any matrix $A$, each eigenvalue is included in at least a disc of center $A_{ii}$ and radius $R_i = \sum_{i \neq j} |A_{ij}|$

$$\begin{bmatrix} \vdots & & \\ A_{i1} & \cdots & A_{ii} \cdots A_{iL} \end{bmatrix}$$



# HW 8 - Problem 2

**Problem 2.** Consider an $N \times \ell$ matrix $\mathbf{X}$, with normalized columns, i.e., $\|\mathbf{X}_i\|_2^2 = 1$, $\forall i = 1, 2, \cdots, \ell$. Supoose that $N < \ell$ and that rank of $\mathbf{X}$ is $N$. Recall that th mutual coherence of $\mathbf{X}$ is given by:

$$\mu(\mathbf{X}) := \max_{\substack{i,j \\ i \neq j}} \left| \mathbf{X}_i^\top \mathbf{X}_j \right|$$

(a) Let $\lambda_1, \lambda_2, \cdots, \lambda_N$ be the non-zero eigenvalues of $\mathbf{X}^\top \mathbf{X}$. Show that $\sum_{i=1}^N \lambda_i = \ell$.

(b) Using the condition $\sum_{i=1}^N \lambda_i = \ell$, show that $\sum_{i=1}^N \lambda_i^2 \geq \frac{\ell^2}{N}$.
*Hint: you can use the method of Largrange multipliers to minimize $\sum_{i=1}^N \lambda_i^2$ subject to $\sum_{i=1}^N \lambda_i = \ell$.*

(c) Using the identity $\sum_{i=1}^N \lambda_i^2 = \sum_{i=1}^\ell \sum_{j=1}^\ell \left| \mathbf{X}_i^\top \mathbf{X}_j \right|^2$ and the result of part (b) show that

$$\mu(\mathbf{X}) \geq \sqrt{\frac{\ell - N}{N(\ell - 1)}}.$$

Welch Bound

@ Since $X$ is a $N \times \ell$ matrix. $X^T X$ is a $\ell \times \ell$ matrix. $(i,i)$ position of this $X^T X$ is obtained by multiplying $i^{th}$ column of $X^T$ and $i^{th}$ raw of $X$ (which are same).

$$\therefore (X^T X)_{i,i} = \underline{X}_i^T \underline{X}_i = \|\underline{X}_i\|_2^2 = 1 \quad \cdots\cdots (1)$$

Since trace of a square matrix is sum of it's diagonal elements,

$$Tr(X^T X) = \sum_{i=1}^{\ell} (X^T X)_{i,i}$$

$$= \sum_{i=1}^{\ell} 1 \quad \cdots\cdots \text{from } (1)$$

$$= \ell \quad \cdots\cdots (2)$$

Trace of a square matrix is also the sum of it's eigenvalues.

$$Tr(X^T X) = \sum_{i=1}^{\ell} \lambda_i$$

$$= \sum_{i=1}^{N} \lambda_i + \sum_{i=N+1}^{\ell} 0 \quad \cdots\cdots$$

Since $X^T X$ is $\ell \times \ell$ and rank $(X) = N$ $(N < \ell)$, it has only $N$ nonzero eigenvalues

$$= \sum_{i=1}^{N} \lambda_i \quad \cdots (3)$$

$$(2), (3) \Rightarrow \sum_{i=1}^{N} \lambda_i = \ell$$

■

(b) Let's find the minimum value of $\sum_{i=1}^{N} \lambda_i^2$ under the constraint $\sum_{i=1}^{N} \lambda_i = L$ shown above.

$$\text{Objective function} = f(\underline{\lambda}) = \sum_{i=1}^{N} \lambda_i^2$$

$$\text{Constraints} = g(\underline{\lambda}) = \sum_{i=1}^{N} \lambda_i - L = 0 \quad \cdots (4)$$

$\therefore$ If $\mu$ is the lagrange multiplyer,

$$\Rightarrow \text{Lagrangian} = \mathcal{L}(\underline{\lambda}, \mu) = f(\underline{\lambda}) - \mu\, g(\underline{\lambda})$$

$$= \sum_{i=1}^{N} \lambda_i^2 - \mu \left( \sum_{i=1}^{N} \lambda_i - L \right)$$

$$\Rightarrow \nabla_{\underline{\lambda}} \mathcal{L} = \frac{\partial}{\partial \underline{\lambda}} \left( \sum_{i=1}^{N} \lambda_i^2 - \mu \left( \sum_{i=1}^{N} \lambda_i - L \right) \right)$$

At the optimal point $\nabla_{\underline{\lambda}} \mathcal{L}$ should be $\underline{0}$. i.e. all the partial derivative in terms of $\lambda_i$'s $i = 1, \ldots, N$ should be $\underline{0}$.

$$\Rightarrow \frac{\partial}{\partial \lambda_i} \left( \sum_{i=1}^{N} \lambda_i^2 - \mu \left( \sum_{i=1}^{N} \lambda_i - L \right) \right) = 0$$

$$\Rightarrow 2\lambda_i - \mu(1) = 0$$

$$\Rightarrow \lambda_i = \mu/2 \qquad i = 1, \ldots, N \quad \cdots (5)$$

Plug this $\lambda_i$'s to (2) $\Rightarrow \sum_{i=1}^{N} (\mu/2) - L = 0$

$$\Rightarrow N\mu/2 = L \Rightarrow \mu = 2L/N \quad \cdots (6)$$

(5), (6) $\Rightarrow \lambda_i = L/N \qquad i = 1, \ldots, N$

Hence these are the values of $\lambda_i$ where $f(\underline{\lambda})$ get's minimum under the constraints.

$$\therefore \min\left(f(\underline{\lambda})\right) = \min\left(\sum_{i=1}^{N} \lambda_i^2\right) = \sum_{i=1}^{N} \left(l/N\right)^2 = l^2/N$$

$$\Rightarrow \sum_{i=1}^{N} \lambda_i^2 \geq l^2/N$$

■

(c) From the given identity

$$\Rightarrow \sum_{i=1}^{l} \sum_{j=1}^{l} |\underline{x}_i^T \underline{x}_j|^2 = \sum_{i=1}^{N} \lambda_i^2$$

$$\geq l^2/N \quad \cdots\cdots \text{ from part (b)}$$

$$\Rightarrow \sum_{i=1}^{l} |\underline{x}_i^T \underline{x}_1|^2 + \sum_{\substack{i \neq j \\ i,j}} |\underline{x}_i^T \underline{x}_j|^2 \geq l^2/N \quad \cdots\cdots \text{ Seperating diagonal and off diagonal terms}$$

$$\Rightarrow \sum_{i=1}^{l} |\|\underline{x}_i\|_2^2|^2 + \sum_{\substack{i \neq j \\ i,j}} |\underline{x}_i^T \underline{x}_j|^2 \geq l^2/N \quad \cdots\cdots \text{ Same columns are multiplied in } x_3\, x_3^T$$

$$\Rightarrow \sum_{i=1}^{l} 1^2 + \sum_{\substack{i \neq j \\ i,j}} |\underline{x}_i^T \underline{x}_j|^2 \geq l^2/N \quad \cdots\cdots \text{ from (1)}$$

$$\Rightarrow \sum_{\substack{i \neq j \\ i,j}} |\underline{x}_i^T \underline{x}_j|^2 \geq l^2/N - l$$

From the definition coherance $= \mu(x) = \max_{i \neq j} |\underline{x}_i^T \underline{x}_j|$ hence $\mu(x) \geq |\underline{x}_i^T \underline{x}_j| \quad \forall\, i \neq j$

$$\Rightarrow \sum_{\substack{i \neq j \\ i,j}} \mu(x)^2 \geq \sum_{\substack{i \neq j \\ i,j}} |\underline{x}_i^T \underline{x}_j|^2 \geq l^2/N - l$$

$$\Rightarrow l(l-1) \cdot \mu(x)^2 \geq l^2/N - l \quad \cdots\cdots \text{ } l(l-1) \text{ combinations for } i \neq j$$

$$\Rightarrow \mu(x)^2 \geq \frac{l^2 - Nl}{N} \frac{1}{l(l-1)} = \sqrt{\frac{l-N}{N(l-1)}}$$

■

Logic

When coherance $\mu(x)$ get lower, columns of $X$ become more and more independant. Hence spark $(x)$ should increase.

**Lemma:** $\text{Spark}(x) \geq 1 + \frac{1}{\mu(x)}$

Proof at class is not solid. Check the discussion 14.

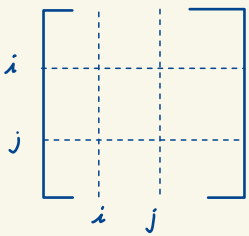
**Proof:** Let's consider the matrix $X$ with normalized columns $\|x_i\|_2^2 = 1$

$$G = X^T X$$

Consider a $p \times p$ submatrix of $G$ such that

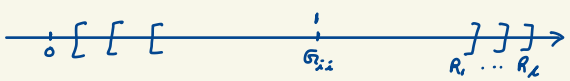
$$p < 1 + 1/\mu(x)$$

$$\Rightarrow (p-1)\mu(x) < 1 \quad \cdots\cdots (1)$$




Then from Gershgorin's Theorem,

$$G_{ii} = 1 > (p-1)\mu(x) \geq \sum_{i \neq j} |G_{ij}| = R_i$$




radius < centroid

$\Rightarrow$ Any set of $P$ columns of $X$ are linear independant.

$\Rightarrow$ $\text{Spark}(x) \geq 1 + \frac{1}{\mu(x)}$

■

- Using this theorem we can loosen the constraints on uniquenes. But this is not combinatorially complex anymore.

**Corollary:** If a solution $\underline{\alpha}_0$ exists such that,

$$\|\underline{\alpha}_0\|_0 < \frac{1}{2}\left(1 + \frac{1}{\mu(x)}\right)$$

Then it is the unique sparsest solution. to $L_1$ norm ie $\min \|\underline{\alpha}\|_1$ st $y = X\underline{\alpha}$

This bound based on $\mu(x)$ restricts the sparcity than that based on spark $(x)$.

2. Let $X$ be a $N \times l$ matrix. Then prove that its spark is bounded as

$$\text{spark}(X) \geq 1 + \frac{1}{\mu(X)},$$

where $\mu(X)$ is the mutual coherence of the matrix.
Hint: Consider the Gram matrix $X^T X$ and the following theorem, concerning positive definite matrices: An $m \times m$ matrix $A$ is positive definite if

$$|A(i,i)| > \sum_{j=1, j\neq i}^{m} |A(i,j)|, \ \forall i = 1, 2, \ldots, m,$$

see, e.g., [Horn 85].

*Solution*: Let the matrix $X$ be

$$X = [\boldsymbol{x}_1^c, \boldsymbol{x}_2^c, \ldots, \boldsymbol{x}_l^c].$$

Assume that the columns are normalized to unity. Otherwise, we have to normalize them, which does not affect the spark and the mutual coherence. Then the corresponding Gram matrix is

$$G = X^T X = \begin{bmatrix} \boldsymbol{x}_1^{cT}\boldsymbol{x}_1^c & \ldots & \boldsymbol{x}_1^{cT}\boldsymbol{x}_l^c \\ \boldsymbol{x}_2^{cT}\boldsymbol{x}_1^c & \ldots & \boldsymbol{x}_2^{cT}\boldsymbol{x}_l^c \\ \vdots & \ddots & \vdots \\ \boldsymbol{x}_l^{cT}\boldsymbol{x}_1 & \ldots & \boldsymbol{x}_l^{cT}\boldsymbol{x}_l^c \end{bmatrix}. \tag{16}$$

Observe that all diagonal elements are equal to one. Moreover, all the off-diagonal elements have absolute value less than or equal to the mutual coherence, by its definition. Let $1 \leq p \leq N$ be the maximum number for which the following holds:

$$(p-1)\mu(X) < 1. \tag{17}$$

This means that any submatrix $G$ of $X^T X$, that results by combining any $p$ columns of $X$, will be positive definite, according to the matrix theorem, stated in the hint, since the diagonal elements are 1 and the $p-1$ off-diagonals are bounded by $\mu(X)$ in absolute value. Moreover, (17) implies that this will also be true for any $m \leq p$. Thus, if this is valid, then any $m \leq p$ columns are linearly independent, hence

$$\text{spark}(X) \geq p + 1, \tag{18}$$

However, since $p$ is the maximum value for which (17) is valid, then

$$1 \leq p\mu(X). \tag{19}$$

Combining (19) with (17) results in

$$\text{spark}(X) \geq 1 + \frac{1}{\mu(X)}.$$

Note that if for $p = N$ (17) is valid, then spark attains its maximum value of $N + 1$. If $p = 1$, this necessarily means that the mutual coherence is 1, hence the spark is bounded by two (only one column can be guaranteed to be linearly independent)

Proof from Discussion 14

# Equivalence of $l_0$ and $l_1$ Minimizers

Donoho and Elad (2003)

i.e $l_1$ Solution $\equiv$ $l_0$ Solution

**Theorem:-** If a solution $\underline{\alpha}_0$ exists such that,

$$\|\underline{\alpha}_0\|_0 < \frac{1}{2}\left(1 + \frac{1}{\mu(x)}\right)$$

Then it is the unique sparsest to

$$\min_{\underline{\alpha}} \|\underline{\alpha}\|_1 \quad \text{such that} \quad \underline{y} = \underline{X}\,\underline{\alpha}$$

**Proof:** Suppose that there is another solution to

$$\left\{ \min_{\underline{\alpha}} \|\underline{\alpha}\|_1 \quad s.t \quad \underline{X}\,\underline{\alpha} = \underline{y} \right\}$$

given by $\underline{\alpha}'$ such that $\|\underline{\alpha}'\|_1 \leq \|\underline{\alpha}_0\|_1$ ..... ①

- Assume that $\underline{e} = \underline{\alpha}' - \underline{\alpha}_0$ then assumption ① can be written as, $\exists\ \underline{e} \neq 0$ such that

$$\|\underline{\alpha}'\|_1 = \|\underline{\alpha}_0 + \underline{e}\|_1 \leq \|\underline{\alpha}_0\|_1$$

Why can we do this assumption? Let $\|\underline{\alpha}_0\|_0 = S$ and suppose that nonzero elements are in the first $S$ entries,

$$\Rightarrow \|\underline{\alpha}_0 + \underline{e}\|_1 = \sum_{i=1}^{\lambda} |\alpha_{0,i} + e_i| \leq \sum_{i=1}^{S} |\alpha_{0,i}|$$

$$\Rightarrow \sum_{i=1}^{S} |\alpha_{0,i} + e_i| + \sum_{i=S+1}^{\lambda} |e_i| \leq \sum_{i=1}^{S} |\alpha_{0,i}|$$

triangle inequality $\Rightarrow \sum_{i=1}^{S} \left(|\alpha_{0,i}| - |e_i|\right) + \sum_{i=S+1}^{\lambda} |e_i| \leq \sum_{i=1}^{S} |\alpha_{0,i}|$

$$\Rightarrow \sum_{i=S+1}^{\lambda} |e_i| \leq \sum_{i=1}^{S} |e_i|$$

$$\Rightarrow \sum_{i=1}^{\lambda} |e_i| \leq 2\sum_{i=1}^{S} |e_i|$$

$$\Rightarrow \|\underline{e}\|_1 \leq 2\sum_{i=1}^{S} |e_i| \quad \text{..... } \circledast$$

- Consider the following algebraic result,

$$
\begin{aligned}
-\underline{e} &= -\underline{e} + \underline{0} \\
\Rightarrow \quad &= -\underline{e} + x^T \underline{0} \\
\Rightarrow \quad &= -\underline{e} + x^T (y - y) \\
\Rightarrow \quad &= -\underline{e} + x^T (x\underline{\theta}_0 - x\underline{\theta}') \quad \cdots \quad (\text{Since } x\underline{\theta}_0 \text{ and } x\underline{\theta}' \text{ boath assumed to be solutions}) \\
\Rightarrow \quad &= -\underline{e} + x^T x \underline{e} \\
\Rightarrow \quad -\underline{e} &= (x^T x - I)\,\underline{e}
\end{aligned}
$$

Hence $\Rightarrow \quad |e_i| = \left| \sum_{i \neq j} G_{ij}\, e_j + \underbrace{(G_{ii} - 1)}_{0} e_i \right|$ ... why?

$$
\Rightarrow |e_i| \leqslant \sum_{i \neq j} |G_{ij}|\, |e_j|
$$

$$
\Rightarrow \leqslant \mu(x) \sum_{i \neq j} |e_j| \quad \cdots \quad G_{ij} \leqslant \mu(x)
$$

$$
\Rightarrow \leqslant \mu(x) \left( \|\underline{e}\|_1 - |e_i| \right)
$$

$$
\Rightarrow |e_i| \leqslant \frac{\mu(x)}{1 + \mu(x)} \|\underline{e}\|_1 \quad \cdots \quad (*)(*)
$$

From (*) and (*)(*)

$$
\Rightarrow \|\underline{e}\|_1 \leqslant 2 \sum_{i=1}^{s} |e_i| \leqslant \frac{2\mu(x)}{1 + \mu(x)} \sum_{i=1}^{s} \|\underline{e}\|_1
$$

$$
\Rightarrow \quad 1 \quad \leqslant \quad \frac{2\mu(x) \cdot s}{1 + \mu(x)}
$$

$$
\Rightarrow \quad s \quad \geqslant \quad \frac{1}{2} \left( 1 + \frac{1}{\mu(x)} \right)
$$

Contradiction since we assumed that $\|\underline{\theta}_0\|_0 = s < \frac{1}{2}\left(1 + \frac{1}{\mu(x)}\right)$

hence $\underline{e} = \underline{0} \Rightarrow \underline{\theta}' = \underline{\theta}_0$ ■

Logic - If we meet these criteria's for our input sensing values $X$, we can have best of boath worlds,

1) Nice convex optimizable properties of $L_1$
2) Sparcity preserving nature of $L_0$

# Robust Sparse Signal Recovery from Noisy X

Homework 8 - Problem 3

**Problem 3.** In this problem, we want to show that $\ell_1$-minimization in the presence of observation noise does not suffer from the noise amplification issue of LS estimator, when the true parameter vector $\boldsymbol{\theta}$ is sufficiently sparse.

Consider an $N \times \ell$ matrix $\mathbf{X}$ with normalized columns and a regression model $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \mathbf{v}$, such that $N < \ell$ and $\|\mathbf{v}\|_2^2 \leq \epsilon^2$, for some constant $\epsilon > 0$. Let $\|\boldsymbol{\theta}\|_0 = s$ and

$$s < \frac{1}{4}\left(1 + \frac{1}{\mu(\mathbf{X})}\right).$$

We want to estimate $\boldsymbol{\theta}$ via the following optimization problem:

$$\min_{\widehat{\boldsymbol{\theta}}} \quad \|\widehat{\boldsymbol{\theta}}\|_1 \qquad \text{s.t.} \qquad \|\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\theta}}\|_2 \leq \delta,$$

for some constant $\delta \geq \epsilon$. Let $\widehat{\boldsymbol{\theta}}$ be the solution to this problem.

(a) Let $\mathbf{e} := \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}$ and suppose that the $s$ non-zero elements of $\boldsymbol{\theta}$ are in its first $s$ entries. Show that:

$$\|\mathbf{e}\|_1 \leq 2\sum_{i=1}^{s} |e_i|$$

(b) Show that the constraint $\|\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\theta}}\|_2 \leq \delta$ implies $\|\mathbf{X}\mathbf{e} - \mathbf{v}\|_2 \leq \delta$ and conclude that:

$$\|\mathbf{X}\mathbf{e}\|_2^2 \leq (\epsilon + \delta)^2.$$

(c) Using the equality $\|\mathbf{X}\mathbf{e}\|_2^2 = \|\mathbf{e}\|_2^2 + \mathbf{e}^\top(\mathbf{X}^\top\mathbf{X} - \mathbf{I})\mathbf{e}$, show that:

$$(1 + \mu(\mathbf{X}))\|\mathbf{e}\|_2^2 \leq (\epsilon + \delta)^2 + \mu(\mathbf{X})\|\mathbf{e}\|_1^2.$$

(d) Show that

$$\sum_{i=1}^{s} |e_i| \leq \sqrt{s}\|\mathbf{e}\|_2$$

*Hint: you can use the Cauchy-Schawrz inequality.*

(e) By combining the results of parts (a), (c), and (d), show that:

$$\|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|_2^2 \leq \frac{(\epsilon + \delta)^2}{1 - \mu(\mathbf{X})(4s - 1)}$$

@ We know the solution for the $L_1$ norm is unique. Hence if $\hat{\underline{\theta}}$ is the solution $\|\hat{\underline{\theta}}\|_1$ is the smallest possible $L_1$ norm

$$\Rightarrow \|\hat{\underline{\theta}} + \underline{e}\|_1 \leq \|\hat{\underline{\theta}}\|_1 \qquad \forall \underline{e}$$

$$\Rightarrow \sum_{i=1}^{L} |\hat{\theta}_i + e_i| \leq \sum_{i=1}^{L} |\hat{\theta}_i| \quad \cdots \hat{\underline{\theta}} \in \mathbb{R}^L \text{ since } X \text{ is } N \times L \text{ dimension}$$

$$\Rightarrow \sum_{i=1}^{S} |\hat{\theta}_i + e_i| + \sum_{i=S+1}^{L} |e_i| \leq \sum_{i=1}^{S} |\hat{\theta}_i| \quad \cdots \text{ assuming first } S \text{ elements are the non-zero values of } \underline{\theta}$$

$$\Rightarrow \sum_{i=1}^{S} |\hat{\theta}_i| - \sum_{i=1}^{S} |e_i| + \sum_{i=S+1}^{L} |e_i| \leq \sum_{i=1}^{S} |\hat{\theta}_i| \quad \cdots \text{ triangle inequality}$$

$$\Rightarrow \quad \sum_{i=s+1}^{l} |e_i| \leq \sum_{i=1}^{s} |e_i|$$

$$\Rightarrow \sum_{i=1}^{s} |e_i| + \sum_{i=s+1}^{l} |e_i| \leq 2 \sum_{i=1}^{s} |e_i|$$

$$\Rightarrow \quad \sum_{i=1}^{l} |e_i| \leq 2 \sum_{i=1}^{s} |e_i|$$

$$\Rightarrow \quad \|\underline{e}\|_1 \leq 2 \sum_{i=1}^{s} |e_i|$$

■

(b) Assume that $\|\underline{y} - X\hat{\underline{a}}\|_2 \leq \delta$

$\Rightarrow \|(X\underline{a} + \underline{v}) - X\hat{\underline{a}}\|_2 \leq \delta$ ..... replace $\underline{y} = X\underline{a} + \underline{v}$

$\Rightarrow \|X(\underline{a} - \hat{\underline{a}}) + \underline{v}\|_2 \leq \delta$

$\Rightarrow \|-X\underline{e} + \underline{v}\|_2 \leq \delta$ ..... $\because \underline{e} = \hat{\underline{a}} - \underline{a}$

$\Rightarrow \|X\underline{e} - \underline{v}\|_2 \leq \delta$

Hence this conclude that $\|\underline{y} - X\hat{\underline{a}}\|_2 \leq \delta \Rightarrow \|X\underline{e} - \underline{v}\|_2 \leq \delta$

$\Rightarrow \|X\underline{e}\|_2 - \|\underline{v}\|_2 \leq \delta$ ..... triangle inequality

$\Rightarrow \|X\underline{e}\|_2 - \delta \leq \|\underline{v}\|_2 \leq \varepsilon$ ..... since $\|\underline{v}\|_2^2 \leq \varepsilon^2$

$\Rightarrow \|X\underline{e}\|_2 \leq \delta + \varepsilon$

$\Rightarrow \|X\underline{e}\|_2^2 \leq (\delta+\varepsilon)^2$

■

(c) Let $G$ be the Gram matrix $X^TX$ i.e. $G = X^TX$

Claim: $|\underline{e}^T(X^TX - I)\cdot\underline{e}| = \mu(X)\|\underline{e}\|_1^2 - \mu(X)\|\underline{e}\|_2^2$

Proof: Assume $G_{i,j}$ is the $i,j^{th}$ entry. It is calculated using $i^{th}$ row of $X^T$ and $j^{th}$ column of $X$

$$G_{i,j} = \underline{x}_i^T \underline{x}_j \quad \cdots (1)$$

$$G_{i,i} = \underline{x}_i^T \underline{x}_i = \|\underline{x}_i\|^2 = 1 \quad \cdots (2) \quad \text{due to normalized columns}$$

Since from definition, coherance $\mu(x) = \max_{i \neq j} |\underline{x}_i^T \underline{x}_j|$ for a matrix with normalized columns

from (1) $\Rightarrow \quad \mu(x) \geq |\underline{x}_i^T \underline{x}_j| = |G_{i,j}| \quad \cdots (3)$

Hence,

$$|\underline{e}^T (x^T x - I) \underline{e}| = \left| \sum_{i=1}^{\lambda} \sum_{j=1}^{\lambda} e_i (G_{i,j} - I_{i,j}) e_j \right|$$

$$= \left| \sum_{\substack{1 \leq i,j \leq \lambda \\ i \neq j}} e_i G_{i,j} e_j + \sum_{i=1}^{\lambda} e_i^2 \underbrace{(G_{i,i} - 1)}_{1 \text{ from } (2)} \right| \quad \text{since } I_{i,j} = 0 \text{ when } i \neq j$$

$$= \left| \sum_{\substack{1 \leq i,j \leq \lambda \\ i \neq j}} e_i G_{i,j} e_j + \sum_{i=1}^{\lambda} e_i^2 \cdot 0 \right| \quad \cdots \text{from (2)}$$

$$= \left| \sum_{\substack{1 \leq i,j \leq \lambda \\ i \neq j}} e_i G_{i,j} e_j \right|$$

$$\leq \sum_{\substack{1 \leq i,j \leq \lambda \\ i \neq j}} |e_i| |G_{i,j}| |e_j| \quad \cdots \text{triangle inequality}$$

$$\leq \sum_{\substack{1 \leq i,j \leq \lambda \\ i \neq j}} |e_i| \cdot \mu(x) \cdot |e_j| \quad \cdots \text{from (3)}$$

$$= \mu(x) \sum_{i=1}^{\lambda} \sum_{j=1}^{\lambda} |e_i| |e_j| - \mu(x) \sum_{i=1}^{\lambda} |e_i|^2$$

$$= \mu(x) \cdot \|\underline{e}\|_1^2 - \mu(x) \|\underline{e}\|_2^2$$

Claim proved.

using the given condition,

$$\Rightarrow \|X\underline{e}\|_2^2 = \|\underline{e}\|_2^2 + \underline{e}^T (x^T x - I) \underline{e}$$

$$\Rightarrow \|\underline{e}\|_2^2 = \|X\underline{e}\|_2^2 - \underline{e}^T (x^T x - I) \underline{e}$$

take absolute value $\Rightarrow \left| \|\underline{e}\|_2^2 \right| = \left| \|X\underline{e}\|_2^2 - \underline{e}^T (x^T x - I) \underline{e} \right|$

triangle inequality $\Rightarrow \left| \|\underline{e}\|_2^2 \right| \leq \left| \|X\underline{e}\|_2^2 \right| + \left| \underline{e}^T (X^T X - I)\underline{e} \right|$

$\Rightarrow \|\underline{e}\|_2^2 \leq \|X\underline{e}\|_2^2 + \left| \underline{e}^T (X^T X - I)\underline{e} \right|$

$\Rightarrow \leq (\delta+\varepsilon)^2 + \left| \underline{e}^T (X^T X - I)\underline{e} \right|$ ... from part (b)

$\leq (\delta+\varepsilon)^2 + \mu(x)\cdot \|\underline{e}\|_1^2 - \mu(x)\|\underline{e}\|_2^2$ .. from claim

$\Rightarrow [1+\mu(x)]\cdot\|\underline{e}\|_2^2 \leq (\delta+\varepsilon)^2 + \mu(x)\cdot\|\underline{e}\|_1^2$ ■

(d) From the Cauchy - Schwartz inequality

$$|\underline{a}^T \underline{b}| \leq \|\underline{a}\|_2 \|\underline{b}\|_2$$

Substituting $\underline{a} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \Rightarrow \|\underline{a}\|_2 = \sqrt{1^2 + \ldots + 1^2} = \sqrt{S}$

$$\underline{b} = \begin{bmatrix} |e_1| \\ \vdots \\ |e_S| \end{bmatrix} \Rightarrow \|\underline{b}\|_2 = \|\underline{e}\|_2$$

we get,

$$\left| \begin{bmatrix} 1 & 1 & \ldots & 1 \end{bmatrix} \begin{bmatrix} |e_1| \\ \vdots \\ |e_S| \end{bmatrix} \right| \leq \sqrt{S}\cdot\|\underline{e}\|_2$$

$$\Rightarrow \sum_{i=1}^{S} |e_i| \leq \sqrt{S}\|\underline{e}\|_2$$ ■

(e) From @ and (d)

$$\|\underline{e}\|_1 \leq 2\sum_{i=1}^{S} |e_i| \leq 2\sqrt{S}\|\underline{e}\|_2$$

$$\Rightarrow \|\underline{e}\|_1^2 \leq 4S\|\underline{e}\|_2^2$$

Substituting this to @

$$\Rightarrow [1+\mu(x)]\cdot\|\underline{e}\|_2^2 \leq (\delta+\varepsilon)^2 + \mu(x)\cdot\|\underline{e}\|_1^2 \leq (\delta+\varepsilon)^2 + \mu(x)4S\|\underline{e}\|_2^2$$

$$\Rightarrow \quad \|\underline{e}\|_2^2 \left[ 1 + \mu(x) - 4s\mu(x) \right] \leq (\delta + \varepsilon)^2$$

$$\Rightarrow \quad \|\underline{e}\|_2^2 \leq \frac{(\delta+\varepsilon)^2}{1 - \mu(x)(4s-1)}$$

$$\Rightarrow \quad \|\hat{\underline{a}} - \underline{a}\|_2^2 \leq \frac{(\delta+\varepsilon)^2}{1 - \mu(x)(4s-1)}$$

■

# THEO - 10

# Sparsity Aware Learning

## Algorithms and Applications

### Recap

- $\underline{y} = X\,\underline{\theta}_o \qquad \underline{y} \in \mathbb{R}^N \quad \underline{\theta}_o \in \mathbb{R}^l \quad N < l$

  Assume that $\underline{\theta}_o$ is sparse meaning that the number of nonzero elements $\ll l$

- Theorem:- If a solution $\underline{\theta}_o$ exists such that,

$$\|\underline{\theta}_o\|_0 < \frac{1}{2}\left(1 + \frac{1}{\mu(X)}\right)$$

  Then it is the unique sparsest to

$$\min_{\underline{\theta}} \|\underline{\theta}\|_1 \quad \text{such that} \quad \underline{y} = X\,\underline{\theta}$$

- $$\mu(X) = \max_{\substack{i,j \\ i \neq j}} \frac{|\underline{x}_i^T \underline{x}_j|}{\|\underline{x}_i\|^2 \|\underline{x}_j\|^2}$$

### Gaussian Random Design

$$X_{i,j} \overset{iid}{\sim} N\left(0, 1/N\right)$$

then $\mu(X) \leq \sqrt{\frac{15 \cdot \log l}{N}}$ with probability $\geq 1 - 2/l$

and $N \geqslant 60 \log l$.

### Partial DFT matrices

$$W^H = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & e^{j\frac{2\pi}{l}} & e^{j\frac{4\pi}{l}} & \cdots & e^{j\frac{2\pi(l-1)}{l}} \\ 1 & e^{j\frac{4\pi}{l}} & e^{j\frac{8\pi}{l}} & \cdots & e^{j\frac{2\pi 2(l-1)}{l}} \\ \vdots & & & & \vdots \\ \cdots & & (k,m) & & \cdots \\ \vdots & & & & \vdots \\ 1 & e^{j\frac{2\pi(l-1)}{l}} & \cdots & & e^{j\frac{2\pi(l-1)l}{l}} \end{bmatrix}$$

$(k,m)^{th}$ element $= e^{j\frac{2\pi}{l}km}$

$k, m \quad 1, 2, \ldots, l-1$

- Assume that $\underline{\theta}$ is sparse, we want to measure $\underline{\theta}$ from random sample indices in the time domain.

$$X[k] = \sum_{n=0}^{L-1} x[n] \cdot e^{-j\,2\pi/L\, nk}$$

$$x[n] = \frac{1}{L} \sum_{k=0}^{L-1} X[k] \cdot e^{j\,2\pi/L\, nk}$$

$$\underline{\theta} = \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[L-1] \end{bmatrix} \quad \cdots \text{DFT} \qquad \underline{x} = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[L-1] \end{bmatrix}$$

$$\underline{\theta} = W \underline{x}$$

$$\underline{x} = \frac{1}{L} W^{H} \underline{\theta}$$

- Consider a sequence of $L$ independant coin flips with probability $N/L$ each for $n = 0, 1, \ldots, L-1$ at time $n$. If the answer is head, measure $x[n]$. Otherwise skip. This provides $M$ measurement such that

$$E[M] = N$$

corresponding to time $n_1, n_2, \ldots, n_M$

$w = e^{j\,2\pi/L}$

$$\begin{bmatrix} x[n_1] \\ x[n_2] \\ \vdots \\ x[n_M] \end{bmatrix} = \frac{1}{L} \underbrace{\begin{bmatrix} 1 & w^{n_1} & w^{2n_1} & \cdots & w^{(L-1)n_1} \\ 1 & w^{n_2} & w^{2n_2} & \cdots & w^{(L-1)n_2} \\ \vdots & & & & \vdots \\ 1 & w^{n_M} & w^{2n_M} & \cdots & w^{(L-1)n_M} \end{bmatrix}}_{X} \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[L-1] \end{bmatrix}$$

$X$: Submatrix of $W$ corresponding to rows $n_1, n_2 \ldots n_M$

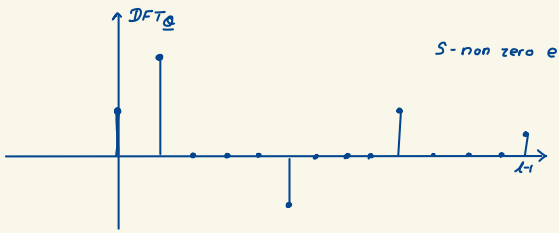
$$\Rightarrow \quad \underline{x}_M = X \underline{\theta}$$

Question - Can we recover $\underline{\alpha} \in \mathbb{R}^{\ell}$ from $\underline{x}_M \in \mathbb{R}^M$

- Randomly sample IDFT matrix $\mu(X) \leq \sqrt{\frac{118(\ell - N)\log \ell}{\ell N}}$ with probability $\geq 1 - 4/\ell - 1/\ell^2$ if $N \geq 16 \log \ell$

- If $\|\underline{\alpha}_o\| < \frac{1}{2}\left(1 + \frac{1}{\mu(X)}\right)$ then $\underline{\alpha}_o$ can be recovered from $\min_{\underline{\alpha}} \|\underline{\alpha}\|_1$ s.t $\underline{x}_M = X\underline{\alpha}$

- Matrices $X$ with random i.i.d. zero mean elements are shown to have small $\mu(X)$




If we consider a random selection of the raws of the DFT matrix of size $N \sim S \cdot \log^4$, then sub-DFT matrix of dimention $N \times \ell$ has small $\mu(X)$

$$W = \begin{matrix} k_1 \\ \vdots \\ k_j \\ \vdots \\ k_N \end{matrix} \begin{bmatrix} & & j & & \\ & & \vdots & & \\ \cdots & \cdots & w^{k_j} & \cdots & \cdots \end{bmatrix} \qquad w = e^{-j2\pi/\ell}$$

$$X = \begin{matrix} N \end{matrix} \begin{bmatrix} & & \ell & & \\ & & & & \\ & & & & \end{bmatrix}$$

Only knowing $x[k_1], \ldots, x[k_N]$ suffices to recover the DFT

$$y = \begin{bmatrix} x[k_1] \\ \vdots \\ x[k_N] \end{bmatrix} = X^H \underline{\alpha}$$ where $X$ is submatrices of $W$

# 1) Greedy Algorithms

Homework 8 - Problem 4

**Problem 4.** Suppose we know that the sparsest solution to $\mathbf{X}\boldsymbol{\theta} = \mathbf{y}$ has a sparsity $s$ and $s < \frac{1}{2}\left(1 + \frac{1}{\mu(\mathbf{X})}\right)$. Assume that the $N \times \ell$ matrix $\mathbf{X}$ has normalized columns, i.e., $\|\mathbf{X}_i\|_2^2 = 1,\ \forall i = 1, 2, \cdots, \ell$.

Furthermore, assume (without loss of generality) that the $s$ non-zero elements of the sparsest solution are in the first $s$ elements of $\boldsymbol{\theta}$, i.e., $\theta_k = 0, \forall k > s$. We thus have:

$$\mathbf{y} = \sum_{i=1}^{s} \mathbf{X}_i \theta_i. \tag{1}$$

Also, suppose (without loss of generality) that the non-zero elements of $\boldsymbol{\theta}$ are sorted in decreasing absolute value: $|\theta_1| \geq |\theta_2| \geq \cdots \geq |\theta_s|$.

The Orthogonal Matching Pursuit algorithm for greedily finding the solution with sparsity $s$ is given as follows:

> Initialization: $\boldsymbol{\theta}^{(0)} = \mathbf{0}$, $\mathbf{r}^{(0)} = \mathbf{y}$, $S^{(0)} = \emptyset$.
> For $m = 1, 2, \cdots$ do:
> 1) Compute correlation values $c_j := \left|\mathbf{X}_j^\top \mathbf{r}^{(m-1)}\right|$, for $j = 1, 2, \cdots, \ell$.
> 2) Find $j_m = \underset{j}{\operatorname{argmax}}\, c_j$, and let $S^{(m)} = S^{(m-1)} \cup \{j_m\}$.
> 3) Find $\boldsymbol{\theta}^{(m)}$ supported on the set $S^{(m)}$ as $\boldsymbol{\theta}^{(m)} := \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \left\|\mathbf{y} - \sum_{i \in S^{(m)}} \mathbf{X}_i \theta_i\right\|_2^2$
> 4) Update the residual $\mathbf{r}^{(m)} \leftarrow \mathbf{y} - \mathbf{X}\boldsymbol{\theta}^{(m)}$.
> 5) Go to step 1, unless $\mathbf{r}^{(m)} = \mathbf{0}$.
> Output: $\boldsymbol{\theta}^{(m)}$.

(a) Show that at the first iteration ($m = 1$), for the algorithm to pick one of the indices of the true support $1 \leq j_1 \leq s$, it is sufficient to have:

$$\left|\mathbf{X}_1^\top \mathbf{y}\right| > \left|\mathbf{X}_k^\top \mathbf{y}\right|, \quad \forall k > s.$$

(b) Using the definition of $\mathbf{y}$ from Eq. (1), show that

$$\left|\mathbf{X}_1^\top \mathbf{y}\right| \geq |\theta_1|\Big(1 - \mu(\mathbf{X})(s-1)\Big) \quad \text{and} \quad \left|\mathbf{X}_k^\top \mathbf{y}\right| \leq |\theta_1|\mu(\mathbf{X})s, \ \forall k > s.$$

(c) Argue that the assumption of $s < \frac{1}{2}\left(1 + \frac{1}{\mu(\mathbf{X})}\right)$ and the result of part (b) imply that the inequality of part (a) holds indeed, and as a result the first step of OMP picks one of the correct indices $1 \leq j_1 \leq s$.

(d) Argue that at the second iteration, the residual $\mathbf{r}^{(1)} = \mathbf{y} - \mathbf{X}_{j_1}\theta_{j_1}^{(1)}$ can be expressed as $\sum_{i=1}^{s} \mathbf{X}_i \tilde{\theta}_i$, for some $\tilde{\theta}_i$'s. By rearranging the columns of $\mathbf{X}$ and elements of $\mathbf{y}$, the elements $\tilde{\theta}_i$ can be sorted in decreasing absolute value. By repeating the arguments of parts (a)-(c), show that the second step chooses another $j_2$ such that $1 \leq j_2 \leq s$, and $j_2 \neq j_1$ (*Hint: show that* $\theta_{j_1}^{(1)} = \mathbf{X}_{j_1}^\top \mathbf{y}$, *and therefore at the second iteration,* $c_{j_1} = 0$).

(e) By repeating the procedure of part (d), argue that at iteration $m$, an index $j_m$ in $\{1, 2, \cdots, s\}$ is selected, such that $j_m \neq j_n, \forall n < m$ (*Hint: use the orthogonal projection property of least squares!).*

(f) Argue that at iteration $s$, the support $S^{(s)}$ is equal to $\{1, 2, \cdots, s\}$, i.e., all the correct components of $\boldsymbol{\theta}$ are picked. Then, show that $\mathbf{r}^{(s)} = \mathbf{0}$, and $\boldsymbol{\theta}^{(s)} = \boldsymbol{\theta}$, i.e., the OMP exactly recovers $\boldsymbol{\theta}$ after $s$ iterations!

@ In the start of the first iteration,

$$\theta^{(0)} = 0$$
$$r^{(0)} = y$$
$$S^{(0)} = \phi$$
$$c_j = |x_j^T r^{(m-1)}| \quad j = 1, 2, \ldots, d$$
$$j_m = \underset{j}{\text{argmax}}\ c_j$$

$m = 1$

$$\therefore j_1 = \underset{j}{\text{argmax}}\ c_j = \underset{j}{\text{argmax}}\ |x_j^T r^{(1-1)}|$$
$$= \underset{j}{\text{argmax}}\ |x_j^T r^{(0)}|$$
$$= \underset{j}{\text{argmax}}\ |x_j^T y| \quad \cdots\cdots \text{since } |x_1^T y| > |x_k^T y| \ \forall\ k > s$$

We can't pick a $j$ that is higher than $s$ to get the argmax

$$1 \leq \underset{j}{\text{argmax}}\ |x_j^T y| \leq s$$
$$1 \leq j_1 \leq s$$

(b) ▸

$$|x_1^T y| = \left| x_1^T \left( \sum_{i=1}^{s} x_i \theta_i \right) \right|$$
$$= \left| \sum_{i=1}^{s} x_1^T x_i \theta_i \right|$$
$$= \left| \sum_{i=2}^{s} (x_1^T x_i)\theta_i + \theta_1 \|x_1\|^2 \right|$$
$$= \left| \sum_{i=2}^{s} (x_1^T x_i)\theta_i + \theta_1 \right|$$
$$\geq |\theta_1| - \left| \sum_{i=2}^{s} (x_1^T x_i)\theta_i \right| \quad \cdots \text{triangle inequality}$$
$$\geq |\theta_1| - \sum_{i=2}^{s} |x_1^T x_i \theta_i| \quad \cdots \text{triangle inequality}$$
$$\geq |\theta_1| - \sum_{i=2}^{s} |x_1^T x_i| |\theta_i|$$
$$\geq |\theta_1| - \max_{i \neq 1} |x_1^T x_i| \cdot \sum_{i=2}^{s} |\theta_i|$$

from the definition of the coherance $\mu(x) = \max_{i \neq j} |x_i^T x_j|$

hence $\mu(x) \geq x_i^T x_i$

$$\therefore |x_1^T y| \geq |\alpha_1| - \mu(x) \cdot \sum_{i=2}^{S} |\alpha_i|$$

$$\geq |\alpha_1| - \mu(x) \cdot (S-1) \cdot |\alpha_1| \quad \text{.... Since } |\alpha_1| \text{ is the highest para. and } |\alpha_1| \geq |\alpha_k| \; \forall k$$

$$|x_1^T y| \geq |\alpha_1| \left(1 - \mu(x) \cdot (S-1)\right)$$

▶ $$|x_k^T y| = \left| x_k^T \left( \sum_{i=1}^{S} x_i \alpha_i \right) \right|$$

$$= \left| \sum_{i=1}^{S} x_k^T x_i \alpha_i \right|$$

$$\leq \sum_{i=1}^{S} |x_k^T x_i \alpha_i| \quad \text{.... triangle inequality}$$

$$\leq \sum_{i=1}^{S} |x_k^T x_i| |\alpha_i|$$

$$\leq \max_{k \neq i} |x_k^T x_i| \cdot \sum_{i=1}^{S} |\alpha_i|$$

$$\leq \mu(x) \sum_{i=1}^{S} |\alpha_i| \quad \text{.... from definition of coherance}$$

$$\leq \mu(x) \sum_{i=1}^{S} |\alpha_1| \quad \text{... Since } |\alpha_1| \geq |\alpha_k| \; \forall k$$

$$\leq |\alpha_1| \cdot \mu(x) \cdot S$$

(c) From the assumption

$$\Rightarrow S < \frac{1}{2}\left(1 + \frac{1}{\mu(x)}\right)$$

$$\Rightarrow 2\mu(x) S < \mu(x) + 1$$

$$\Rightarrow \mu(x) \cdot S < 1 - \mu(x) \cdot (S-1)$$

$$\Rightarrow |\alpha_1| \mu(x) S < |\alpha_1| \cdot \left(1 - \mu(x)(S-1)\right)$$

from (b) Second equation

$$\Rightarrow |\alpha_1| \mu(x) S < |\alpha_1| \cdot \left(1 - \mu(x)(S-1)\right) \leq \|x_1^T y\|$$

from ⓑ first equation $\Rightarrow \|x_k^T y\| \leq |a_1| \mu(x) s < |a_1| \cdot (1 - \mu(x)(s-1)) \leq \|x_1^T y\|$

$\Rightarrow \|x_1^T y\| \geq \|x_k^T y\|$

hence the assumption required for part ⓒ is true. Hence the condition proved in ⓒ is true. ∴ first step of OMP picks one correct indices in $1 \leq j_1 \leq s$.

ⓓ For $m = 2$ case

From the previous iteration $\longrightarrow r^{(1)} = y - X\underline{a}^{(1)} \Rightarrow r^{(1)} = y - x_{j_1} a_{j_1}^{(1)}$

hence,

$$r^{(1)} = y - x_{j_1} a_{j_1}^{(1)}$$

$$= \left(\sum_{i=1}^{s} x_i a_i\right) - x_{j_1} a_{j_1}^{(1)}$$

$$= \sum_{\substack{i=1 \\ i \neq j_1}}^{s} x_i a_i + \sum_{\substack{i=1 \\ i=j_1}}^{s} x_i (a_i - a_{j_1}^{(1)})$$

$$c_j = |x_j^T r^{(2-1)}| = |x_j^T r^{(1)}| = |x_j^T (y - X\underline{a}^{(1)})|$$

Hence,

$$\left\| y - \sum_{i \in S^{(1)}} x_i a_i \right\|_2^2 = \|y - x_{j_1} a_{j_1}^{(1)}\|_2^2$$

$$= y^T y - 2 y^T x_{j_1} a_{j_1}^{(1)} + x_{j_1}^T x_{j_1} (a_{j_1}^{(1)})^2$$

$$\Rightarrow \frac{\partial}{\partial a_{j_1}^{(1)}} = -2 y^T x_{j_1} + 2 x_{j_1}^T x_{j_1} (a_{j_1}^{(1)})$$

At the optimal point partial differentiation should be zero

$$\Rightarrow -2 y^T x_{j_1} + 2 x_{j_1}^T x_{j_1} (a_{j_1}^{(1)}) = 0$$

$$\Rightarrow a_{j_1}^{(1)} = x_{j_1}^T y$$

The residual at the next iteration is,

$$\Rightarrow r^{(1)} = y - X_{j_1} \theta_{j_1}^{(1)}$$

$$= y - X_{j_1} X_{j_1}^T y$$

$$\Rightarrow c_{j_1} = |X_{j_1}^T r^{(1)}| = |X_{j_1}^T r^{(1)}|$$

$$= |X_{j_1}^T (y - X_{j_1} X_{j_1}^T y)|$$

$$= |X_{j_1}^T y - X_{j_1}^T y|$$

$$= \underline{0}$$

$\therefore$ from the end of the second iteration $c_{j_1} = 0$ $\therefore$ in the second iteration $j_1$ is not $\underset{j}{\text{argmax}}\ c_j$. It picks another $j_2$ that is not $j_1$.

(a) $r^{(m-1)} = y - X \underline{\theta}^{(m-1)}$

$$\underline{\theta}^{m-1} = \underset{\theta}{\text{arg min}} \left\| y - \sum_{i \in S^{(m-1)}} X_i \theta_i \right\|_2^2$$

Taking the derivative with respect to $\theta_j$ and setting it to '0' we get

$$\frac{\partial}{\partial \theta_j} \theta^{(m-1)} = -X_j^T \left( y - \sum_{i \in S^{(m-1)}} X_i \theta_i \right) = 0 \quad \forall j \in S^{(m-1)}$$

$$c_j = |X_j^T r^{(m-1)}| = \left| X_j^T \left( y - \sum_{i \in S^{(m-1)}} X_i \theta_i \right) \right| = 0$$

So all $c_{j_1}, c_{j_2}, \ldots, c_{j_{m-1}}$ won't be picked since all of them are '0' in the next round.

(f) Since distinct supports are selected in first $s$ steps according to part (e), after $s$ iterations, the set of selected indices is $\{1, 2, \ldots, s\}$

$$\alpha^s = \underset{\alpha}{\operatorname{argmin}} \left\| y - \sum_{i \in s^s} x_i \tilde{\alpha}_i \right\|_2^2$$

Since $y = \sum_{i=1}^{s} x_i \alpha_i$

$$\alpha^s = \underset{\alpha}{\operatorname{argmin}} \left\| \sum_{i=1}^{s} x_i \alpha_i - \sum_{i=1}^{s} x_i \tilde{\alpha}_i \right\|_2^2$$

$$= \underset{\alpha}{\operatorname{argmin}} \left\| \sum_{i=1}^{s} x_i (\alpha_i - \tilde{\alpha}_i) \right\|_2^2$$

Since norm is non negative $\Rightarrow \tilde{\alpha}_i = \alpha_i \quad \forall i$

$$\Rightarrow \alpha^{(s)} = \underline{\alpha}$$

$r^{(s)} = y - X\alpha^{(s)} = X\alpha - X\alpha = 0$ //

The OMP recovers $\alpha$ after $s$ iterations

# 2) Iterative Shrinkage Thresholding (IST) Algorithms

$$\underline{y} = X\underline{\theta}_0 + \underline{e} \qquad \text{where} \quad \underline{e} - \text{noise}$$

Now the three optimization problems are

## 1) Basis Pursuit Denoising

$\delta$ - constant

$$\min_{\underline{\theta}} \|\underline{\theta}\|_1 \qquad s.t \qquad \|\underline{y} - X\underline{\theta}\| \leq \delta$$

## 2) LASSO

$\rho$ - constant

$$\min_{\underline{\theta}} \|\underline{y} - X\underline{\theta}\|_2^2 \qquad s.t \qquad \|\underline{\theta}\|_1 \leq \rho$$

## 3) $L_1$ Regularized LS

$\lambda$ - constant

$$\min_{\underline{\theta}} \|\underline{y} - X\underline{\theta}\|^2 + \lambda \|\underline{\theta}\|_1$$

## $N \gg l$

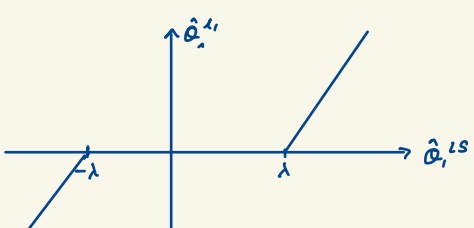
- For $N \geq l$ assume $X^T X = I$

$$\hat{\underline{\theta}}^{LS} = X^T \underline{y}$$

$$\hat{\underline{\theta}}^{L_1} = \arg\min_{\underline{\theta}} \frac{1}{2} \|\underline{y} - X\underline{\theta}\|^2 + \lambda \|\underline{\theta}\|_1$$

$$\hat{\underline{\theta}}^{L_1}_{\lambda} = sgn(\hat{\theta}_i^{LS}) \cdot \max(|\hat{\theta}_i^{LS}| - \lambda, 0) = \mathcal{S}_{\lambda}(\hat{\theta}_i^{LS})$$

Soft thresholding at level $\lambda$



# $N \ll d$

- Question :- Can we do something similar for $N < d$ ?

Motivation :- ▸ How can we solve the following without matrix inversion.

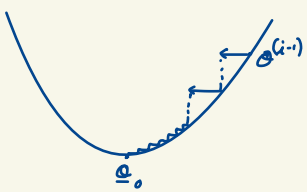
$$\min_{\underline{\theta}} \quad \frac{1}{2} \| \underline{y} - X\underline{\theta} \|^2 = J(\underline{\theta})$$

- Use Gradient descent.

$$\underline{\theta}^{(i)} = \underline{\theta}^{(i-1)} - \mu \frac{\partial}{\partial \underline{\theta}} J \Big|_{\underline{\theta}^{(i-1)}}$$

($\mu$: Step noise)

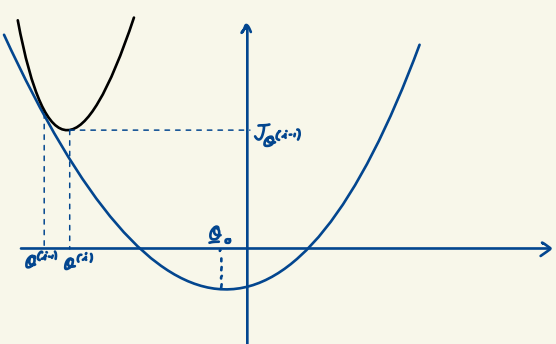
$$= \underline{\theta}^{(i-1)} + \underbrace{\mu X^T}_{\text{Gain}} \underbrace{(\underline{y} - X\underline{\theta}^{(i-1)})}_{\text{Current innovation}} \quad \cdots\cdots \quad \frac{\partial J}{\partial \underline{\theta}} = \nabla_{\underline{\theta}} J = X^T(\underline{y} - X\underline{\theta})$$




$$\underline{\theta}^{(i)} = \underline{\theta}^{(i-1)} + \mu X^T (\underline{y} - X\underline{\theta}^{(i-1)})$$

$$= \arg\min_{\underline{\theta}} \left\{ J(\underline{\theta}^{(i-1)}) + (\underline{\theta} - \underline{\theta}^{(i-1)})^T \nabla_{\underline{\theta}} J \Big|_{\underline{\theta}^{(i-1)}} + \frac{1}{2\mu} \| \underline{\theta} - \underline{\theta}^{(i-1)} \|_2^2 \right\}$$

$$\Rightarrow \quad \nabla_{\underline{\theta}} J \Big|_{\underline{\theta}^{(i-1)}} + \frac{1}{\mu} (\underline{\theta} - \underline{\theta}^{(i-1)}) = \underline{0}$$



**Goal** :- Solve

$$\min_{\underline{\alpha}} \quad \frac{1}{2} \underbrace{\| \underline{y} - X\underline{\alpha} \|^2}_{J(\underline{\alpha})} + \lambda \| \underline{\alpha} \|_1$$

Historically, given $\underline{\alpha}^{(i-1)}$

$$\underline{\alpha}^{(i)} = \arg\min_{\underline{\alpha}} \left\{ J(\underline{\alpha}^{(i-1)}) + (\underline{\alpha} - \underline{\alpha}^{(i-1)}) \nabla_{\underline{\alpha}} J \Big|_{\underline{\alpha}^{(i-1)}} + \frac{1}{2\mu} \| \underline{\alpha} - \underline{\alpha}^{(i-1)} \|^2 + \lambda \| \underline{\alpha} \|_1 \right\}$$

$$= \arg\min_{\underline{\alpha}} \left\{ \frac{1}{2\mu} \| \underline{\alpha} - \underline{\alpha}^{(i-1)} + \mu \nabla_{\underline{\alpha}} J \Big|_{\underline{\alpha}^{(i-1)}} \|_2^2 - \frac{\mu}{2} \| \nabla_{\underline{\alpha}} J \Big|_{\underline{\alpha}^{(i-1)}} \|^2 + \lambda \| \underline{\alpha} \|_1 \right\}$$

$$\underline{\alpha}^{(i)} = \underbrace{\mathcal{S}_{\lambda\mu}}_{\text{Deterministic}} \left( \underbrace{\underline{\alpha}^{(i-1)} + \mu X^T (\underline{y} - X\underline{\alpha}^{(i-1)})}_{\text{LS update}} \right)$$

The error to the true solution $\underline{\alpha}_0$, $\| \underline{\alpha}^{(i)} - \underline{\alpha}_0 \|_2^2 \sim O(1/i)$

If $\mu < \frac{1}{\lambda_{max}(X^T X)}$ the algorithm converges to the true solution.

End - exam breakdow

5 Question ⟶
- 1 - Sparcity
- 1 - Spectral
- 2 - Linear predictions / Kalman / AR(ma) / Yule walker Levinson durbin
- 1 - mix of auto-correlation. / pole zeros / Nothing from filter banks / QMF / PR etc